MINIMAL SYMPLECTIC 4-MANIFOLDS ARE SMOOTHLY IRREDUCIBLE

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Abstract. We show that every smoothly embedded 3-sphere in a symplectic 4-manifold bounds a blowup of the standard 4-ball on one side. To do this, we use a symplectic 1-cocycle on a relative trisection to construct a weak symplectic filling with S^3-boundary. This implies that every homotopy 4-ball that embeds into a symplectic 4-manifold is standard. Consequently, we establish the smooth 4-dimensional Schoenflies conjecture and prove that many potential counterexamples to the smooth 4-dimensional Poincare conjecture, such as Gluck twists and Cappell-Shaneson homotopy spheres, are standard.

1. Introduction

A remarkable feature of closed symplectic 4-manifolds (X, ω) is that their connect summands have very restricted topology. Recall that a symplectic 4-manifold (X, ω) is *minimal* if it has no symplectic (-1) spheres. Taubes proved that symplectic 4-manifolds have nontrivial Seiberg-Witten invariants. An argument of Witten therefore implies that if (X, ω) splits as a smooth connected sum, then one summand must have negative-definite intersection form and Donaldson's Diagonalization theorem implies this summand has the same intersection form as $n\overline{\mathbb{CP}}^2$ for some $n \geq 0$. Work of Kotschick in the case of $b_2^+(X) > 1$ [Kot95] and Hamilton-Kotschick in the case $b_2^+(X) = 1$ [HK06] has established that this summand can be blown-down until it is an integer homology 4-sphere (with additional constraints on the fundamental group).

A *homotopy 4-ball* is a compact, contractible, smooth 4-manifold with boundary homeomorphic to S^3. By Freedman, every homotopy 4-ball B is homeomorphic to the *standard 4-ball* $B^4 \subset \mathbb{R}^4$ and it remains an open question whether there exist *exotic* homotopy 4-balls that are not diffeomorphic to the standard 4-ball. The related smooth, 4-dimensional Schoenflies problem asks whether every homotopy 4-ball that smoothly embeds in S^4 is standard. A smooth 4-manifold X is *topologically irreducible* if every smoothly embedded S^3 bounds a homotopy 4-ball on one side.

Conjecture 1.1 (Gompf [Gom95])**.** *Every minimal, closed symplectic 4-manifold is topologically irreducible.*

Suggestive evidence for a much stronger result is the following fundamental result in symplectic topology. Let $(\widetilde{B}, \omega)$ be a symplectic manifold with boundary Y and let (Y, ξ) be a contact structure on the boundary. Then $(\widetilde{B}, \omega)$ is a *weak filling* of (Y, ξ) if the restriction of ω to the hyperplane field $\xi \subset TY \subset T\widetilde{B}|_Y$ is a symplectic structure; in this case, we also say that ω_S *dominates* ξ.

Theorem 1.2 (Gromov [Gro85], Eliashberg [Eli90])**.** *Let (S^3, ξ) be a contact structure and let $(\widetilde{B}, \omega)$ be a weak symplectic filling of (S^3, ξ). Then there is a diffeomorphism*

$$\widetilde{B} \cong B^4 \# n\overline{\mathbb{CP}}^2$$

for some $n \geq 0$. In particular, if $\widetilde{B}$ is contractible, then it is a standard 4-ball.

In this paper, we prove a stronger version of Gompf's conjecture. A smooth manifold X is *smoothly irreducible* if every smoothly embedded S^3 bounds a standard 4-ball on one side. In other words, if a smooth 4-manifold is smoothly irreducible, then it has no nontrivial smooth connected sum decompositions.

Theorem 1.3. *Every smoothly embedded 3-sphere S in a closed symplectic 4-manifold bounds a blow-up of the standard 4-ball on one side. In particular,*

(1) if (X, ω) is minimal then it is smoothly irreducible.

(2) if a homotopy 4-ball $\widetilde{B}$ embeds into a closed symplectic 4-manifold, then it is diffeomorphic to B^4.

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1.1. **Convexity and the Gompf invariant.** Given some separating $S \subset (X, \omega)$, the conclusion of Theorem 1.3 can clearly only hold for at most one of the two connected components of $X \setminus S$. There must be some geometric asymmetry between the two summands. For example, if S is *contact-type*, meaning there exists a Liouville vector field transverse to S, then one summand is *symplectically convex* and called a *symplectic filling* of S, while the other summand is *symplectically concave* and called a *symplectic cap*. The topology of symplectic fillings are much more heavily constrained than the topology of symplectic caps.

Given a *smooth* decomposition along S, how does one determine which summand should be symplectically convex and which summand should be symplectically concave? The Gompf invariant of an almost-complex structure provides the answer.

Let (X, ω, J) be a closed symplectic 4-manifold with compatible almost-complex structure. Let $S \subset (X, \omega, J)$ be a smoothly embedded, separating, cooriented 3-sphere and let $\widetilde{B}$ be the compact 4-manifold it bounds (with respect to the given coorientation). Gompf [Gom98] defined a $\mathbb{Z}$-valued homotopy invariant of $\xi = TS \cap J(TS)$, the hyperplane field of J-complex lines along S, as

$$\Theta_G(\xi) = c_1(\widetilde{B}, J)^2 - 2\chi(\widetilde{B}) - 3\sigma(\widetilde{B})$$

where $c_1(\widetilde{B}, J)$ denotes the restriction of $c_1(J)$ to $H^2(\widetilde{B}; \mathbb{Z})$. This invariant is a well-defined invariant of the 2-plane field ξ, independent of the bounding almost-complex 4-manifold. Moreover, given any two 2-plane fields ξ, ξ', the relative Gompf invariant is divisible by 4:

$$\Theta_G(\xi', \xi) = \Theta_G(\xi') - \Theta_G(\xi) \in 4\mathbb{Z}$$

Since $c_1(J)$ is determined by the symplectic form, the Gompf invariant is an invariant of the cooriented 3-sphere S up to symplectic deformation of (X, ω). If a fixed ω is understood, we use $\Theta_G(S)$ to denote the Gompf invariant determined by any compatible J. It follows from the Hirzebruch index theorem that $\Theta_G(-S) = -\Theta_G(S)$, where $-S$ denotes reversing the coorientation on S.

For the standard tight contact structure (S^3, ξ_{std}), we have

$$\Theta_G(\xi_{std}) = -2$$

because it arises as the boundary of the round B^4 in $(\mathbb{C}^2, \omega_{std}, J_{std})$. Therefore,

$$\Theta_G(S) \in -2 + 4\mathbb{Z}$$

for any smoothly embedded, separating, cooriented 3-sphere and given any smooth, separating 3-sphere, there exists a coorientation with negative Gompf invariant.

1.2. **Constructive approach.** Let $S \subset (X, \omega)$ be a smoothly embedded, cooriented 3-sphere in a symplectic 4-manifold bounding $\widetilde{B} \subset X$. The main work of this paper is the following theorem; Theorem 1.3 follows by combining it with Theorem 1.2.

Theorem 1.4. *Let (X, ω) be a closed symplectic 4-manifold and let $S \subset (X, \omega)$ be a smoothly embedded, cooriented 3-sphere bounding $\widetilde{B}$. If $\Theta_G(S) < 0$, then there exists a symplectic structure $(\widetilde{B}, \widetilde{\omega})$ that weakly fills a contact structure (S^3, ξ)*

Our approach to Theorem 1.4 is to *construct* a symplectic form $\widetilde{\omega}$ on $\widetilde{B}$ that weakly fills (S^3, ξ), rather than finding an ambient isotopy of the 3-sphere S within the original (X, ω) or deforming the symplectic structure ω until the hypersurface is contact-type. Our technical approach is through symplectic 1-cocycles on relative trisections (see Subsections 1.5,1.7, and 2 below). But we first give the reader a non-technical, 'big picture' overview.

Constructing symplectic forms from a handle decomposition is well-known to be a difficult problem. Weinstein handles can be attached along isotropic spheres, but only up to the middle dimension, and an arbitrary homotopy 4-ball may require 3-handles in a handle decomposition. Moreover, the 2-handle framings must be realized by a Legendrian link. To sidestep this issue, we use the fact that $\widetilde{B}$ embeds in a closed symplectic 4-manifold. Auroux constructed branched coverings from (X, ω) to $(\mathbb{CP}^2, \omega_{FS})$ [Aur00]. We can turn this data into a (potentially very, very large) handle decomposition that is compatible with both the symplectic structure and the connected sum decomposition. We keep the induced symplectic form on

the 2-skeleton; then the 3- and 4-handles can be re-attached as if they are upside-down Weinstein 1- and 0-handles.

The handle decomposition is obtained by pulling back the standard trisection of $\mathbb{CP}^2$ to X. The projective plane has an efficient handle decomposition given by the three affine charts; however, it admits a larger, "inside-out" handle decomposition coming from a trisection. After sufficiently many stabilizations, which can be accomplished via an isotopy of the branch locus in $\mathbb{CP}^2$, the trisection of X is compatible with ω and also reducible with respect to a given connect sum decomposition. The trisection of X restricts to a relative trisection of $\widetilde{B}$. The relative trisection induces a stratification on the boundary, which is a cell decomposition consisting of a single 0-handle, a single 1-handle, three 2-handles and three 3-handles, regardless of the number of interior handles. This is drastically different than, say, the approach in [Sch84, Sch08, AF15] where large handle decompositions of an arbitrary $S^3 \subset S^4$ are obtained by restricting a perfect Morse function on S^4.

Moreover, this construction gives significant control over the symplectic form along the boundary, which is the important constraint when constructing a weak filling. Specifically, the three 3-handles along the boundary are attached *before* the interior 3- and 4-handles. Hence, we have sufficient freedom to control the symplectic geometry along the boundary.

1.3. **Relative trisections.** We review the relevant definitions.

Definition 1.5. A *trisection* $\mathcal{T}$ of a closed, oriented 4-manifold is a decomposition

$$X = X_1 \cup X_2 \cup X_3$$

such that

(1) each sector X_λ is diffeomorphic to B^4 or $\natural_k S^1 \times B^3$ for some $k \geq 1$,
(2) each double intersection $H_\lambda = X_\lambda \cap X_{\lambda+1}$ is a 3-dimensional, genus g 1-handlebody, and
(3) the triple intersection $\Sigma = X_1 \cap X_2 \cap X_3$ is a closed surface of genus g.

We refer to Σ as the *central surface* of the trisection and the union $\mathcal{S} = H_1 \cup H_2 \cup H_3$ as the *spine* of the trisection.

To extend this definition to 4-manifolds with nonempty boundary, we first fix some notation. Let $Z_0 = B^4$ and $Z_k = \natural_k S^1 \times B^3$ and let $Y_k = \partial Z_k$. For any integers r, s let $\Sigma_{s,t}$ denote the compact, oriented surface of genus s and t boundary components. For any integers r, s, t, let $H_{r,s,t}$ denote a *compression body* cobordism between $\Sigma_{r,t}$ and $\Sigma_{s,t}$. In particular, if $r = s$, then $H_{r,s,t} = \Sigma_{r,t} \times [0,1]$ and if $r > s$, then $H_{r,s,t}$ is obtained from $H_{r,r,t}$ by attaching $(r-s)$ 3-dimensional 2-handles.

Let g, p, b be nonnegative integers with $g \geq p$ and $b \geq 1$. There is a decomposition

$$Y_k = Y^-_{g,k;p,b} \cup Y^0_{g,k;p,b} \cup Y^+_{g,k;p,b}$$

where

$$\begin{aligned} Y^+_{g,k,p,b} &\cong H_{g,p,b} \\ Y^0_{g,k,p,b} &\cong \Sigma_{p,b} \times [0,1] \\ Y^-_{g,k,p,b} &\cong H_{g,p,b} \end{aligned}$$

Definition 1.6. Let X be a compact, connected, oriented 4-manifold with connected boundary Y. A $(g, k_1, k_2, k_3; p, b)$-*relative trisection* of X is a decomposition

$$X = X_1 \cup X_2 \cup X_3$$

where

(1) for each $\lambda = 1, 2, 3$, there is a diffeomorphism $\phi : X_\lambda \to Z_{k_\lambda} = \natural_{k_\lambda} S^1 \times B^3$
(2) for each $\lambda = 1, 2, 3$ and taking indices mod 3, we have

$$\begin{aligned} \phi(X_\lambda \cap X_{\lambda-1}) &= Y^-_{g,k_\lambda,p,b} \\ \phi(X_{\lambda+1} \cap X_\lambda) &= Y^+_{g,k_\lambda,p,b} \\ \phi(X_\lambda \cap \partial X) &= Y^0_{g,k_\lambda,p,b} \end{aligned}$$

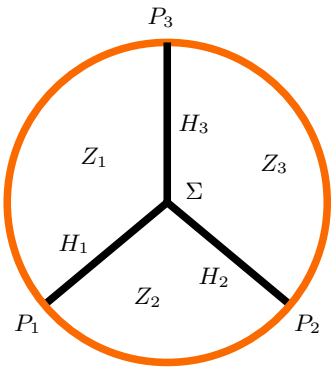


FIGURE 1. A diagrammatic representation of a relative trisection of a manifold with non-empty boundary ∂X (in orange).

Analogously to the closed case, we define $H_\lambda = X_\lambda \cap X_{\lambda+1}$, which is a compression body. Moreover, we refer to $\Sigma = X_1 \cap X_2 \cap X_3$ as the *central surface* of the relative trisection and $\mathcal{S} = H_1 \cup H_2 \cup H_3$ as the *spine* of the trisection.

Definition 1.7. The *capped spine* of a relative trisection is the union

$$\widehat{\mathcal{S}} = \mathcal{S} \cup \partial X = H_1 \cup H_2 \cup H_3 \cup \partial X$$

The boundary of a tubular neighborhood $\nu(\widehat{\mathcal{S}})$ of the capped spine has four connected components, which can be partitioned into two subsets: the *exterior* boundary component ∂X, and the *interior* boundary consisting of three connected components $\{\widehat{Y}_\lambda \cong \partial X_\lambda\}$. See Figure 2.

A relative trisection $(X, \mathcal{T})$ induces a decomposition

$$\partial X = Q_1 \cup Q_2 \cup Q_3$$

such that

(1) the triple intersection $B := Q_1 \cap Q_2 \cap Q_3 = \partial\Sigma$ is a fibered link in ∂X,
(2) each double intersection $P_\lambda := Q_\lambda \cap Q_{\lambda+1}$ is a fiber surface for B, and
(3) each sector $Q_\lambda \cong P_\lambda \times [0,1]$ is a thickened surface.

In other words, a relative trisection induces an open book decomposition of ∂X [CGPC18, Lemma 11]. We refer to B as the *binding* and P_λ as a *page* of this open book decomposition. In addition, we also denote

$$\mathcal{S}_\partial := P_1 \cup P_2 \cup P_3 = \mathcal{S} \cap \partial X$$

and refer to it as the *boundary spine* of the relative trisection.

A tubular neighborhood $\nu(\mathcal{S})$ of the spine is a manifold with corners. There are three *interior faces* $Y_\lambda^* = H_\lambda \cup -H_{\lambda-1}$ and one *boundary face* $\nu(\mathcal{S}_\partial) = \nu(\partial H_1 \cup \partial H_2 \cup \partial H_3)$ that are 3-manifolds with boundary. There are three corner surfaces $S_\lambda = P_\lambda \cup -P_{\lambda-1}$.

1.4. **Trisections and handlebody decompositions.** To guide the reader, we review how a (relative) trisection induces a handle decomposition appropriate to constructing symplectic forms.

1.4.1. *Standard perspective.* Gay and Kirby proved that every closed, smooth, oriented 4-manifold X admits a trisection $X = X_1 \cup X_2 \cup X_3$ [GK16]. Roughly speaking, a trisection can be obtained from a handle decomposition according to the rule:

(1) X_1 contains the 0- and 1-handles,
(2) X_2 contains the 2-handles, and
(3) X_3 contains the 3- and 4-handles.

This is the 'standard' topological perspective of a trisected 4-manifold.

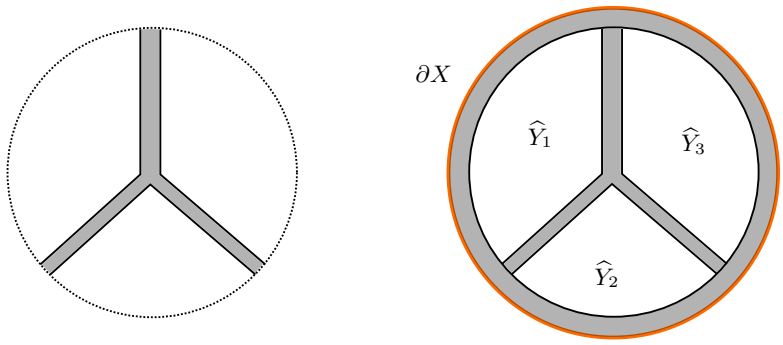


FIGURE 2. (*Left*) A tubular neighborhood of the spine, which is a manifold with corners. (*Right*) A tubular neighborhood of the capped spine, which has one exterior boundary component ∂X and three interior boundary components $\widehat{Y}_1, \widehat{Y}_2, \widehat{Y}_3$.

1.4.2. *Inside-out perspective.* However, a trisection $\mathcal{T}$ also determines an 'inside-out' handle decomposition of X as follows:

(1) $\Sigma \times D^2$ consists of a 0-handles, $2g$ 1-handles, and a 2-handle;
(2) each $H_\lambda \times I$ (viewed as an upside-down 3-dimensional 1-handlebody), consists of g 2-handles and a 3-handle,
(3) each X_λ (viewed as an upside-down 4-dimensional 1-handlebody), consists of k_λ 3-handles and a 4-handle.

Note that the spine $\mathcal{S} = H_1 \cup H_2 \cup H_3$ contains the 2-skeleton of the 'inside-out' handle decomposition determined by $\mathcal{T}$.

From this 'inside-out' handlebody perspective, it is apparent how this construction in Theorem 2.2 avoids the standard difficulties in attaching 2- and 3-handles to a symplectic 4-manifold with boundary. First, from the existence of a symplectic 1-cocycle we obtain a symplectic form over the 2-skeleton. Moreover, the boundary of the spine is strongly concave with respect to this symplectic structure. Therefore, the 3- and 4-handles of X_λ can be attached symplectically, by viewing them as upside-down Weinstein handles of index 1 and 0, respectively.

1.4.3. *Relative inside-out perspective.* A relative trisection also determines an 'inside-out' handle decomposition as follows:

(1) $\Sigma_{g,b} \times D^2$ consists of a 0-handle and $(2g + b - 1)$ 1-handles,
(2) each compression body H_λ consists of $(g - p)$ 2-handles,
(3) each thickened page $P_\lambda \times D^2$ (viewed as an upside-down 3-dimensional 1-handlebody) consists of $(1 - \chi(P_\lambda))$ 2-handles and a 3-handle, and
(4) each 4-dimensional sector X_λ consists of k_λ 3-handles and a 4-handle.

In the relative case, the capped spine $\widehat{\mathcal{S}}$ contains the 2-skeleton of this inside-out handle decomposition.

The existence of a symplectic 1-cocycle yields a symplectic structure on the spine $\mathcal{S}$. The new crucial step is therefore to extend this symplectic structure across the capped spine, such that the interior boundary is strongly concave. Then as in the closed case, the 3- and 4-handles can be attached symplectically by viewing them as upside-down Weinstein handles.

1.4.4. *An orange.* In order to give the reader an intuitive idea of how one builds X from a relative trisection, we drop down one dimension and describe the standard relative trisection of B^3 by analogy with an orange:

(1) start with the central core (made of fibrous white pith) that is the axis around which the orange slices (carpels) are arranged;

(2) attach several septa, also made of white pith, to the central core so as to divide the interior into several chambers,
(3) attach the exterior rind,
(4) fill in the interior with the orange slices (carpels).

Remark 1.8. From the perspective of building weak symplectic fillings, the key insight is that, when building an orange in this way, the exterior rind is attached *before* the interior carpels. Therefore, we can control ω along the exterior boundary of the capped spine and ensure it is weakly convex. The final step, extending across the 3- and 4-handles, then depends only on identifying the induced contact structures on the strongly concave interior boundary.

1.5. **Relative trisections and symplectic 1-cocycles.** We now turn our attention to ensuring that the symplectic structure $(\widetilde{B},\widetilde{\omega})$ has weakly convex boundary. The first major step in the proof is constructing a relative trisection $\mathcal{T}$ of $\widetilde{B}$ and a corresponding symplectic 1-cocycle $\widetilde{\beta}$ for $(\widetilde{B},\mathcal{T})$. We proceed in the following way:

(1) We can choose a *reducible* (smooth) trisection $\mathcal{T}_{\#}$ of X compatible with the connect sum decomposition (Section 4).
(2) We can find a trisection $\mathcal{T}_{symp}$ of X given by pulling back $(\mathbb{CP}^2,\mathcal{T}_{std})$ via a symplectic branched covering $\pi : X \to \mathbb{CP}^2$ [LC26]. This (singular) branch locus can be isotoped into transverse bridge position (see [LCMS21, LC23])
(3) There exists a common trisection stabilization $\mathcal{T}$ of both $\mathcal{T}_{\#}$ and $\mathcal{T}_{symp}$ [GK16, Theorem 11]. Stabilization preserves reducibility and can also be accomplished by an isotopy of the branch locus in $\mathbb{CP}^2$ [LC26, Theorem 2.20]. Therefore, we can obtain a reducible trisection $\mathcal{T}$ of X with a symplectic 1-cocycle $\widetilde{\beta}$ by pulling back from $\mathbb{CP}^2$ (see [LC26, Section 3.4])
(4) Restricting to $\widetilde{B}$ gives the required trisection and symplectic 1-cocycle.

Secondly, we modify the symplectic 1-cocycle so that it is *pseudoconvex* (Definition 2.14) along the boundary spine and has *Legendrian binding* (Definition 5.16). These conditions simplify the construction of a weak filling along the binding and ensure that along the binding, we can choose the contact structure of the weak filling to look like a standard Legendrian neighborhood.

Summarizing the first step, we have the following theorem.

Theorem 1.9 (Existence)**.** *Let M be a connect summand of a closed, symplectic 4-manifold (X,ω). There exists*

(1) a relative trisection $\mathcal{T}$ of $\widetilde{B} := M \setminus B^4$ inducing the D^2 open book decomposition on $\partial\widetilde{B} \cong S^3$, and
(2) a tight, pseudoconvex symplectic 1-cocycle with Legendrian binding on $(\widetilde{B},\mathcal{T})$.

1.6. **Homotopy implies isotopy.** From Theorem 1.9, we have a tight, pseudoconvex symplectic 1-cocycle with Legendrian binding. In order to extend ω across the capped spine and obtain a weak symplectic filling, we may first need to modify the symplectic structure near the boundary spine $\mathcal{S}_\partial = P_1 \cup P_2 \cup P_3$. The Gompf invariant $\Theta_G(\partial\widetilde{B})$ obstructs this in general, as Theorem 1.9 applies to both summands of a decomposition. We will explain how a symplectic 1-cocycle encodes the Gompf invariant $\Theta_G(\partial\widetilde{B})$. Then we will prove a 'Homotopy implies isotopy' result, namely that if $\Theta_G(\partial\widetilde{B}) = -2$ then we can modify the symplectic structure to construct a weak filling.

Let (M,ξ) be a contact 3-manifold, let L be a fibered link in M that is ξ-Legendrian, and let P_1, P_2, P_3 be three disjoint, convex fiber surfaces. First, we have a decomposition

$$(M,\xi) = (\nu(L),\xi_{std}) \cup (\widehat{M},\xi_\phi)$$

into a standard Legendrian neighborhood of the binding and the mapping torus of the monodromy of the open book decomposition. Cutting the mapping torus along the three pages, we have a further decomposition

$$(\widehat{M},\xi_\phi) = (P_1 \times [0,1],\xi_1) \cup (P_2 \times [0,1],\xi_2) \cup (P_3 \times [0,1],\xi_3)$$

into three contact structures on a product 3-manifold.

1.6.1. *Contact groupoid.* We can reinterpret these contact structures as morphisms in the *contact category* $\mathcal{C}(P)$ of the fiber surface. Roughly speaking, the contact category of a compact surface is the category whose objects are dividing sets on P and whose morphisms are isotopy classes of contact structures on $P \times [0,1]$.

Returning to (S^3, ξ_{std}), let U be a Legendrian unknot with $(tb, rot) = (-n, r)$. While the corresponding open book decomposition has trivial monodromy, the geometric monodromy on the dividing set is rotation by $-\frac{2\pi}{n}$. We denote this by the shift map S and therefore the monodromy corresponds to the map

$$\phi : \Gamma \longrightarrow S(\Gamma) \tag{1.1}$$

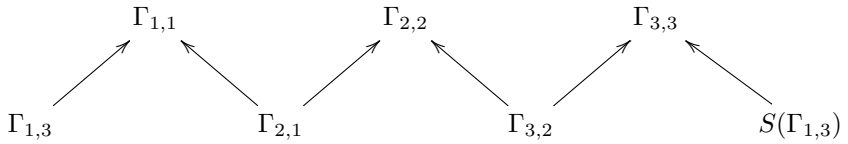
in the contact category $\mathcal{C}(D^2, n, r)$ of the disk. Cutting the mapping torus along three Seifert disks, we obtain a factorization

$$\Gamma_1 \to \Gamma_2 \to \Gamma_3 \to S(\Gamma_1) \tag{1.2}$$

in the contact category $\mathcal{C}(D^2, n, r)$ of the disk.

Now, given a symplectic 1-cocycle $\widetilde{\beta} = (\widetilde{\beta}_1, \widetilde{\beta}_2, \widetilde{\beta}_3)$, we have three Liouville vector fields (ρ_1, ρ_2, ρ_3) that point inward along the three internal boundary components of the spine. The corresponding contact forms $\alpha_i = \omega(\rho_i, -)$ induce the dividing set $\Gamma_{i,j}$ on the Seifert disk V_j in the boundary spine $\mathcal{S}_\partial$. This yields the following diagram

(1.3)



in the contact category $\mathcal{C}(D^2, n, r)$.

By formally inverting morphisms, we can obtain the *contact groupoid* $\mathcal{G}(D^2, n, r)$ of the disk from the contact category. Passing to the nerve $N\mathcal{G}(D^2, n, r)$, we can view Diagrams 1.2 and 1.3 as paths in $N\mathcal{G}(D^2, n, r)$. These paths are homotopic precisely when the Gompf invariant determined by $(\widetilde{B}, \mathcal{T}, \widetilde{\beta})$ agrees with the Gompf invariant of (S^3, ξ_{std}).

Proposition 1.10. *Let $|r| < n - 1$.*

1. *The nerve of the contact groupoid $\mathcal{G}(D^2, n, r)$ is a $K(\mathbb{Z}, 1)$.*
2. *Given two tight dividing sets Γ_0, Γ_1, there is a 1-1 correspondence between homotopy classes of paths in $N\mathcal{G}(D^2, n, r)$ from Γ_0 to Γ_1 and homotopy classes of 2-plane field $(D^2 \times [0,1], \xi)$ such that Γ_i divides the characteristic foliation on $D^2 \times \{i\}$.*

1.6.2. *Strong contact groupoid.* Therefore, we want to use the homotopy from Diagram 1.3 to Diagram 1.2 to modify ω. However, working with dividing sets does not suffice for constructing weak fillings. The dividing set controls the contact structure up to a C^0-small ambient isotopy, while the weak filling condition $\lambda \wedge \omega > 0$ is only open under C^1-small ambient isotopies of the contact form λ. In particular, we need to work directly with characteristic foliations and realize Diagram 1.2 by a sequence of foliations:

$$\mathcal{F}_1 \to \mathcal{F}_2 \to \mathcal{F}_3 \to S(\mathcal{F}_1) \tag{1.4}$$

To this end, we introduce the *strong contact category* $\mathcal{SC}$ and *strong contact groupoid* $\mathcal{SG}$, whose objects are characteristic foliations. There is a 'forgetful functor'

$$\Gamma : \mathcal{SG} \to \mathcal{G}$$

that sends a characteristic foliation to its dividing set. For a disk, this map is a fibration.

Proposition 1.11. *The functor*

$$\Gamma : \mathcal{SG}(D^2, n, r) \longrightarrow \mathcal{G}(D^2, n, r)$$

that sends a characteristic foliation $\mathcal{F}$ to its dividing set $\Gamma(\mathcal{F})$ is a fibration of groupoids. Consequently, the induced map

$$\Gamma : N\mathcal{SG}(D^2, n, r) \to N\mathcal{G}(D^2, n, r)$$

is a (Serre) fibration and satisfies the homotopy lifting property for paths.

Returning the symplectic 1-cocycle, the contact form α_i induces a characteristic foliation $\mathcal{F}_{i,j}$ on the disk V_j. We obtain the diagram

(1.5)
$$\begin{array}{ccccccc} & \mathcal{F}_{1,1} & & \mathcal{F}_{2,2} & & \mathcal{F}_{3,3} & \\ \nearrow & & \nwarrow \quad \nearrow & & \nwarrow \quad \nearrow & & \nwarrow \\ \mathcal{F}_{1,3} & & \mathcal{F}_{2,1} & & \mathcal{F}_{3,2} & & S(\mathcal{F}_{1,3}) \end{array}$$

in the strong contact groupoid $\mathcal{SG}$, which lifts Diagram 1.3. The homotopy in the contact groupoid from Diagram 1.3 to Diagram 1.2 lifts to a homotopy in the strong contact groupoid from Diagram 1.5 to our desired Diagram 1.4.

1.7. **Overview of construction.** The weak symplectic filling $(\widetilde{B}, \omega)$ is constructed by gluing together 10 building blocks, each of which is a symplectic manifold with corners. These are:

(1) the spine $(\nu(\mathcal{S}), \omega_{\widetilde{\beta}})$, constructed from a tight, pseudoconvex symplectic 1-cocycle with Legendrian binding. (Section 2)
(2) three collars $([0,1] \times Y_\lambda^*, \omega)$ for the interior faces of the spine, on which we have a weak filling interpolating between characteristic foliations on S^2 (Sections 7 and 8.5)
(3) three 3-handles $(B^3 \times [0,1], d(e^t \alpha_{std}))$, which are symplectizations of the standard tight contact structure on B^3 and plumbed into the (thickened) spine along the corner $S^2 = \partial Y_\lambda^*$
(4) three Weinstein 1-handlebodies (Z_λ, ω), glued into the interior, strongly concave boundary of the capped spine.

These building blocks are glued together in one of two ways:

(1) using a transverse Liouville vector field to identify faces
(2) plumbing (Section 6)

Moreover, if two building blocks both have faces that become part of the exterior boundary $\partial \widetilde{B}$, we can also glue together weakly-filled contact structures on the boundary, provided that they induce the same characteristic foliation along the corner surface (Proposition 7.7).



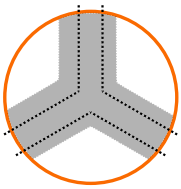
FIGURE 3. The 10 symplectic building blocks.

1.8. **Homotopy 4-spheres.** Many well-known potential counterexamples to the smooth 4-dimensional Poincare conjecture are smooth connect summands of some rational complex surface $\mathbb{CP}^2 \# k\mathbb{CP}^2$ for $k \geq 0$, hence by Theorem 1.3 must be diffeomorphic to S^4. In this subsection, we list some of the specific implications.

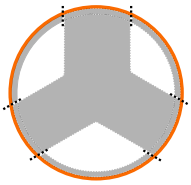
Remark 1.12. To prove the general result, we require Auroux's branched covering result [Aur00], along with its extension by Auroux-Katzarkov [AK00]. However, if $X \subset \mathbb{CP}^N$ is a complex projective surface, then projecting onto a generic hyperplane $\mathbb{CP}^2 \subset \mathbb{CP}^N$ is the required branched covering. Moreover, since the branch locus $\mathcal{D} \subset \mathbb{CP}^2$ is holomorphic, it is straightforward to extract a factorization of the full twist $\Delta^2 \subset B_n$ encoding the (nodal, cuspidal) branch curve $\mathcal{D}$. All of the homotopy 4-spheres considered below are



(A) Step 1: Start with the spine



(B) Step 2: Interpolate foliations along collars of the interior faces



(C) Step 3: Plumb in the boundary 3-cells

FIGURE 4. The steps in building up a weak symplectic filling. Step 4 is to fill in the three interior, strongly concave boundary components with Weinstein 1-handlebodies.

summands of some rational, hence projective, surface and hence can be proved standard without appealing to [Aur00, AK00].

1.8.1. *Smooth 4d Schoenflies conjecture.* The first application is the solution to smooth 4-dimensional Schoenflies problem, since if a homotopy 4-ball embeds in S^4, it embeds into every closed symplectic 4-manifold.

Theorem 1.13 (Smooth 4-dimensional Schoenflies)**.** *Let $\widetilde{B}$ be a homotopy 4-ball embedded in S^4. Then $\widetilde{B}$ is diffeomorphic to the standard 4-ball.*

1.8.2. *Gluck twists.* Second, we conclude that Gluck twists on knotted 2-spheres in S^4 are standard. Given an embedded 2-sphere $\mathcal{K}$ in S^4, Gluck introduced a surgery operation that produces a homotopy 4-sphere $S_{\mathcal{K}}$ [Glu62]. Melvin later showed that $\mathbb{CP}^2 \# S_{\mathcal{K}}$ is diffeomorphic to $\mathbb{CP}^2$ [Mel77]. Since the latter is symplectic, this implies that $S_{\mathcal{K}}$ is standard.

Theorem 1.14 (Gluck Twists)**.** *Let $\mathcal{K}$ be a knotted 2-sphere in S^4 and let $S_{\mathcal{K}}$ be the homotopy 4-sphere obtained by a Gluck twist on $\mathcal{K}$. Then $S_{\mathcal{K}}$ is diffeomorphic to S^4.*

1.8.3. *Cappell-Shaneson homotopy spheres.* Third, a similar argument is relevant to the homotopy 4-spheres constructed by Cappell-Shaneson [CS76]. Let $A \in SL(3,\mathbb{Z})$ be a matrix satisfying $det(A - I) = 1$. Then A determines a homeomorphism $\phi_A : T^3 \to T^3$ whose induced map on $H_1(T^3, \mathbb{Z})$ is precisely A. Therefore, we can use ϕ_A to construct M_A, a T^3-bundle over S^1 with monodromy ϕ_A, resulting in a homology $S^1 \times S^3$. After choosing a framing $\epsilon \in \mathbb{Z}/2\mathbb{Z}$, we can surger along an essential curve γ representing the S^1-factor to obtain a homotopy 4-sphere Σ_A^ϵ.

Proposition 1.15. *Let $A \in SL(3,\mathbb{Z})$ be a Cappell-Shaneson matrix and $\epsilon \in \mathbb{Z}/2\mathbb{Z}$ a framing. There is a diffeomorphism*

$$\Sigma_A^{\epsilon_0} \# S^2 \times S^2 \cong S^2 \times S^2$$

Proof. The is an exercise in Kirby calculus, starting from a Kirby diagram as in [AK79]. □

We then obtain the following corollary.

Theorem 1.16 (Cappell-Shaneson spheres)**.** *Every Cappell-Shaneson homotopy 4-sphere Σ_A^ϵ is diffeomorphic to S^4.*

1.8.4. *Andrews-Curtis presentations.* Let $P = \langle x_1, \dots, x_n | s_1, \dots, s_m \rangle$ be a presentation of the trivial group. The presentation is *balanced* if $m = n$ and is *trivial* if $x_i = s_{j_i}$ for $i = 1, \dots, n$. Given a balanced presentation P, we can obtain a 5-dimensional 2-handlebody

$$W_P = B^5 \cup \{n \text{ 1-handles}\} \cup \{n \text{ 2-handles}\}$$

whose boundary is a homotopy 4-sphere. Sliding 2-handles over 2-handles, we obtain a new presentation of the trivial group. Handlesliding therefore induces the so-called *Andrews-Curtis equivalence relation* on balanced presentations of the trivial group. If the presentation P is Andrews-Curtis equivalent to the trivial presentation, then W_P is diffeomorphic to B^5 and therefore its boundary is a standard S^4. The Andrews-Curtis conjecture states that all balanced presentations of the trivial group are Andrews-Curtis equivalent to the trivial presentation. However, there exist many potential counterexamples, which give rise to potential exotic homotopy 4-spheres. The best known potential counterexamples consist of 2 generators and 2 relations.

Proposition 1.17. *Let P be a balanced presentation of the trivial group and let $\Sigma_P = \partial W_P$ be the corresponding homotopy 4-sphere. Suppose that after adding k extra relations, the presentation is Andrews-Curtis equivalent to the trivial presentation. Then Σ_P is a connect summand of $k\mathbb{CP}^2\#k\overline{\mathbb{CP}}^2$ or $\#kS^2 \times S^2$.*

Proof. Handlesliding 5-dimensional 2-handles corresponds to Andrews-Curtis equivalence in the presentation of π_1. Therefore, after sliding, we can cancel n 1-/2- handle pairs, leaving k 5-dimensional 2-handles attached to B^5. The boundary is therefore as specified (depending on the framings on the 2-handles) and $(\Sigma \smallsetminus B^4) \subset W \subset \partial(W \times [0,1])$. □

Corollary 1.18. *Let P be a balanced presentation of the trivial group. Suppose that after adding one relation to P, it becomes Andrews-Curtis equivalent to the trivial presentation. Then Σ_P is diffeomorphic to S^4.*

Corollary 1.19. *If $P = \langle x_1, x_2 | s_1, s_2 \rangle$ has two generators and two relations, then Σ_P is diffeomorphic to S^4.*

Proof. Let $P' = \langle x_1, x_2 | s_1, s_2, x_2 \rangle$. Then we can eliminate all instances of x_2 in the relations s_1, s_2 via Andrews-Curtis moves to obtain

$$P'' = \langle x_1, x_2 | x_1^{k_1}, x_1^{k_2}, x_2 \rangle$$

for some integers k_1, k_2. Since P'' is a presentation of the trivial group, the exponents k_1 and k_2 must be relatively prime, hence by further Andrews-Curtis moves we can obtain a presentation P''' where $k_1 = k_2 = 1$, which is a trivial presentation. □

1.8.5. *0-surgery pairs.* Another approach (e.g.[MP23]) to constructing potential exotic 4-spheres involves pairs of knots with the same 0-surgery. Let K be a knot in S^3 and let $S^3_0(K)$ denote 0-framed surgery on K. Let K, K' be distinct knots with the same 0-surgery. If K is slice, then we can obtain a homotopy 4-sphere by first carving out a slice disk $\mathcal{D}$ for K from B^4, then gluing this complement to the 0-trace of K'.

Proposition 1.20. *Let W denote a homotopy 4-sphere obtained from a pair K, K' of knots and slice disk $\mathcal{D}_K$. Then W is a connect summand of $S^2 \times S^2$.*

Proof. Let $W^\circ = W \smallsetminus \{4 - handle\}$. Turning the 0-trace of K' upside-down, it consists of a 0-framed 2-handle and a 4-handle. By an isotopy, we can assume the attaching 1-sphere of this 2-handle is disjoint from the boundary of the removed slice disk. Therefore, we can reattach $\mathcal{D}_K \times D^2$ along ∂W°. But this is diffeomorphic to B^4 union the upside-down 2-handle from the 0-trace of K'; hence it is just the 0-trace of some K'' in S^3. The double of the 0-trace of any knot is diffeomorphic to $S^2 \times S^2$, hence W° embeds in $S^2 \times S^2$. □

Theorem 1.21. *Let W denote a homotopy 4-sphere obtained from a pair K, K' of knots and slice disk $\mathcal{D}_K$. Then W is diffeomorphic to S^4.*

Akbulut and Kirby originally conjectured that if two knots K, K' have homeomorphic 0-surgeries, then they are concordant (Problem 1.19 in Kirby's problem list [Kir97]). This was disproved in general by Yasui [Yas26], however it is true for slice knots.

Theorem 1.22 (Slice Akbulut-Kirby conjecture)**.** *Let K, K' be knots in S^3 satisfying $S^3_0(K) \cong S^3_0(K')$. Then K is smoothly slice if and only if K' is smoothly slice.*

Proof. Suppose that K is slice. Then we can build a homotopy 4-sphere by removing a slice disk $\mathcal{D}_K$ and attaching the 0-trace of K'. Since the resulting homotopy 4-ball embeds into $S^2 \times S^2$, it is standard. Therefore, the 0-trace of K' embeds into S^4 and the Trace Embedding Lemma implies that K' is slice. □

1.8.6. *Stabilization number.* Sixth, given a homotopy 4-sphere $\widetilde{S}$, there exists a 5-dimensional cobordism from S^4 to $\widetilde{S}$ consisting of n 2-handles and n 3-handles, for some $n \geq 0$. Cutting this cobordism between the 2- and 3-handles shows that $\widetilde{S}\#_n S^2 \times S^2$ is diffeomorphic to $S^4\#_n S^2 \times S^2 = \#_n S^2 \times S^2$. In this context, connecting summing with $S^2 \times S^2$ is known as *stabilization.*

Definition 1.23. Let $\widetilde{B}$ be a homotopy 4-ball. The *size* $s(\widetilde{B})$ of $\widetilde{B}$ is the minimum $k \geq 0$ such that $\widetilde{B}$ embeds into $S^4\#_k S^2 \times S^2$.

Corollary 1.24 (Single stabilization)**.** *Let $\widetilde{S}$ be a homotopy 4-sphere and let $\widetilde{B} = \widetilde{S} \smallsetminus B^4$ be the corresponding homotopy 4-ball.*

(1) If $s(\widetilde{B}) \leq 1$, then $\widetilde{B}$ is standard.
(2) Let $\widetilde{S}$ be a homotopy 4-sphere. If $\widetilde{S}$ becomes equivalent to S^4 after a single stabilization, then it is diffeomorphic to S^4.

Seven, another common method for constructing exotic smooth structures is via (integral) log transforms on embedded tori with trivial normal bundle. However, an integral log transform on a single T^2 in S^4 cannot produce an exotic 4-sphere.

Corollary 1.25 (Log transforms)**.** *Let $\{X_i\}$ be a family of homotopy 4-spheres for $i = 0, 1, \dots$ such that*

(1) X_0 is diffeomorphic to S^4, and
(2) X_{i+1} is obtained from X_i by an integral log transform.

Then each X_i is diffeomorphic to S^4.

Proof. Baykur and Sunukjian proved [BS13, Lemma 4] that given these hypotheses, there are diffeomorphisms

$$X_i \# S^2 \times S^2 \cong X_{i+1} \# S^2 \times S^2$$

However, since $X_0 \cong S^4$, this implies that each $X_i \# S^2 \times S^2$ is diffeomorphic to $S^2 \times S^2$, so the previous theorem implies that each X_i is standard. □

1.9. **Towards SPC4.** The smooth 4-dimensional Poincare conjecture asks whether S^4 has a unique smooth structure or whether there exist exotic smooth structures.

Conjecture 1.26 (SPC4)**.** *S^4 has a unique smooth structure.*

Theorem 1.3 leads to the following natural question.

Question 1.27. Does every homotopy 4-ball embed into some closed symplectic 4-manifold?

Alternatively, we may be able to adapt our arguments here to a general homotopy 4-ball as we have the following version of Theorem 1.9.

Theorem 1.28. *Let $\widetilde{B}$ be a homotopy 4-ball. There exists*

(1) a relative trisection $\mathcal{T}$ of $\widetilde{B}$ inducing the D^2 open book decomposition along $S^3 = \partial\widetilde{B}$, and
(2) a pseudoconvex symplectic 1-cocycle with Legendrian binding on $(\widetilde{B}, \mathcal{T})$.

Proof. (Sketch) $\widetilde{B}$ embeds into $\#_k S^2 \times S^2$ for k sufficiently large. We have a holomorphic double-branched covering

$$\pi : \mathbb{CP}^1 \times \mathbb{CP}^1 \to \mathbb{CP}^2$$

over a degree-2 rational curve $\mathcal{C}_2$. Adding trivial handles to the branch locus in $\mathbb{CP}^2$ corresponds to connect-summing with $S^2 \times S^2$ upstairs. We can assume the branch locus is in transverse bridge position and pull back the symplectic 1-cocycle. However, it will be overtwisted in at least one sector. After bridge perturbations of the branch locus, we can assume that the trisection is reducible and restrict as in Theorem 2.13. Moreover, the proof of Theorem 1.9 does not require tightness, so we can achieve pseudoconvexity and Legendrian binding for this symplectic 1-cocycle on $\widetilde{B}$. □

The restriction of the symplectic 1-cocycle to $\widetilde{B}$ *may* be tight, in which case we can repeat our construction here and deduce that $\widetilde{B}$ is standard.

In a different direction, it would be interesting to generalize trisection and branched-cover constructions to near-symplectic structures. Given a homotopy 4-sphere Σ and symplectic 4-manifold (X, ω), their connect sum $X\#\Sigma$ admits a near-symplectic structure with non-vanishing Seiberg-Witten invariants. There might be exploitable structure in the corresponding pseudoholomorphic divisors that allows one to adapt our construction to an arbitrary homotopy 4-ball (see [Tau98, Ger21, Ger22]).

1.10. **Outline.** Section 2 introduces symplectic cocycles on trisected and relatively trisected 4-manifolds. Section 3 describes the characteristic foliations and dividing sets induced on the corners of the relative spine as well as the induced maps in the contact category. Section 4 discusses reducible trisections and decompositions along 3-spheres. Section 5 describes how to enlarge and Legendrian stabilize a symplectic 1-cocycle, in order to ensure that it is pseudoconvex and the binding can be standardized. Section 6 introduces strongly minimally twisting contact structures and describes how to plumb two symplectizations. Section 7 proves several results about gluing weak fillings along surfaces and flexibility of the characteristic foliation for tight S^2's. Section 8 covers the proof of Theorem 1.4. Section 9 introduces the contact category, strong contact category, their groupoid versions and computes the fundamental group of the nerve. Section 10 proves several results relating the Gompf invariant and strongly minimally twisting contact structures on S^2.

1.11. **Acknowledgements.** I am grateful to Dave Gay and Gordana Matic for many conversations over several years on this project and to Jeff Meier for helpful comments on a draft. I also thank the Max Planck Institute for Mathematics in Bonn for hosting me while part of this work was conducted.

2. Symplectic 1-cocycles

2.1. **Inspiration.** A key aspect of our approach is that the symplectic structure can be constructed from geometric data, called a *symplectic 1-cocycle*, on the codimension-1 spine of a trisection. This approach is developed in [LC24, LC26] and is inspired by the following observations in complex geometry.

Let X be a projective surface with very ample line bundle H and holomorphic embedding

$$i : X \to \mathbb{CP}^N$$

such that $i^*(\mathcal{O}(1)) = H$. The bundle H admits a Hermitian connection whose curvature (1,1)-form ω is a symplectic form on X. Let $\mathcal{U} = \{U_i\}$ be an open cover of X by Stein domains. Then we have isomorphisms

$$H^{1,1}_{\overline{\partial}}(X) \cong H^1(X, \Omega^1) \cong \check{H}^1(\mathcal{U}, \Omega^1)$$

between the Dolbeault cohomology of X, the sheaf cohomology of the sheaf of holomorphic 1-forms Ω^1, and Čech cohomology of Ω^1 with respect to $\mathcal{U}$. Under this isomorphism, the (1,1)-form ω is identified with a 1-cocycle $\beta = (\beta_{i,j})$ of closed, holomorphic 1-forms, where each $\beta_{i,j}$ is defined over $U_i \cap U_j$.

A generic triple (s_0, s_1, s_2) of holomorphic sections of H determine a branched covering

$$p : X \to \mathbb{CP}^2$$

such that $p^*(\mathcal{O}(1)) = H$. Using the complex modulus, consider the decomposition $X = X_0 \cup X_1 \cup X_2$ where

$$\begin{aligned} X_i &= \{x \in x : |s_{i-1}(x)|, |s_{i+1}(x)| \leq |s_i(x)|\} \\ H_i &= X_i \cap X_{i-1} \\ &= \{x \in x : |s_{i+1}(x)| \leq |s_{i-1}(x)| = |s_i(x)|\} \\ \Sigma &= X_1 \cap X_2 \cap X_3 \\ &= \{x \in x : |s_0(x)| = |s_1(x)| = |s_2(x)|\} \end{aligned}$$

Generically, Σ is a (real) 2-dimensional surface and each H_i is a 3-manifold with boundary Σ. Each X_i is a compact Stein set (although not a 1-handlebody in general, so this is not a trisection decomposition). By slightly enlarging, we obtain an open cover $\mathcal{U}$ by Stein domains and the symplectic form $\omega \in H^{1,1}_{\overline{\partial}}(X)$ is identified with a triple $\beta = (\beta_0, \beta_1, \beta_2) \in \check{H}^1(\mathcal{U}, \Omega^1)$ of holomorphic 1-forms, where β_i is defined on a neighborhood of H_i.

Since the modulus function is pluriharmonic, the hypersurface H_i is Levi-flat and foliated by holomorphic curves. From the perspective of 3-dimensional topology, this foliation by holomorphic curves has properties similar to a taut foliation, even though $\Sigma = \partial H_i$ is generally not a collection of tori. Moreover, the imaginary part of the closed, holomorphic 1-form β_i actually *defines* this foliation by holomorphic disks, in that its kernel 2-plane field is tangent to the foliation.

2.2. **Closed 4-manifolds.** We first review symplectic 1-cocycles on closed 4-manifolds. Let X be a closed, oriented, smooth 4-manifold and let $\mathcal{T}$ be a trisection of X, where Σ_g is the central surface and H_1, H_2, H_3 are the 3-dimensional sectors.

Definition 2.1. A *symplectic 1-cocycle* on $(X, \mathcal{T})$ is a triple $\widetilde{\beta} = (\widetilde{\beta}_1, \widetilde{\beta}_2, \widetilde{\beta}_3)$ of closed, nonvanishing 1-forms on the handlebody sectors of the trisections, such that the restrictions $\beta_\lambda = \widetilde{\beta}_\lambda|_\Sigma$ satisfy the following conditions:

(1) (1-cocycle condition)
$$\beta_1 + \beta_2 + \beta_3 = 0$$
(2) (pairwise positivity)
$$\beta_\lambda \wedge \beta_{\lambda+1} \geq 0$$
with equality only on the set $\overline{b} = \{b_1, \ldots, b_{2g-2}\} = \beta_\lambda^{-1}(0)$.
(3) all three 1-forms have saddle Morse singularities at the points of $\overline{b}$ and there exists a smooth function $f : \Sigma \to \mathbb{R}$ and neighborhoods $\{b_i \in U_i\}$ such that
   (a) $f|_{U_i} = \pm 1$ for each $i = 1, \ldots, 2g-2$,

(b) $f(b_1) + \cdots + f(b_{2g-2}) = 0$,
(c) $\widetilde{\beta}_\lambda = \pi^*(\beta_\lambda) + d(s\,\pi^*(f))$ on a collar neighborhood of $\partial H_\lambda = \Sigma$, where

$$\tau_\lambda : \nu(\Sigma) \cong [-\delta, 0] \times \Sigma$$

is a fixed trivialization of a collar neighborhood and s, π are the projections onto the factors.

A symplectic 1-cocycle determines the germ of a symplectic structure along the spine $\mathcal{S} = H_1 \cup H_2 \cup H_3$ of the trisection. Recall that a vector field v is a *Liouville vector field* for a symplectic form ω if it satisfies

$$\mathcal{L}_v \omega = \omega \tag{2.1}$$

Cartan's formula implies that $\alpha = \omega(v, -)$ is a primitive for the symplectic structure and if Y is a hypersurface transverse to the flow of v, then α restricts to a contact form along Y.

Theorem 2.2 ([LC24])**.** *Let $\widetilde{\beta}$ be a symplectic 1-cocycle on $(X, \mathcal{T})$. There exists*
(1) a symplectic structure $(\nu(\mathcal{S}), \omega_{\widetilde{\beta}})$ on a neighborhood of the spine, and
(2) an inward-pointing Liouville vector field ρ along the boundary $\partial\nu(\mathcal{S})$.

Consequently, the spine has the structure of a *strong symplectic cap*. To obtain a global symplectic structure on X, we would like to glue this strong symplectic cap to a *strong symplectic filling*. Each sector of a trisection is a 4-dimensional 1-handlebody, which admits a Weinstein structure that is unique up to deformation. Moreover, the induced contact structure on the boundary ∂X_i is the unique tight contact structure on that 3-manifold (either S^3 or $\#_k S^1 \times S^2$). Therefore, we can easily state the necessary and sufficient condition for a symplectic 1-cocycle on $(X, \mathcal{T})$ to determine a global symplectic structure (X, ω).

Definition 2.3. A symplectic 1-cocycle $\widetilde{\beta}$ on $(X, \mathcal{T})$ is *tight* if the contact structure $(\partial\nu(\mathcal{S}), \ker(\omega(\rho, -)))$ induced by the inward-pointing Liouville vector field is tight.

Theorem 2.4. *Let $\widetilde{\beta}$ be a tight symplectic 1-cocycle on $(X, \mathcal{T})$. Then the symplectic structure $(\nu(H_1 \cup H_2 \cup H_3), \omega_{\widetilde{\beta}})$ extends to a global symplectic structure $(X, \omega_{\widetilde{\beta}})$.*

Remark 2.5. In Definition 2.1, it is necessary to choose auxiliary data $(f, \tau_1, \tau_2, \tau_3)$. However, (a) the space of such choices is path connected, (b) every 1-parameter family of choices yields a 1-parameter family of symplectic 1-cocycles, therefore a 1-parameter of symplectic forms, (c) since $(\beta_1, \beta_2, \beta_3)$ are fixed, this family of symplectic forms are cohomologous and therefore isotopic.

2.2.1. *Simplifying assumption.* The construction of $(\nu(\mathcal{S}), \omega_{\widetilde{\beta}})$ and the Liouville vector fields from a symplectic 1-cocycle requires choosing a second triple of 1-forms $(\kappa_1, \kappa_2, \kappa_3)$ over the 3-dimensional handlebodies such that

$$\kappa_\lambda|_\Sigma = \beta_{\lambda+1} \tag{2.2}$$

$$\widetilde{\beta}_\lambda \wedge d\kappa_\lambda > 0 \tag{2.3}$$

for each $\lambda = 1, 2, 3$. In [LC24] it is shown that such a triple can always be constructed from a symplectic 1-cocycle. In addition, as in Remark 2.5 any 1-parameter family of primitives extends to an isotopy of the symplectic form. For our purposes (in particular Subsection 8.3) we require more explicit control over these primitives near Σ. Therefore, we restrict to symplectic 1-cocycles that satsify the following assumption

Assumption 2.6. Fix a collar neighborhood of $\partial H_\lambda = \Sigma$ with coordinate $s \leq 0$. For any $\delta > 0$, there exists some 1-form κ_λ on H_λ such that

(1) $\beta_\lambda \wedge d\kappa_\lambda > 0$ everywhere.
(2) κ_λ is C^∞-small on the complement of $[-2\delta, 0] \times \Sigma$, and
(3) $\kappa_\lambda = \Phi_\lambda(s)\widetilde{\beta}_{\lambda+1}$ on $[-2\delta, 0] \times \left(\Sigma \smallsetminus \bigcup_{i=1}^{2g} U_i\right)$ where
  (a) $\Phi'_\lambda > 0$, and
  (b) $\Phi_\lambda(s) = 1 + s$ for $s \in [-\delta, 0]$,
  and $\{U_i\}$ are as in Definition 2.1.

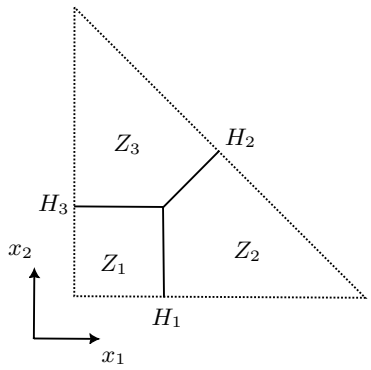


FIGURE 5. A label decomposition of the moment map of $\mathbb{CP}^2$ into the standard trisection

As demonstrated in the following subsection, there exist such primitives for the standard symplectic 1-cocycle on $(\mathbb{CP}^2, \mathcal{T}_{std})$. Moreover, they exists for any symplectic 1-cocycle obtained as in [LC26] via a symplectic branched covering $(X, \omega) \to (\mathbb{CP}^2, \omega_{FS})$.

2.3. **Building ω on $\mathbb{CP}^2$.** To understand the symplectic structure near the spine $\nu(\mathcal{S}) \subset \partial \widetilde{B}$, we review the construction of the symplectic structure on $\nu(H_1 \cup H_2 \cup H_3)$. Since the symplectic structure on $(X, \widetilde{\omega})$ is obtained by pulling back along a branched covering

$$\pi : (X, \omega) \to (\mathbb{CP}^2, \omega_{FS})$$

we start with the trisection construction of $(\mathbb{CP}^2, \omega_{FS})$.

$$\mu : \mathbb{CP}^2 \to \mathbb{RP}^2$$

given by

$$[z_0 : z_1 : z_2] \mapsto [|z_0|^2 : |z_1|^2 : |z_2|^2]$$

2.3.1. *Coordinates.* Let (θ_1, θ_2) be angular coordinates on T^2 and let (x_1, x_2) be rectangular coordinates on $\mathbb{R}^2$. Our symplectic 1-cocycle is given by

$$\beta_1 = d\theta_1 \qquad \beta_2 = d\theta_2 - d\theta_1 \qquad \beta_3 = -d\theta_2$$

which satisfies the relation

$$\beta_\lambda \wedge \beta_{\lambda+1} = d\theta_1 \wedge d\theta_2$$

for all $\lambda = 1, 2, 3$. By changing angular coordinates

$$\widetilde{\theta}_1 = \theta_1 \qquad \widetilde{\theta}_2 = -\theta_1 + \theta_2 \qquad \widetilde{\theta}_3 = -\theta_2$$

we have that $\beta_\lambda = d\widetilde{\theta}_\lambda$.

In a neighborhood of the central surface, we can assume that locally in $T^2 \times D^2$ the pieces of the trisection are given by:

$$\begin{aligned} H_1 &:= \{x_2 \leq 0, x_1 = 0\} & Z_1 &= \{x_1 \leq 0, x_2 \leq 0\} \\ H_2 &:= \{-x_1 + x_2 = 0, x_2 \geq 0\} & Z_2 &= \{x_1 \geq 0, x_1 - x_2 \geq 0\} \\ H_3 &:= \{x_2 = 0, x_1 \leq 0\} & Z_3 &= \{x_2 \geq 0, x_1 - x_2 \leq 0\} \end{aligned}$$

In particular, if we set

$$\phi_1 = x_1 \qquad \phi_2 = x_2 - x_1 \qquad \phi_3 = -x_2$$

then

$$H_\lambda = \{\phi_{\lambda-1} \geq 0, \phi_\lambda = 0\} \qquad Z_\lambda = \{\max(-\phi_{\lambda-1}, \phi_\lambda) \leq 0\}$$

and in particular, Z_λ is locally the sublevel set of $m_\lambda = \max(\phi_\lambda, -\phi_{\lambda-1})$.

We attach $H_\lambda \times I$ to $T^2 \times D^2$ as follows. Let $H_\lambda = S^1 \times D^2$ with coordinates (ψ_1, s, ψ_2) for $\psi_1, \psi_2 \in S^1$ and $s \in [0,1]$ and let t be a normal coordinate on I. We choose the identification

$$\begin{aligned} t &= \phi_\lambda & \psi_1 &= \widetilde{\theta}_\lambda \\ s &= 1 - \phi_{\lambda-1} & \psi_2 &= \widetilde{\theta}_{\lambda+1} \end{aligned}$$

In our coordinates, the 1-form $d\psi_1$ generates $H^1_{DR}(H_\lambda)$ and is identified with $\beta_\lambda = d\widetilde{\theta}_\lambda$ under the gluing. Therefore β_λ extends across the solid torus H_λ.

Finally, we obtain a smooth manifold $\nu(H_1 \cup H_2 \cup H_3)$ with three smooth boundary components as follows. Recall that the sector Z_λ is locally defined near Σ by $m_\lambda = 0$. We can extend m_λ across $H_\lambda \times I \cup H_{\lambda-1} \times I$. Now let $\widehat{m}_\lambda$ denote some C^0-close smoothing of m_λ (as a function of x_1, x_2), that agrees with m_λ outside of a tubular neighborhood of Σ. Let $\widehat{Y}_\lambda$ be the level set of $\widehat{m}_\lambda^{-1}(-\delta')$ for some small $\delta' > 0$. We can assume that the smoothing is chosen so that the form

$$\zeta_\lambda = d\phi_\lambda - d\phi_{\lambda-1} = dt + ds$$

coorients $\widehat{Y}_\lambda$ in $\Sigma \times D^2$.

2.3.2. *The symplectic form.* We now describe a symplectic form on $\mathbb{CP}^2$ that is triply-symmetric with respect to the trisection.

By construction, we have the 1-cocycle relations

$$\begin{aligned} \beta_1 + \beta_2 + \beta_3 &= 0 \\ \phi_1 + \phi_2 + \phi_3 &= 0 \end{aligned}$$

It follows that the 2-form

$$\omega = d\phi_\lambda \wedge \beta_\lambda - d\phi_{\lambda-1} \wedge \beta_{\lambda+1}$$

and its primitive

$$\mu_* = \phi_\lambda \beta_\lambda - \phi_{\lambda-1}\beta_{\lambda+1}$$

are independent of λ as well. In complex polar coordinates, this form is

$$\omega = dx_1 \wedge d\theta_1 - (-dx_2) \wedge (d\theta_2 - d\theta_1)$$

Squaring we obtain

$$\omega \wedge \omega = 2dx_1 \wedge d\theta_1 \wedge dx_2 \wedge d\theta_2$$

Therefore it is a symplectic form inducing the standard complex orientation.

To extend the symplectic form across the handlebodies of the trisection, we define primitives

$$\begin{aligned} \mu_\lambda &= \mu_* + \beta_\lambda - \beta_{\lambda-1} \\ &= \mu_* + 2\beta_\lambda + \beta_{\lambda+1} \end{aligned}$$

The 1-cocycle relation implies that

$$\mu_\lambda - \mu_{\lambda+1} = 3\beta_\lambda$$

We extend the primitive μ_λ across the handlebody H_λ as well. Fix some small $\delta, \epsilon' > 0$. We can choose an increasing function $\Phi(s)$ satisfying

$$\Phi(s) = \begin{cases} \epsilon' s^2 & \text{for } s \in [0, 1-2\delta] \\ s & \text{for } s \in [1-\delta, 1] \end{cases}$$

and define $\kappa_\lambda = \Phi(s) d\psi_2$.

Then we can extend μ_λ over H_λ as

$$\mu_\lambda = (t+2)d\psi_1 + \kappa_\lambda \tag{2.4}$$

Moreover

$$\omega = d\mu_\lambda = dt \wedge d\psi_1 + \Phi' ds \wedge d\psi_2$$

extends as a symplectic form over $H_\lambda \times I$ as well. Note that $\mu_\lambda = \mu_{\lambda-1} - 3\beta_{\lambda-1}$, so it extends across $H_{\lambda-1}$ as well. Therefore, μ_λ is defined over the entirety of $\widehat{Y}_\lambda$.

For concreteness, assume $\lambda = 1$ and consider μ_1 along $\widehat{Y}_1 = H_1 \cup -H_3$. Set

$$\begin{aligned} \Phi_1 &= \Phi(1-\phi_3) & \Phi_2 &= \Phi(1-\phi_1) & \Phi_3 &= \Phi(1-\phi_2) \\ &= \Phi(1+x_2) & &= \Phi(1-x_1) & &= \Phi(1+x_1-x_2) \end{aligned}$$

Unpacking Equation 2.4, over H_1 we have

$$\begin{aligned} \mu_1 &= (\phi_1+2)\beta_1 + \Phi_1\beta_2 \\ &= (x_1+2)\beta_1 + \Phi_1 \cdot (-\beta_1 - \beta_3) \\ &= (2 + x_1 - \Phi_1)\beta_1 - \Phi_1\beta_3 \end{aligned}$$

and over H_3 we have

$$\begin{aligned} \mu_1 &= \mu_3 - 3\beta_3 \\ &= (\phi_3+2)\beta_3 + \Phi_3\beta_1 - 3\beta_3 \\ &= (2-x_2)\beta_3 + \Phi_3\beta_1 - 3\beta_3 \\ &= \Phi_3\beta_1 + (-1-x_2)\beta_3 \end{aligned}$$

In particular, we can assume that

$$\mu_1 = \Phi_3\beta_1 - \Phi_1\beta_3 \tag{2.5}$$

everywhere in a sufficiently small neighborhood of $H_1 \cup -H_3$.

Lemma 2.7. *The 1-form μ_λ defines the standard tight contact structure along $\widehat{Y}_\lambda \cong S^3$.*

Proof. We can see from Equation 2.5 that μ_1 defines (S^3, ξ_{join}) as in [LC20, Section 3.1], hence is tight by [LC20, Lemma 3.1]. □

Consequently, we obtain a symplectic structure on $\nu(H_1 \cup H_2 \cup H_3)$ with three inward-pointing Liouville vector fields. Each primitive μ_λ defines the standard tight contact structure on $\widehat{Y}_\lambda$; therefore this boundary component can be filled in symplectically with (B^4, ω_{std}). The result is a symplectic structure $(\mathbb{CP}^2, \omega)$.

2.4. **Symplectic structure on (X, ω).** The branched covering is a local diffeomorphism away from the ramification locus $\mathcal{R}$. After a perturbation, we can assume that the reducing curve $\gamma \subset \Sigma$ is disjoint from $\mathcal{R}$ and that each compressing disk V_λ intersects it transversely in a finite number of points. Therefore, away from these points, the local structure of (X, ω) is identical to the local structure of the symplectic form constructed in Subsection 2.3.

We have a *symplectic 1-cocycle* $\{\beta_1, \beta_2, \beta_3\}$ and 1-cocycle of functions $\{\phi_1, \phi_2, \phi_3\}$ on $(X, \mathcal{T})$ after pulling back this data from $\mathbb{CP}^2$ via the branched covering. The 1-forms satisfy

$$\beta_\lambda \wedge \beta_{\lambda+1} \geq 0$$

with equality precisely at the ramification points.

In a neighborhood of the central surface Σ, we can assume that locally in $\Sigma \times D^2$ the pieces of the trisection are given by

$$H_\lambda = \{\phi_{\lambda-1} \geq 0, \phi_\lambda = 0\} \qquad Z_\lambda = \{\max(-\phi_{\lambda-1}, \phi_\lambda) \leq 0\}$$

and in particular that Z_λ is locally the sublevel set of $m_\lambda = \max(\phi_\lambda, -\phi_{\lambda-1})$.

We attach $H_\lambda \times I$ to $\Sigma \times D^2$ as follows. Let s be a normal coordinate near ∂H_λ and t a normal coordinate on I. We choose the identification

$$t = \phi_\lambda \qquad s = 1 - \phi_{\lambda-1}$$

The 1-form β_λ extends across H_λ as a closed, nonvanishing 1-form.

We obtain a smooth manifold $\nu(H_1 \cup H_2 \cup H_3)$ with three smooth boundary components as follows. We can extend m_λ across $H_\lambda \times I \cup H_{\lambda-1} \times I$ and let $\widetilde{m}_\lambda$ be some C^0-close smoothing of m_λ that agrees with m_λ outside a tubular neighborhood of Σ. Let $\widehat{Y}_\lambda$ be the level set of $\widetilde{m}_\lambda^{-1}(-\epsilon)$ for some small $\delta' > 0$. We can assume that the smoothing is chosen so that the form

$$\zeta_\lambda = d\phi_\lambda - d\phi_{\lambda-1} = dt + ds$$

coorients $\widehat{Y}_\lambda$ in $\Sigma \times D^2$.

The 2-form

$$\omega = d\phi_\lambda \wedge \beta_\lambda - d\phi_{\lambda-1} \wedge \beta_{\lambda+1}$$

is symplectic away from the ramification locus. This 2-form and its primitive

$$\mu_* = \phi_\lambda \beta_\lambda - \phi_{\lambda-1}\beta_{\lambda+1}$$

are well-defined independent of λ. There exists a C^∞-small perturbation of μ_*, ω, supported near the ramification locus, so that ω is a symplectic form everywhere.

To extend the symplectic form across the handlebodies of the trisection, we define primitives

$$\begin{aligned}\mu_\lambda &= \mu_* + \widetilde{\beta}_\lambda - \widetilde{\beta}_{\lambda-1} \\ &= \mu_* + 2\widetilde{\beta}_\lambda + \widetilde{\beta}_{\lambda+1}\end{aligned}$$

The 1-cocycle relation implies that

$$\mu_\lambda - \mu_{\lambda+1} = 3\widetilde{\beta}_\lambda$$

We can extend μ_λ over H_λ as

$$\mu_\lambda = (t+2)\widetilde{\beta}_\lambda + \Phi(s)\widetilde{\beta}_{\lambda+1} \tag{2.6}$$

and extend $\mu_{\lambda+1}$ over H_λ as

$$\mu_{\lambda+1} = (t-1)\widetilde{\beta}_\lambda + \Phi(s)\widetilde{\beta}_{\lambda+1} \tag{2.7}$$

Define

$$\Phi_\lambda \coloneqq \Phi(1 - \phi_{\lambda-1})$$

The triply-symmetric version of the contact form is

$$\mu_\lambda = \Phi_{\lambda-1}\widetilde{\beta}_\lambda - \Phi_\lambda \widetilde{\beta}_{\lambda-1} \tag{2.8}$$

everywhere in a sufficiently small neighborhood of $H_\lambda \cup -H_{\lambda-1}$.

There exists an inward-pointing Liouville vector field ρ_λ along $\widehat{Y}_\lambda$ satisfying

$$\mu_\lambda = \omega(\rho_\lambda, -)$$

Consequently, we obtain a symplectic structure on $\nu(H_1 \cup H_2 \cup H_3)$ with three strongly concave boundary components. By construction, the induced contact structures are tight and can be symplectically filled in with Weinstein 1-handlebodies. The result is a symplectic structure on (X, ω).

2.4.1. *Local model near singular points.* The 1-forms $\{\beta_\lambda\}$ have Morse singularities at the $2g-2$ singular points on Σ. Using the Morse lemma, we can assume that, up to topological conjugacy, near a singular point they have the form

$$\widetilde{\beta}_1 = xdx - ydy \pm ds \qquad \beta_2 = xdy + ydx \pm ds$$

Then

$$\omega = d\phi_1 \wedge \widetilde{\beta}_1 - d\phi_3 \widetilde{\beta}_2$$

becomes

$$\omega = d\phi_1 \wedge (\beta_1 \mp d\phi_3) - d\phi_3 \wedge (\beta_2 \mp d\phi_2)$$

so

$$\omega \wedge \omega = d\phi_1 \wedge d\phi_3 \wedge \beta_1 \wedge \beta_2 = (x^2 + y^2) d\phi_1 \wedge d\phi_3 \wedge dx \wedge dy$$

which vanishes along the disk normal to the singular point.

In each handlebody H_λ, we can choose a collection of $g-1$ arcs $\{\gamma_{i,\lambda}\}$ transverse to the foliation $\ker(\beta_\lambda)$ whose oriented boundary is $B = \beta^{-1}(0) \subset \Sigma$. Let ϕ_B be a bump function on Σ in an ϵ-neighborhood of B. The 1-form

$$\mu_B = \phi_B f\,(xdy - ydx)$$

on Σ extends to a 1-form on a tubular neighborhood of each arc $\{\gamma_{i,\lambda}\}$ such that

$$\widetilde{\beta}_\lambda \wedge d(\mu_B) > 0$$

along each arc. Therefore, for $\delta > 0$ sufficiently small the 2-form

$$\omega = d\mu_\lambda + \delta d\mu_B$$

is symplectic. (cf. [LC24, Proposition 4.4])

2.5. **Symplectic 1-cocycles on relative 4-manifolds.** Now, let X be a compact, oriented, smooth 4-manifold with boundary and let $\mathcal{T}$ be a relative trisection of X. Each 3-dimensional sector H_λ is a compression body, whose boundary decomposes as

$$\partial H_\lambda = \Sigma \cup_B P_\lambda$$

where the central surface Σ is a compact surface with boundary, the multicurve $B = \partial\Sigma$ is the binding of an open book decomposition of the boundary ∂X, and P_λ is a page of the open book decomposition on the boundary.

2.5.1. *Corner coordinates.* The 3-dimensional sector H_λ is a manifold with corners along B. Condition (3) in Definition 2.1 requires choosing an identification $\tau_\lambda : \nu(\Sigma) \cong [-\delta, 0] \times \Sigma$ of a collar neighborhood with a trivial fibration. However, near the corner we do not want to restrict this identification to agree in the obvious way with the local structure of a manifold with corners.

First, let $(\widetilde{r}, \theta, s)$ be standard corner coordinates near B: specifically, let θ be a coordinate along B, let $\widetilde{r}$ be a collar coordinate of B in Σ and let s be a collar coordinate of Σ in H_λ, so that

$$\Sigma = \{s = 0; \widetilde{r} \leq 0\} \qquad P_\lambda = \{s \leq 0; \widetilde{r} = 0\} \qquad B = \{\widetilde{r} = s = 0\}$$

Let $\widehat{H}_\lambda = H_\lambda \cup P_\lambda \times [0, \infty)$ and $\widehat{\Sigma} = \Sigma \cup B \times [0, \infty)$ be cylindrical extension of H_λ and Σ.

Definition 2.8. A trivialization $\tau : \nu(\widehat{\Sigma}) \cong [-\delta, 0] \times \widehat{\Sigma}$ is *boundary-admissible* if there exists some function $g : B \to \mathbb{R}$ such that

$$\pi : \nu(\widehat{\Sigma}) \to \widehat{\Sigma} = (\widetilde{r} - g(\theta)s^2, \theta)$$

near B.

If τ is a boundary-admissible collar trivialization, we define

$$r = \widetilde{r} - g(\theta)s^2$$

and we will work with respect to the new collar coordinates (r, θ, s) near the binding B.

2.5.2. *Symplectic 1-cocycles.*

Definition 2.9. A *symplectic 1-cocycle* on $(X, \mathcal{T})$ is a triple $\widetilde{\beta} = (\widetilde{\beta}_1, \widetilde{\beta}_2, \widetilde{\beta}_3)$ of closed, nonvanishing 1-forms on the handlebody sectors of the trisections, such that the restrictions $\beta_\lambda = \widetilde{\beta}_\lambda|_\Sigma$ satisfy the following conditions:

(1) (1-cocycle condition)
$$\beta_1 + \beta_2 + \beta_3 = 0$$
(2) (pairwise positivity)
$$\beta_\lambda \wedge \beta_{\lambda+1} \geq 0$$
(3) all three 1-forms vanish at the same m points $b_1, \ldots, b_m$ of Σ and there exists a smooth function $f : \Sigma \to \mathbb{R}$ and neighborhoods $\{b_i \in U_i\}$ such that
   (a) $f|_{U_i} = \pm 1$ for each $i = 1, \ldots, m$,
   (b) $f = 0$ near $\partial\Sigma$, and
   (c) $\widetilde{\beta}_\lambda = \pi^*(\beta_\lambda) + d(s\,\pi^*(f))$ on a collar neighborhood of $\partial H_\lambda = \Sigma$, where
   $$\tau_\lambda : \nu(\widehat{\Sigma}) \cong [-\delta, 0] \times \widehat{\Sigma}$$
   is a boundary-admissible trivialization of a collar neighborhood and s, π are the projections onto the factors.
(4) the restriction $\beta_\lambda|_B$ is the differential of a Morse function on the binding B,
(5) the restriction $\widetilde{\beta}_\lambda|_{P_\lambda}$ is locally Morse,

2.5.3. *Symplectic structures.* As in the closed case, a symplectic 1-cocycle determines the germ of a symplectic 1-cocycle on the spine $\mathcal{S} = H_1 \cup H_2 \cup H_3$ of a relative trisection.

Recall that since $(X, \mathcal{T})$ is a relative trisection, the tubular neighborhood $\nu(\mathcal{S})$ is a manifold with corners. Its boundary can be partitioned into the *exterior boundary* $\nu_{\partial X}(\partial \mathcal{S})$, which is a tubular neighborhood of $\partial \mathcal{S}$ in the exterior boundary of X, and an *internal boundary* consisting of three subsets Y_1, Y_2, Y_3. There are homeomorphisms

$$Y_\lambda^* = H_\lambda \cup -H_{\lambda-1} \qquad \partial Y_\lambda^* = S_\lambda = P_\lambda \cup -P_{\lambda-1}$$

Repeating the proof of Theorem 2.2 from [LC24], we deduce

Theorem 2.10. *Let $\widetilde{\beta}$ be a symplectic 1-cocycle on $(X, \mathcal{T})$. There exists*

(1) a symplectic structure $(\nu(S), \omega_{\widetilde{\beta}})$ on a neighborhood of the spine, and

(2) for each $\lambda = 1, 2, 3$, a Liouville vector field ρ_λ for ω that is transverse to the interior boundary component $Y_\lambda^* = H_\lambda \cup -H_{\lambda+1}$.

To extend Definition 2.3, we only require tightness along the interior boundary of $\nu(\mathcal{S})$.

Definition 2.11. A symplectic 1-cocycle $\widetilde{\beta}$ on a relative trisection $(X, \mathcal{T})$ is *tight* if each contact structure $(Y_\lambda^*, \ker(\omega(\rho_\lambda, -)))$ is tight.

One simple construction of tight symplectic 1-cocycles on relative trisections is via restriction to a codimension-0 submanifold.

Proposition 2.12. *Let $(X, \mathcal{T})$ be a (relative) trisection of a 4-manifold and let $\widetilde{\beta}$ be a tight symplectic 1-cocycle. Suppose that $X' \subset X$ is a codimenion-0 submanifold such that $\mathcal{T}$ restricts to a relative trisection $(X', \mathcal{T}')$. Then, after possibly a C^∞-small perturbation of X' near the boundary spine $\mathcal{S}'_\partial$ of $\mathcal{T}'$, the restriction $\widetilde{\beta}'$ of $\widetilde{\beta}$ to $(X', \mathcal{T}')$ is a tight symplectic 1-cocycle.*

Proof. Conditions (1),(2), and (3ab) in Definition 2.9 are immediate. Conditions (4) and (5) can be achieved by a generic perturbation and the boundary-admissible trivialization for (3c) can be achieved by an isotopy of P'_λ in an arbitrary neighborhood of its boundary. Finally, the restriction of a tight contact structure to a codimension-0 submanifold remains tight. □

Theorem 2.13. *Let $M \subset (X, \omega)$ be a compact, codimenion-0 submanifold of a closed symplectic 4-manifold. Let $(M, \mathcal{T})$ be any relative trisection. Then there exists*

(1) a trisection $(X, \mathcal{T}')$ such that $(M, \mathcal{T}) \subset (X, \mathcal{T}')$, and

(2) a tight, symplectic 1-cocycle $\widetilde{\beta}'$ on $(X, \mathcal{T}')$

Consequently, there exists a tight symplectic 1-cocycle $\widetilde{\beta}$ with respect to $(M, \mathcal{T})$.

Proof. First, choose some trisection $(X \setminus M, \mathcal{T}^\circ)$ of the complement of M that induces the mirror open book decomposition on the boundary $\partial(X \smallsetminus M) \cong -\partial M$. We can then glue these together to obtain some trisection $(X, \widehat{\mathcal{T}})$ such that $(M, \mathcal{T}) \subset (X, \widehat{\mathcal{T}})$. Second, we can find some trisection $(X, \mathcal{T}_s)$ that admits a tight symplectic 1-cocycle. There is a common stabilization $(X, \mathcal{T}')$ of $\widehat{\mathcal{T}}$ and $\mathcal{T}_s$. Stabilizations can be performed in the complement of $(M, \mathcal{T})$, hence we still have a stratified embedding $(M, \mathcal{T}) \subset (X, \mathcal{T}')$. Moreover, $(X, \mathcal{T}')$ also admits a tight symplectic 1-cocycle.

The theorem now follows from the previous proposition. □

2.6. **Pseudoconvex spine.** In order to construct (X, ω) with strongly convex boundary, it helps to add some boundary conditions to a symplectic 1-cocycle on a relative trisection. In our setting, we can choose a compatible almost-complex structure J on $(\nu(S), \omega)$ so that the closed 1-form $\widetilde{\beta}_\lambda$ induces a foliation on H_λ of J-holomorphic leaves. We therefore demand that this foliation is appropriately 'pseudoconvex' near $P_\lambda = \partial H_\lambda \cap \partial X$.

Let X be a complex manifold. Recall that a domain $G \subset X$ is *pseudoconvex* if ∂G is locally defined by a plurisubharmonic function. One consequence is that G satisfies the *Continuity Principle*. Specifically, consider a smooth family $u_t : \mathbb{D} \times [0, 1] \to X$ of holomorphic disks. Suppose that

(1) $u(\partial\mathbb{D}\times[0,1])\subset\partial G$, and
(2) $u(\mathbb{D}\times\{0\})\subset G$

Then the image of each holomorphic disk u_t is contained in G for all $t\in[0,1]$.

We can mimic this property as follows. Let $p\in P_\lambda$ be an interior point. Fix a small neighborhood $U_p\subset H_\lambda$ and an embedding $\phi_p: U_p\to\mathbb{C}\times\mathbb{R}$, with coordinates (w,t) and $\phi_p(p)=(0,0)$, such that

$$\widetilde{\beta}_\lambda=\phi_p^*(dt)$$

Definition 2.14. We say that a symplectic 1-cocycle $(\mathcal{T},\widetilde{\beta})$ on $(X,\mathcal{T})$ is *pseudoconvex at p in P_λ* if

(1) p is an interior point and for U_p sufficiently small, the set $\phi_p(U_p)\cap(\mathbb{C}\times\{t\})$ is polynomially convex (i.e. each connected component is simply-connected) or empty for all $t\in\mathbb{R}$.
(2) p is a boundary point in B and $g_\lambda(p)>0$, where $g_\lambda: B\to\mathbb{R}$ is the function induced by the boundary-admissible collar trivialization of Σ near B (Definition 2.8).

Furthermore, we say that $(\mathcal{T},\beta)$ has a *pseudoconvex spine* if it is pseudoconvex at every point in P_λ for $\lambda=1,2,3$.

2.6.1. *Pseudoconvexity and cooriented singularities.* We can give a simple reinterpretation of pseudoconvexity in terms of the cooriented Morse singularities of $\widetilde{\beta}_\lambda$ along P_λ. The 3-dimensional 1-form $\widetilde{\beta}_\lambda$ coorients each Morse singularity of $\widetilde{\beta}_\lambda$ along P_λ, with respect the boundary orientation induced on $P_\lambda\subset\partial H_\lambda$.

Lemma 2.15. *A symplectic 1-cocycle is pseudoconvex at an interior point $p\in P_\lambda$ unless p is either (a) a negatively cooriented local maximum, or (b) a positively cooriented local minimum.*

Proof. First, suppose that $p\in P_\lambda$ is an interior point. If p is not a Morse singularity, then $\widetilde{\beta}$ is pseudoconvex at p. Moreover, if p is a saddle singularity, then $\widetilde{\beta}$ is also pseudoconvex at p. Finally, suppose that p is a local extremum. Via the Morse Lemma, we can choose local coordinates (x,y,z) and a defining function $\widetilde{r}$ such that

$$\widetilde{\beta}_\lambda=dz \qquad \widetilde{r}=\sigma_z z+\sigma(x^2+y^2)$$
$$P_\lambda=\widetilde{r}^{-1}(0) \qquad H_\lambda=\widetilde{r}^{-1}((-\infty,0])$$

where $\sigma,\sigma_z\in\{\pm1\}$. The singularity p is pseudoconvex if and only if $\sigma=\sigma_z$. If $\sigma=\sigma_z=+1$, then it is a positively cooriented local maximum, while if $\sigma=\sigma_z=-1$, then it is a negatively cooriented local minimum. □

Lemma 2.16. *Let $(\mathcal{T},\widetilde{\beta})$ be a symplectic 1-cocycle such that all local maxima of $\widetilde{\beta}_\lambda$ in P_λ are positively cooriented and all local minima are negatively cooriented. There exists an isotopy of $\widetilde{\beta}$, supported in a neighborhood of B, to a pseudoconvex symplectic 1-cocycle.*

Proof. By the previous lemma, $\widetilde{\beta}$ is pseudoconvex at the interior points of each P_λ; therefore it suffices to check at the boundary points. Again, we can choose local coordinates (x,y,z) and a defining function $\widetilde{r}$ such that

$$B=\{\widetilde{r}=y=0\} \qquad P_\lambda=\{\widetilde{r}=0;y\le0\}$$
$$H_\lambda=\{\widetilde{r}\le0;y\le0\} \qquad \widetilde{\beta}_\lambda=dz$$

Moreover, since we have a boundary-admissible collar trivialization, we there are functions g,r such that

$$\widetilde{r}=r+gy^2$$

If p is not a singularity of $\beta_\lambda|_B$, then we can locally isotope P_λ rel boundary until $g>0$. Pulling $\widetilde{\beta}_\lambda$ back by a corresponding ambient isotopy, we obtain a 1-parameter family of symplectic 1-cocycles. Finally, suppose that p is a singular point of $\beta_\lambda|_B$. Then via the Morse Lemma we can assume

$$\widetilde{r}=\sigma_z z+\sigma_x x^2+\sigma_y y^2$$

for $\sigma_x,\sigma_y,\sigma_z\in\{\pm1\}$. The symplectic 1-cocycle $\widetilde{\beta}$ is pseudoconvex at p if and only if $\sigma_y=\sigma_z$. If $\sigma_x=\sigma_y$ as well, then p is a local extremum; if $\sigma_z=+1$, then it is a positively cooriented local maximum, while if

$\sigma_z = -1$, then it is a negatively cooriented local minimum. However, if $\sigma_x \neq \sigma_y$, then p is a boundary saddle point and not a local extremum.

Suppose instead that $\sigma_y \neq \sigma_z$. If σ_x, σ_y are equal, then p is local extremum. Therefore

$$\widetilde{r} = \sigma_z(z - x^2 - y^2)$$

Consequently, if $\sigma_z = +1$, then p is a positively cooriented local minimum, while if $\sigma_z = -1$, then p is a negatively cooriented local maximum.

If σ_x and σ_y are not equal, (so that $\sigma_x = \sigma_z$), then p is a boundary saddle point. By a local isotopy of P_λ rel boundary, we can introduce an interior sadddle point and modify p to be a local extremum. We now have $\sigma_y = \sigma_x = \sigma_z$, and therefore $\widetilde{\beta}$ is pseudoconvex at p. □

2.7. **Darboux ball.** To illustrate the construction, we start with an introductory example where the reducing curve B is contractible in Σ. This corresponds to a Darboux ball.

We can locally fix coordinates $\widetilde{\theta}_3, \widetilde{\theta}_1$ such that

$$\omega = dx_1 \wedge d\widetilde{\theta}_1 - dx_2 \wedge d\widetilde{\theta}_3$$

so that

$$\omega \wedge \omega = 2dx_1 \wedge dx_2 \wedge d\widetilde{\theta}_1 \wedge d\widetilde{\theta}_3$$

is a positive volume form on the 4-manifold. Therefore if $dx_1 \wedge dx_2$ orients the normal bundle to Σ, the 2-form $\Omega = d\widetilde{\theta}_1 \wedge d\widetilde{\theta}_3$ orients Σ.

Let

$$r = \widetilde{\theta}_1^2 + \widetilde{\theta}_3^2 - \epsilon^2$$

and let $\gamma = r^{-1}(0)$ be a contractible reducing curve that is a circle of radius $\epsilon > 0$. Then dr determines the coorientation on the 3-sphere.

Let z be an angular coordinate on γ such that

$$\widetilde{\theta}_3 = -\epsilon\cos(z) \qquad \widetilde{\theta}_1 = \epsilon\sin(z)$$
$$d\widetilde{\theta}_3 = \epsilon\sin(z)dz \qquad d\widetilde{\theta}_1 = \epsilon\cos(z)dz$$

Then

$$\epsilon^2 dz = \widetilde{\theta}_1 d\widetilde{\theta}_3 - \widetilde{\theta}_3 d\widetilde{\theta}_1$$

and so

$$\begin{aligned} dr \wedge \epsilon^2 dz &= 2(\widetilde{\theta}_1 d\widetilde{\theta}_1 + \widetilde{\theta}_3 d\widetilde{\theta}_3) \wedge \epsilon^2(\widetilde{\theta}_1 d\widetilde{\theta}_3 - \widetilde{\theta}_3 d\widetilde{\theta}_1) \\ &= 4\epsilon^2(\widetilde{\theta}_1^2 + \widetilde{\theta}_3^2) d\widetilde{\theta}_1 \wedge d\widetilde{\theta}_3 \\ &= 4\epsilon^4 \Omega \end{aligned}$$

which is the positive area form on Σ. Consequently, letting dr coorient the hypersurface, we have that

$$dvol_{S^3} = dz \wedge dx_1 \wedge dx_2$$

is the induced orientation on the 3-sphere.

We now restrict to the solid torus $S^1 \times D^2 = \nu(\gamma)$, with z an angular coordinate on S^1 and (x_1, x_2) rectangular coordinates on D^2. Then

$$\omega = dx_1 \wedge (\epsilon\sin(z)dz) + dx_2 \wedge (\epsilon\cos(z)dz)$$

The kernel line field $\Lambda = \ker(\omega)$ is spanned by the vector field

$$v = \cos(z)\partial_{x_1} - \sin(z)\partial_{x_2}$$

which is tangent to the meridional disks of the solid torus. The 1-form

$$\lambda = \cos(z)dx_1 - \sin(z)dx_2$$

is a contact form satisfying

$$\lambda \wedge \omega = \epsilon(\sin^2(z) + \cos^2(z))dz \wedge dx_1 \wedge dx_2 = \epsilon dvol_{S^3} > 0.$$

3. Induced characteristic foliation

In this section, we explicitly describe the characteristic foliation on the boundary ∂Y_λ^* induced by the Liouville vector field ρ_λ of a symplectic 1-cocycle $\widetilde{\beta}$. Then we specialize to the case where $\widetilde{\beta}$ is pseudoconvex with Legendrian boundary and $\partial Y_\lambda^* = S^2$. We describe how to determine the dividing set $\Gamma_{i,j}$ induced by α_i on V_j and prove that the induced map $\Gamma_{\lambda+1,\lambda} \to \Gamma_{\lambda,\lambda}$ in the contact groupoid is nontrivial.

3.1. **The boundary surface S_λ.** Let $0 < \delta' \ll \delta$ be small positive constants to be determined later. In coordinates $(\phi_\lambda, -\phi_{\lambda-1})$ around the central surface, we let

(1) 2δ be the radius of a tubular neighborhood of the central surface Σ. In particular, we fix an identification
$$\nu(\Sigma) \cong \Sigma \times [-2\delta, 2\delta]^2$$
(2) δ' be the radius of a tubular neighborhood of each handlebody H_λ. In particular, we fix an identification
$$\nu(H_\lambda) \cong H_\lambda \times [-\delta', \delta']$$

Let $m_\lambda = \max(\phi_\lambda, -\phi_{\lambda-1})$ and let $\widetilde{m}_\lambda$ be a C^0-small smoothing. Set
$$Y_\lambda^* \coloneqq \widetilde{m}_\lambda^{-1}(-\delta')$$
We can parametrize the level set $\widetilde{m}_\lambda^{-1}(-\delta')$ by a path $\overline{y}(t) = (y_1(t), y_2(t))$ such that
$$\overline{y}(t) = \begin{cases} (t, -\delta') & \text{for } t < -2\delta' \\ (-\delta', -t) & \text{for } t > 2\delta' \end{cases}$$
and $y_1' \geq 0$ and $y_2' \leq 0$. We also can choose the smoothing $\widetilde{m}_\lambda$ such that $x_1^2 + x_2^2$ has a unique mininum along the level curves. This corresponds to a unique point $\overline{y}(0)$ at which $\overline{y}' \cdot \overline{y} = 0$.

Now we restrict our attention to $S_\lambda = \partial Y_\lambda^*$. We have a decomposition
$$S_\lambda = -\widetilde{P}_{\lambda-1} \cup \mathcal{A}_\lambda \cup \widetilde{P}_\lambda \tag{3.1}$$
where
$$-\widetilde{P}_{\lambda-1} \coloneqq (P_{\lambda-1} \times \{\delta'\}) \smallsetminus \nu(\Sigma) \qquad \mathcal{A}_\lambda \coloneqq S_\lambda \cap \nu(\Sigma) \qquad \widetilde{P}_\lambda \coloneqq (P_\lambda \times \{-\delta'\}) \smallsetminus \nu(\Sigma)$$
In particular, $\mathcal{A}_\lambda$ is the annular region of S_λ in a 2δ-neighborhood of the binding while $\widetilde{P}_{\lambda-1}, \widetilde{P}_\lambda$ are the interiors of the pages.

3.1.1. *Coordinates.* On Σ near $\partial\Sigma$, we have coordinates (r, z) where r is the outward normal to $\partial\Sigma$ in Σ and z is a coordinate along $\partial\Sigma$. Furthermore, we have coordinates (t, z) on the annular region $\mathcal{A}_\lambda$ of S_λ near the binding

3.1.2. *Orientations.* As S_λ is a codimension-2 corner of $\nu(\mathcal{S})$, it naturally sits as the boundary of two distinct 3-dimensional boundary faces. Therefore, the 'conventional' orientation depends on perspective. First, set
$$dr \wedge dz \wedge dx_1 \wedge dx_2$$
as the positive volume form. The parameterization $\overline{y}(t)$ of $\widetilde{m}^{-1}(-\delta')$ above implies that
$$dx_1 \wedge dx_2 \sim dt \wedge d\widetilde{m}_\lambda$$
define the same coorientation on the central surface Σ.

(1) The boundary spine $\nu(\mathcal{S}_\partial)$ eventually lies in the exterior boundary of $\widetilde{B}$ and dr coorients $\partial\widetilde{B}$. Viewing S_λ as part of the *negative* boundary of $\nu(\mathcal{S}_\partial)$, we see that $d\widetilde{m}_\lambda$ coorients S_λ in $\nu(\mathcal{S}_\partial)$. Therefore $dr \wedge d\widetilde{m}_\lambda$ coorients S_λ and $dz \wedge dt$ orients S_λ.
(2) The face $\widetilde{Y}_\lambda^*$ eventually forms part of the interior of the capped spine, which has an inward-pointing Liouville vector field. Therefore $d\widetilde{m}_\lambda$ (co)orients this hypersurface. Viewing S_λ as the positive boundary of $\widetilde{Y}^*$, then dr (co)orients S_λ in $\widetilde{Y}_\lambda^*$. Consequently $d\widetilde{m}_\lambda \wedge dr$ coorients S_λ and $dt \wedge dz$ orients S_λ.

We adopt the first convention throughout this paper as the main technical problem is to construct a contact structure along $\partial\widetilde{B}$ weakly filled by ω.

Remark 3.1. However, our orientation conventions have the unfortunate property that ω is a *negative* area form on P_λ near the convex local maxima of $\widetilde{\beta}_\lambda$ and a *positive* area form on P_λ near the convex local minima of $\widetilde{\beta}_\lambda$.

To see this, choose local coordinates (ϕ_λ, x, y, z) with $\widetilde{\beta}_\lambda = dz$. At a convex local maximum, the surface P_λ is locally defined by

$$r = z + x^2 + y^2$$

We can further assume that

$$\kappa_\lambda = \epsilon(xdy - ydx)$$

so that

$$d\phi_\lambda \wedge \widetilde{\beta}_\lambda \wedge d\kappa_\lambda = \epsilon d\phi_\lambda \wedge dz \wedge dx \wedge dy > 0$$

However,

$$dr \wedge dx \wedge dy = dz \wedge dx \wedge dy$$

Therefore, with $dr \wedge d\phi_\lambda$ coorienting P_λ we see that $\Omega = dy \wedge dx$ orients it and $d\kappa_\lambda = -2\epsilon\Omega$.

Remark 3.2. For some fixed λ, we have that

$$\widetilde{P}_\lambda \subset S_\lambda \cap P_\lambda \times \{-\delta'\} \qquad\qquad -\widetilde{P}_\lambda \subset S_{\lambda+1} \cap P_\lambda \times \{+\delta'\}$$

Note that while we have a clear identification

$$S_\lambda \cap P_\lambda \times \{-\delta'\} \cong S_{\lambda+1} \cap P_\lambda \times \{+\delta'\}$$

(away from the binding), the (co)orientations given by $d\widetilde{m}_\lambda$ and $d\widetilde{m}_{\lambda+1}$ are opposite. Hence the notation $\widetilde{P}_\lambda$ and $-\widetilde{P}_\lambda$ encodes this discrepancy.

3.2. **The characteristic foliation.** We split our description of the characteristic foliation according to the decomposition of Line 3.1 – that is, inside and outside a 2δ-neighborhood of the binding.

First, recall that near the binding we have

$$\mu_\lambda = \Phi_{\lambda-1}\widetilde{\beta}_\lambda - \Phi_\lambda\widetilde{\beta}_{\lambda-1} \tag{3.2}$$

where

$$\Phi_\lambda = \Phi(1 - \phi_{\lambda-1}) \qquad\qquad \Phi(s) = \begin{cases} s & \text{for } s \geq -\delta \\ \epsilon' s & \text{for } s \leq -2\delta \end{cases}$$

Moreover the 1-forms

$$\kappa_\lambda = \Phi_\lambda\widetilde{\beta}_{\lambda-1} \qquad\qquad \kappa_{\lambda-1} = \Phi_{\lambda-1}\widetilde{\beta}_\lambda$$

extend across H_λ and $H_{\lambda-1}$, respectively, are $O(\epsilon')$, and satisfy

$$d\phi_\lambda \wedge \widetilde{\beta}_\lambda \wedge d\kappa_\lambda > 0 \qquad\qquad d\phi_{\lambda-1} \wedge \widetilde{\beta}_{\lambda-1} \wedge d\kappa_{\lambda-1} > 0$$

3.2.1. *Interior of pages.* Along $\widetilde{P}_\lambda \subset \{-\delta'\} \times P_\lambda$ we have

$$\mu_\lambda = (2 - \delta')\widetilde{\beta}_\lambda + \kappa_\lambda$$

and along $\widetilde{P}_{\lambda-1} \subset \{-\delta'\} \times P_{\lambda-1}$, we have

$$\mu_\lambda = (-1 - \delta')\widetilde{\beta}_{\lambda-1} + \kappa_{\lambda-1}$$

where $\kappa_\lambda, \kappa_{\lambda-1}$ are $O(\epsilon')$. Furthermore, as in Remark 3.1, we can assume Morse coordinates (ϕ_λ, x, y, z) near the critical points with

$$\widetilde{\beta}_\lambda = dz \qquad\qquad r = \pm(z + x^2 + \sigma y^2) \qquad\qquad \kappa_\lambda = \epsilon'(xdy - ydx)$$

for $\sigma \in \{\pm 1\}$. Then $\Omega = \mp dy \wedge dx$ locally orients P_λ. For $\sigma \in \{\pm 1\}$ we have

$$\begin{aligned}\mu_\lambda &= (2-\delta')dz + \kappa_\lambda \\ &= (2-\delta')(-2xdx - 2\sigma y dy) + \epsilon'(xdy - ydx) \\ &= (-2(2-\delta')x - \epsilon y)dx + (-2\sigma(2-\delta')y + \epsilon' x)dy\end{aligned}$$

The unique vector field satisfying $\iota_{v_\lambda}\Omega = \mu_\lambda$ is

$$v = (-2(2-\delta')x - \epsilon' y)\partial_y - (-2\sigma(2-\delta')y + \epsilon' x)\partial_x$$

As

$$\widetilde{\beta}_\lambda = -2xdx - 2\sigma y dy$$

along P_λ, we see that

$$\widetilde{\beta}_\lambda(v_\lambda) = \epsilon'(x^2 + \sigma y^2)$$

An analogous computation along $-P_{\lambda-1}$ shows that

$$\widetilde{\beta}_{\lambda-1}(v_\lambda) = \epsilon'(x^2 + \sigma y^2)$$

as well. Summarizing this discussion, we have

Lemma 3.3. *Outside a 2δ-neighborhood of the binding,*

1. *the characteristic foliation can be made C^∞-close to $\ker(\beta_\lambda)$ and $\ker(\beta_{\lambda-1})$,*
2. *there is a 1-1 correspondance between the singular points of μ_λ and the singular points of β_λ and $\beta_{\lambda-1}$ in the interiors of $\widetilde{P}_\lambda$ and $-\widetilde{P}_{\lambda-1}$, respectively.*
3. *In neighborhoods of the local extrema, the characteristic foliation is directed by an upward gradient-like vector field for $\widetilde{\beta}_\lambda$.*

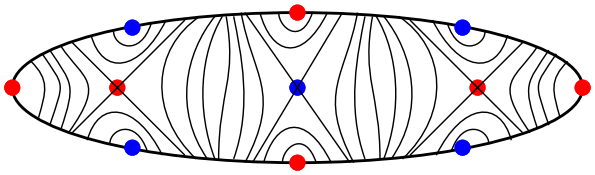


FIGURE 6. An example of a $\widetilde{\beta}_\lambda$ on a compressing disk V_λ when $\widetilde{\beta}$ is pseudoconvex with Legendrian binding. Positive singularities are in red and negative singularities are in blue.

3.2.2. *Comparing $\mu_\lambda, \mu_{\lambda+1}$ along P_λ.* In the subsequent discussion, we will need to compare the foliation induced on $\widetilde{P}_\lambda$ by μ_λ and the foliation induced on $-\widetilde{P}_\lambda$ by $\mu_{\lambda+1}$. In particular, we have

$$\mu_\lambda = (2-\delta')\widetilde{\beta}_\lambda + \kappa_\lambda \qquad\qquad \mu_{\lambda+1} = (-1-\delta')\widetilde{\beta}_\lambda + \kappa_\lambda$$

Note that μ_λ is a perturbation of $+2\widetilde{\beta}_\lambda$ and $\mu_{\lambda+1}$ is a perturbation of $-\widetilde{\beta}_\lambda$. So while both foliations are almost-tangent to the level curves of $\widetilde{\beta}_\lambda$, the vector fields v_λ and $v_{\lambda+1}$ directing these foliations are (almost) antipodal.

Remark 3.4. While we refrain at the moment from making rigorous statements, we can view v_λ as the *right-most* upward gradient vector field and $v_{\lambda-1}$ as the *left-most* upward gradient vector field.

To see this, consider local coordinates $(d\phi_\lambda, x, y, z)$ with $\widetilde{\beta}_\lambda = dz$ and $r = y$. Then we have local coordinates (x, z) on P_λ and $\Omega = dx \wedge dz$ is a positive area form. Therefore

$$\iota_{+\partial_x}\Omega = dz$$

so we can assume v_λ is a perturbation of $+\partial_x$ while $v_{\lambda+1}$ is a perturbation of $-\partial_x$.

3.2.3. *Binding.* We now restrict our attention to a 2δ-neighborhood of the binding and describe the characteristic foliation of $\xi_\lambda = \ker(\mu_\lambda)$ induced on the annular piece $\mathcal{A}_\lambda \subset \partial Y_\lambda^0$. For concretenesss, we take $\lambda = 1$ and $(\phi_\lambda, -\phi_{\lambda-1}) = (x_1, x_2)$ in the following section.

Along the central surface, we have collar coordinates (r, z) near $B = \partial\Sigma$. In this coordinates, we have

$$\beta_1 = p_1 dr + q_1 dz$$
$$\beta_3 = p_3 dr + q_3 dz$$

Define the functions

$$\theta(z) = \arctan(q_3/q_1) \qquad\qquad C(z) = \sqrt{q_1^2 + q_3^2}$$

so that

$$q_1 = C\cos(\theta) \qquad\qquad q_3 = C\sin(\theta)$$

For some small $\epsilon > 0$, define

$$\widetilde{r} = r + g(z)(x_1^2 + x_2^2)$$

and consider the level set $\{\widetilde{r} = 0\}$.

In these coordinates, we have

$$\mu_1 = g(z)\,(\overline{y}' \cdot \overline{y})\,Pdt + Qdz \tag{3.3}$$

where

$$P = \Phi_3 p_1 - \Phi_1 p_3 \qquad\qquad Q = \Phi_3 q_1 - \Phi_1 q_3$$

The term $\overline{y}' \cdot \overline{y}$ is $O(\delta)$, so the characteristic foliation is almost horizontal for δ sufficiently small and we can determine its singularities.

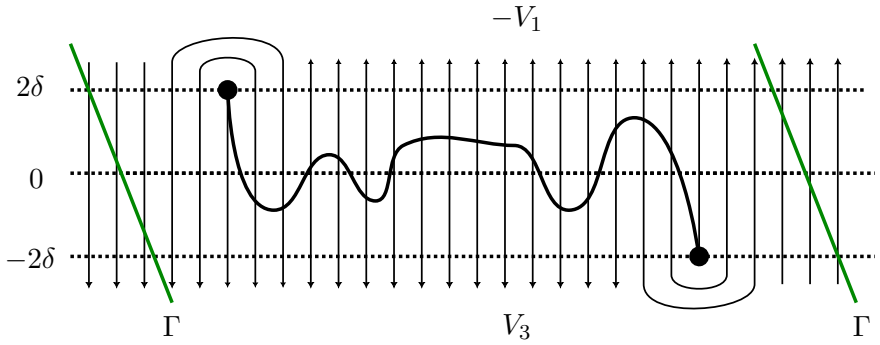


FIGURE 7. The characteristic foliation in a 2δ-neighborhood of the binding. A component of ζ^+ connects a point where $q_3 = 0$ (*bottom right*) to a point where $q_1 = 0$ (*top left*)

Proposition 3.5. *Suppose that $q_1(z) - q_3(z)$ and $g(z)$ are transverse to 0 with no common zeros. Then for δ, δ' sufficiently small, the singularities of the characteristic foliation occur at the common zeros of*

$$Q = 0 \qquad\qquad g \cdot \log(\tan(\theta)) = 0$$

The singularity is elliptic if

$$\frac{d}{dz}\left(g \cdot \log(\tan(\theta))\right) > 0$$

and hyperbolic otherwise. In particular,

1. *for each zero of $\theta = \pi/4$, there exists a unique negative singular point; it is elliptic if $g\theta' > 0$ and hyperbolic otherwise.*
2. *for each zero of $\theta = 5\pi/4$, there exists a unique positive singular point; it is elliptic if $g\theta' > 0$ and hyperbolic otherwise.*

(3) *for each zero of* $g(z)$ *in* $[0, \pi/4]$*, there exists a unique negative singular point; it is elliptic if* $g' > 0$ *and hyperbolic otherwise.*
(4) *for each zero of* $g(z)$ *in* $[\pi/4, \pi/2]$*, there exists a unique negative singular point; it is elliptic if* $g' < 0$ *and hyperbolic otherwise.*
(5) *for each zero of* $g(z)$ *in* $[\pi, 5\pi/4]$*, there exists a unique positive singular point; it is elliptic if* $g' > 0$ *and hyperbolic otherwise.*
(6) *for each zero of* $g(z)$ *in* $[5\pi/4, 3\pi/2]$*, there exists a unique positive singular point; it is elliptic if* $g' < 0$ *and hyperbolic otherwise.*

Proof. The zeros of μ_1 occur when either

$$\overline{y}' \cdot \overline{y} = 0 \tag{3.4}$$

or

$$g(z) = 0 \tag{3.5}$$

or

$$\frac{\Phi_3}{\Phi_1} = \frac{p_3}{p_1} \tag{3.6}$$

and

$$\frac{\Phi_3}{\Phi_1} = \frac{q_3}{q_1} \tag{3.7}$$

First, when $Q = 0$, note that Φ_1, Φ_3 and therefore $\frac{\Phi_3}{\Phi_1}$ and $\frac{q_3}{q_1}$ are all positive; therefore q_3, q_1 have the same sign. This implies that either $0 \leq \theta(z) \leq \frac{\pi}{2}$ or $\pi \leq \theta(z) \leq \frac{3\pi}{2}$. Define

$$\zeta^+ = \theta^{-1}([0, \pi/2]) \qquad \zeta^- = \theta^{-1}([\pi, 3\pi/2])$$

Second, since

$$\beta_1 \wedge \beta_3 = (p_1 q_3 - p_3 q_1) dr \wedge dz > 0$$

and the binding is compact, we get a positive lower bound

$$\frac{p_1}{p_3} - \frac{q_1}{q_3} > M > 0$$

and therefore we cannot simultaneously solve $P = 0$ and $Q = 0$. This implies that the zeros of μ_1 occur at the common zeros of Equation 3.7 and either 3.4 or 3.5.

To compute the sign, we have that

$$\omega = d\mu_1 = -Q_t dz \wedge dt + O(\epsilon')$$

To determine whether a singular point is elliptic or hyperbolic, we consider the vector field $dz \wedge dt$-dual to μ_1, linearize at the singular point, and compute the determinant. If the sign agrees with the sign of ω, then the singular point is elliptic; otherwise it is hyperbolic.

The dual vector field is

$$v = Q\partial_t - \left(g(z)\left(\overline{y}' \cdot \overline{y}\right)\right) P\partial_z$$

When $\overline{y}' \cdot \overline{y} = 0$, the linearization is

$$L = \begin{bmatrix} Q_t & -g(z)\left(\overline{y}' \cdot \overline{y}\right)' P \\ Q_z & 0 \end{bmatrix}$$

whose determinant is

$$det(L) = \epsilon Q_z g(\overline{y}' \cdot \overline{y})' P$$

When $g(z) = 0$, the linearization is

$$L = \begin{bmatrix} Q_t & 0 \\ Q_z & -g'(z)\left(\overline{y}' \cdot \overline{y}\right) P \end{bmatrix}$$

whose determinant is

$$det(L) = -Q_t g'(\overline{y}' \cdot \overline{y})P$$

We have that

$$Q = \Phi_3 q_1 - \Phi_1 q_3 = C\left(\Phi_3 \sin(\theta) - \Phi_1 \cos(\theta)\right)$$

so

$$\begin{aligned} Q_t &= C(\Phi_3' \cdot (y_1' - y_2') \sin(\theta) - \Phi_1' \cdot y_2' \cos(\theta) \\ Q_z &= \theta' C(\Phi_3 \cos(\theta) - \Phi_1 \sin(\theta)) + C'(\Phi_3 \sin(\theta) - \Phi_1 \cos(\theta)) \\ &= \theta' C(\Phi_3 \cos(\theta) + \Phi_1 \sin(\theta)) \end{aligned}$$

when $Q = 0$.

Since $C, \Phi_1', \Phi_3', y_1' > 0 > y_2'$, the sign of Q_t equals the sign of $\cos(\theta)$ (which agrees with the sign of $\sin(\theta)$. Likewise, the sign of Q_z is the sign of $\theta' \cos(\theta)$.

The sign of P is locally constant along the arcs $\theta(\zeta^{\pm})$. To compute this sign, we can investigate it near the endpoints. First, suppose that $\theta(z) \sim 0$, so that $q_3 \sim 0$ and $q_1 > 0$. This implies that $p_3 < 0$ and $\Phi_3 \sim 0$. Therefore

$$\Phi_3 p_1 - \Phi_1 p_3 \sim -\Phi_1 p_3 > 0$$

Conversely, if $\theta(z) \sim \pi$, so that $q_3 \sim 0$ and $q_1 < 0$, then $p_3 < 0$ while we still have $\Phi_3 \sim 0$. Therefore

$$\Phi_3 p_1 - \Phi_1 p_3 \sim -\Phi_1 p_3 < 0$$

In other words, the sign of P is positive along $\theta(\zeta^+)$ and negative along $\theta(\zeta^-)$.

Finally, the function $\overline{y}' \cdot \overline{y}$ is increasing with a unique zero at $t = 0$. If $Q = 0$ and $t < 0$, we have that $\Phi_3 < \Phi_1$ so therefore $q_3 < q_1$; similarly for $Q = 0$ and $t > 0$, we have that $q_3 > q_1$.

□

3.3. **Gradient-like characteristic foliations.** We now assume that the symplectic 1-cocycle is pseudoconvex with Legendrian binding and that V_λ is a topological disk. The 1-form $\widetilde{\beta}_\lambda$ is closed, hence its restriction to V_λ is exact. Therefore, by abuse of terminology we make

Definition 3.6. A characteristic foliation $\mathcal{F}$ on V_λ is *gradient-like* for $\widetilde{\beta}_\lambda$ if there exists a vector field v directing $\mathcal{F}$ such that

$$\widetilde{\beta}_\lambda(v) \geq 0$$

with equality at the critical points of $\widetilde{\beta}_\lambda$.

In this subsection, we replace the foliation $\ker(\alpha_i|_{V_j})$ by a gradient-like foliation $\mathcal{F}_{i,j}$ with the same dividing set $\Gamma_{i,j}$. This simplifies both the construction of the weakly filled contact structure as well as the proof that there is a tight morphism $\Gamma_{\lambda,\lambda} \to \Gamma_{\lambda-1,\lambda}$ in the contact category of the disk.

First, the characteristic foliation is relative simple if $\widetilde{\beta}$ is pseudoconvex with Legendrian binding. The foliation is C^∞-close to the level curves of $\widetilde{\beta}$ along the pages as well as C^∞-close to perpendicular to the binding near the binding. Furthermore, we can exactly describe the singularities of the singular foliation.

Lemma 3.7. *Suppose that the symplectic 1-cocycle $\widetilde{\beta}$ is pseudoconvex with Legendrian binding. Then for the characteristic foliation $\mathcal{F}_\lambda$ induced by α_λ on $S_\lambda = P_\lambda \cup -P_{\lambda+1}$:*

1. *there is a 1-1 correspondence between singular points of $\mathcal{F}_\lambda$ in the interior of P_λ and zeros of $\widetilde{\beta}_\lambda$ in the interior of P_λ,*
2. *there is a 1-1 correspondence between singular points of $\mathcal{F}_\lambda$ in the interior of $-P_{\lambda+1}$ and zeros of $\widetilde{\beta}_{\lambda+1}$ in the interior of $-P_{\lambda+1}$,*
3. *for each component $B_i \subset B = \partial P_\lambda$, there exist*
   (a) *$n_i + k_i$ positive elliptic and k_i positive hyperbolic points*
   (b) *$n_i + l_i$ negative elliptic and l_i negative hyperbolic points*

of $\mathcal{F}_\lambda$ along B_i, where

$$2n_i = \#(\Gamma_\lambda \pitchfork B_i)$$

is the minimal intersection number of the dividing set with B_i.

Proof. We can apply Proposition 3.5 to determine the zeros near the binding. The integer n_i is equal to the number of components of $\zeta^+ = \theta^{-1}([0,\pi/2])$ and the number of components of $\zeta^- = \theta^{-1}([\pi,3\pi/2])$. Since $\widetilde{\beta}$ is pseudoconvex, we have that $g > 0$ and $\theta' > 0$ when $q_1 = 0$ or $q_3 = 0$. This implies that

$$\lim_{\theta\to 0} g\log(\tan(\theta)) = -\infty \qquad \lim_{\theta\to \pi/2} g\log(\tan(\theta)) = +\infty$$

□

Lemma 3.8. *For each fixed neighborhood U of $\zeta^\pm$ and V of the saddle points in the interior of $\widetilde{P}_\lambda$ and $-\widetilde{P}_{\lambda-1}$, there exists a foliation $\mathcal{F}'_\lambda = ker(\tau_\lambda)$ along S_λ such that*

(1) $(\mu_\lambda, \tau_\lambda)$ meet convexly,
(2) $\mathcal{F}'_\lambda$ is gradient-like in $\widetilde{P}_\lambda$ and $-\widetilde{P}_{\lambda-1}$,
(3) there exists some Γ_λ that divides both $\mathcal{F}_\lambda = ker(\mu_\lambda)$ and $\mathcal{F}'_\lambda$, and
(4) $\mathcal{F}'_\lambda$ is C^∞-close to $\pm\partial_z$ in $\mathcal{A}^\pm_\lambda \setminus U$, where $\mathcal{A}^\pm_\lambda$ are the positive and negative components of $\mathcal{A}_\lambda \smallsetminus \Gamma_\lambda$.

Proof. Let (t,z) be coordinates near the binding, let v_λ be a vector field directing the characteristic foliation, let $\phi(t)$ be a bump function that equals 1 on $[-\delta,\delta]$ and supported in $[-2\delta,2\delta]$. Let $\epsilon \gg \epsilon' > 0$ and let u_λ be an $O(\epsilon)$ gradient-like vector field for $\widetilde{\beta}$ tangent to the boundary of $\widetilde{V}_\lambda$ and $-\widetilde{V}_{\lambda+1}$. Define

$$v'_\lambda = v_\lambda + CQ\partial_z + u_\lambda$$

for $C \gg 0$ and define $\mathcal{F}'_\lambda = \ker(v'_\lambda)$. Then

$$\mu_\lambda(v') = CQ^2\phi + \widetilde{\beta}_\lambda(u) + \kappa_\lambda(u) > 0$$

Hence the foliations meet convexly.

Now, if u is $O(\epsilon)$ then it still remains almost tangent to the level curves of $\widetilde{\beta}$ and does not change the characteristic foliation up to topologicaly conjugacy. Moreover, we can assume that the characteristic foliation remains transverse to Γ_λ. Similarly, the vector field $Q\partial_z$ is positively transverse to Γ so v' is transverse to Γ_λ.

The foliation $\ker(\tau_\lambda)$ is transverse to Γ_λ, which divides $\ker(\mu_\lambda)$. It will divide $\ker(\tau_\lambda)$ as well if there are no periodic orbits. There cannot be any periodic orbits of this foliation: there are none contained in the interior of $\widetilde{V}_\lambda$ or $-\widetilde{V}_{\lambda-1}$ because the flow is transverse to the level curves of a Morse function. Moreover, every flowline that intersects the annular $\mathcal{A}_\lambda$ region will have either its α- or ω-limit one of the elliptic points contained in $\mathcal{A}_\lambda$, as the flow is almost tangent to $\ker(\partial_z)$. □

Proposition 3.9. *The strongly minimally-twisting contact structure $(S^2 \times [0,1], \xi)$ determined by the convex pair $\mu_\lambda, \tau_\lambda$ is tight.*

Proof. This follows by coming the previous lemma with Corollary 10.12. □

3.4. **The Giroux graph of $\mathcal{F}'_\lambda$.** We now return to Remark 3.4 and make it precise, in order to describe the Giroux graph of the foliation $\mathcal{F}'_\lambda$ on S_λ.

Let β be a closed 1-form on a surface and let h be a saddle point. In Morse coordinates, we can assume that $\beta = d(xy)$. Denote

$$X_\pm = [0,\pm\infty) \times \{0\} \qquad Y_\pm = \{0\} \times [0,\pm\infty)$$

Then

(1) the *right unstable separatrices* of the upward flow are $X_\pm$
(2) the *left unstable separatrices* of the upward flow are $Y_\pm$
(3) the *right stable separatrices* of the upward flow are $Y_\pm$
(4) the *left stable separatrices* of the upward flow are $X_\pm$.

Definition 3.10. Let $\widetilde{\beta}_\lambda$ be a closed 1-form on the disk $\widetilde{V}_\lambda$ in Roussarie-Thurston general position (i.e. no interior maxima or minima)

(1) a *right* unstable separatrix R_u consists of flowing along a right unstable separatrix of h until hitting the boundary $\partial\widetilde{V}_\lambda$, then turning left and flowing upward along the boundary to the maximum point.
(2) a *right* stable separatrix R_s consists of flowing (backwards) along a right stable separatrix of h until hitting the boundary $\partial\widetilde{V}_\lambda$, then turning left and flowing downward along the boundary to the minimum point.
(3) a *left* unstable separatrix L_u consists of flowing along a left unstable separatrix of h until hitting the boundary $\partial\widetilde{V}_\lambda$, then turning right and flowing upward along the boundary to the maximum point.
(4) a *left* stable separatrix L_s consists of flowing (backwards) along a left stable separatrix of h until hitting the boundary $\partial\widetilde{V}_\lambda$, then turning right and flowing downward along the boundary to the minimum point.

Lemma 3.11. *For the foliation $\mathcal{F}'_\lambda$ on S_λ:*

(1) Over $\widetilde{V}_\lambda$, the separatrices of a hyperbolic point are C^0-close to the right separatrices of $\widetilde{\beta}_\lambda$.
(2) Over $-\widetilde{V}_{\lambda-1}$, the separatrices of a hyperbolic point are C^0-close to
  (a) the right separatrices of $\widetilde{\beta}_{\lambda-1}$, with respect to the (negative) orientation on $\widetilde{V}_\lambda$ induced by S_λ, and
  (b) the left separatrices of $\widetilde{\beta}_{\lambda-1}$, with respect to the positive orientation on $\widetilde{V}_\lambda$

Proof. Near a hyperbolic point, we can see that the flow is close to the right separatrices. Then as it reaches the boundary, it turns to be almost-tangent to the boundary of $\widetilde{V}_\lambda$. □

Summarizing, we have the following qualitative description of $\mathcal{F}_\lambda, \mathcal{F}_{\lambda+1}$ along $\widetilde{V}_\lambda$:

(1) $\mathcal{F}_\lambda$ is the 'right-most' gradient-like vector field for $\widetilde{\beta}_\lambda$ on $\widetilde{V}_\lambda$: the upward flow leaves a hyperbolic point to the right, hugging the separatrix until it gets close to the boundary then hugs the boundary until is reaches a maximum.
(2) $\mathcal{F}_{\lambda+1}$ is the 'left-most' gradient-like vector field for $\widetilde{\beta}_\lambda$ on $\widetilde{V}_\lambda$: the upward flow leaves a hyperbolic point to the left, hugging the separatrix until is gets close to the boundary then hugs the boundary until it reaches a maximum.

Let $\mathcal{F}'_{\lambda,\lambda}$ denote the restriction of $\mathcal{F}'_\lambda$ to $\widetilde{V}_\lambda$ and $\mathcal{F}'_{\lambda+1,\lambda}$ the restriction of $\overline{\mathcal{F}}'_{\lambda+1}$ to $\widetilde{V}_\lambda$.

Lemma 3.12. *The pair $\mathcal{F}'_{\lambda,\lambda}, \mathcal{F}'_{\lambda+1,\lambda}$ meet convexly along $\widetilde{V}_\lambda$.*

Proof. Both foliations are both gradient-like and positively transverse to $\ker(\widetilde{\beta}_\lambda)$. In the interior of $\widetilde{V}_\lambda$ they are small perturbations of antipodal vector fields, therefore they never agree in the interior. Along the boundary, they are both almost-tangent to the boundary, except that $\mathcal{F}_{\lambda,\lambda}$ is always moving slightly 'right' (with respect to the upward orientation on $\ker(\widetilde{\beta}_\lambda)$ while $\mathcal{F}_{\lambda+1,\lambda}$ is always moving slightly 'left'. □

Furthermore, we can describe the dividing set $\Gamma_{\lambda,\lambda}$ along $\widetilde{V}_\lambda$ as follows,

(1) for each negative hyperbolic point in the interior, take the boundary of neighborhood of the 'right' separatrices
(2) if two separatrices hit the same ascending edge (i.e between a maximum and minimum of $\widetilde{\beta}_\lambda$) of $\partial\widetilde{V}_\lambda$, join the dividing set between them,
(3) for each ascending edge, connect the upper edge of the dividing set on highest negative separatrix to the boundary just past the maximum point.

See Figure 8. The dividing set $\Gamma_{\lambda+1,\lambda}$ along $\widetilde{V}_\lambda$ can be determined similarly, except using the 'left' separatrices.

3.5. **The morphism $\Gamma_{\lambda+1,\lambda} \to \Gamma_{\lambda,\lambda}$.** By Lemma 3.12, the pair $\mathcal{F}'_{\lambda,\lambda}, \mathcal{F}'_{\lambda+1,\lambda}$ meet convexly along $\widetilde{V}_\lambda$, hence there is a strongly minimally twisting contact structure $(D^2 \times [0,1], \xi_{\lambda+1,\lambda})$ interpolating betwen them and a corresponding morphism $\Gamma_{\lambda+1,\lambda} \to \Gamma_{\lambda,\lambda}$ in the contact category of the disk. In this subsection, we establish

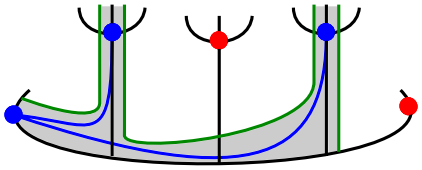
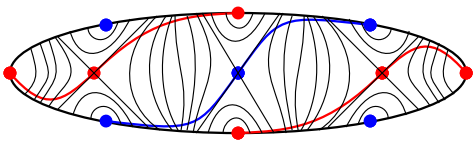
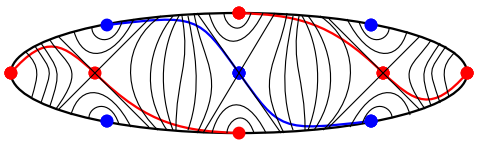


FIGURE 8. To determine the dividing set $\Gamma_{\lambda,\lambda}$, take pushoffs of the 'right' unstable separatrices of the negative hyperbolic points in the interior and then connect them along the boundary of V_λ.

that this contact structure is tight and given an explicit presentation in terms of bypass attachments. We assume that V_λ is in Roussarie-Thurston general position (Definition 5.2).



(A) The Giroux graph of the 'right-most' gradient-like foliation $\mathcal{F}_{\lambda,\lambda}$.



(B) The Giroux graph of the 'left-most' gradient-like foliation $\mathcal{F}_{\lambda+1,\lambda}$.



(C) The change in dividing set from $\Gamma_{\lambda+1,\lambda}$ (*left*) to $\Gamma_{\lambda,\lambda}$ (*right*), along with the two bypass arcs generating the elementary contact cobordism.

FIGURE 9. The two characteristic foliations $\mathcal{F}_{\lambda+1,\lambda}$ and $\mathcal{F}_{\lambda,\lambda}$ for the symplectic 1-cocycle in Figure 6 and their corresponding dividing sets.

Recall that a morphism $\Gamma \xrightarrow{I} \Gamma'$ is *elementary* if it can be realized by attaching bypasses along a set I of disjoint bypass arcs on Γ. Moreover, an elementary morphism is tight if and only if no subset P of I forms a *pinwheel* [HKM05]. Note that if there does exist a pinwheel, then we can factor the morphism as

$$\Gamma \xrightarrow{P} \Gamma_{OT} \xrightarrow{I \smallsetminus P} \Gamma'$$

where Γ_{OT} contains a contractible component.

Proposition 3.13. *Suppose that $\widetilde{\beta}$ is pseudoconvex with Legendrian binding and $(tb, rot) = (-n, r)$ and suppose that V_λ is in Roussarie-Thurston general position.*

(1) the contact structure $(D^2 \times [0,1], \xi_{\lambda+1,\lambda})$ is elementary and generated by a set I_λ of at most $n-1-|r|$ disjoint bypasses,

(2) there are no pinwheels formed by the bypasses in I_λ

Therefore, the contact structure $(D^2 \times [0,1], \xi_{\lambda+1,\lambda})$ is tight and there are nonzero morphisms

$$\Gamma_{\lambda+1,\lambda} \to \Gamma_{\lambda,\lambda} \qquad\qquad \mathcal{F}'_{\lambda+1,\lambda} \to \mathcal{F}_{\lambda,\lambda}$$

is the contact category $\mathcal{C}(D^2, n, r)$ and strong contact category $\mathcal{SC}(D^2, n, r)$, respectively.

Proof. Since $\widetilde{\beta}$ is pseudoconvex with Legendrian binding, there exists $2n$ boundary center points along the boundary; if V_λ is in Roussarie-Thurston general position, then there are no interior center points and therefore for Euler characteristic reasons there must be exactly $n-1$ saddle points. We will construct a bypass sequence consisting of at most one bypass for each unstable separatrix of each negative saddle point; by a symmetric argument, there exists a bypass sequence consisting of at most one bypass for each stable separatrix of each positive saddle point. Letting s_+, s_- denote the number of positive and negative saddles, respectively, we have

$$\begin{aligned} n &= s_+ + s_- + 1 & n - 1 - r &= 2s_- \\ r &= s_+ - s_- & n - 1 + r &= 2s_+ \\ & & n - 1 - |r| &= \min(2s_-, 2s_+) \end{aligned}$$

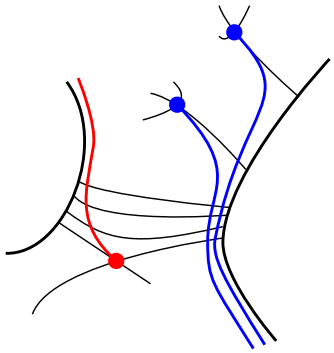
We can obtain a bypass sequence in the following way. Take an unstable separatrix of the left-most upward flow for some negative hyperbolic point h. There is a leaf of $\ker(\widetilde{\beta}_\lambda)$ just above h_- that intersects this separatrix near the boundary ∂V_λ. At the other end of the leaf, it intersects several stable separatrices of the flow that limit to some positive hyperbolic points $h_{+,1}, \dots, h_{+,k}$. See Figure 10a. By handlesliding h_- over each of these positive saddle points, we get a sequence of k bypass moves. However, each bypass is a rotation of the successive bypass. See Figure 10b. Therefore, the entire sequence can be achieved just by adding this last bypass. Of course, if the leaf of $\ker(\widetilde{\beta}_\lambda)$ does not intersect the positive Giroux graph, then no bypass is required. Moreover, as there is at most one bypass per separatrix and these are all disjoint, we can choose the bypass arcs disjointly and the morphism is elementary.

Now, assume that V_λ is in Roussarie-Thurston general position, so that there are no center singularities of $\widetilde{\beta}_\lambda$ in the interior of V_λ; equivalently, there are no elliptic singularities of the characteristic foliation in the interior, as they all lie along the boundary of V_λ. For any subset $P \subset I_\lambda$, we can perform the bypasses in P. If the bypasses in P form a pinwheel, then after performing the handle slides, the Giroux graph of the corresponding foliation has a connected component completely contained in the interior of V_λ. This component must have at least one elliptic singularity, which is a contradiction. □

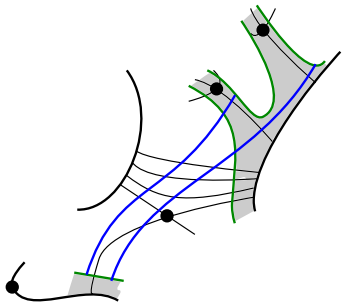
Proposition 3.14. *There exists foliations $\mathcal{F}''_{\lambda,\lambda}$ and $\mathcal{F}''_{\lambda+1,\lambda}$ on $\widetilde{V}_\lambda$ such that*

1. *$\mu_\lambda, \mathcal{F}''_{\lambda,\lambda}$ meet convexly on $\widetilde{V}_\lambda$ and the corresponding strongly minimally-twisting contact structure is tight,*
2. *$\mu_{\lambda+1}, \mathcal{F}''_{\lambda+1,\lambda}$ meet convexly on $-\widetilde{V}_\lambda$ and the corresponding strongly minimally-twisting contact structure is tight, and*
3. *the morphism $\eta_\lambda : \mathcal{F}''_{\lambda+1,\lambda} \to \mathcal{F}''_{\lambda,\lambda}$ in the strong contact category can be realized as a sequence of elementary handleslides.*

Proof. Take $\mathcal{F}''_{\lambda+1,\lambda} = \mathcal{F}'_{\lambda+1,\lambda}$. Then we can realize the morphism $\Gamma_{\lambda+1,\lambda} \to \Gamma_{\lambda,\lambda}$ by a sequence of handle slides along arcs tangent to $\ker(\widetilde{\beta}_\lambda)$. Let $\mathcal{F}''_{\lambda,\lambda}$ be the resulting gradient-like foliation; it is topologically isotopic through gradient-like foliations to $\mathcal{F}'_{\lambda,\lambda}$ so that $\mathcal{F}'_{\lambda,\lambda}, \mathcal{F}''_{\lambda,\lambda}$ meet convexly and therefore by transitivity the pairs $\mu_\lambda, \mathcal{F}''_{\lambda,\lambda}$ meet convexly as well. □



(A) For each stable separatrix of a positive hyperbolic point h_+, there is a bypass handleslide for each unstable separatrix of a negative hyperbolic point that intersects the same leaves of $\widetilde{\beta}_\lambda$ just above h_+. The bypass arcs can be assumed tangent to the foliation $\ker(\widetilde{\beta}_\lambda)$



(B) The corresponding bypass arcs with respect to the dividing set. Note that the left bypass arc is a rotation of the right bypass arc.

FIGURE 10. The bypasses corresponding to a stable separatrix of a positive hyperbolic point in the interior of V_λ.

4. Reducible trisections

4.1. **Trisections of spheres.** The n-sphere has a standard trisection, obtained by pulling back a trisection on the unit disk D in $\mathbb{R}^2$.

Definition 4.1. The *standard trisection* $D = D_1^s \cup D_2^s \cup D_3^s \subset \mathbb{R}^2$ of the unit disk is defined by choosing the following subsets (in radial coordinates):

$$D_1^s = \left\{0 \leq \theta \leq \frac{2\pi}{3}\right\}, \qquad D_2^s = \left\{\frac{2\pi}{3} \leq \theta \leq \frac{4\pi}{3}\right\}, \qquad D_3^s = \left\{\frac{4\pi}{3} \leq \theta \leq 2\pi\right\}.$$

It is symmetric under rotation by $2\pi/3$.

Definition 4.2 ([LCM21])**.** The *standard trisection* of S^k for $k \geq 2$ is the decomposition $\mathcal{T}_{std} = \{\pi^{-1}(D_i^s) \cap S^k\}$ where $\pi : \mathbb{R}^{k+1} \to \mathbb{R}^2$ is a coordinate projection and $D_1^s \cup D_2^s \cup D_3^s$ is the standard trisection of the unit disk in $\mathbb{R}^2$.

Example 4.3. *The standard trisection of S^2.* The triple intersection of $\mathcal{T}_{std}$ consists of the North and South poles; the double intersections are three meridians connecting the poles; and the sectors are the three bigons bounded by the pairs of meridians.

Example 4.4. *The standard trisection of S^3.* The triple intersection consists of a single S^1, unknotted in S^3; each double intersection is a D^2 Seifert disk for the unknot; and each sector of the trisection is a 3-ball. In addition, we can give two other useful interpretations:

(1) The lens space $L(1,3)$. Here, S^3 is decomposed into three lenses: i.e take $D^2 \times [0,1]$ and collapse the vertical boundary to a circle. These lenses are glued sequentially along their boundaries.
(2) The genus 0 open book decomposition of S^3. Recall that S^3 has an open book decomposition (B, ϕ) where the binding B is the unknot and the fibration $\phi : S^3 \setminus \nu(B) \cong S^1 \times D^2 \to S^1$ is trivial. This induces a trisection as follows. Choose any three distinct pages $\{\phi^{-1}(\theta_1), \phi^{-1}(\theta_2), \phi^{-1}(\theta_3)\}$ of the open book decomposition. Their union becomes the spine of the trisection and the binding B is the codimension 2 stratum of the trisection. The complement of the spine consists of three open 3-balls.

The following lemma is immediate from the definition.

Lemma 4.5. *Let $\mathcal{T}_{std} = (S_1, S_2, S_3)$ be the standard trisection of S^k. Then each S_i is diffeomorphic to B^k; each double intersection $S_i \cap S_j$ is diffeomorphic to B^{k-1} and the central surface is diffeomorphic to S^{k-2}.*

In particular, the standard trisection of S^4 is a $(0;0)$-trisection.

Example 4.6. S^4 also admits an unbalanced $(1;1,0,0)$-trisection. It can be described by a trisection diagram on T^2 with attaching curves α_1, α_2 parallel and the third attaching curve α_3 geometrically dual to both. This implies that $Y_1 = H_1 \cup -H_2$ is homeomorphic to $S^1 \times S^2$ and bounds $S^1 \times B^3$, while $Y_2 = H_2 \cup -H_3$ and $Y_3 = H_3 \cup -H_1$ are homeomorphic to S^3 and bound B^4. By permuting the indices, we also obtain unbalanced $(1;0,1,0)$ and $(1;0,0,1)$-trisections of S^4.

4.2. **Reducible Heegaard splittings.** First, we recall the definition of reducibility in dimension 3 and state an easy result for parallelism.

Definition 4.7. A Heegaard splitting $Y = H_\alpha \cup_\Sigma H_\beta$ is *reducible* if there is an essential simple closed curve $B \subset \Sigma$ that bounds a disk in both handlebodies.

Note that if a Heegaard splitting is reducible, then the union of the two disks bounded by B is an embedded 2-sphere bisected by the Heegaard splitting. Moreover, this 2-sphere is uniquely determined up to isotopy by the curve B.

Conversely, given any 2-sphere we can find a Heegaard splitting that bisects it.

Lemma 4.8. *Suppose that $S \subset Y$ is a separating 2-sphere. There exists a reducible Heegaard splitting $Y = H_\alpha \cup_\Sigma H_\beta$ such that S intersects Σ along a separating, essential simple closed curve B.*

Proof. The 2-sphere S determines a connected sum decomposition $Y = Y_1 \#_S Y_2$. We can construct the required Heegaard splitting of Y by finding Heegaard splittings $Y_1 = H_{\alpha,1} \cup_{\Sigma_1} H_{\beta,1}$ and $Y_2 = H_{\alpha,2} \cup_{\Sigma_2} H_{\beta_2}$ and taking their connected sum. Up to homeomorphism, the separating 2-sphere S will intersect Σ along the curve γ that determines the connected sum $\Sigma = \Sigma_1 \#_B \Sigma_2$. □

Remark 4.9. Note that Haken's Lemma implies something stronger: if S is an essential 2-sphere in Y, then *every* Heegaard splitting of Y is reducible. Currently, there is no 4-dimensional generalization of Haken's lemma to trisections of 4-manifolds.

4.3. **Reducible trisections.**

Definition 4.10. Let $S \subset \#_k S^1 \times S^2$ be a smoothly embedded 2-sphere and let $D \subset \natural_k S^1 \times B^3$ be a properly embedded, smooth 3-ball with boundary S. We say that D is *standard* if $\#_k S^1 \times S^2 \setminus D$ has a handle decomposition consisting of only 0- and 1-handles.

Lemma 4.11. *Let $S \subset \#_k S^1 \times S^2$ be a smoothly embedded 2-sphere.*

(1) S bounds a standard B^3 in $\natural_k S^1 \times B^3$.
(2) Let D_1, D_2 be standard 3-balls with boundary S. Then there exists a diffeomorphism of pairs $(\natural_k S^1 \times B^3, D_1) \cong (\natural_k S^1 \times B^3, D_2)$ that is the identity on the boundary.

Proof. Identify $\#_k S^1 \times S^2$ with $\partial \natural_k S^1 \times B^3$. The latter has a handle decomposition consisting of one 0-handle and k 1-handles. By [Lau72], we can perform a sequence of handle slides and isotopies so that S is disjoint from the belt spheres of the 1-handles. In particular, we can assume S lies in the boundary of the 0-handle. Now we can add a canceling pair of 0/1-handles and isotope S to agree with the belt sphere of the 1-handle. The standard B^3 it bounds is the cocore of this 1-handle.

Part (2) is a restatement of the well-known theorem of Laudenbach and Poénaru [LP72] that the 3- and 4-handles of a smooth 4-manifold are attached uniquely, up to diffeomorphism. □

Definition 4.12 ([MSZ16])**.** A trisection $\mathcal{T}$ of X is *reducible* if there exists an essential simple closed curve B in Σ that bounds a disk V_λ in each 3-dimensional handlebody H_λ of the trisection.

Applying Lemma 4.11 in each sector of a reducible trisection, we obtain the following proposition.

Proposition 4.13. *Let $\mathcal{T}$ be a reducible trisection. Let B denote a reducing curve and let $V_\lambda \subset H_\lambda$ denote compressing disks bounded by B. Then*

(1) The union $S_\lambda = V_\lambda \cup -V_{\lambda-1}$ is an embedded 2-sphere that bounds a standard 3-ball $D_\lambda \subset X_\lambda$.
(2) The union $D_1 \cup D_2 \cup D_3$ is homeomorphic to S^3 and is exactly the standard trisection of S^3

Moreover this 3-sphere is unique up to diffeomorphism.

Conversely, given any 3-sphere in a 4-manifold, we can find a reducible trisection that trisects it.

Lemma 4.14. *Let X be a closed, smooth, oriented 4-manifold and S denote a smoothly embedded, separating S^3 in X. There exists a reducible trisection $\mathcal{T}$ of X such that S intersects:*

(1) the central surface along an essential separating simple closed curve B,
(2) each 3-dimensional handlebody H_λ along a single disk V_λ with boundary B,
(3) each 4-dimensional handlebody X_λ along a single standard 3-ball D_λ with boundary $V_\lambda \cup -V_{\lambda-1}$.

In particular, $\mathcal{T}$ induces the standard trisection on S.

Proof. As in the proof of Lemma 4.8, we take the connected sum decomposition $X = X_1 \#_S X_2$, then trisect each factor and take the connected sum of trisections. □

Definition 4.15. A smoothly embedded, separating 3-sphere $S \subset X$ is *standard with respect to a trisection* $\mathcal{T}$ if it intersects the trisection as in Lemma 4.14.

Lemma 4.16. *Let $\mathcal{T}$ be a reducible trisection*

(1) Every stabilization of $\mathcal{T}$ is reducible.

(2) *If a 3-sphere S is standard with respect to $\mathcal{T}$, then up to isotopy we can assume it is standard with respect to every stabilization of $\mathcal{T}$.*

Proof. Stabilization is well-defined and can be assumed to happen in the neighborhood of some point away from the curve B (see the discussion after [GK16, Definition 8]). □

Finally, standard 3-spheres are determined up to diffeomorphism by the isotopy class of their intersection with the spine of the trisection.

Proposition 4.17. *Suppose that M_1, M_2 are standard S^3's in X with respect to a trisection $\mathcal{T}$ and suppose that $M_1 \cap \Sigma$ and $M_2 \cap \Sigma$ are isotopic in Σ to the same simple closed curve B. There is a diffeomorphism $\phi : X \to X$ fixing $\mathcal{T}$ and such that $\phi(M_1) = M_2$.*

Proof. By assumption, the curve B bounds a disk V_λ in H_λ that is unique up to isotopy. Thus, by an isotopy, we can assume that M_1 and M_2 have the same intersection with the spine of the trisection and apply Proposition 4.13. □

4.4. **Reducible and symplectic.** Suppose that we have a smoothly embedded, separating 3-sphere $S \subset (X, \omega)$. We can find some trisection $\mathcal{T}$ that is compatible with both the connect sum decomposition and the symplectic structure.

First, we have the following key existence result for symplectic 1-cocycles.

Theorem 4.18 ([LC26])**.** *Let (X, ω) be a closed, symplectic 4-manifold. Then*

(1) *there exists a trisection $\mathcal{T}$ and tight symplectic 1-cocycle for $(X, \mathcal{T})$, and*
(2) *every stabilization of $\mathcal{T}$ admits a tight symplectic 1-cocycle.*

Now, we can combine this with the discussion of reducible trisections to obtain the following.

Theorem 4.19. *Let (X, ω) be a closed, symplectic 4-manifold and $S \subset X$ a smoothly embedded 3-sphere. Then there exists a trisection $\mathcal{T}$ of X and tight, symplectic 1-cocycle $\widetilde{\beta}$ for $(X, \mathcal{T})$ such that S is standard with respect to $\mathcal{T}$.*

Proof. By Lemma 4.14, we can choose some trisection $\mathcal{T}_{red}$ that is reducible with respect to S. By Theorem 4.18, we can choose some trisection $\mathcal{T}_{symp}$ that admits a tight symplectic 1-cocycle. By [GK16, Theorem 11], there is some common trisection stabilization $\mathcal{T}$ of $\mathcal{T}_{red}$ and $\mathcal{T}_{symp}$. By Lemma 4.16, the 3-sphere S is standard with respect to $\mathcal{T}$. Moreover, by Theorem 4.18, the trisection $\mathcal{T}$ admits a tight symplectic 1-cocycle. □

5. Pseudoconvexity and Legendrian stabilization

In this section, we complete the proof of Theorem 1.9. Our setup is as follows. Suppose we have a smoothly embedded, separating, cooriented 3-sphere $S \subset (X, \omega)$ bounding $\widetilde{B}$. In order to obtain a relative trisection of $\widetilde{B}$ inducing the D^2-open book on the 3-sphere S, we start by constructing a reducible trisection $\mathcal{T}$ of X. Then we can find a (separating) reducing curve $B \subset \Sigma$ in the trisection surface. Cutting along this curve we obtain two relative trisections (of $\widetilde{B}$ and its complement in X), where the reducing curve B becomes the unknotted binding of the induced open book decomposition on S. Furthermore, if we have a tight symplectic 1-cocycle $\widetilde{\beta}$ for $\mathcal{T}$, it restricts to a tight symplectic 1-cocycle on the relative trisections of the two components. We then ensure pseudoconvexity in two stages. First, we can isotope the boundary spine $\mathcal{S}_\partial$ relative to $\mathcal{T}$ until it is in Roussarie-Thurston general position (Subsection 5.2). We then eliminate concave extrema along the binding δ by enlarging across saddle points of Σ. By a similar enlargment, we can Legendrian stabilize the binding.

5.1. **Pseudoconvexity by isotoping the spine.** Let $S \subset (X, \omega)$ be a smoothly embedded, cooriented 3-sphere bounding $\widetilde{B}$. We can construct a tight symplectic 1-cocycle $(\mathcal{T}, \widetilde{\beta})$ where S is standard with respect to $\mathcal{T}$. By restriction, we obtain a relative trisection of $\widetilde{B}$ and a tight symplectic 1-cocycle. However, to ensure that the resulting symplectic 1-cocycle is pseudoconvex, we first isotope the reducing curve B and compressing disks $\{V_\lambda\}$ with respect to $\widetilde{\beta}$.

Proposition 5.1. *Let $B \subset \Sigma$ be a reducing curve. There exists an isotopy of B such that*

(1) B is disjoint from the singularities of β, and

(2) the restriction of each $\widetilde{\beta}_\lambda$ to B is the exterior derivative of a Morse function on B.

Moreover, for each $\lambda = 1, 2, 3$ there exists an isotopy of the compressing disk V_λ such that

(3) the symplectic 1-cocycle $(\mathcal{T}, \widetilde{\beta})$ is pseudoconvex at each interior point of V_λ, and

(4) each boundary critical point along $B = \partial V_\lambda$ is a local extremum.

Proof. The first two properties can be achieved by a generic perturbation. Then, we can choose an isotopy of V_λ rel boundary to Roussarie general position (Proposition 5.3) so that there are no extrema in the interior of V_λ. Then $(\mathcal{T}, \widetilde{\beta})$ is pseudoconvex in the interior of V_λ. Finally, we can replace each boundary saddle point with an interior saddle point and a boundary extremum. □

We now complete the proof of Theorem 1.9. Using Propositions 5.1 and 5.10, we can go from a reducing curve in a tight symplectic 1-cocycle and obtain a tight, pseudoconvex symplectic 1-cocycle on a relative trisection. Then by Proposition 5.18 we can enlarge this to ensure that we also have Legendrian binding.

5.2. **Roussarie-Thurston general position.** Roussarie [Rou74] and Thurston [Thu72, Thu86] independently proved a generalization of Alexander's Theorem (i.e. the 3-dimensional smooth Schoenflies conjecture). Specifically, a surface S can be isotoped with respect to a taut foliation $\mathcal{F}$ until the induced foliation $\mathcal{F}|_S$ has no center singularities. The foliations $\mathcal{F}_\lambda$ induced by the symplectic 1-cocycle $\widetilde{\beta}_\lambda$ on the handlebody H_λ is analogous to being taut [LC24, Section 3.2]. Therefore, by a slight modification of this classical result, we can remove center singularities along the compressing disks V_1, V_2, V_3.

Let $\mathcal{F}_\lambda = \ker(\widetilde{\beta}_\lambda)$ denote the foliation on H_λ and let $\mathcal{F}'_\lambda$ denote the induced foliation on V_λ. Since V_λ is contractible, the 1-form $\widetilde{\beta}_\lambda|_{V_\lambda} = df_\lambda$ is exact when restricted to the compressing disk. After a generic perturbation, the singularities of $\mathcal{F}'_\lambda$ in the interior of V_λ are either *centers* (which correspond to index 0/2 critical points of $f_\lambda|_{V_\lambda}$) or *saddles* (which correspond to index 1 critical points of $f_\lambda|_{V_\lambda}$). In addition, there are also critical points of $f_\lambda|_{\partial V_\lambda}$, which can be divided into *boundary centers* and *boundary saddles*. These can be described the following local models, where V_λ is identified with the upper-half plane in $\mathbb{R}^2$ and ∂V_λ is identified with the x-axis. A *boundary center* is conjugate to the foliation induced by $d(x^2 + y^2)$ and a *boundary saddle* is conjugate to the foliation induced by $d(x^2 - y^2)$.

Definition 5.2. We say that the compressing disk V_λ is in *Roussarie general position* if the foliation $\mathcal{F}'$ has no center singularities in the interior of V_λ.

Proposition 5.3. *Suppose that $f_\lambda|_{\partial V_\lambda}$ is Morse. There exists an ambient isotopy of V_λ, fixed pointwise along ∂V_λ, such that it is in Roussarie general position.*

This proposition is essentially an application of the Roussarie-Thurston result, although we require slightly different hypotheses than is usually stated in the literature. In particular, we do not assume the boundary ∂V_λ is either contained in a leaf or everywhere transverse to the foliation. The proof strategy is the same: inductively cancel each center singularity with a saddle. However, we must also account for 'half' singularities on the boundary. Consequently an interior center can be cancelled with a boundary saddle to produce a boundary center (Figure 11). For concreteness, we follow the proof of [CC03, Theorem 9.5.5] and emphasize the important differences.

First, we need two 3-manifold lemmas.

Lemma 5.4. *There exists a fibered 3-manifold $(M_\lambda, d\pi_\lambda)$ and an embedding $i : H_\lambda \hookrightarrow M_\lambda$ such that*

$$i^*(d\pi_\lambda) = \widetilde{\beta}_\lambda$$

and i_* is π_1-injective on the leaves of the foliation.

Proof. The pair $(H_\lambda, \widetilde{\beta}_\lambda)$ is obtained as the simple branched cover of $(S^1 \times D^2, d\theta)$ along a transverse tangle τ_λ. We can embed $S^1 \times D^2 \hookrightarrow S^1 \times S^2$ and extend τ_λ to a link L_λ transverse to the foliation by 2-spheres. If the homomorphism $\pi_1(S^1 \times D^2 \setminus \tau_\lambda) \to S_n$ extends to a homorphism $\pi_1(S^1 \times S^2 \setminus L) \to S_n$, then we can take M_λ to be the branched covering over L determined by this homomorphism.

To construct L_λ, choose any transverse closure of τ_λ. To this, we can add components of the form $S^1 \times \{x\}$. Suppose we want to connect two components of τ_λ labeled by distinct transpositions (ij) and (kl). If i, j, k, l are all distinct, we can link with components $S^1 \times \{x\}$ labeled by (li) and (jk) since

$$(ij)(jk)(li) = (jk)(li)(kl)$$

If, say, $j = l$, then we can link with a component labeled by (ik) since

$$(ij)(ik) = (ik)(kj)$$

This produces a link L'_λ in $S^1 \times D^2$ and a well-defined branched covering. Let $\mu = (i_1 j_1) \cdots (i_m j_m) \in S_n$ be the monodromy around ∂D^2. We can then add m components to L'_λ, labeled by the transpositions $(i_1 j_1), \ldots, (i_m j_m)$. This produces the desired branched covering.

Finally, the inclusion is π_1-injective on leaves if no boundary component of a leaf in H_λ bounds a disk in M_λ. To ensure this, we can add set of components labeled by (ij) for $1 \leq i < j < n$ just outside $S^1 \times \partial D^2$. This ensures that no boundary component of a leaf is capped off by a disk in M_λ. □

Then, the key fact which makes Proposition 5.3 possible is the following.

Lemma 5.5. *Let $\mathcal{F}_t$ be a leaf and let $\delta = V_\lambda \cap \mathcal{F}_t$ be a connected, simple closed curve. Then δ bounds an embedded disk in V_λ and an embedded disk in the leaf $\mathcal{F}_t$.*

Proof. Since V_λ is homeomorphic to D^2, the 2-dimensional Jordan-Schoenflies Theorem implies that δ bounds an embedded disk. Consequently, δ is contractible in H_λ. The leaf $\mathcal{F}_\lambda$ embeds into a fiber surface in M_λ, which is incompressible in M_λ by Novikov's theorem; hence the leaf $\mathcal{F}_\lambda$ is also incompressible in $H_\lambda \subset M_\lambda$. Therefore, since δ is contractible it must bound an embedded disk in the leaf $\mathcal{F}_t$. □

Proof of Proposition 5.3. Roussarie divides center singularities into Type (a) and Type (b), according to the boundary of a so-called *adequate neighborhood* [Rou74, Lemma 1] (cf. [CC03, Exercise 9.5.7]). Emanating from a center singularity is a sequence of concentric, closed leaves of $\mathcal{F}'$. An adequate neighborhood is the closure of the largest open neighborhood foliated by such leaves. For Type (a), the boundary of an adequate neighborhood consists of a single saddle singularity, connected by a single separatrix; For Type (b), the boundary consists of a single saddle singularity connected by two separatrices. However, in the case with boundary singularities of $\mathcal{F}'$, we have a third case, Type (a)', where the boundary of the adequate neighborhood contains a single boundary saddle singularity and a single separatrix. See Figure 11.

Applying [CC03, Lemma 9.5.9], we can cancel all Type (a) and Type (a)' center singularities by an ambient isotopy. We will now show that Lemma 5.5 excludes Type (b) centers. The adequate neighborhood of a Type

(b) center is a pinched annulus. After a small thickening, this annulus is bounded by two simple closed curves c, c' in V_λ. These each bound disks D, D', respectively. We can further assume that $D \subset D'$ and D is disjoint from the adequate neighborhood of the Type (b) center. For Euler characteristic reasons, there must be a center singularity of $\mathcal{F}'$ in D. Furthermore, as in [CC03, Lemma 9.5.10], there must be at least one Type (a) center, which is a contradiction.

Consequently, we have removed all center singularities from the interior of V_λ. □

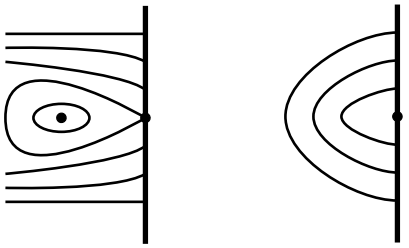


FIGURE 11. *(Left)* A Type-(a)' adequate neighborhood of a center. *(Right)* Via an isotopy, this can be replaced by a boundary center.

5.3. **Twisting and turning.** When constructing a weak symplectic filling, our aim is to realize the binding as a Legendrian link. The symplectic 1-cocycle determines the contact framing (i.e. Thurston-Bennequin invariant) of the binding components.

Along the central surface, we have collar coordinates (r, z) near $B = \partial\Sigma$. In this coordinates, we have

$$\beta_1 = p_1 dr + q_1 dz$$
$$\beta_3 = p_3 dr + q_3 dz$$

Restricting to $\{r = 0\}$, we have coordinates (z, x_1, x_2) on a surface-framed tubular neighborhood $\nu(B)$ in the boundary $\partial\widetilde{B}$. In these coordinates,

$$\omega|_B = dx_1 \wedge q_1 dz - dx_2 \wedge (q_1 + q_3) dz$$

Set

$$\phi(z) = \arctan(q_3/q_1) \qquad q_1 = A\cos(\phi)$$
$$A(z) = \sqrt{q_1^2 + q_3^2} \qquad q_3 = A\sin(\phi)$$

where $\phi : S^1 \to S^1$. We can express the symplectic form as

$$\omega = A(z)\,(dx_1 \wedge \sin(\phi(z))dz - dx_2 \wedge (\sin(\phi) + \cos(\phi))dz)$$

The kernel line field $\Lambda = \ker(\omega)$ is spanned by

$$v = (\sin(\phi) + \cos(\phi))\partial_{x_1} - \sin(\phi(z))\partial_{x_2}$$

It is tangent to the meridional disks of the solid torus $\nu(B)$. Take the 1-form

$$\alpha = (\sin(\phi) + \cos(\phi))dx_1 - \sin(\phi)dx_2 \tag{5.1}$$

which satisfies

$$\alpha \wedge \omega = A(z)((\sin(\phi) + \cos(\phi))^2 + \sin^2(\phi))dz \wedge dx_1 \wedge dx_2 > 0$$

Therefore, the 2-plane field $\xi = \ker(\alpha)$ is dominated by ω. However, we have that

$$\alpha \wedge d\alpha = \phi' dvol_S$$

so the derivative of ϕ determines whether $\ker(\alpha)$ is positive or negative contact.

Lemma 5.6. *Let $B_i \subset B$ be a component of the binding. The following integers coincide:*

(1) the turning number of B_i with respect to the (singular) trivialization of $T\Sigma$ determined by $\widetilde{\beta}$,
(2) the twisting of ξ along B_i with respect to a page P_λ, and
(3) the degree of the map $\phi_i : S^1 \to S^1$.

Proof. All three can be computed as the signed count of points where $q_1 = 0$ and $q_3 > 0$, where the sign is determined by the sign of q_1'. □

We define

$$tw(B_i, \widetilde{\beta})$$

to be the integer in Lemma 5.6

5.4. **Pseudoconvexity and twisting.** Pseudoconvexity of a symplectic 1-cocycle is defined by the condition that extremal points along the binding are convex (Lemma 2.15). This corresponds exactly to the condition that the weakly filled contact structure defined in Equation 5.1 is a positive (as opposed to negative) contact structure.

Lemma 5.7. *Let $\widetilde{\beta}$ be a symplectic 1-cocycle, let x be an extremum of $\widetilde{\beta}_\lambda$ along the binding B, and let $\xi = ker(\alpha)$ be the 2-plane field defined in Equation 5.1.*

(1) if x is a convex extremum, then ξ is a positive contact structure near x.
(2) if x is a concave extremum, then ξ is a negative contact structure near x.

Proof. Via the Morse lemma, choose local coordinates (u, v) on Σ such that

$$\beta_3 = dv \qquad r = u - \sigma v^2$$
$$\beta_1 = du \qquad z = v$$

for $\sigma \in \{\pm 1\}$. Then

$$\beta_3 = dz \qquad \phi = \arctan(\sigma z)$$
$$\beta_1 = dr + 2\sigma z dz \qquad \phi' = \sigma \frac{1}{1 + \sigma^2 z^2}$$

So the sign of ϕ' is exactly given by σ. □

More generally, we have

$$q_1 = A(z)\cos(\phi) \qquad q_2 = A(z)(-\cos(\phi) - \sin(\phi))$$
$$q_3 = A(z)\sin(\phi)$$

When the sympletic 1-cocycle is pseudoconvex, the extremal points along the binding occur in a well-defined sequence.

Lemma 5.8. *If $\phi' > 0$, then we will have the following singularities in sequence:*

(1) a minimum of $\widetilde{\beta}_3$ (at $\phi = 0$)
(2) a maximum of $\widetilde{\beta}_1$ (at $\phi = \frac{\pi}{2}$)
(3) a minimum of $\widetilde{\beta}_2$ (at $\phi = \frac{3\pi}{4}$)
(4) a maximum of $\widetilde{\beta}_3$ (at $\phi = \pi$)
(5) a minimum of $\widetilde{\beta}_1$ (at $\phi = \frac{3\pi}{2}$
(6) a maximum of $\widetilde{\beta}_2$ (at $\phi = \frac{7\pi}{4}$)

5.5. **Enlarging.** Not every required modification of a symplectic 1-cocycle can be achieved by isotoping the boundary spine $\mathcal{S}_\partial$ within a larger trisection. In order to achieve pseudoconvexity along the binding and to Legendrian stabilize the binding, we *enlarge* the symplectic 1-cocycle; this preserves the underlying relative trisection while locally modifying the geometric data. In both cases, this corresponds to extending across a hyperbolic singularity of the sympectic 1-cocycle along the central surface.

Adapting [Gay02, Definition 1.6], we make



FIGURE 12. A neighborhood of a concave maximum (*Left*) can be identified with an arc in the local model of a positive hyperbolic point of Σ (*Right*). Then the reducing curve can be isotoped across the hyperbolic point, which replaces the concave maximum with a convex maximum.

Definition 5.9. By *enlarging* a symplectic 1-cocycle $(\mathcal{T}, \widetilde{\beta})$ on $\widetilde{B}$, we mean the following process. Choose a smooth function $h : \mathcal{S}_\partial \to [0, \infty)$ on the boundary spine and let $C = \{(t, p) : 0 \leq t \leq h(p)\} \subset \mathbb{R} \times \mathcal{S}_\partial$. We attach C to the spine $\mathcal{S}$ by the obvious identification of $\{0\} \times \mathcal{S}_\partial$ with $\mathcal{S}_\partial \subset \mathcal{S}$.

Then $\mathcal{S} \cup C = (H_1 \cup C_1) \cup (H_2 \cup C_2) \cup (H_3 \cup C_2)$ is the spine of a relative trisection of an enlargement of $\widetilde{B}$. If $\widetilde{\beta}'$ is a symplectic 1-cocycle on $\mathcal{S} \cup C$ whose restriction to $\mathcal{S}$ is $\widetilde{\beta}$, then we say that $\widetilde{\beta}'$ is an *enlargement* of $\widetilde{\beta}$.

Given some $N \subset \mathcal{S}_\partial$, we say that the enlargement is *supported* in N if the support of h is contained in N.

We have two key local results to prove.

Proposition 5.10. *Let $\widetilde{\beta}$ be a tight symplectic 1-cocycle and let x be a concave extremum of β_λ along B. Fix a sufficiently small neighborhood N of x. There exists an tight enlargement $\widetilde{\beta}'$ of $\widetilde{\beta}$, supported in N, such that $\widetilde{\beta}'$ is pseudoconvex in N.*

Proposition 5.11. *Let $\widetilde{\beta}$ be a tight, pseudoconvex symplectic 1-cocycle. Fix a point $x \in B$ and a sufficiently small neighborhood N of x. There exists a tight, pseudoconvex enlargement $\widetilde{\beta}'$ of $\widetilde{\beta}$, supported in N, that induces a positive/negative Legendrian stabilization of the binding δ.*

5.6. **Local model of hyperbolic point.** A local model for a hyperbolic singularity of $\widetilde{\beta}$ along the central surface Σ is described in [LC26, Section 3.4], which we repeat here. Hyperbolic points correspond to ramification points of a branched covering

$$\pi : X \to \mathbb{CP}^2$$

along the trisection surface. We can describe this locally as a holomorphic branched covering $\pi : \mathbb{C}^2 \to \mathbb{C}^2$ as follows.

With respect to complex coordinates $\{z_j = x_j + iy_j\}$, set

$$t = x_1 - \sigma x_2 \qquad\qquad s = -y_1 - \sigma y_2$$

for $\sigma \in \{\pm 1\}$ and consider the trisection decomposition is given by

$$\begin{aligned} H_1 &= \{s \geq 0, t = 0\} & Z_1 &= \max(-s, t) \leq 0\} \\ H_2 &= \{t \geq 0, -t - s = 0\} & Z_2 &= \max(-t, -t - s) \leq 0\} \\ H_3 &= \{-t - s \geq 0, s = 0\} & Z_3 &= \max\{(t + s, s) \leq 9\} \end{aligned}$$

In particular, the central surface is given by

$$\Sigma = \{t = s = 0\} = \{z_2 = \sigma \overline{z}_1\}$$

Set

$$\widetilde{\beta}_1 = dy_1 - \sigma dy_2 \qquad\qquad \widetilde{\beta}_2 = -dx_1 - \sigma dx_2$$

Then

$$dt \wedge \widetilde{\beta}_1 - ds \wedge \widetilde{\beta}_2 = 2(dx_1 \wedge dy_1 + dx_2 \wedge dy_2) = 2\omega_{std}$$

Consequently,

$$dt \wedge ds \wedge (\widetilde{\beta}_1 \wedge \widetilde{\beta}_2) > 0$$

so $\widetilde{\beta} = (\widetilde{\beta}_1, \widetilde{\beta}_2, -\widetilde{\beta}_1 - \widetilde{\beta}_2)$ is a symplectic 1-cocycle that determines the standard symplectic structure.

Take the branched covering $\pi(w_1, w_2) = (w_1, w_2^2)$ and pull back the trisection. Then

$$\pi^*(t) = x_1 + \sigma(-x_2^2 + y_2^2) \qquad\qquad \pi^*(s) = -y_1 - \sigma(2x_2y_2)$$

The symplectic 1-cocycle pulls back as

$$\pi^*(\widetilde{\beta}_1) = dy_1 - 2\sigma d(x_2y_2) \qquad\qquad \pi^*(\widetilde{\beta}_2) = -dx_1 + 2\sigma d(y_2^2 - x_2^2)$$

Restricting to $\pi^*(\Sigma) = \{\pi^*(s) = \pi^*(t) = 0\}$, we have that

$$\pi^*(\widetilde{\beta}_1)|_{\pi^*(\Sigma)} = -4\sigma\, d(x_2y_2) \qquad\qquad \pi^*(\widetilde{\beta}_2)|_{\Sigma} = -4\sigma\, d(y_2^2 - x_2^2)$$

which have hyperbolic singularities at the origin.

Restricting to $H_1 = \{s \geq 0, t = 0\}$, we can choose local coordinates $(x_2, y_2, z = y_1 - 2\sigma(x_2y_2))$ so that

$$\pi^*(\widetilde{\beta}_1) = dz \qquad\qquad \pi^*(s) = -z - 4\sigma\, x_2y_2$$

Consequently, H_1 can be described as the superlevel set

$$H_1 = \{z \geq -4\sigma(x_2y_2)\}$$

Remark 5.12. In Definition 2.9, we assumed that

$$\widetilde{\beta}_\lambda = \pi^*(\beta_\lambda) + d(s\pi^*(f))$$

for some function f that vanishes along the binding $B = \partial\Sigma = \partial V_\lambda$. However, the function f must be nonzero at the hyperbolic point. We can modify the symplectic form as in Subsection 2.4.1 using a bump function whose support is disjoint from the initial and terminal positions of the binding curve B.

Lemma 5.13. *Enlarging a symplectic 1-cocycle is locally tight: the restriction of each contact structure along an interior boundary component Y_λ^* to its intersection with N is tight. In particular, within N the induced characteristic foliation has no periodic orbits.*

Proof. We can identify the local model of a hyperbolic point with a subset of any tight symplectic 1-cocycle obtained by branching over a holomorphic curve in $\mathbb{CP}^2$. Within this local model, enlarging is equivalent to enlarging within a tight contact structure, hence its restriction to the local neighborhood is tight. □

5.7. **Pseudoconvexity.**

Proof of Proposition 5.10. Without loss of generality, assume that x is a concave maximum of $\widetilde{\beta}_1$. Using the Morse lemma, we can identify a neighborhood of x in N with an arc in a local model of a positive hyperbolic point as in Figure 12. We can then isotope B across the positive hyperbolic point. This isotopy extends to all three compressing disks without introducing any new critical points in the interior. We can assume that in place of the concave maximum of $\widetilde{\beta}_1$, we obtain by Lemma 5.8 the following (all convex) singularities in order along B: (i) a maximum of $\widetilde{\beta}_2$, (ii) a minimum of $\widetilde{\beta}_3$, (iii) a maximum of $\widetilde{\beta}_1$, (iv) a minimum of $\widetilde{\beta}_2$, and (v) a maximum of $\widetilde{\beta}_3$. The minima of $\widetilde{\beta}_2$ and $\widetilde{\beta}_3$ are saddle points, hence we can isotope these saddles into the interior leaving a convex minimum along the boundary. See Figure 13.

To check that this enlargement is tight, we describe the effect on the characteristic foliations of $V_\lambda \cup -V_{\lambda+1}$. First, consider $V_1 \cup -V_2$. In Figure 14, we depict $V_1 \cup -V_2$ before and after the enlargement. Before, the concave maximum along V_1 combines with the previous singular point to form a negative elliptic point; the preceeding singularity is either a convex minimum on V_1 or a concave minimum on V_2, (which changes sign in $-V_2$); these are depicted in the left panels of Figures 14a and 14b, respectively. Afterward, the negative

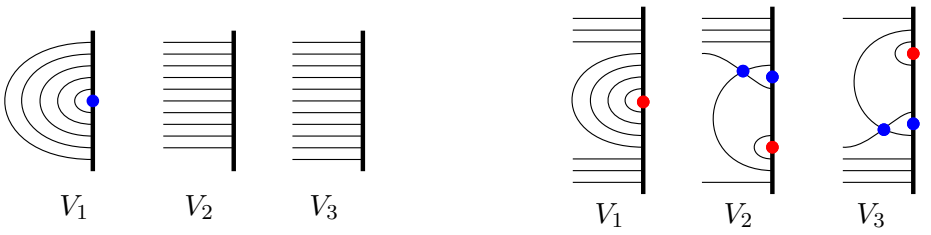


FIGURE 13. The effect of enlarging near a concave maximum of V_1 on the characteristic foliation of the three compressing disks. On V_1, the concave maximum is exchanged for a convex maximum. On both V_2 and V_3, one boundary maximum, one boundary minimum, and one negative hyperbolic point are introduced.

elliptic moves and a positive elliptic/hyperbolic pair in elimination position are introduced; see the right panels of Figures 14a and 14b. Therefore, the effect on the characteristic foliation is a topological isotopy followed by introducing a pair of singular points in elimination position, which is contactomorphic to a vertically-invariant contact structure. In particular, we have enlarged (Y_1, ξ_1) by the tight $(S^2 \times [0,1], \xi_{std})$; this preserves tightness.

Second, the case of $V_3 \cup -V_1$ is essentially identical to the first case, except for changing the signs and rotating the figures by 180°. Finally, the case of $V_2 \cup -V_3$ is depicted in Figure 15. Here, both a positive and negative elliptic/hyperbolic pair in elimination position are introduced. Again, this effect preserves tightness of the contact structure on (Y_2, ξ_2). □



(A) Case 1: the preceeding singularity is a convex minimum on V_1.

(B) Case 2: the preceeding singularity is a concave maximum on V_2.

FIGURE 14. The effect of enlarging near a concave maximum of V_1 on the characteristic foliation of $V_1 \cup -V_2$. In both cases, the negative elliptic point moves and a positive elliptic/hyperbolic pair is introduced.

5.8. **Legendrian stabilization.**

Proof of Proposition 5.11. We describe how to Legendrian stabilize the binding at a given point $x \in B$ disjoint from the extrema of $\widetilde{\beta}$. In principle, there are 12 distinct cases, according to the signs of $\beta_1, \beta_2, \beta_3$ near x and the sign of of the stabilization. Without loss of generality, we assume that $\beta_1, \beta_2 > 0 > \beta_3$ and describe both a positive and a negative Legendrian stabilization.

As in the proof of Proposition 5.10, we can identify a local neighborhood of x with a subset of the local model of a hyperbolic singularity, then enlarge by isotoping across the hyperbolic point and extending to each of the three compressing disks. If we assume that $\beta_1, \beta_2 > 0$ at x, then x lies between a minimum of β_1

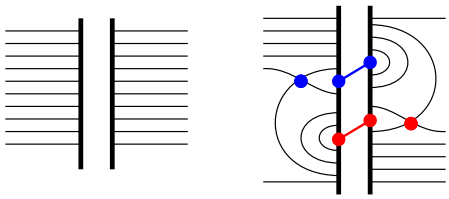
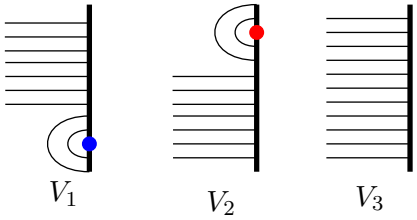
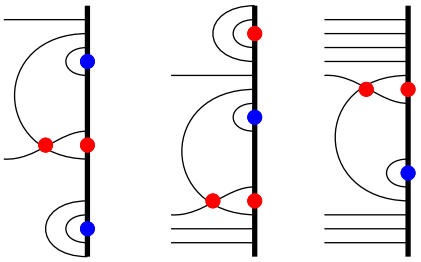


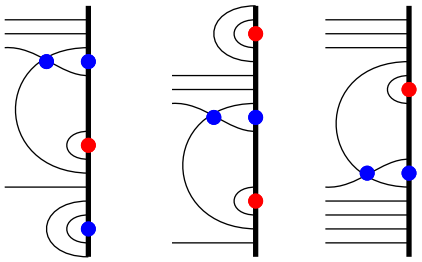
FIGURE 15. The effect of enlarging near a concave maximum of V_1 on the characteristic foliation of $V_2 \cup -V_3$. Both a positive and negative elliptic/hyperbolic pair are introduced.



(A) The foliations before stabilization



(B) Positive stabilization. A boundary maximum, a boundary minimum, and an interior positive saddle are added to each foliation.

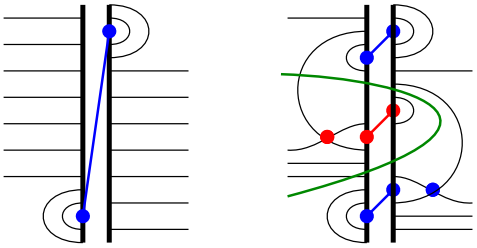


(C) Negative stabilization. A boundary maximum, a boundary minimum, and an interior negative saddle are added to each foliation.

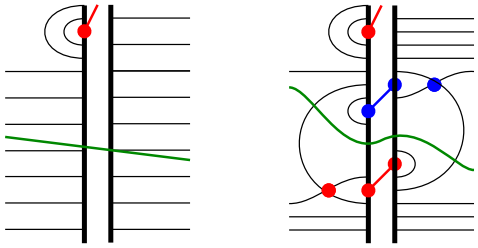
FIGURE 16. The effect of enlarging near a generic point to achieve a Legendrian stabilization. We assume $\beta_1, \beta_2 > 0 > \beta_3$.

and a maximum of β_2. Then by Lemma 5.8, we introduce six new extrema in a neighborhood of x: (i) a maximum of β_2, (ii) a minimum of β_3, (iii) a maximum of β_1, (iv) a minimum of β_2; (v) a maximum of β_3; (vi) a minimum of β_1. For a positive stabilization, we also obtain a positive saddle in the interior of each compressing disk, each of which pairs with the newly introduced maximum; for a negative stabilization, each compressing disk contains a negative saddle paired with the newly introduced minimum. See Figure 16.

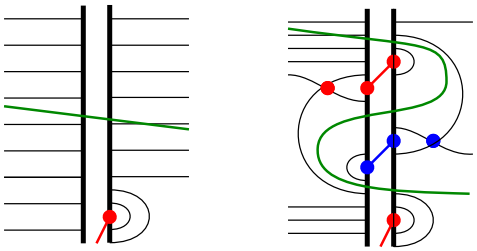
This Legendrian stabilization preserves tightness. On the surfaces $V_2 \cup -V_3$ and $V_3 \cup -V_1$, both a positive and negative elliptic/hyperbolic pair in elimination position are introduced; as they can be removed by a C^0-small isotopy, this operation preserves tightness in these two sectors. The effect on the characteristic foliation of $V_1 \cup -V_2$ is slightly more interesting. In both positive and negative Legendrian stabilization, the



(A) $V_1 \cup -V_2$. The negative elliptic splits into two negative elliptics and one negative hyperbolic, while a positive elliptic/hyperbolic pair is introduced in between.



(B) $V_2 \cup -V_3$. Both a positive and negative elliptic/hyperbolic pair are introduced. Note that after the stabilization, we can still choose the dividing set to intersect the binding exactly once.



(C) $V_3 \cup -V_1$. Both a positive and negative elliptic/hyperbolic pair are introduced.

FIGURE 17. The Before (*Left*) and After (*Right*) effect of positive Legendrian stabilization on the three characteristic foliations. The dividing set before and after the stabilization is also depicted.

single negative elliptic point splits into two negative elliptic points and a negative hyperbolic point, while a positive elliptic/hyperbolic pair in elimination position are introduced. See Figure 17. □

5.9. **Legendrian binding.** The motivation for a symplectic 1-cocycle having *Legendrian binding* is the following standard fact in convex surface theory.

Proposition 5.14. *Let L be a Legendrian curve on a convex surface Σ. Then*

$$tw(L,\Sigma) = -\frac{1}{2}\#(\Gamma \pitchfork L)$$

Definition 5.15. Let $\widetilde{\beta}$ be a symplectic 1-cocycle and let $B_i \subset B$ be a connected component of the binding. Let

$$\#(\Gamma_\lambda \pitchfork B_i)$$

denote the minimal intersection number between $B_i \subset \partial V_\lambda, \partial V_{\lambda+1}$ and a dividing set Γ_λ for the characteristic foliation on $V_\lambda \cup -V_{\lambda+1}$ induced by α_λ.

Definition 5.16. Let $\widetilde{\beta}$ be a pseudoconvex symplectic 1-cocycle with binding $B = \partial\Sigma$. We say that $\widetilde{\beta}$ has *Legendrian binding* if for each component B_i of B and for each sector $\lambda = 1, 2, 3$, we have that

$$tw(B_i, \widetilde{\beta}) = -\frac{1}{2}\#(\Gamma_\lambda \cap B_i)$$

Definition 5.17. Given a pseudoconvex symplectic 1-cocycle $\widetilde{\beta}$, we define the *sector-λ defect*

$$d_\lambda := -\left(\frac{1}{2}\#(\Gamma_\lambda \pitchfork B) + \sum tw(B_i, \widetilde{\beta})\right)$$

and the *total defect*

$$d := d_1 + d_2 + d_3$$

The defect satisfies $d \geq 0$ is nonnegative and $\widetilde{\beta}$ has Legendrian binding if and only if $d = 0$.

Proposition 5.18. *Let $\widetilde{\beta}$ be a tight, pseudoconvex symplectic 1-cocycle. There exists an enlargement $\widetilde{\beta}'$ that is tight, pseudoconvex, and with Legendrian binding.*

Proof. To prove the proposition, we neet to inductively decrease the total defect. We will show that via Legendrian stabilization, we can strictly decrease some d_λ while preserving $d_{\lambda-1}, d_{\lambda+1}$.

Consider a point $x \in B_i$ is some boundary component that is not a local extremum of $\widetilde{\beta}$. Exactly two of $\beta_1, \beta_2, \beta_3$ have the same sign. Suppose first that $\beta_1, \beta_2 > 0$, so that x lies between a minimum y_1 of β_1 and a maximum y_2 of β_2. There exists a neighborhood of the arc from y_1 to y_2 that contains a single attracting elliptic point that is the ω-limit of every point in the neighborhood. Therefore, we can assume that the dividing set Γ_2 is disjoint from the segment from y_1 to y_2. Similarly, we can assume that Γ_2 does not intersect any arc where $\beta_1, \beta_2 < 0$ either. See the left frame of Figure 17a.

Now, without loss of generality, consider an arc of B_i where $\beta_2 > 0 > \beta_3$, which starts at a maximum of β_3 and ends at a maximum of β_2. The dividing set Γ_3 intersects this arc an odd number of times, because the maxima of β_3 are contained in the negative region and the maxima of β_2 are contained in the positive region of $(V_2 \cup -V_3) \smallsetminus \Gamma_3$. If the dividing set intersects at least 3 times, let x denote the second point. We can perform a positive Legendrian stabilization here, which stricly decreases $tw(B_i, \widetilde{\beta})$ by 1. To check that the total defect decreases, it suffices to choose dividing sets such that $\#(\Gamma_3 \cap B_i)$ remains the same and $\#(\Gamma_1 \cap B_i)$ and $\#(\Gamma_2 \cap B_i)$ increase by 2. See Figure 17.

If the signs are reversed and $\beta_3 > 0 > \beta_2$, we perform a negative stabilization instead. Up to cyclic permutations of the indices, this covers all cases. We can therefore inductively reduce the defect until it vanishes. □

6. Convex corners and plumbing

In this section, we give an explicit model for plumbing together two symplectizations. Let (Y,ξ) be a contact 3-manifold such that Y has nonempty, convex boundary; equivalently, there exists an outward-pointing contact vector field ρ along ∂Y. Given a contact form α, we can construct the symplectization $(\mathbb{R}\times Y, d(e^v\alpha))$ where $\nu=\partial_v$ is a Liouville vector field. Now, given two contact structures (Y,ξ_1) and (P,ξ_2) and a homeomorphism $\phi:\partial Y\to\partial P$, we can plumb their symplectizations together by identifying the Liouville vector field of one to the outward contact vector field of the other.

6.1. **Convex corners.** Plumbing symplectizations requires a geometric assumption on the characteristic foliations induced by ξ_1,ξ_2 on $S=\partial Y\cong\partial P$.

Definition 6.1. Let S be a closed surface and let α_0,α_1 be contact forms on $\nu(S)$ such that S is convex with respect to both $\xi_0=\ker(\alpha_0)$ and $\xi_1=\ker(\alpha_1)$. We say that α_1,α_0 *meet convexly* along S or that (α_1,α_0) are a *convex pair* if

(1) α_0,α_1 have the same signed singularity set, and
(2) $\alpha_1\wedge\alpha_0|_S\geq 0$, with equality at the singularity set.

Remark 6.2. The 'convexity' in Definition 6.1 is 4-dimensional and not a reference to convex surface theory in 3-dimensions. For example, let (X,J) be a complex surface and let ϕ_0,ϕ_1 be smooth plurisubharmonic functions so that $\alpha_i=d^{\mathbb{C}}\phi_i$ is a contact form along $Y_i=\phi^{-1}(0)$. Then $\phi=\max(\phi_0,\phi_1)$ is a continuous plurisubharmonic function and $Y=\phi^{-1}(0)$ is a manifold with corners at $S=Y_0\cap Y_1$. In this case, positivity of intersections for complex lines implies that

$$\alpha_1\wedge\alpha_0|_S\geq 0$$

with equality precisely at the J-complex points of S.

When J is integrable, a continuous plurisubharmonic function can be approximated by a smooth plurisubharmonic function. Therefore, there is a plurisubharmonic smoothing $\widetilde{\phi}$ of ϕ and $\widetilde{Y}=\widetilde{\phi}^{-1}(0)$ is a contact-geometric smoothing of Y_0,Y_1 along S. Smoothing this corner introduces a contact structure on $(-\epsilon,\epsilon)\times S$ that interpolates between ξ_0 and ξ_1. This interpolating contact structure is *strongly minimally twisting* and is unique up to isotopy rel boundary.

6.2. **Strongly minimally twisting contact structures.** Honda introduced the notion of a *minimally twisting* contact structure on $T^2\times[0,1]$, which are defined in terms of the slopes of the dividing curves in $H_1(T^2;\mathbb{Z})$ [Hon00]. We extend this to *strongly minimally twisting* contact structures on $\Sigma\times[0,1]$, which are defined analogously but in terms of the characteristic foliation.

Definition 6.3. A contact structure $(\Sigma\times[0,1],\xi)$ is *strongly minimally twisting* if, up to an ambient isotopy of ξ, there exists a contact form $\alpha=gdz+\beta_z$ and finite subset $B\subset\Sigma$ such that

(1) $\beta_z^{-1}(0)=B$ for all $z\in[0,1]$, and
(2) $\beta_t\wedge\beta_s\geq 0$ for all $0\leq s<t\leq 1$, with equality only on the boundary of Σ and at B.

Example 6.4. Honda's *basic slices* are both minimally-twisting and strongly minimally twisting. Every minimally twisting $(T^2\times[0,1],\xi)$ can be decomposed into a concatenation of basic slices. However, such a contact structure is strongly minimally twisting if and only if all of these basic slices have the same sign.

There is a 1-1 correspondance between convex pairs and strongly minimally twisting contact structures.

Proposition 6.5. *Let (α_1,α_0) be a convex pair along Σ. Then*

(1) there exists a strongly minimally-twisting contact structure $(\Sigma\times[0,1],\xi)$ such that

$$\xi=\begin{cases}\xi_0 & \text{near } t=0\\ \xi_1 & \text{near } t=1\end{cases}$$

(2) if $(\Sigma\times[0,1],\xi')$ is another strongly-minimally twisting contact structure satisfying (1), then ξ' is isotopic to ξ rel boundary.

Proof. Since α_0, α_1 have the same cooriented tangency set, we can choose neighborhood U of the tangenices on which

$$\alpha_0 \wedge d\alpha_1 > 0 \qquad \text{and} \qquad \alpha_1 \wedge d\alpha_0 > 0$$

Furthermore, after multiplying the contact forms by positive functions, we can assume that

$$d\beta_0 = d\beta_1 = 0$$

on the complement of U.

Now, for $\epsilon > 0$ set

$$\widehat{\alpha}_i = \epsilon\, u_i dt + \beta_i$$

and

$$\alpha = \phi\widehat{\alpha}_1 + (1-\phi)\widehat{\alpha}_0$$

where ϕ is a smooth cutoff function satisfying $\phi' > 0$ on the open interval $(0,1)$.

We have

$$d\alpha = \phi' dt \wedge (\beta_1 - \beta_0) + \epsilon\phi(du_1 \wedge dz + d\beta_1) + \epsilon(1-\phi)(du_0 \wedge dz + d\beta_0)$$

and

$$\begin{aligned} \alpha \wedge d\alpha = {} & \epsilon\phi^2 \alpha_1 \wedge d\alpha_1 + \epsilon(1-\phi)^2 \alpha_0 \wedge d\alpha_0 + \phi' dt \wedge \beta_1 \wedge \beta_0 \\ & + \epsilon\phi(1-\phi)(\alpha_1 \wedge d\alpha_0 + \alpha_0 \wedge d\alpha_1) \end{aligned}$$

The first two terms are positive, the third is nonnegative and only vanishes if $\phi' = 0$ or $\beta_1 = \beta_0 = 0$. The fourth term may be negative, but it is dominated by the first line if ϕ is sufficiently close to 0 or 1. Specifically, we can choose some $\delta > 0$ such that α is a positive contact form on $[0,\delta]$ and $[1-\delta, 1]$. Moreover, since α_0, α_1 have the same cooriented singularity set, there is some open neighborhood U of the singularity set in which the fourth term is positive. Consequently, over the compact set $\Sigma \setminus U \times [\delta, 1-\delta]$, the third term is bounded below by a positive constant and by choosing $\epsilon > 0$ sufficiently small we obtain a positive contact form.

The second part follows from Lemma 6.7 below. □

We start first with a preparatory lemma.

Lemma 6.6. *Let $\mathcal{F} = ker(\beta_0)$ be a Morse-Smale singular foliation on Σ. There exists a Morse-Bott function $f : \Sigma \to \mathbb{R}$ such that*

(1) $\beta_0 \wedge df \geq 0$, with equality precisely at the singularities and periodic orbits of $\mathcal{F}$, and
(2) the regular level set $f^{-1}(0)$ divides $\mathcal{F}$

.

We now prove a second fundamental lemma that allows us to modify minimally-twisting contact structures.

Lemma 6.7. *Let $(\Sigma \times [0,1], \xi)$ be a strongly minimally twisting contact structure with such that $\Sigma \times \{0,1\}$ have Morse-Smale characteristic foliations. Let $\mathcal{F}_z = ker(\xi|_{\Sigma \times \{z\}})$ denote the family of characteristic foliations induced by ξ. Let β_0 be a 1-form defining $\mathcal{F}_0$ and let f be a function as in the previous lemma. Furthermore let*

$$B = (\beta_0 \wedge df)^{-1}(0)$$

be the subset consisting of singularities and periodic orbits of $\mathcal{F}_0$. Then

(1) There exists a function

$$\phi : \Sigma \times [0,1] \to [0, \infty)$$

satisfying

$$\phi(x,0) = 0 \qquad\qquad \frac{\partial \phi}{\partial z} > 0$$
$$\phi(U, z) < \frac{\pi}{2}$$

where U is a neighborhood of $B = \beta_0^{-1}(0)$, such that

$$\mathcal{F}_z = ker(\cos(\phi)\beta_0 + \sin(\phi) df)$$

(2) Given any function ϕ satisfying the condition in Part (1), there exists a strongly minimally twisting contact structure $(\Sigma \times [0,1], \xi_\phi)$ whose characteristic foliations satisfy

$$\mathcal{F}_z = ker(\cos(\phi)\beta_0 + \sin(\phi)df)$$

(3) Given any 1-parameter family ϕ_t for $t \in [0,1]$ of functions satisfying the conditions in Part (1), there exists an isotopy through strongly minimally twisting contact structures from $(\Sigma \times [0,1], \xi_{\phi_0})$ to $(\Sigma \times [0,1], \xi_{\phi_1})$.

Proof. Since each $\mathcal{F}_z$ has the singularities precisely at B, we can find a 1-parameter family β_z satisfying $\beta_z^{-1}(0) = B$ and such that $\mathcal{F}_z = \ker(\beta_z)$. Since β_0, df span $T^*\Sigma$ except at B, then

$$\beta_z = P_z\beta_0 + Q_z df$$

over $\Sigma \setminus B$. The function $\phi_z : (\Sigma \smallsetminus B) \times [0,1] \to S^1$ defined by

$$\phi_z = \arctan(Q_z/P_z)$$

lifts uniquely to a function $\phi : (\Sigma \smallsetminus B) \times [0,1] \to \mathbb{R}$ satisfying $\phi(x,0) = 0$.

Since $\Sigma \times [0,1]$ is compact, we can choose a constant such that

$$\frac{\partial \phi}{\partial z} > \delta > 0$$

Fix an open neighborhood U of the singularity set in which $d\beta_0 \neq 0$. Let Ω be an area form on Σ. The complement of U is compact, so we can choose a constant $C > 0$ such that

$$df \wedge \beta_0 > C\Omega$$

on the complement of U. Finally, let u be any smooth function on Σ such that

$$d\beta_0|_U = u\Omega$$

Set

$$\alpha = \epsilon u dz + \cos(\phi)\beta_0 + \sin(\phi)df$$

Then

$$d\alpha = \epsilon du dz - \sin(\phi)(\phi' dz + d\phi_z) \wedge \beta_0 + \cos(\phi)d\beta_0 + \cos(\phi)(\phi' dz + d\phi_z) \wedge df$$

So

$$\begin{aligned}\alpha \wedge d\alpha = {} & \epsilon u dz \wedge (-\sin(\phi)d\phi_z \wedge \beta_0 + \cos(\phi)d\beta_0 + \cos(\phi)d\phi_z \wedge df) \\ & + \epsilon\,(\cos(\phi)dz \wedge \beta_0 \wedge du + \sin(\phi)dz \wedge df \wedge du) \\ & + \phi'(\cos^2(\phi) + \sin^2(\phi))dz \wedge df \wedge \beta_0\end{aligned}$$

Since β_0, df vanish at the singularities, after shrinking U, we can assume that $d\beta_0$ dominates all of the $O(\epsilon)$-terms; hence we have constructed a contact form.

To obtain Part (3), we note that we can let ϵ vary in t to obtain a 1-parameter family of contact forms. By Gray stability, this can be realized by an isotopy. □

We immediately have the following corollary, which allows us to realize any intermediate singular foliation as the characteristic foliation of an intermediate surface, without sacrificing the minimally-twisting condition.

Lemma 6.8. *Let $(\Sigma \times [0,1], \xi)$ be a strongly minimally-twisting contact structure. Suppose that*

$$\mathcal{F}_0 = ker(\beta_0) \qquad \mathcal{F}_1 = ker(\beta_1)$$

For any β satisfying

$$\beta^{-1}(0) = B \qquad\qquad \beta \wedge \beta_0 \geq 0$$
$$\beta_1 \wedge \beta \geq 0$$

there exists an isotopy of $(\Sigma \times [0,1], \xi)$ through strongly minimally-twisting contact structures, fixed near $z = \{0,1\}$, such that

$$\mathcal{F}_{1/2} = ker(\beta)$$

Example 6.9. Let $(\Sigma \times [0,1], \xi)$ be a minimally twisting contact structure. Let $U \subset \Sigma$ be an open subset. Then we can 'wind back the clock' on the twisting foliations just in the set U as follows.

By integrating locally, since $\beta_1 \wedge \beta_0 > 0$ away from the singularities, we can find coordinates such that

$$\beta_0 = dy \qquad \beta_1 = dx$$

Then by Lemma 6.8, if we have any foliation

$$\mathcal{F} = \ker(f dx + g dy)$$

for $f, g > 0$, then up to isotopy we can realize the singular foliation $\mathcal{F}$ along an intermediate cross-section.

6.3. **Plumbing.** We now give an explicit model for plumbing two symplectizations together along a convex surface S, provided that the contact forms meet convexly along S, and extending a weak filling.

Let $(Y, \xi_\rho = \ker(\alpha_\rho))$ and $(P, \xi_\nu = \ker(\alpha_\nu))$ be contact structures and let $\phi : \partial Y \to \partial P$ be a diffeomorphism such that $\alpha_\rho, \phi^*(\alpha_\nu)$ meet convexly along $S = \partial Y \cong \partial P$. We can extend ϕ to identify the symplectizations

$$([-\epsilon, \epsilon] \times Y, \omega_1 = d(e^v \alpha_\rho)) \qquad ([-\epsilon, 0] \times P, \omega_2 = d(e^r \alpha_\nu))$$

near S so that v is a collar coordinate on P near S and r is a collar coordinate on Y near S. Let

$$X = [-\epsilon, \epsilon] \times Y \cup_\phi [-\epsilon, 0] \times P$$

The *interior boundary* $\widehat{Y}$ of X is a smoothing of the level set $\{\max(r - \epsilon, v) = -\epsilon_0\}$ for some $\epsilon > \epsilon_0 > 0$ sufficiently small, and the *exterior boundary* of X is the level set $r^{-1}(0)$.

Proposition 6.10. *Suppose that α_ρ, α_ν meet convexly along S.*

(1) There exists a symplectic structure (X, ω) such that $\omega = \omega_1$ for $r - v < -\epsilon$.

(2) There exists an inward-pointing Liouville vector field along the interior boundary $\widehat{Y}$ and a decomposition

$$(\widehat{Y}, \xi) = (Y_0, \xi_\rho) \cup (S \times [0,1], \xi_{\rho,\nu}) \cup (P, \xi_\nu)$$

(3) Let α be a contact form along $[-\epsilon, \epsilon] \times \partial Y$ such that
   (a) $\alpha \wedge \omega_1 > 0$, and
   (b) $\alpha|_S = \alpha_\nu$
   Then α extends to a contact form over the exterior boundary satisfying $\alpha \wedge \omega > 0$.

Proof. Let Ω be an area form on S and let (r, v) be normal coordinates on $S \times D^2$. We choose our orientation convention so that

$$dvol := dr \wedge dv \wedge \Omega > 0 \tag{6.1}$$

is a positive volume form.

First, we can write

$$\alpha_\rho = u dr + \beta_\rho$$
$$d\alpha_\rho = du \wedge dr + d\beta_\rho + dr \wedge \beta'_\rho$$

where β_ρ is an r-family of 1-forms on S. Since α_ρ is a contact form, this implies that

$$dv \wedge \alpha_0 \wedge d\alpha_0 = dv \wedge dr \wedge (u d\beta_0 + \beta_0 \wedge (du - \beta'_0)) > 0$$

By assumption, we have that $d\beta_0 = \omega|_S$. Let p be a singularity of the characteristic foliation $\mathcal{F}_0$ on S; equivalently it is a point where $\beta_0 = 0$. Then

$$dv \wedge dr \wedge u d\beta_0 = (-u) dr \wedge dv \wedge d\beta_0 > 0$$

Therefore, given our orientation convention in Line 6.1, at a *positive* singularity, where $\omega|_S = d\beta_0$ is positive, the function u is *negative* in some neighborhood of the singularity; similarly, the function u is positive in some neighborhood of each negative singularity.

Second, take a vertically-invariant contact form

$$\alpha_1 = hdv + \beta_1$$

where 1-form $\beta_1|_S$ defines the singular foliation $\mathcal{F}_1$. Since $\mathcal{F}_0, \mathcal{F}_1$ meet convexly we know that

$$\beta_1 \wedge \beta_0 \geq 0$$

This is an open condition, so we can assume it remains true for ϵ sufficiently small.

Without loss of generality, we can assume that there exists neighborhoods U_p of the singularities such that

(1) β_1 is closed on the complement of $\bigcup_p U_p$
(2) $h, (-u)$, and $d\beta_1$ are positive on U_p if p is a positive singularity, and
(3) $h, (-u)$, and $d\beta_1$ are negative on U_p if p is a negative singularity.
(4) $dh = 0$ on some smaller neighborhood $V_p \subset U_p$ for each p.

We then replace α_1 with

$$\alpha_1 = \epsilon' hdv + \beta_1$$

for some small $\epsilon' > 0$. Then

$$d\alpha_1 = \epsilon' dh \wedge dv + d\beta_1$$

Third Define

$$\lambda = e^v \alpha_0 + M\phi(r - v)e^{Ar}\alpha_1$$

where ϕ is a smooth cutoff function such that $\phi(\epsilon) = 1$ and $\phi(-\epsilon) = 0$; and $M, A \gg 0$ are constants satisfying Inequalities 6.3-6.5 below. Then

$$d\lambda = e^v(dv \wedge \alpha_0 + d\alpha_0) + M\phi e^{Ar}(Adr \wedge \alpha_1 + d\alpha_1) + M\phi' e^{Ar}(dr - dv) \wedge \alpha_1 \tag{6.2}$$

Outside the support of ϕ, we have $d\lambda = \omega$. Over the interior of P along the level set $r = -\epsilon_0$, we have that

$$\lambda = e^v \alpha_0 + Me^{-A\epsilon_0}\alpha_1$$

For fixed A, ϵ_0, as M grows sufficiently large the 2-plane field $\ker(\lambda)$ is a perturbation of the contact structure $\ker(\alpha_1)$. In particular, it is isotopic to ξ. Hence we can extend λ as a contact form over the interior of M so that it defines a contact structure isotopic to ξ.

Fourth, fix some small $\epsilon > \delta > 0$ and let $\psi(r, v)$ denote a smoothing of $\max(r, v)$ that equals $\max(r, v)$ outside the region $\{|r - v| < \delta\}$. We can further assume that the partial derivatives ψ_r, ψ_v are nonnegative. Consider a level set $\widehat{Y} = \psi^{-1}(-\epsilon_0)$. We can parametrize is by a path $(v(t), r(t))$ for a parameter t such that

$$(v(t), r(t)) = \begin{cases} (-\epsilon_0, t) & \text{for } t \leq -\delta \\ (-t, -\epsilon_0) & \text{for } t \geq \delta \end{cases}$$

and v is monotonically nonincreasing while r is monotonically nondecreasing.

Along the surface S, we have

$$\lambda_t = e^{v(t)}\beta_{r(t)} + M\phi(r(t) - v(t))e^{Ar(t)}\beta_1$$

For $s < t$, we have

$$\lambda_t \wedge \lambda_s = M\left(e^{v(s)+Ar(t)}\phi(r(t) - v(t))) - e^{v(t)+Ar(s)}\phi(r(s) - v(s))\right)\beta_1 \wedge \beta_0 \geq 0$$

since

$$v(s) + Ar(t) \geq v(t) + Ar(s)$$
$$\phi(r(t) - v(t)) \geq \phi(r(s) - v(s))$$

Consequently, assuming Claim 2 below that λ restricts to a contact form along $\widehat{Y}$, its restriction to this product neighborhood $S \times (-\delta, \delta)$ is (after a small perturbation) strongly minimally-twisting.

Before proving that $d\lambda$ is a symplectic form and λ a contact form along $\widehat{Y}$, we observe the following inequalities. First,

$$dr \wedge dv \wedge \alpha_0 \wedge \alpha_1 = dr \wedge dv \wedge \beta_0 \wedge \beta_1 \geq 0$$

Since α_0, α_1 are contact forms, we see that

$$dv \wedge \alpha_0 \wedge d\alpha_0 > 0$$
$$dr \wedge \alpha_1 \wedge d\alpha_1 > 0$$

but since the contact forms are defined along 3-manifolds, we also get

$$dr \wedge \alpha_0 \wedge d\alpha_0 = 0$$
$$dv \wedge \alpha_1 \wedge d\alpha_1 = 0$$

For the mixed pairings, we have

$$\begin{aligned} dv \wedge \alpha_0 \wedge d\alpha_1 &= dv \wedge \alpha_0 \wedge d\beta_1 \\ &= dr \wedge dv \wedge (-ud\beta_1) \geq 0 \\ dv \wedge \alpha_1 \wedge d\alpha_0 &= dv \wedge \beta_1 \wedge dr \wedge (\beta_0' - du) \\ dr \wedge \alpha_0 \wedge d\alpha_1 &= dr \wedge \beta_0 \wedge (\epsilon' dh \wedge dv) \\ &= \epsilon' dr \wedge dv \wedge \beta_0 \wedge dh \\ dr \wedge \alpha_1 \wedge d\alpha_0 &= dr \wedge \alpha_1 \wedge d\beta_0 \\ &= \epsilon' dr \wedge dv \wedge (hd\beta_0) \geq 0 \end{aligned}$$

and finally

$$d\alpha_0 \wedge d\alpha_1 = -\epsilon' dr \wedge dv \wedge dh \wedge (\beta_0' - du)$$

We can also choose positive constants such that

$$-D_1 < dv \wedge \alpha_1 \wedge d\alpha_0 < C_1$$

and

$$dr \wedge \alpha_0 \wedge d\alpha_1 > -D_2 \qquad\qquad d\alpha_0 \wedge d\alpha_1 > -D_3$$

Claim 1. After possibly shrinking the neighborhoods $\{U_p\}$ of the singularities, we can assume that on these neighborhoods

$$dv \wedge \alpha_1 \wedge d\alpha_0 + dv \wedge \alpha_0 \wedge d\alpha_1 > 0$$
$$dr \wedge \alpha_1 \wedge d\alpha_0 + dr \wedge \alpha_0 \wedge d\alpha_1 > 0$$

The first inequality follows since β_1 vanishes at the singularities, while $-ud\beta_1$ is strictly positive there. The second follows since β_0 vanishes at the singularities, while $hd\beta_0$ is strictly positive.

Claim 2 $d\lambda$ is a symplectic form.

Multiplying out we obtain

$$\begin{aligned} d\lambda \wedge d\lambda = {} & 2e^{2v} dv \wedge \alpha_0 \wedge d\alpha_0 + 2M^2 \phi^2 e^{2Ar} A dr \wedge \alpha_1 \wedge d\alpha_1 \\ & + 2M\phi e^{Ar+v} \left(dv \wedge \alpha_0 \wedge A dr \wedge \alpha_1 + dv \wedge \alpha_0 \wedge d\alpha_1 + A dr \wedge \alpha_1 \wedge d\alpha_0 + d\alpha_0 \wedge d\alpha_1 \right) \\ & + 2M\phi' e^{Ar+v} \left(dv \wedge \alpha_0 \wedge dr \wedge \alpha_1 + dr \wedge \alpha_1 \wedge d\alpha_0 - dv \wedge \alpha_1 \wedge d\alpha_0 \right) \\ & + 2M^2 e^{2Ar} \phi \phi' dr \wedge \alpha_1 \wedge d\alpha_1 \end{aligned}$$

Outside the support of ϕ, ϕ' we have that

$$d\lambda \wedge d\lambda = 2e^{2v} dv \wedge \alpha_0 \wedge d\alpha_0 > 0$$

Since symplecticity is an open condition, we can choose some $-\epsilon < \delta' < \epsilon$ such that $\phi(\delta') > 0$ and $d\lambda \wedge d\lambda$ is symplectic for $r - v < \delta'$.

The first line on the right is strictly positive and the fourth is nonnegative. In the second line, the last term may be negative. However, it vanishes on each small neighborhood V_p of a singularity. On the complement of these neighborhoods, we can choose some constant such that

$$dr \wedge dv \wedge \alpha_0 \wedge \alpha_1 > Q_1 > 0$$

Therefore the second line is positive for

$$A > D_3/Q_1 \tag{6.3}$$

In the third line, the last term may be negative. However, by choosing M sufficiently large, the second term in the first line will dominate it. To see this, choose $P > 0$ such that

$$dr \wedge \alpha_1 \wedge d\alpha_1 > P$$

Then we require that

$$M > e^{v - Ar} \frac{C_1 \phi'}{AP\phi^2} \tag{6.4}$$

over the compact interval $\delta' \leq r - v \leq \epsilon$.

This proves that $d\lambda$ is symplectic.

Claim 3 There is an inward-pointing Liouville vector field along $\widehat{Y} = \psi^{-1}(\epsilon_0)$.

Since $d\lambda$ is a symplectic form, the 1-form primitive λ is $d\lambda$-dual to a Liouville vector field

$$d\lambda(\rho, -) = \lambda$$

To be inward-pointing, we require that $d\psi(\rho) > 0$, which is equivalent to the condition

$$d\psi \wedge \lambda \wedge d\lambda > 0$$

As $d\psi$ is a nonnegative linear combination of dr, dv, we can check positivity for just these component 1-forms.

First,

$$\begin{aligned}
dv \wedge \lambda \wedge d\lambda &= dv \wedge \left(e^v \alpha_0 + M\phi e^{Ar} \alpha_1\right) \wedge \left(e^v d\alpha_0 + M\phi e^{Ar} \left(A dr \wedge \beta_1 + d\beta_1\right) + M\phi' e^{Ar} dr \wedge \alpha_1\right) \\
&= e^{2v} dv \wedge \alpha_0 \wedge d\alpha_0 + M\phi' e^{Ar+v} dr \wedge dv \wedge \alpha_0 \wedge \alpha_1 \\
&+ M\phi A e^{Ar+v} dv \wedge \alpha_0 \wedge dr \wedge \alpha_1 + M\phi e^{Ar+v} dv \wedge \alpha_0 \wedge d\alpha_1 \\
&+ M\phi e^{Ar+v} dv \wedge \alpha_1 \wedge d\alpha_0 \\
&= e^{2v} dv \wedge \alpha_0 \wedge d\alpha_0 + M\phi e^{Ar+v} dv \wedge \left(\left(\frac{\phi'}{\phi} + A\right) \alpha_1 \wedge \alpha_0 \wedge dr + \alpha_0 \wedge d\alpha_1 + \alpha_1 \wedge d\alpha_0\right)
\end{aligned}$$

By Claim 1, we know that we can choose open neighborhoods $\{U_p\}$ of the singularities in which

$$dv \wedge \alpha_0 \wedge d\alpha_1 + dv \wedge \alpha_1 \wedge d\alpha_0 > 0$$

Outside these neighborhoods, we can find a constant Q_2 such that

$$dr \wedge dv \wedge \alpha_0 \wedge \alpha_1 > Q_2$$

so the 4-form is positive if

$$A > D_1/Q_2 \tag{6.5}$$

Secondly,

$$\begin{aligned}
dr \wedge \lambda \wedge d\lambda &= dr \wedge e^v \alpha_0 \wedge d\lambda + dr \wedge M\phi e^{Ar} \alpha_1 \wedge d\lambda \\
&= dr \wedge e^v \beta_0 \wedge \left(e^v (dv \wedge \alpha_0 + d\alpha_0) + M\phi e^{Ar} d\alpha_1 + M\phi' e^{Ar} (-dv \wedge \alpha_1) \right) \\
&+ M\phi e^{Ar} (dr \wedge \alpha_1 \wedge (e^v (dv \wedge \alpha_0 + d\alpha_0) + M\phi e^{Ar} d\alpha_1) \\
&= M\phi e^{Ar+v} dr \wedge \alpha_0 \wedge d\alpha_1 + M\phi' e^{Ar+v} dr \wedge dv \wedge \alpha_0 \wedge \alpha_1 \\
&+ M\phi e^{Ar+v} dr \wedge dv \wedge \alpha_0 \wedge \alpha_1 + M\phi e^{Ar+v} dr \wedge \alpha_1 \wedge d\alpha_0 + M^2 \phi^2 e^{2Ar} dr \wedge \alpha_1 \wedge d\alpha_1 \\
&= M\phi e^{Ar} dr \wedge \left(e^v \left(\alpha_0 \wedge d\alpha_1 + \alpha_1 \wedge d\alpha_0 + \left(1 + \frac{\phi'}{\phi} \right) dv \wedge \alpha_0 \wedge \alpha_1 \right) + M\phi e^{Ar} \alpha_1 \wedge d\alpha_1 \right)
\end{aligned}$$

By Claim 1, we know that we can choose open neighborhoods $\{U_p\}$ of the singularities in which

$$dr \wedge \alpha_0 \wedge d\alpha_1 + dr \wedge \alpha_1 \wedge d\alpha_0 > 0$$

Hence, the above form is positive in those neighborhoods. Furthermore, the term $\frac{\phi'}{\phi}$ tends to $+\infty$ as ϕ goes to 0. Therefore, we can assume that this 4-form is positive for $0 < \phi \leq \delta''$. Then for $\phi > \delta''$ the final term dominates for M sufficiently large.

Finally, we prove Statement (3). Along the hypersurface $\{r = 0\}$, Equation 6.2 becomes

$$d\lambda = e^v (dv \wedge \alpha_0 + d\alpha_0) + M\phi(-v) d\alpha_1 - M\phi' dv \wedge \alpha_1$$

Hence

$$\begin{aligned}
dr \wedge \alpha_1 \wedge d\lambda &= e^v (dr \wedge \alpha_0 \wedge dv \wedge \alpha_1 + dr \wedge \alpha_1 \wedge d\alpha_0) + M\phi dr \wedge \alpha_1 \wedge d\alpha_1 \\
&= dr \wedge \alpha_1 \wedge \omega + M\phi dr \wedge \alpha_1 \wedge d\alpha_1
\end{aligned}$$

which is positive.

□

7. Interpolating weak fillings

The main aim of this section is to describe how to interpolate between weakly-filled contact structures along the boundary.

7.1. **Dividing sets, characteristic foliations, and equivalence of contact structures.** We remind the reader that here are notions of equivalence for two contact structures ξ_1, ξ_2 with convex boundary on a 3-manifold Y with nonempty boundary. The stronger equivalence is defined in terms of the characteristic foliation $\mathcal{F}$ while the weaker is defined in terms of the dividing set Γ.

We first state a lemma about convex surfaces, dividing sets, and characteristic foliations.

Lemma 7.1. *Let $\Gamma \subset \Sigma$ be a multicurve. There exists a contact structure $(\Sigma \times \mathbb{R}, \xi_\Gamma)$ defined by a vertically-invariant contact form*

$$\alpha = udz + \beta$$

where $\Gamma = u^{-1}(0)$. Moreover, for any two foliations $\mathcal{F}_1, \mathcal{F}_2$ divided by Γ, there exist functions $\Psi_1, \Psi_2 : \Sigma \to \mathbb{R}$ such that

(1) $\Phi_1 < \Phi_2$ everywhere
(2) ξ induces the characteristic foliation $\mathcal{F}_1$ on the graph of Φ_1,
(3) ξ induces the characteristic foliation $\mathcal{F}_2$ on the graph of Φ_2.

Given an identification

$$\Sigma \times [0,1] \cong \{(x,z) \in \Sigma \times \mathbb{R} : \Phi_1(x) \leq z \leq \Phi_2(x)\}$$

let $(\Sigma \times [0,1], \xi_{1,2})$ denote the restriction of ξ_Γ. It is well-defined, up to ambient isotopy, in terms of $\mathcal{F}_1, \mathcal{F}_2$.

Definition 7.2. Let ξ_1, ξ_2 be two contact structures on Y.

(1) *Strong equivalence.* There is a diffeomorphism $\Phi : Y \to Y$ such that

$$d\Phi(\xi_1) = \xi_2$$

(2) *Weak equivalence.* There exists strong equivalences

$$\Phi_{1,2} : (Y, \xi_1) \cup (\partial Y \times [0,1], \xi_{1,2}) \to (Y, \xi_2)$$
$$\Phi_{2,1} : (Y, \xi_2) \cup (\partial Y \times [0,1], \xi_{2,1}) \to (Y, \xi_1)$$

7.2. **Germs of weak fillings.**

Definition 7.3. Fix a closed, oriented surface Σ and an (isotopy class of) multicurve $\Gamma \subset \Sigma$

(1) Let $\mathfrak{C}(\Sigma, \Gamma)$ denote the set of germs of contact *forms* on Σ whose characteristic foliations are Morse-Smale and divided by Γ up to isotopy.
(2) Let $\mathfrak{F}(\Sigma, \Gamma)$ denote the set of Morse-Smale characteristic foliations on Σ divided by Γ up to isotopy.

Lemma 7.4. *Let $\beta \in \mathfrak{C}(\Sigma, \Gamma)$ be the germ of a contact form on Σ and let $\mathcal{F} = ker(\zeta) \in \mathfrak{F}(\Sigma, \Gamma)$ be a Morse-Smale foliation. Let Ω be an area form on Σ and $d\beta = f\Omega$.*

There exists a contact germ $(\Sigma \times (-\delta, \delta), \xi)$ inducing $\mathcal{F}$ on $\Sigma \times \{0\}$ and dominated by

$$\omega = e^t(dt \wedge \beta + d\beta)$$

if and only if

(1) $fd\zeta > 0$ near the zeros of ζ, and
(2) along $B = f^{-1}(0)$.

$$\beta \wedge \zeta > 0$$

Proof. Since ζ is Morse-Smale, we can choose a closed 1-form q satisfying $q \wedge \zeta \geq 0$ with equality at the zeros of ζ. Set

$$\alpha = fdt + \zeta + tCq$$

where C is a positive constant. Then

$$\alpha \wedge d\alpha = (fd\zeta + \zeta \wedge df + Cq \wedge \zeta + tCq \wedge df) \wedge dt$$

is a positive contact form along $\{t = 0\}$ for C sufficiently large. In addition

$$\alpha \wedge \omega = \left(\beta \wedge \zeta + tC\beta \wedge q + f^2\Omega\right) \wedge dt$$

is a positive volume form along $\{t = 0\}$. Therefore, there exists some $\delta > 0$ such that $\xi = \ker(\alpha)$ is a contact structure dominated by ω. □

7.3. **Gluing weak fillings together.** We start with some general results about interpolating 2-plane fields and contact structures across a surface.

Definition 7.5. Let $\mathcal{F}_0, \mathcal{F}_1$ be singular foliations and let ω be a 2-form on Σ. We say that ω *dominates the pair* $\mathcal{F}_0, \mathcal{F}_1$ if there exists open sets U, V with $\overline{U} \subsetneq V \subset \Sigma$ such that

(1) ω is nonzero in V, and
(2) $\mathcal{F}_0 = \mathcal{F}_1$ on $\Sigma \setminus U$.

First, we start with an easy lemma to interpolate between 2-plane fields dominated by ω.

Lemma 7.6. *Let $(\Sigma \times [-\delta, 0], \xi_-)$ and $(\Sigma \times [0, \delta])$ be 2-plane fields, let ω be a 2-form on $\Sigma \times [-\delta, \delta]$, and let $\mathcal{F}_\pm = \xi_\pm|_{\Sigma\times\{0\}}$. If $\omega|_{\Sigma\times\{0\}}$ dominates $\mathcal{F}_-, \mathcal{F}_+$, then there exists a 2-plane field $(\Sigma \times [-\delta, \delta], \xi)$ dominated by ω and agrees with $\xi_\pm$ near $\pm\delta$.*

Proof. Set

$$\lambda_\pm = u_\pm dz + \beta_\pm$$

Choose a bump function ϕ, supported in V, that equals ± 1 on U, according to the sign of ω and $c_\pm(t)$ cutoff functions that equal 1 at $t = 0$ and 0 at $t = \pm\delta$. Then the kernel of

$$\widetilde{\lambda}_\pm = (u_\pm + Mc_\pm(t)\phi)dz + \beta_\pm$$

satisfy $\widetilde{\lambda}_\pm \wedge \omega > 0$ for $M > 0$. Moreover, for M sufficiently large, the straight-line interpolation between $\widetilde{\lambda}_\pm$ along $t = 0$ remains dominated by ω. □

Second, we prove that weak fillings can be glued together if their characteristic foliations match up along the interface surface.

Proposition 7.7 (Gluing weak fillings)**.** *Let ω be a 2-form on $\Sigma \times [-\delta, \delta]$ and suppose that*

(1) $\lambda_- = u_-dz + \beta_+$ is a contact form on $\Sigma \times [-\delta, 0]$ and $\lambda_- \wedge \omega > 0$
(2) $\lambda_+ = u_+dz + \beta_-$ is a contact form for $\Sigma \times [0, \delta]$ and $\lambda_+ \wedge \omega > 0$

with $\beta_+ = \beta_-$ along $\Sigma \times \{0\}$ and defines a Morse-Smale characteristic foliation. There exists a contact form

$$\lambda = udz + \beta$$

where $\lambda = \lambda_-$ near $z = -\delta$ and $\lambda = \lambda_+$ near $z = +\delta$ and $\lambda \wedge \omega$ for $z \in [-\delta, \delta]$.

Proof. The proof is slightly subtle, because although $\beta_+ = \beta_-$ along $\Sigma \times \{0\}$, to glue these contact forms together we need to further assume that β_+, β_- agree to first-order at $z = 0$. Therefore, we first need to perturb $\lambda_\pm$ before gluing. Also, without loss of generality, we can extend $\lambda_\pm$ slight across $z = 0$ as contact forms dominated by ω and assume that $\lambda_\pm$ are defined on $[-\delta, \delta]$.

We can choose some ζ_0 such that

$$\zeta_0 \wedge \beta_0 > -\min(u_\pm d\beta_0 + \beta_0 \wedge du_\pm)$$

and apply Lemma 7.8 to assume that

$$\lambda_\pm = u_\pm dz + \beta_0 + z\zeta_0 + O(z^2)$$

For $0 < \delta'$, let $c(z)$ be a smooth cutoff function on $[-\delta', \delta']$, and let

$$\begin{aligned}\lambda &= c\lambda_+ + (1 - c)\lambda_- \\ &= (cu_+ + (1 - c)u_-)\, dz + (c\beta_+ + (1 - c)\beta_-)\end{aligned}$$

so

$$\begin{aligned}\lambda \wedge d\lambda = c^2\lambda_+ \wedge d\lambda_+ + (1-c)^2\lambda_- \wedge d\lambda_- \\ + c(1-c)(\lambda_+ \wedge d\lambda_- + \lambda_- \wedge d\lambda_+) \\ + c' dz \wedge \lambda_+ \wedge \lambda_-\end{aligned}$$

Furthermore

$$\lambda_\pm \wedge d\lambda_\mp = (u_\pm d\beta_\mp + \beta_\pm \wedge du_\mp + \beta'_\mp \wedge \beta_\pm) \wedge dz$$

Since $\beta_+ = \beta_-$ along $z = 0$, we have that

$$\lambda_+ \wedge d\lambda_- + \lambda_- \wedge d\lambda_+ = \lambda_+ \wedge d\lambda_+ + \lambda_- \wedge d\lambda_- > 0$$

along $z = 0$, hence also in some neighborhood of $z = 0$. Finally c' is $O(1/\delta')$ while $dz \wedge \lambda_+ \wedge \lambda_-$ is $O((\delta')^2)$, so the final term is $O(\delta')$. Therefore, λ is a contact form for $\delta' > 0$ sufficiently small. □

Lemma 7.8. *Let*

$$\xi = ker(udz + \beta_0 + z\beta_1 + O(z^2))$$

be a contact structure on $\Sigma \times [0, \delta]$ dominated by ω.

(1) Suppose that

$$ud\beta_0 + \beta_0 \wedge du + \zeta \wedge \beta_0 > 0$$

There exists a perturbation ξ' of ξ that is dominated by ω and equals

$$\xi' = ker(udz + \beta_0 + z\zeta + O(z^2))$$

for $z \in [0, \delta']$ for some $0 < \delta' < \delta$.

(2) Suppose that

$$\begin{aligned}vd\beta_0 + \beta_0 \wedge dv + \beta_1 \wedge \beta_0 > 0 \\ (vdz + \beta_0) \wedge \omega > 0\end{aligned}$$

along $z = 0$. Then there exists a perturbation ξ' of ξ that is dominated by ω and equals

$$\xi' = ker(vdz + \beta_0 + z\beta_1 + O(z^2))$$

for $z \in [0, \delta']$ for some $0 < \delta' < \delta$.

(3) Suppose that

$$\begin{aligned}vd\beta_0 + \beta_0 \wedge dv + \zeta_1 \wedge \beta_0 > 0 \\ (vdz + \beta_0) \wedge \omega > 0\end{aligned}$$

There exists a perturbation ξ' of ξ that is dominated by ω and equals

$$\xi' = ker(vdz + \beta_0 + z\zeta + O(z^2))$$

for $z \in [0, \delta']$ for some $0 < \delta' < \delta$.

Proof. Let $\lambda = udz + \beta_0 + z\beta_1 + O(z^2)$ be the contact form for ξ. Then

$$\lambda \wedge d\lambda = (ud\beta_0 + \beta_0 \wedge du + \beta_1 \wedge \beta_0) \wedge dz + O(z)$$

To prove part (1), we first claim that for any $\epsilon > 0$, we can choose a function $g : [0, \delta] \to \mathbb{R}$ such that

$$\begin{array}{lll} g(0) = 0 & |g| < \epsilon & g'(0) = 1 \\ g(\delta) = 0 & g' > -\epsilon & \end{array}$$

Now set $\lambda' = \lambda + g \cdot (\zeta - \beta_1)$. Then

$$\begin{aligned}\lambda' \wedge d\lambda' = (ud\beta_0 + \beta_0 \wedge du + \zeta \wedge \beta) \wedge dz + O(z) + O(\epsilon) \\ \lambda' \wedge \omega = \lambda \wedge \omega + O(\epsilon)\end{aligned}$$

Therefore, choosing ϵ, δ' sufficiently small, we can obtain the desired contact structure dominated by ω.

For part (2), let $c(z)$ be a cutoff function such that $c(0) = 1$ and $c(\delta') = 0$. Define $\lambda' = \lambda + c \cdot (v - u)dz$. Then

$$\lambda' \wedge d\lambda' = (c \cdot (vd\beta_0 + \beta_0 \wedge dv) + (1-c)(ud\beta_0 + \beta_0 \wedge du) + \beta_1 \wedge \beta_0) \wedge dz + O(z)$$
$$\lambda' \wedge \omega = c \cdot (vdz + \beta_0) \wedge \omega + (1-c)(udz + \beta_0) \wedge \omega + O(z)$$

which are both positive for $\delta' > 0$ sufficiently small.

Finally, for part (3), since $\ker(\beta_0)$ is Morse-Smale, choose a function $f : \Sigma \to \mathbb{R}$ such that $\mathcal{F}$ is directed by a gradient-like vector field for f, so that $df \wedge \beta_0 \geq 0$. We can choose a function $h : \Sigma \to \mathbb{R}$ such that

$$hdf \wedge \beta_0 > -\min(ud\beta_0 + \beta_0 \wedge du, vd\beta_0 + \beta_0 \wedge dv)$$

We can then apply Parts (1) and (2) to obtain sequences of contact structures

$$\xi_0 = \ker(udz + \beta_0 + z\beta_1 + O(z^2))$$
$$\xi_1 = \ker(udz + \beta_0 + z\, hdf + O(z^2))$$
$$\xi_2 = \ker(vdz + \beta_0 + z\, hdf + O(z^2))$$
$$\xi_3 = \ker(vdz + \beta_0 + z\,\zeta + O(z^2))$$

dominated by ω. □

Third, we state another easy gluing result.

Proposition 7.9 (Gluing along convex pairs)**.** *Let ω be a 2-form on $\Sigma \times [-\delta, \delta]$ and suppose that*

(1) $\lambda_- = u_- dz + \beta_+$ is a contact form on $\Sigma \times [-\delta, 0]$ and $\lambda_+ \wedge \omega > 0$
(2) $\lambda_+ = u_+ dz + \beta_-$ is a contact form for $\Sigma \times [0, \delta]$ and $\lambda_- \wedge \omega > 0$

with $\mathcal{F}_\pm = ker(\beta_\pm)$ along $\Sigma \times \{0\}$.

If $\mathcal{F}_-, \mathcal{F}_+$ are a convex pair, there exists a contact form

$$\lambda = udz + \beta$$

where $\lambda = \lambda_-$ near $z = -\delta$ and $\lambda = \lambda_+$ near $z = +\delta$ and $\lambda \wedge \omega$ for $z \in [-\delta, \delta]$.

Proof. Define

$$\lambda = c\lambda_+ + (1-c)\lambda_-$$

as in the previous proposition, where c is a cutoff function on $[-\delta', \delta']$. Near the singular points, we have

$$\lambda_- \wedge d\lambda_+ > 0 \qquad\qquad \lambda_+ \wedge d\lambda_- > 0$$

and furthermore we have that

$$c'dz \wedge \beta_+ \wedge \beta_-$$

is positive (except at the singular points) and $O(1/\delta')$ while the other two terms are bounded. Hence by sending δ' to 0 the interpolating 2-plane field is contact. □

7.4. **Flexibility of characteristic foliation.** Given a convex surface Σ with dividing set Γ, Giroux Flexibility ensures that we can realize any characteristic foliation divided by Γ via a C^0-small ambient isotopy of Σ. In general, we cannot expect the same flexibility and glue two weak fillings together as in Proposition 7.7 simply because they induce the same dividing set. However, in the case of tight contact structures on S^2, we can glue together weak fillings inducing inequivalent characteristic foliations under some restricted conditions. In particular, we have enough flexilibity to prove our main resolt.

Definition 7.10. Let $\mathcal{F}$ be a characteristic foliation on S^2 and let $\gamma : [0,1] \to S^2$ be a handleslide arc (Definition 9.14).

(1) The handleslide is *positive* if $\gamma(0)$ lies in the stable manifold of some h_0 and $\gamma(1)$ lies in the unstable manifold of some h_1
(2) the handleslide is *negative* if $\gamma(0)$ lies in the unstable manifold of some h_0 and $\gamma(1)$ lies in the stable manifold of some h_1.

(3) An *elementary neighborhood* of γ is a neighborhood $U \cong [-\delta, 1+\delta] \times [-\delta, \delta]$ of γ with coordinates (x, y) such that
$$\gamma(t) = (t, 0) \qquad\qquad \mathcal{F}|_U = \ker(dx)$$
(4) An *positive (resp. negative) elementary handle-slide* along γ is the 1-parameter family of foliations
$$\mathcal{F}_t = \ker(dx \pm \phi f dy)$$
where
(a) ϕ is a bump function on some elementary neighborhood U of γ that equals 1 along γ,
(b) $f(t, y)$ is a nonpositive function satisfying
(i) $f(0, y) = 0$
(ii) $\frac{\partial f}{\partial t} > 0$, and
(iii) $-\int_{-\delta}^{\delta} f(1/2, y)dy = 1$
(5) An elementary handle slide along γ is *dominated* by ω if $\omega|_U$ is nonzero.

Positive and negative handle-slides need to be treated differently. If $\mathcal{F}'$ is obtained from $\mathcal{F}$ by a positive handle slide, then the pair $\mathcal{F}', \mathcal{F}$ is convex and we can apply Proposition 7.9 to modify the characteristic foliation. However, if $\mathcal{F}'$ is obtained from $\mathcal{F}$ by a negative handle-slide, then it requires more work to modify the characteristic foliation. Since the handleslide is trivial, after eliminating singularities in pairs (by a C^0-small isotopy) we can eliminate the handleslide and obtain a 1-parameter family of foliations with a negative shear. Giroux's realization lemma [Gir00, Lemma 2.3] implies that we can find a contact structure that induces this 1-parameter family of foliations. However, this requires tilting the contact planes to be almost-tangent to the surface. Consequently, we have the following restricted version of the realization lemma.

Lemma 7.11 (Realization lemma for weak fillings)**.** *Let $(\Sigma \times [0, 1], \xi)$ be a 2-plane field dominated by a 2-form ω and inducing the $\mathcal{F}_z = ker(\beta_z)$ of singular foliations on Σ. Let Σ^* be a subsurface. Suppose that*
(1) ξ is contact near $\Sigma \times \{0, 1\}$ and on $(\Sigma \setminus \Sigma^*) \times [0, 1]$.
(2) there exists a family of functions $u_z : \Sigma^* \to \mathbb{R}$ such that
(a) $u^{-1}(0) = \partial\Sigma^*$ and u is nonzero in the interior of Σ^*
(b) $u_z\, d\beta_z + \beta_z \wedge du_z > 0$ on Σ^* for all $z \in [0, 1]$, and
(c) $u_z\, dz \wedge \omega$ is a postive volume form on $\Sigma^* \times [0, 1]$.

There exists
(1) a contact structure $(\Sigma \times [0, 1], \xi')$ dominated by ω and inducing the singular foliations $\mathcal{F}_z$
(2) a homotopy from ξ to ξ', supported in $\Sigma^* \times [0, 1]$, through 2-plane fields dominated by ω and inducing the singular foliations $\mathcal{F}_z$.

Proof. Let $\alpha = vdz + \beta_z$ be a defining 1-form for ξ. For $C(z)$ sufficiently large, then $\widetilde{\alpha} = \alpha + C(z)u_z\, dz$ is a contact form satisfying $\widetilde{\alpha} \wedge \omega > 0$. □

We can appy this to obtain the following gluing result.

Proposition 7.12. *Let ω be a 2-form on $\Sigma \times [-\delta, \delta]$ and suppose that*
(1) $\lambda_- = u_- dz + \beta_+$ is a contact form on $\Sigma \times [-\delta, 0]$ and $\lambda_- \wedge \omega > 0$
(2) $\lambda_+ = u_+ dz + \beta_-$ is a contact form for $\Sigma \times [0, \delta]$ and $\lambda_+ \wedge \omega > 0$

Let $\Sigma^* \subset \Sigma$ be a subsurface such that
(1) $\beta_+ = \beta_-$ along $(\Sigma \setminus \Sigma^*) \times \{0\}$
(2) $\Sigma^* \times \{0\}$ is ω-symplectic, and
(3) there exists a function $u : \Sigma^* \to [0, \infty)$ satisfying
(a) $u\, d\beta_\pm + \beta_\pm \wedge du > 0$
(b) $u(\partial\Sigma^*) = 0$

There exists a contact form
$$\lambda = udz + \beta$$
where $\lambda = \lambda_\pm$ near $z = \pm\delta$ and $\lambda \wedge \omega$ for $z \in [-\delta, \delta]$.

7.5. **Negative handleslides.** Modifying the foliation by a negative handleslide requires some control over the symplectic form. To simplify the task, we assume that ω is *almost-positive*, which means that ω is a positive area, except in some ϵ-neighborhoods of the negative singular points. With the assumption, we further split negative handleslide into two cases: whether the 'over' hyperbolic point is positive or negative.

7.5.1. *Positive hyperbolics.* When we have a negative handleslide of some positive hyperbolic over another hyperbolic, we can apply Proposition 7.12 as follows. Let h_+ be the positive hyperbolic point. It is connected to two (distinct) positive elliptic points e_1, e_2. We can find a subsurface $\Sigma^* \cong D^2$ containing h_+, e_1, e_2 and the handleslide arc γ but no negative singular points and whose boundary is transverse to the foliation. See Figure 18. If ω is almost-positive, we can assume Σ^* is ω-symplectic.

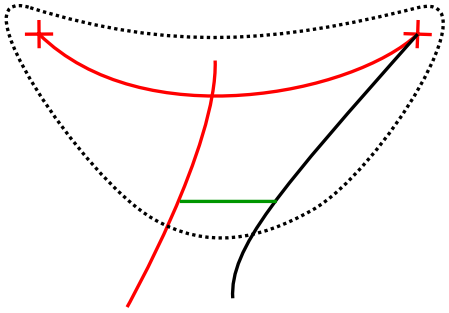


FIGURE 18. When handlesliding a positive hyperbolic over another hyperbolic, if the symplectic form is almost-positive then we can choose a symplectic surface containing the handleslide arc γ.

Proposition 7.13. *Let ω be a 2-form on $S^2 \times [-\delta, \delta]$ that is almost-positive along $S^2 \times \{0\}$ and suppose that*

(1) $\lambda_- = u_- dz + \beta_+$ is a contact form on $\Sigma \times [-\delta, 0]$ and $\lambda_- \wedge \omega > 0$
(2) $\lambda_+ = u_+ dz + \beta_-$ is a contact form for $\Sigma \times [0, \delta]$ and $\lambda_+ \wedge \omega > 0$

where $\mathcal{F}_+ = ker(\beta_+)$ is obtained from $\mathcal{F}_- = ker(\beta_-)$ by an elementary negative trivial handleslide of a positive hyperbolic over some other hyperbolic. There is a contact form λ where $\lambda = \lambda_\pm$ near $z = \pm\delta$ and $\lambda \wedge \omega'$ for $z \in [-\delta, \delta]$.

7.5.2. *Negative hyperbolics.* The most difficult case is a negative handleslide of a negative hyperbolic over some other hyperbolic. In this case, we will first need to modify ω by enlarging over a 4-dimensional cobordism (see Subsections 7.6 and 7.7) before gluing the two contact structures together. Moreover, we will need to satisfy a quantitative lower bound in order to modify ω sufficiently.

The first step is to eliminate some of the negative singularities in pairs. We need the following basic lemma about tight characteristic foliations on S^2. Recall that an *elimination arc* in a characteristic foliation $\mathcal{F}$ is a separatrix connecting an elliptic and hyperbolic point of the same sign.

Lemma 7.14. *Let $\mathcal{F}$ be a tight characteristic foliation on S^2 and let γ be a trivial handle slide arc.*

(1) If γ is a positive handleslide arc, let e_+ denote the α-limit of $\gamma(1+\delta)$ and let e_- denote the ω-limit of $\gamma(-\delta)$ for $\delta > 0$ sufficiently small.
(2) If γ is a negative handleslide arc, let e_+ denote the α-limit of $\gamma(-\delta)$ and let e_- denote the ω-limit of $\gamma(1+\delta)$ for $\delta > 0$ sufficiently small.

One can choose eliminations arcs $\{C_i^+\}$ and $\{C_j^+\}$, one for each positive and negative, respectively, hyperbolic point in the foliation, such that

(1) each C_i^+ and C_j^- is disjoint from the arc γ
(2) the (closures of the) separatrices $\{C_i^+\}$ are mutually disjoint and disjoint from e_+, and
(3) the (closures of the) separatrices $\{C_j^-\}$ are mutually disjoint and disjoint from e_-.

Proof. Let h_0 denote the ω-limit of $\gamma(0)$; it is a positive hyperbolic point. Let E denote the edge of the positive Giroux graph C_+ containing $\gamma(0)$. Since G_+ is a tree, the graph $G_+ \smallsetminus E$ has two connected components. Moreover, since the bypass is trivial, we know that (a) h_0 and e_+ lie in different components. (b) the component containing h_0 has an even number of vertices, equally elliptic and hyperbolic, that can be paired uniquely, and(c) the component containing e_+ is a tree with an odd number of vertices; after picking any elliptic point, we can uniquely pair the remaining vertices. The same argument holds for the negative Giroux graph. □

Let h_- be the negative hyperbolic point. It has two stable separatrices σ_1^-, σ_2^- that connect it to one or two positive elliptic points. If the positive elliptic points are distinct, there exist a sequence of separatrices $\sigma_1^+, \dots, \sigma_{2k}^+$ in the positive Giroux graph connecting them. The union of all $2k+2$ separatrices is a simple closed curve in S^2 that bounds a disk D disjoint from the handleslide arc γ. By slightly enlarging D, we obtain a disk D' whose boundary is transverse to the characteristic foliation $\mathcal{F}$. Inside this disk D', there is a unique pairing of the negative singularities (including h_-). Using the Elimination Lemma [Gir91, Eli93] (see also [Gei08, Lemma 4.6.26]), we can choose a C^0-small ambient isotopy of the surface (supported in D') to eliminate these singularities in pairs.

After eliminating the singularities in pairs, consider the characteristic foliation $\mathcal{F}'$ and a vector field v directing it. For a point x not in the positive Giroux graph, the flow of v takes it to $\partial D'$. We can choose a set $U \subset (D' \smallsetminus D)$ such that if $x \in D'$ and ω is nonpositive at x, the flowline through x intersects U before reaching $\partial D'$. See Figure 19.

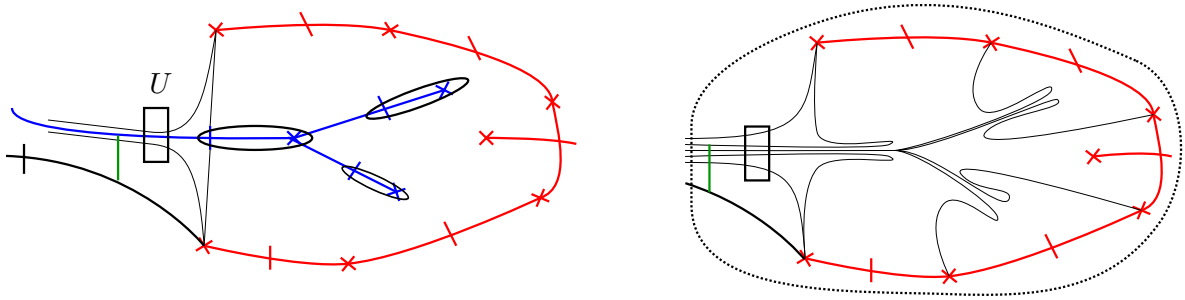


FIGURE 19. For a negative handleslide of a negative hyperbolic over some other hyperbolic, we eliminate the negative singularities in pairs. Afterwards, every flowline in the interior must either be contained in the positive Giroux graph or path through the set U above the handleslide arc γ.

Proposition 7.15. *Let ω be a 2-form on $S^2 \times [-\delta, \delta]$ that is almost-positive along $S^2 \times \{0\}$ and suppose that*

(1) $\lambda_- = u_- dz + \beta_+$ is a contact form on $\Sigma \times [-\delta, 0]$ and $\lambda_- \wedge \omega > 0$
(2) $\lambda_+ = u_+ dz + \beta_-$ is a contact form for $\Sigma \times [0, \delta]$ and $\lambda_+ \wedge \omega > 0$

where $\mathcal{F}_+ = ker(\beta_+)$ is obtained from $\mathcal{F}_- = ker(\beta_-)$ by an elementary negative trivial handleslide of a negative hyperbolic over some other hyperbolic.

Fix an area form Ω on U. There exists a constant C such that if $\omega = f\Omega$ with $f(x, y) > C$ on U, then there exists an enlargement from ω to ω' over a symplectic cobordism and a contact form

$$\lambda = u dz + \beta$$

where $\lambda = \lambda_\pm$ near $z = \pm\delta$ and $\lambda \wedge \omega'$ for $z \in [-\delta, \delta]$.

Proof. We can choose some $D' \subset D''$ such that the elementary neighborhood of the handle slide in contained in D'' and ω is a positive area form on $D'' \smallsetminus D'$. See Figure 19. In order to apply Proposition 7.12, we need to ensure that D' is ω-symplectic as well. By assumption, the two foliations agree on the disk D' and via a

C^0-small isotopy supported in D' we can cancel the negative singularities. As this is an ambient isotopy, we can pull back ω as well. Now inside D', every point with nonpositive ω-area flows through U before reaching the boundary of D'; hence if ω is sufficiently large everywhere in U, we have sufficient area to redistribute inward along the leaves of $\mathcal{F}'$ using Proposition 7.19. □

7.6. **Enlarging.** We prove two lemmas that allow us to modify ω along the boundary by enlarging ([Gay02, Definition 1.6] and compare with Definition 5.9) while also preserving the weak filling condition along the boundary.

First, we describe enlarging in the interior of the boundary. Recall that a *confoliation* on a 3-manifold is a 2-plane field defined by a 1-form μ satisfying $\mu \wedge d\mu \geq 0$.

Lemma 7.16. *Let (W, ω) be a symplectic 4-manifold with boundary.*

1. *Let $U \subset \partial W$ be an open set, let μ be a confoliation 1-form defined on U satisfying $\mu \wedge \omega > 0$. Let ϕ be a nonnegative function supported in U. There exists an enlargement of (W, ω) supported in U such that ω, restricted to the boundary, is $\omega' = \omega + d(\phi\mu)$.*
2. *Let λ be a positive contact form along ∂W satisfying $\lambda \wedge \omega$. Let ϕ be a nonnegative function. There exists an enlargment of (W, ω) whose restriction to the boundary is $\omega' = \omega + d(\phi\lambda)$. Moreover, we have $\lambda \wedge \omega' > 0$.*

Proof. Fix a transverse vector field ∂_r along ∂W. With respect to the flow of this vector field, we obtain a decomposition

$$\omega = \omega_r + dr \wedge q_r$$

Since ω is nondegenerate, there is a unique pair of positive function f and vector field v_r satisfying

$$\mu = \omega_r(v_r, -) + f q_r$$

Replacing ∂_r with $f_r \partial_r + v_r$, we can assume that $q_0 = \mu$. Moreover, we can perturb ω near the boundary and assume that

$$\omega = \omega_0 + d(r\mu)$$

for $-\delta \leq r \leq \delta$ for some small $\delta > 0$. Since μ is a confoliation, this 2-form is symplectic for $r \in [0, \infty)$. Hence we can attach an infinite collar $U \times [0, \infty)$ to the boundary of ∂W and extend ω across this infinite collar. We can enlarge so that the new boundary is defined by $r = \phi$ and the restriction of ω to this hypersurface is as required.

For part (2), note that if ϕ is nonnegative and λ is a positive contact form, then $\phi\lambda$ is a confoliation as

$$\phi\lambda \wedge d(\phi\lambda) = \phi^2 \lambda \wedge d\lambda \geq 0$$

Moreover, we have that

$$\lambda \wedge \omega' = \lambda \wedge \omega + \lambda \wedge d(\phi\lambda) = \lambda \wedge \omega + \phi(\lambda \wedge d\lambda) > 0$$

□

Secondly, we describe enlarging near the corner surface of a symplectic manifold (W, ω). We restrict to a local model

$$(W, \omega) = ([0, \infty) \times (-\infty, 0] \times \Sigma, d(e^t \alpha))$$

with coordinates (t, r) normal to Σ and contact form

$$\alpha = u dr + \beta$$

We let $[0, \infty) \times \{0\} \times \Sigma$ be the *exterior boundary* and let $\{0\} \times (-\infty, 0] \times \Sigma$ be the *interior boundary*. The vector field ∂_t is Liouville for $\omega = d(e^t\alpha)$ and points inward along the interior boundary. We further assume that there is a contact structure $\xi_\partial = \ker(v dt + \zeta)$ along the exterior boundary that is dominated by ω. Therefore, the interior boundary is strongly concave and the exterior boundary is weakly convex. Let $U \subset \Sigma$ be an open set. We say that the corner is *convex* along $U \subset \Sigma$ if (ζ, β) are a convex pair along U.

Lemma 7.17. *Let U be an open set in Σ and let μ_0 be a closed 1-form on U such that in U*

$$\zeta \wedge \beta > 0 \qquad \mu_0 \wedge \beta > 0 \qquad \mu_0 \wedge \zeta > 0$$

with respect to the orientation induced by $dt \wedge dr$. Let ψ be a compactly supported, nonnegative function in U. For $\delta > 0$ sufficiently small, then exists a symplectic enlargement of (W, ω) supported in $[0, \delta] \times \{0\} \times U$ such that

(1) $\omega = d(e^t(\alpha + r\mu_0))$ along the interior boundary,
(2) the corner surface is defined by the equations $\{t = 0, r = \psi\}$.
(3) $\lambda \wedge \omega > 0$ along the exterior boundary. In particular, the exterior boundary is weakly convex.

Proof. Since $\zeta \wedge \beta > 0$ on U, we can apply Lemma 7.4 and assume that over some interval $[0, \delta]$, the contact structure along the exterior boundary is defined by

$$\lambda = \zeta + tq$$

in U for some 1-form q. We can assume that ∂_r is a contact vector field and that

$$\alpha = udr + \beta$$

for some function u on Σ. We can replace ∂_r along the interior boundary with a different contact vector field and assume that

$$\alpha = \epsilon udr + \beta$$

for any $\epsilon > 0$. We can enlarge as in Lemma 7.16 using the confoliation $\mu = e^t\mu_0$ on $U \times [0, \delta]$ since

$$\begin{aligned} dr \wedge \mu \wedge \omega &= dr \wedge e^t\mu_0 \wedge (dt \wedge \beta + d\beta) \\ &= e^t(dt \wedge dr \wedge \mu_0 \wedge \beta) \end{aligned}$$

Near the interior boundary, our symplectic form is

$$\omega = d(e^t\alpha) + d(re^t\mu_0) = d(e^t(\alpha + r\mu_0))$$

Therefore, the interior boundary is strongly concave if and only if $\alpha' = \alpha + r\mu_0$ is a contact form. We have

$$d\alpha' = \epsilon du \wedge dr + d\beta + dr \wedge \mu_0$$

and

$$\begin{aligned} \alpha' \wedge d\alpha' &= \alpha \wedge d\alpha + dr \wedge \mu_0 \wedge \beta + r\mu_0 \wedge (\epsilon du \wedge dr + d\beta) \\ &= \alpha \wedge d\alpha + dr \wedge \mu_0 \wedge (\beta + \epsilon r du) \end{aligned}$$

Therefore, if $\mu_0 \wedge \beta > 0$, then since ψ is compactly supported in $V \subset U$, we can choose ϵ sufficiently small so that α' is a contact form.

Now, let $c(t)$ be a smooth cutoff function on $[0, \delta]$ such that $c(0) = 1$ and $c(\delta) = 0$. Now we enlarge to the level set

$$r = c(t)\psi$$

Along the exterior boundary, the symplectic form restricts to

$$\omega = e^t(dt \wedge \beta + d\beta + (c'\psi dt + cd\psi) \wedge \mu_0)$$

and

$$\begin{aligned} dr \wedge \lambda \wedge \omega &= e^t(dr \wedge dt \wedge \beta \wedge (\zeta + tq) + c'\psi\, dr \wedge dt \wedge \mu_0 \wedge \zeta) \\ &= e^t dt \wedge dr \wedge ((\zeta + tq) \wedge \beta - c'\psi\, \mu_0 \wedge \zeta) \end{aligned}$$

which is positive since $c' \leq 0 \leq \psi$; therefore the enlarged symplectic form ω still dominates $\xi_\partial = \ker(\lambda)$. □

As a consequence of the previous lemma, we can 'smooth out' the symplectic form along the corner surface.

Lemma 7.18. *Let e be an elliptic point of the singular foliation $ker(\beta)$ on Σ. Suppose there exists a neighborhood $U \cong D^2$ of e such that*

(1) β, ζ meet convexly in U, and

(2) there exist polar coordinates (r,θ) on U such that $\beta = \phi(r,\theta)d\theta$

For any $\delta > 0$ and function $0 < \psi \leq \phi$ with $\psi = \phi$ in a neighborhood of ∂U, there exists an enlargment supported in $U \times [0,\delta]$ such that

(1) Along the corner surface, the ω-primitive β' induced by the inward-pointing Liouville vector field ∂_t is a C^∞-small perturbation of $\psi(r,\theta)d\theta$
(2) ξ_∂ remains dominated by ω,
(3) β', ζ meet convexly in U.
(4) the contact structure along the enlarged portion of the interior boundary is weakly vertically-invariant.

In particular, if ϕ is $O(\epsilon)$ along ∂U, then we can assume that ψ and $\omega = d\beta' = \psi_r\, dr \wedge d\theta$ are $O(\epsilon)$ in U as well.

Proof. Choose a C^∞-small perturbation μ_0 of $(\psi - \phi)d\theta$ that satifies $\mu_0 \wedge d\theta > 0$ and apply Lemma 7.17. Then the ω-primitive along $\{0\} \times \{r\} \times \Sigma$ is C^∞-close to

$$\beta_r = ((1-r)\phi + r\psi)\, d\theta$$

Therefore, the characteristic foliations induced on Σ are topologically conjugate for $r \in [0,1]$ and the contact structure is weakly vertically-invariant. □

7.7. **Redistributing symplectic area.** Given Giroux's Realization Lemma [Gir00, Lemma 2.3] and Lemma 7.11, the natural question is to ask whether one can modify ω to ensure that the subsurface Σ^* is symplectic. We give a positive criterion for this question that is partly topological and partly quantitative. Before stating the result precisely, we give a rough description of the geometric intuition.

Suppose we have a separating multicurve $\gamma \subset \Sigma$, bounding a subsurface Σ^c, that is transverse to the characteristic foliation for all $z \in [0,1]$ and which contains only positive singularities and repelling periodic orbits. Then it is possible to choose a positive function h, supported in Σ^c, such that $\alpha = hdz + \lambda$ is a contact form. However, we can only ensure that $\alpha \wedge \omega > 0$ when $dz \wedge \omega > 0$ – i.e when Σ^c is ω-symplectic. If λ is contact over some subsurface $\Sigma^b \subset \Sigma^c$, we can use the 1-form $\mu = \phi\lambda$ (where ϕ is a bump function supported in Σ^b) to redistribute symplectic area within Σ^b.

However, if we restrict ourselves to only modifying ω by enlarging along the boundary of a weak filling, we encounter a 'one-way' restriction: we can redistribute along the integral curves of the characteristic foliation, but we can only move symplectic area *inward*. In other words, suppose β_z defines the characteristic foliation, Ω_z is an area form, and v is the vector field Ω_z-dual to β_z whose flow integrates to the characteristic foliation. Then we can only move symplectic area in the $(-v)$-direction.

We therefore have the following obstruction. Fix a point $p \in \Sigma^b$. Let ζ_p denote the oriented arc along the characteristic foliation from p to $\partial\Sigma^b$ and let r be a parameter along ζ_p such that $\Omega_z = dr \wedge \beta_z$. We can express

$$\omega_z = g\Omega_z$$

for some function g. A necessary requirement to enlarge the weak filling and replace ω with some ω' such that ω' is a positive area form along ζ_p is that the integral

$$\int_{\zeta_p} g dr,$$

which represents the total ω-area along ζ_p, must be positive. More generally, to have a symplectic enlargement from ω to some ω' such that Σ^b is ω'-symplectic, the function

$$F(p) = \int_{\zeta_p} g dr \tag{7.1}$$

must be positive for each point $p \in \Sigma^b$.

Now consider the following geometric setup. Topologically, suppose have three subsurfaces

$$\Sigma^a \subset \Sigma^b \subset \Sigma^c \subset \Sigma$$

such that

(1) λ is contact over $\Sigma^b \times [0,1]$, but not necessarily over the entirety of $\Sigma^c \times [0,1]$,
(2) each surface $\Sigma^a \times \{z\}$ and $(\Sigma^c \setminus \Sigma^b) \times \{z\}$ is ω-symplectic,
(3) the characteristic foliation in each $(\Sigma^c \setminus \Sigma^a) \times \{z\}$ is *annular*, meaning that each point lies on a unique flowline ζ that starts in $\partial\Sigma^a \times \{z\}$, intersects $\partial\Sigma^b \times \{z\}$ transversely exactly once, and terminates on $\partial\Sigma^c \times \{z\}$.

In particular, the three subsurfaces are homeomorphic to each other and their boundaries are positively transverse to the characteristic foliation. Moreover, at each point in $\Sigma^c \times [0,1]$, either λ is a positive contact form or ω is a positive area form on the surface. Then, if the function F defined in Equation 7.1 is positive everywhere on $(\Sigma^b \setminus \Sigma^a)$, we can enlarge from ω to ω' and find a contact form $\alpha = hdz + \lambda$ such that $\alpha \wedge \omega' > 0$ on $\Sigma^c \times [0,1]$.

With this setup in mind, we state this more precisely.

Proposition 7.19. *Let $\lambda \wedge \omega > 0$ on $\Sigma \times [0,1]$. Suppose there exists a smooth family of functions*

$$r_z : \Sigma \to (-\infty, 1]$$

parametrized by $z \in [0,1]$ with no critical values in $[-1,1]$ such that

(1) each sublevel set $r_z^{-1}((-\infty,-1])$ is both $d\lambda$- and ω-symplectic,
(2) λ is contact over the sublevel set $r_z^{-1}(-\infty, 0])$
(3) each annulus $r_z^{-1}([0,1])$ is ω-symplectic
(4) λ is contact over the complement of $r_z^{-1}((-\infty,1]))\times[\delta, 1-\delta]$ for some $\delta > 0$.
(5) $\Omega_z = dr_z \wedge \lambda$ is a positive area form on $r_z^{-1}([-1,1])$ for all $z \in [0,1]$.

Let $\omega_z = g_z\Omega_z$ denote the restriction of ω. Suppose there exists a positive function F supported in $\Sigma^b \times [0,1]$ and satisfying

$$dF = g_z dr_z + A\beta_z + Bdz$$

for each $z \in [0,1]$. Then

(1) there exists an enlargement from ω to ω' such that each sublevel set $r_z^{-1}((-\infty,1])$ is ω'-symplectic,
(2) there exists a positive function h, supported in $r_z^{-1}((-\infty,1]))$ such that $\alpha = hdz + \lambda$ is contact everywhere.

Moreover, $\alpha \wedge \omega' > 0$.

Proof. First, we want to modify the contact form such that $d\lambda$ is a positive area form on the pages. Note that the interior of Σ^a is both $d\lambda$- and ω-symplectic. Without loss of generality, we can assume that $dr_z \wedge \lambda \geq 0$ over the interior of $r^{-1}((-\infty,-1])$. Let $H(r)$ be a nonnegative, decreasing function that equals e^{-Cr} for $r \in [-1,1]$ for $C \gg 0$. Then $\alpha_t = \lambda + tHdz$ satisfies

$$\alpha_t \wedge d\alpha_t = \lambda \wedge d\lambda + tHdz \wedge d\lambda - tH'dz \wedge dr \wedge \lambda > 0$$

so this defines a homotopy of 1-forms. We have not assumed that $d\lambda$ is a positive area form on $r_z^{-1}([-1,0])$, however if we choose C sufficiently large, then α is a contact form and $d\alpha$ is a positive area form.

Since F is strictly positive, we can choose some positive $\widetilde{F}$ such that

$$d\widetilde{F} = \widetilde{g}_z dr_z + \widetilde{A}\beta_z + \widetilde{B}dz$$

with $\widetilde{g}_z < g_z$. Let $\psi(z)$ be a bump function supported in $[0,1]$ that equals 1 on $[\delta, 1-\delta]$.

Now consider the 1-form

$$\mu = \psi(z)\widetilde{F}\lambda$$

Since λ is a contact form in the support of $\psi\widetilde{F}$ and the latter is nonnegative, we can apply Lemma 7.16 to enlarge the symplectic 4-manifold while preserving the weak filling.

The new symplectic structure $\omega' = \omega + d\mu$ satisfies

$$\begin{aligned}(\omega + d\mu)_{\Sigma^b \times \{z\}} &= g_z\Omega_z + \psi\widetilde{F}d\lambda + \psi d\widetilde{F}\lambda \\ &= (g_z - \widetilde{g}_z)\Omega_z + \psi\widetilde{F}d\lambda\end{aligned}$$

which is strictly positive. □

8. Weak Filling

In this section, we combine the building blocks to construct the desired weak filling. As described in Subsection 1.6 of the introduction, the symplectic 1-cocycle $\widetilde{\beta}$ yields a path in the strong contact groupoid $\mathcal{SG}(D^2, n, r)$ whose homotopy class is determined by the Gompf invariant $\Theta_G(\widetilde{B}, \widetilde{\beta})$. If the Gompf invariant agrees with that of the standard tight contact structure (S^3, ξ_{std}), then we can take advantage of this homotopy to produce a weak symplectic filling.

8.1. **Dividing sets by sector.** We have to turn paths in the (strong) contact groupoid of the disk D^2 into sequences of tight characteristic foliations on $S^2 = -D^2 \cup D^2$. We first specify a sequence of tight dividing sets on the boundary 2-spheres, then in Subsection 8.2 specify sequences of characteristic foliations.

Recall that if $\Gamma \to \Gamma'$ is a morphism in the contact category of D^2, there is a tight contact structure (B^3, ξ_{std}) inducing the dividing set $\overline{\Gamma}\#\Gamma'$ along $\partial B^3 = S^2 = -D^2 \cup D^2$. Moreover, if there is a factorization

$$\Gamma = \Gamma_0 \to \Gamma_1 \to \cdots \to \Gamma_k = \Gamma'$$

then we get sequences of tight dividing sets

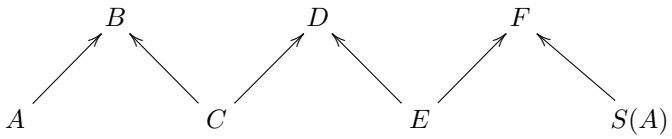
$$\overline{\Gamma}_0\#\Gamma_k, \overline{\Gamma}_0\#\Gamma_{k-1}, \ldots, \overline{\Gamma}_0\#\Gamma_1, \overline{\Gamma}_0\#\Gamma_0$$

and

$$\overline{\Gamma}_0\#\Gamma_k, \overline{\Gamma}_1\#\Gamma_k, \ldots, \overline{\Gamma}_{k-1}\#\Gamma_k, \overline{\Gamma}_k\#\Gamma_k$$

on S^2. The dividing set $\overline{\Gamma}_0\#\Gamma_i$ is obtained from $\overline{\Gamma}_0\#\Gamma_{i+1}$ by carving a trivial bypass along some dual bypass arc in D^2. Similarly statements hold for characteristic foliations as well, where foliations on S^2 are related by elementary trivial handleslides.

Now for exposition purposes, we can relabel the terms of Line 1.3 as



Moreover, recall from Section 3.1 that we have chosen tubular neighborhoods $\nu(\Sigma) = \Sigma \times [-2\delta, 2\delta]^2$ and $\nu(H_\lambda) \cong [-\delta', \delta'] \times H_\lambda$ for some $\delta > \delta' > 0$ to be determined below. In addition, we have a decomposition

$$S_\lambda = \partial Y_\lambda^* = -\widetilde{V}_{\lambda-1} \cup \mathcal{A}_\lambda \cup \widetilde{V}_\lambda$$

where

$$-\widetilde{V}_{\lambda-1} \coloneqq (V_{\lambda-1} \times \{\delta'\}) \smallsetminus \nu(\Sigma) \qquad \mathcal{A}_\lambda \coloneqq S_\lambda \cap \nu(\Sigma) \qquad \widetilde{V}_\lambda \coloneqq (V_\lambda \times \{-\delta'\}) \smallsetminus \nu(\Sigma)$$

8.1.1. *Bypass sequences.* We perform the following sequences of bypasses along the three boundary components of $\nu(\mathcal{S}_\partial)$ of the thickened boundary spine.

Sector 1. Along the boundary $-\widetilde{V}_3 \cup \widetilde{V}_1$ of $\nu(\mathcal{S}_\partial)$, the dividing set is $\overline{A}\#B$. Attach trivial bypasses along $\widetilde{V}_1$ to change B to A. The resulting dividing set is $\overline{A}\#A$.

Sector 2. Along the boundary $-\widetilde{V}_1 \cup \widetilde{V}_2$ of $\nu(\mathcal{S}_\partial)$, the dividing set is $\overline{C}\#D$. We have the following attaching/carving sequence of bypasses and corresponding sequence of dividing sets on S^2.

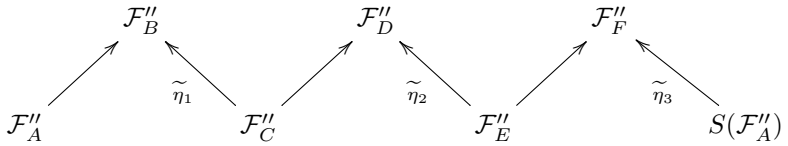
(1) attach trivial bypasses along $\widetilde{V}_2$ to change D to C ($\overline{C}\#C$)
(2) carve trivial bypasses along $\widetilde{V}_2$ to change C to B ($\overline{C}\#B$)
(3) attach trivial bypasses along $-\widetilde{V}_1$ to change $\overline{C}$ to $\overline{B}$ ($\overline{B}\#B$)
(4) carve trivial bypasses along $-\widetilde{V}_1$ to change $\overline{B}$ to $\overline{A}$ ($\overline{A}\#B$)
(5) attach trivial bypasses along $\widetilde{V}_2$ to change B to A. ($\overline{A}\#A$)

Sector 3. Along the boundary $-\widetilde{V}_2 \cup \widetilde{V}_3$ of $\nu(\mathcal{S}_\partial)$, the dividing set is $\overline{E}\#F$. We have the following attaching/carving sequence of bypasses and corresponding sequence of dividing sets on S^2.

(1) attach trivial bypasses along $\widetilde{V}_3$ to change F to E ($\overline{E}\#E$)
(2) carve trivial bypasses along $\widetilde{V}_3$ to change E to D ($\overline{E}\#D$)
(3) attach trivial bypasses along $-\widetilde{V}_2$ to change $\overline{E}$ to $\overline{D}$ ($\overline{D}\#D$)
(4) carve trivial bypasses along $-\widetilde{V}_2$ to change $\overline{D}$ to $\overline{C}$ ($\overline{C}\#D$)
(5) attach trivial bypasses along $\widetilde{V}_3$ to change D to C ($\overline{C}\#C$)
(6) carve trivial bypasses along $\widetilde{V}_3$ to change C to B ($\overline{C}\#B$)
(7) attach trivial bypasses along $-\widetilde{V}_2$ to change $\overline{C}$ to $\overline{B}$ ($\overline{B}\#B$)
(8) carve trivial bypasses along $-\widetilde{V}_2$ to change $\overline{B}$ to $\overline{A}$ ($\overline{A}\#B$)
(9) carve trivial bypasses along $\widetilde{V}_3$ to change B to A ($\overline{A}\#A$)
(10) carve trivial bypasses along $\widetilde{V}_3$ to change A to $S(A)$. ($\overline{A}\#S(A)$)

8.2. **Characteristic foliations along compressing disks.** Now we choose sequences of characteristic foliations to realize the above sequences of trivial bypasses. Importantly, we need to realize mirror characteristic foliations on $\widetilde{V}_\lambda$ and $-\widetilde{V}_\lambda$. Furthermore, note that the dividing set $\overline{A}$ on $-\widetilde{V}_1$ and $S(A)$ on $\widetilde{V}_1$ *are* mirror, because the shift functor S is the (geometric) monodromy of the contact structure with respect the open book decomposition.

From Proposition 3.14, we have foliations $\mathcal{F}''_{\lambda,\lambda}, \mathcal{F}''_{\lambda+1,\lambda}$ along $\widetilde{V}_\lambda$ and a morphism $\widetilde{\eta}_\lambda : \mathcal{F}''_{\lambda+1,\lambda} \to \mathcal{F}''_{\lambda,\lambda}$ generated by elementary handle slides in the interior of $\widetilde{V}_\lambda$. As above, we can relabel the terms of Line 1.5 as



We have the following sequences of foliations along the compressing disks:

(1) $-\widetilde{V}_3$. No bypasses are attached, the foliation remains $\overline{\mathcal{F}}''_{A,3} = \overline{\mathcal{F}}''_A$.

(2) $\widetilde{V}_1$. We can construct a sequence of foliations
$$\mathcal{F}''_B = \mathcal{F}''_{B,1} \to \mathcal{F}''_{A,1}$$
where $\mathcal{F}''_{A,1}$ is some foliation on V_1 realizing the dividing set A.

(3) $-\widetilde{V}_1$. We have a sequence
$$\overline{\mathcal{F}}''_C = \overline{\mathcal{F}}''_{C,1} \xrightarrow{\overline{\widetilde{\eta}}_1} \overline{\mathcal{F}}''_{B,1} \to \overline{\mathcal{F}}''_{A12}$$

(4) $\widetilde{V}_2$. We can construct a sequence of foliations
$$\mathcal{F}''_D = \mathcal{F}''_{D,2} \to \mathcal{F}''_{C,2} \to \mathcal{F}''_{B,2} \to \mathcal{F}''_{A,2}$$
where $\mathcal{F}''_{A,2}$ is some foliation on V_2 realizing the dividing set A.

(5) $-\widetilde{V}_2$. We have a sequence
$$\overline{\mathcal{F}}''_E = \overline{\mathcal{F}}''_{E,2} \xrightarrow{\overline{\widetilde{\eta}}_2} \overline{\mathcal{F}}''_{D,2} \to \overline{\mathcal{F}}''_{C,2} \to \overline{\mathcal{F}}''_{B,2} \to \overline{\mathcal{F}}''_{A,2}$$

(6) Now, along $\widetilde{V}_3$, we have the path

$$F \to E \to D \to C \to B \to A \to S(A) \tag{8.1}$$

which is homotopic to

$$F \xrightarrow{\eta_3^{-1}} S(A) \xrightarrow{\eta_3} F \to E \to D \to C \to B \to A \to S(A) \tag{8.2}$$

and therefore homotopic to

$$F \xrightarrow{\eta_3^{-1}} S(A) \xrightarrow{\Delta^{-k}} S(A) \tag{8.3}$$

where Δ denotes a bypass triangle and $k \in \mathbb{Z}$. We can choose a lift

$$\mathcal{F}''_{F,3} \xrightarrow{\widetilde{\eta_3}^{-1}} S(\mathcal{F}''_{A,3}) \to \mathcal{F}''_{S(A),3}$$

of Line 8.3. The homotopy from Line 8.3 to Line 8.2 lifts to the strong contact groupoid (Proposition 9.38), then truncating we obtain a path

$$\mathcal{F}''_F = \mathcal{F}''_{F,3} \to \mathcal{F}''_{E,3} \to \mathcal{F}''_{D,3} \to \mathcal{F}''_{C,3} \to \mathcal{F}''_{B,3} \to \mathcal{F}_{A,3} \to \mathcal{F}_{S(A),3} \tag{8.4}$$

of foliations.

The choice of $\mathcal{F}''_{S(A),3}$ depends on k:

(a) If $k > 0$, by applying Lemma 10.18 we can assume that the morphism $S(\mathcal{F}''_{A,3}) \to \mathcal{F}''_{S(A),3}$ is a composition of elementary handle slides in the interior of $\widetilde{V}_3$ and that $\mathcal{F}''_{S(A),3}, S(\mathcal{F}''_{A,3})$ meet convexly,
(b) If $k = 0$, we let $\mathcal{F}''_{S(A),3} = S(\mathcal{F}''_{A,3})$,
(c) If $k < 0$, by applying Lemma 10.18 we can assume that the morphism $S(\mathcal{F}''_{A,3}) \to \mathcal{F}''_{S(A),3}$ is a composition of elementary handle slides in the interior of $\widetilde{V}_3$ and that $S(\mathcal{F}''_{A,3}), \mathcal{F}''_{S(A),3}$ meet convexly.

Remark 8.1. The lift in Line 8.4 may require stabilization in the strong contact groupoid. Stabilization

$$\Sigma : \mathcal{SG}_{n,r,k} \to \mathcal{SG}(n, r, k+1)$$

corresponds to adding canceling elliptic/hyperbolic pairs of singularities. This can be accomplished by an isotopy of the symplectic 1-cocycle (or equivalently V_1 within H_1) to introduce a pair of critical points – a convex extremum and a saddle point – of the same coorientation. While this undoes Roussarie-Thurston general position, it does not destroy the pseudoconvexity and Legendrian binding properties of $\widetilde{\beta}$ or the tightness of the map $S(\Gamma_{1,1}) \to \Gamma_{3,1}$.

8.3. **Modifying to be almost-positive.** In order to realize a sequence of trivial handle slides, they must each be dominated by ω (Definition 7.10). Over $\nu(H_\lambda) \cong [-\delta', \delta'] \times H_\lambda$, the symplectic form is

$$\omega = dt \wedge \widetilde{\beta}_\lambda + d\kappa_\lambda$$

so its restriction to $\pm\widetilde{V}_\lambda \subset \{t = \mp\delta'\}$ is determined by a primitive κ_λ as in Subsection 2.2.1. We can modify these primitives so that ω is a positive area form, except in an arbitrarily-small neighborhood of the negative singular points of $\widetilde{\beta}_\lambda$ in V_λ.

Proposition 8.2. *For every fixed neighborhood U of the negative critical points of $\widetilde{\beta}_\lambda$ in V_λ, there exists a primitive κ_λ such that $d\kappa_\lambda$ is a positive area form on $V_\lambda \smallsetminus U$.*

Proof. First, we fix Morse local models for κ_λ near the negative singular points. Second, to start we have that κ_λ is $O(\epsilon')$; without loss of generality we can scale it and assume it is $O((\epsilon')^2)$.

Let p be a point in the interior of V_λ. There exists a path γ_p in V_λ transverse to $\ker(\widetilde{\beta}_\lambda)$ from p to some maximum m of $\widetilde{\beta}_\lambda|_{V_\lambda}$. Since $\widetilde{\beta}$ is pseudoconvex, we have that $d\kappa_\lambda < 0$ near m.

Fix coordinates (t, x, y, z) on $[-\delta', \delta'] \times H_\lambda$ near p such that $V_\lambda = \{y = 0\}$ and $\widetilde{\beta}_\lambda = dz$ and $H_\lambda = \{y \leq 0\}$. Therefore, with our orientation conventions (Subsection 3.1.2) the 2-form $dy \wedge dt$ coorients the surface and $dx \wedge dz$ orients it.

We can extend these coordinates to a neighborhood $\nu(\gamma_p)$ of the path in H_λ. In particular, we have

$$\nu(\gamma_P) = [-2\delta_0, 2\delta_0] \times [-\epsilon_0, 0] \times \gamma_p$$

with coordinates (x, y, z) for some $\epsilon_0, \delta_0 > 0$.

Define

$$\zeta_p = -\phi(y)\psi(z)c(x)dx$$

where

(1) $\phi(y)$ is a cutoff function on $[-\epsilon_0, 0]$ with $\phi(0) = 1$,
(2) $\psi(z)$ is a nonpositive function such that $\phi' = -1$ on $[-\delta_0, \delta_0]$ and such that $\psi' \leq 0$ on the complement of U.
(3) $c(x)$ is a bump function on $[-2\delta_0, 2\delta_0]$ that equals 1 on $[-\delta_0, \delta_0]$.

Then $d\zeta_p$ is nonnegative on the complement of U and equal to $dx \wedge dz$ on a δ_0-neighborhood of p. Since $\widetilde{V}_\lambda$ is compact, we can choose a finite number of points $\{p_i\}$ to cover $V_\lambda \smallsetminus U$ and define

$$\kappa'_\lambda = \kappa_\lambda + \sum \epsilon' \zeta_{p_i}$$

so that $d\kappa'_\lambda$ satisfies the conclusions of the lemma. In addition, we have

$$\widetilde{\beta}_\lambda \wedge d\zeta_p = -\phi' \psi c \, dz \wedge dx \wedge dy \geq 0$$

which implies that $\mu_\lambda, \mu_{\lambda+1}$ remain contact forms along H_λ. □

8.4. **Weak filling over the spine.** We now construct the first of our building blocks: the spine of the relative trisection. The boundary spine $\mathcal{S}_\partial = V_1 \cup V_2 \cup V_3$ is the union of three disjoint Seifert disks along the unknot binding.

Proposition 8.3. *For $0 < \delta' \ll \delta$ sufficiently small, there exists a contact structure $(\nu(\mathcal{S}_\partial), \xi'_\partial)$ dominated by ω that induces the following singular foliations on the three boundary components:*

(1) $\overline{\mathcal{F}}''_{A,3} \cup \mathcal{F}''_{A,1}$ on $-\widetilde{V}_3 \cup \widetilde{V}_1$
(2) $\overline{\mathcal{F}}''_{A,1} \cup \mathcal{F}''_{A,2}$ on $-\widetilde{V}_1 \cup \widetilde{V}_2$
(3) $\overline{\mathcal{F}}''_{A,2} \cup \mathcal{F}''_{S(A),3}$ on $-\widetilde{V}_2 \cup \widetilde{V}_3$.

Moreover,

(4) there exists a 2-plane field $(\nu(\mathcal{S}_\partial), \xi_\partial)$ dominated by ω, satisfying (1) and (2) above but inducing $\overline{\mathcal{F}}''_{A,2} \cup S(\mathcal{F}''_{A,3})$ on $-\widetilde{V}_2 \cup \widetilde{V}_3$.
(5) if $k \geq 0$, then ξ_∂ is a contact structure.
(6) $\mu_\lambda, ker(\xi_\partial)$ meet convexly along S_λ in a 3δ-neighborhood of the binding.

Proof. Our approach is:

(a) In a 2δ-neighborhood of the binding, we have a standard Legendrian neighborhood $(\nu(U), \xi_{std})$ where $-n = tw(\delta, \widetilde{\beta})$ is the Thurston-Bennequin number of the unknotted binding,
(b) Along each compressing disk, we extend ξ_{std} to induce the required foliation along $\widetilde{V}_\lambda \times \{0\}$, then apply Lemma 7.4 to obtain a weak filling in some δ'-neighborhood.
(c) When $k = 0$, we are done. If $k < 0$, we can apply Lemma 7.6 to modify ξ'_∂ to ξ_∂, while if $k > 0$ we apply Proposition 7.9 instead.

Moreover, along the binding we define a function f satisfying $df \wedge \omega > 0$, then perturb the foliation $\ker(df)$ to a contact structure. Via the Lemma 8.4 below, choosing f is equivalent to choosing a vector field transverse to $\ker(\omega)$.

Recall that near the binding, we have coordinates (t, z) on S_λ and

$$\mu_\lambda = \epsilon g(\overline{y}' \cdot \overline{y}) P dt + Q dz$$

Suppose we have a vector field v and corresponding function f from the Lemma, with

$$df|_{S_\lambda} = f_t dt + f_z dz$$

Away from $\zeta = \{Q = 0\}$, we have $|Q| \gg O(\epsilon)$. Moreover, the sign of f_t is determined by whether v is pointing positively or negatively transverse to S_λ in the (x_1, x_2)-plane. Therefore, we can ensure that df meets $\ker(\mu_\lambda)$ convexly just by ensuring v points in the correct normal direction, then choosing δ, δ' sufficiently small.

To define f near ζ, we choose an explicit function near the zeros of β_λ, then extend. The reason is as follows. Let z_0 be a zero of β_λ along the binding. In coordinates $(\phi_\lambda, \phi_{\lambda-1}, z)$, the set $\zeta_\lambda \subset S_\lambda$ determined by μ_λ enters a neighborhood of $(-\delta', 2\delta, z_0)$ while $\zeta_{\lambda+1} \subset S_{\lambda+1}$ determined by $\mu_{\lambda+1}$ enters a neighborhood

of$(+\delta', 2\delta, z_0)$. In particular, these are sufficiently close to one another that we need to check that df meets both foliations convexly here.

Now assume that $\lambda = 1$. Recall that

$$\mu_1 = g(z)(\overline{y}_1' \cdot \overline{y}_1)P_1 dt + Q_1 dz$$
$$\mu_2 = g(z)(\overline{y}_2' \cdot \overline{y}_2)P_2 dt + Q_2 dz$$

Since z_0 is a zero of β_1, then $q_1 = O(z)$ and $p_1 \neq 0$. Moreover, we have $\Phi_1 = O(\epsilon')$ while Φ_2, Φ_3 are close to 1. Therefore Q_1, Q_2 are $O(\epsilon' + \delta)$ and

$$P_1(z_0)P_2(z_0) = (\Phi_3 p_1 - \Phi_1 p_3)(\Phi_1 p_2 - \Phi_2 p_1) = -\Phi_3\Phi_2 p_1^2 + O(\epsilon') < 0$$

Furthermore, note that $\overline{y}_1' \cdot \overline{y}_1 > 0$ but $\overline{y}_2' \cdot \overline{y}_2 < 0$.

Now, set

$$f = P_1(z_0)(\cos(z)\phi_1 + \sin(z)\phi_3)$$

Near $\phi_3 = 2\delta$, the surfaces S_1, S_2 lie in level sets of ϕ_1. Therefore,

$$df|_{S_1} = df|_{S_2} = P_1(z_0)\left(\phi_3 dz + O(z)d\phi_3\right)$$

Consequently

$$df_{S_1} \wedge \mu_1 = \left(g(\overline{y}_1' \cdot \overline{y}_1)P_1^2\phi_3 + O(z + \epsilon')\right) dz \wedge dt$$
$$df_{S_2} \wedge \mu_2 = \left(g(\overline{y}_2' \cdot \overline{y}_2)P_1 P_2 \phi_3 + O(z + \epsilon')\right) dz \wedge dt$$

which are both positive for z, ϵ' sufficiently small.

Having defined f at the endpoints of ζ, we can extend to a function on ζ. To ensure that it meets μ_λ convexly, we just need to choose f such that it has critical points along ζ at exactly the singular points, which lie along ζ by Proposition 3.5. □

Lemma 8.4. *Let $\nu(B)$ denote the 2δ-neighborhood of B in the boundary spine.*

(1) There exists a function $m(x_1, x_2, z)$ on $\nu(B)$ such that

$$\omega = d(m\, dz) = dm \wedge dz$$

(2) Let v be a vector field satisfying

$$dz(v) = 0 \qquad \text{and} \qquad dm(z) > 0$$

Then there exists a function f satisfying

$$df(v) = 0 \qquad \text{and} \qquad df \wedge dm \wedge dz > 0$$

(3) The triple (f, m, z) give coordinates on $\nu(B)$, and

(4) The 1-form $\alpha_\partial = df + g\, dz$ defines a weak filling if $\frac{\partial g}{\partial m} > 0$.

(5) The binding is ker(α_∂)-Legendrian if and only if $\frac{\partial f}{\partial z} + g = 0$ along the binding.

Proof. Define

$$m(x_1, x_2, z) = -x_2 b_3(z) + (x_2 - x_1)b_1(z)$$

and we can extend this using Φ. Then

$$\omega|_B = d(mdz) = dm \wedge dz$$

Since ω is nondegenerate, this implies that dm is nonvanishing. Let v be a nonvanishing vector field such that

$$dz(v) = 0 \qquad dm(v) > 0$$

We can integrate the vector field then choose a function f constant on the integral curves (i.e. so that $df(v) = 0$). In a fixed z-slice, the line field $\ker(\omega)$ is transverse to the integral curves and so up to changing the sign of f we can assume

$$df \wedge dm \wedge dz > 0$$

everywhere. In particular, the triple (f, m, z) gives coordiantes on a tubular neighborhood of the binding. Then take

$$\alpha_\partial = df + g dz$$

If $\frac{\partial g}{\partial m} > 0$, then this is a contact form which further satisfies

$$\alpha_\partial \wedge \omega = df \wedge dm \wedge dz > 0$$

□

8.5. **Collars.** The next step is to modify the characteristic foliations induced by ξ_∂ along the corners of the spine using the sequences of foliations from Section 8.2.

Proposition 8.5. *After modifying the primitives $\{\kappa_\lambda\}$, enlarging along the boundary spine, and modifying the inward-pointing Liouville vector fields $\{\rho_\lambda\}$, there exists a 2-plane field $(\nu(S_\partial), \xi_\partial)$ dominated by ω that induces singular foliations on the three boundary components topologically conjugate to the following:*

(1) $\overline{\mathcal{F}}''_{A,3} \cup \mathcal{F}''_{B,1}$ on $-\widetilde{V}_3 \cup \widetilde{V}_1$,
(2) $\overline{\mathcal{F}}''_{C,1} \cup \mathcal{F}''_{D,2}$ on $-\widetilde{V}_1 \cup \widetilde{V}_2$,
(3) $\overline{\mathcal{F}}''_{E,2} \cup \mathcal{F}''_{F,3}$ on $-\widetilde{V}_2 \cup \widetilde{V}_3$,

Moreover

(4) if $k \geq 0$, then ξ_∂ is a contact structure,
(5) $\mu_\lambda, \xi_\partial$ meet convexly along S_λ

8.5.1. *Realizing sequences of handleslides.* The results of Section 7 concerning *gluing* two weak fillings together, while here we want to change the characteristic foliation induced by ξ_∂ along the boundary.

Consider a contact structure $(S^2 \times [0, \delta], \xi_\partial)$ that is dominated by $\omega = e^t(dt \wedge \beta + d\beta)$. Suppose that ξ_∂ induces the characteristic foliation $\ker(\zeta_+)$ on $S^2 \times \{0\}$ and we want to modify it by an elementary handleslide to $\ker(\zeta_-)$. By Lemma 7.4, we can assume

$$\xi_\partial = \ker(u\, dt + \zeta_+ + tq)$$

for $t \in [0, \delta']$ for some 1-form q and $\delta' > 0$. Without loss of generality, we can choose q and shrink δ' so that $\ker(u\, dt + \zeta_- + tq)$ also defines a contact structure dominated by ω for $t \in [0, \delta']$. We can then glue these two contact structures along $t = \delta'/2$.

8.5.2. *Locating sufficient symplectic area.* The bypass sequences to realize the foliations in Section 8.2 include both positive and negative handleslides (see Definition 7.10). In order to apply Proposition 7.15 and perform negative trivial handleslides of negative hyperbolics while maintaining the weak filling property, we need to locate sufficient positive symplectic area just below the negative hyperbolic point. Since ω is closed, this requires concentrating an equal amount of negative symplectic area near the negative elliptic point at the other end of the unstable separatrix. However, a 'sufficient' amount of positive symplectic area near one negative hyperbolic point depends on the amount of negative symplectic area concentrated near other negative elliptic points, which depends on all of the other negative handleslides in the sequence.

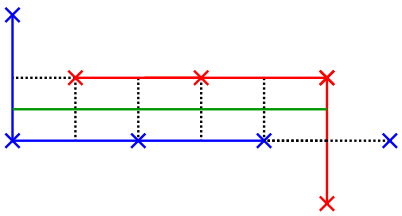
Therefore, in order to achieve the required estimates on symplectic area we proceed as follows:

(1) Start with a 'spiky' symplectic form on the spine with large amounts of positive and negative symplectic area concentrated near pairs of negative hyperbolic and negative elliptic singularities, respectively, in $V_\lambda \times [-\delta', \delta']$.
(2) After a negative handleslide at step i, enlarge the spine to smooth out the symplectic area near this pair of negative hyperbolic/elliptic singular points (using Lemma 7.18) until is satisfies a bound independent of i.

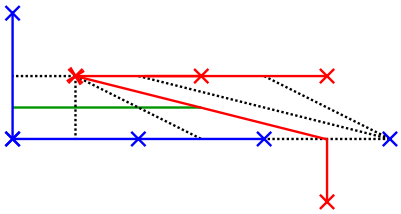
In this way, if we have a sequence of k handleslides, the symplectic area required at step i depends only on step $i+1$ through step k. Therefore, we can work backwards from the final step to the first step to determine how much symplectic area is required for each negative handleslide.

For concreteness, to get a total order on all of the handleslides, we will perform all of the handleslides in Sector 1, then the handleslides in Sector 2, and finally all of the handleslides in Sector 3.

8.5.3. *Disjoint modifications.* As described in Section 9.5.2, we cannot expect that the handleslides are contained in mutually disjoint elementary neighborhoods. However, we are able to assume that the modifications to ω below are disjoint for the following reason. Suppose that the Thurston-Bennequin invariant of the binding is $-n$. We can identify $V_\lambda \cong \mathbb{D}$ such that the Giroux graph in an annular region near the boundary lies in the rays $\{\mathbb{R}e^{\frac{ki\pi}{n}}\}$ for $k = 1, \dots, 2n$ with the negative Giroux graph corresponding to k even and positive Giroux graph corresponding to k odd. We only need to modify ω in a neighborhood of the negative Giroux graph. Moreover, we can realize the universal morphism by $n-1$ 'long handleslides' (Figure 20a). In a long handleslide, we can handleslide all of the positive hyperbolics over the negative hyperbolics first, then handleslide the specificed negative hyperbolic over the remaining hyperbolics. See Figure 20b. Conducted in this way, we are only required to modify ω near *one* unstable separatrix for each $k = 2, 4, \dots, 2n$; hence we can keep these modifications disjoint.



(A) A long bypass arc relative to the Giroux graph.



(B) Midway through the long bypass handlesliding.

FIGURE 20. A long bypass can be decomposed.

8.5.4. *Categorizing the handleslides.* There are several distinct cases of handleslides we need to consider, as this affects how we choose to modify ω to perform negative handleslides. We first start with some observations:

- With the orientation reversed on $-\widetilde{V}_\lambda$, the signs of the handleslide are reversed. In particular, a positive (resp. negative) handleslide on $\widetilde{V}_\lambda$ in sector-$(\lambda-1)$ is mirrored by a negative (resp. positive) handleslide on $-\widetilde{V}_\lambda$ in sector-λ.
- The sign of the symplectic form flips as well; however, while the sign of the singularities change, the symplectic form changes from 'almost-positive' to 'almost-negative'. Therefore, it is the same hyperbolic points that need to be considered with both orientations.
- Since we start with ξ_∂ inducing foliations divided by A (and $S(A)$ on $\widetilde{V}_3$), we perform the bypass sequences in Section 8.2 in the reverse order.

The handleslides in our bypass sequences come in several types:

(1) On $-\widetilde{V}_1$ and $-\widetilde{V}_2$, we have the bypass sequences $\overline{\widetilde{\eta}}_1^{-1}$ and $\overline{\widetilde{\eta}}_2^{-1}$, respectively, and on $\widetilde{V}_3$ we have the bypasses constituting $\widetilde{\eta}_3$. These are comprised of *positive* handleslides. Bypasses are always positive handleslides [Gir00, Lemma 2.14], while mirroring and reversing a handleslide both flip the sign of the handleslide.
(2) The remaining handleslides occur as part of universal morphisms in concentric annular regions near the boundary of $\widetilde{V}_\lambda$ (see Lemma 9.22). In each such annulus, there are six possibilities:
   (a) $\mathcal{F}''_{A,1}$ (resp. $\mathcal{F}''_{A,2}$) is obtained from $\mathcal{F}''_{B,1}$ (resp. $\mathcal{F}''_{D,2}$) via positive handleslides in the annulus and $\overline{\mathcal{F}}''_{A,1}$ (resp. $\overline{\mathcal{F}}''_{A,2}$) is obtained from $\overline{\mathcal{F}}''_{C,1}$ (resp. $\overline{\mathcal{F}}''_{E,2}$) via negative handleslides in the annulus.
   (b) $\mathcal{F}''_{A,1}$ (resp. $\mathcal{F}''_{A,2}$) is obtained from $\mathcal{F}''_{B,1}$ (resp. $\mathcal{F}''_{D,2}$) via negative handleslides in the annulus and $\overline{\mathcal{F}}''_{A,1}$ (resp. $\overline{\mathcal{F}}''_{A,2}$) is obtained from $\overline{\mathcal{F}}''_{C,1}$ (resp. $\overline{\mathcal{F}}''_{E,2}$) via positive handleslides in the annulus.
   (c) $S(\overline{\mathcal{F}}''_{A,3})$ is obtained from $\overline{\mathcal{F}}''_{F,3}$ via positive (resp. negative) handleslides in the annulus.

(d) $\mathcal{F}''_{A,1}$ (resp. $\mathcal{F}''_{A,2}$) agrees with $\mathcal{F}''_{B,1}$ and $\mathcal{F}''_{C,1}$ (resp. $\mathcal{F}''_{D,2}$ and $\mathcal{F}''_{E,2}$) in the annulus. In the bypass sequence along V_1 (resp. V_2), a collection of positive handleslides occur first in the annulus and are then later undone by negative handleslides.

(e) $\mathcal{F}''_{A,1}$ (resp. $\mathcal{F}''_{A,2}$) agrees with $\mathcal{F}''_{B,1}$ and $\mathcal{F}''_{C,1}$ (resp. $\mathcal{F}''_{D,2}$ and $\mathcal{F}''_{E,2}$) in the annulus. In the bypass sequence along V_1 (resp. V_2), a collection of negative handleslides occur first in the annulus and are then later undone by positive handleslides.

(f) $S(\overline{\mathcal{F}}''_{A,3})$ agrees with $\overline{\mathcal{F}}''_{F,3}$ in the annulus. In the bypass sequence along $-V_3$, a collection of positive (resp. negative) handleslides occur first in the annulus, which are later undone by negative (resp. positive) handleslides.

Note that (2a),(2b) and (2c) are equivalent up to mirroring, hence can be treated the same. Similarly (2d),(2e), and (2f) are equivalent. We therefore have two cases to consider: (2abc), and (2def). Moreover, the bypass sequences in distinct annuli are disjoint and can be considered independently; however, *within* a specific annulus, the handleslides may not be mutually disjoint.

8.5.5. *Cases (2abc).* In these cases, after performing the negative handleslide, the foliation induced by ξ_∂ meets μ_λ convexly. Hence, we only need to modify our primitive κ_λ to locate sufficient ω-area from the beginning, then immediately after the handleslide we can apply Lemma 7.18 to even out the area distribution.

Let (X, Y) denote $(B, C), (D, E)$ or $(F, S(A))$, as appropriate for $\lambda = 1, 2, 3$. On $\widetilde{V}_\lambda$, we have a foliation $\mathcal{F}''_{X,\lambda}$ that is gradient-lke with respect to $\ker(\widetilde{\beta}_\lambda)$. The foliation $\ker(\mu_\lambda)$ is directed by a vector field that is the 'right-most' upward gradient vector field, while $\ker(\mu_{\lambda+1})$ is directed by the 'left-most' upward gradient vector field.

We fix an annulus near the boundary of $\widetilde{V}_\lambda$ containing the handleslides. In this annular region, we can assume that $\mathcal{F}''_{X,\lambda}$ on $\widetilde{V}_\lambda$ and $\overline{\mathcal{F}}''_{Y,\lambda}$ agree up to mirroring. Let γ be a handleslide arc with $\gamma(0)$ contained in an unstable separatrix σ for some negative hyperbolic point h. Let e be the negative elliptic point at the other boundary of σ. Since ω is almost-positive, we have C^0-small neighborhoods U_h, U_e of h, e such that ω is a positive in our local model on the complement of U_h, U_e.

We can choose a neighborhood $W \cong [-2\delta_0, 2\delta_0] \times [-1, 1]$ of $\gamma(0)$ with coordinates (x, z) such that

(1) $\gamma(0) = (0, 0)$
(2) $\mathcal{F}''_{X,\lambda} = \ker(-dx)$
(3) $[-2\delta_0, 2\delta_0] \times \{1\}$ is contained in U_e
(4) W is disjoint from U_h
(5) every flowline that passes through U_h and limits to e passes through W

Lemma 8.6. *Given any $C > 0$ there exists a modification of κ'_λ to κ''_λ, supported in W, such that*

(1) $d\kappa''_\lambda > C\, dx \wedge dz$ on $[-\delta_0, \delta_0] \times [-3/4, -1/2]$,
(2) $d\kappa''_\lambda$ is almost-positive.
(3) $\mathcal{F}''_{X,\lambda}$ meets μ_λ convexly,
(4) $\overline{\mathcal{F}}''_{Y,\lambda}$ meets $\mu_{\lambda+1}$ convexly,

Proof. We can modify the construction in Proposition 8.2 and find a primitive

$$\zeta_\sigma = -\phi(y)\psi(z)c(x)dx$$

with $\psi' < -C$ on $[-3/4, -1/2]$. Then take $\kappa''_\lambda = \kappa'_\lambda + \zeta_\sigma$. Since $\mathcal{F}''_{X,\lambda} = \mathcal{F}''_{Y,\lambda} = \ker(\zeta_\sigma)$ on the support of ζ_σ, this preserves the fact that the foliations meet the primitives convexly. □

Therefore, we can locate sufficient positive symplectic area to perform the negative handleslide.

8.5.6. *Case (2def).* The second case is more complicated. For example, consider Case (2d) on the compressing disk $\widetilde{V}_\lambda$. In our annulus, we have the following four steps:

(1) a sequence of positive handleslides in $\widetilde{V}_\lambda$
(2) a sequence of negative handleslides in $\widetilde{V}_\lambda$, inverting the above positive handleslides
(3) a sequence of negative handleslides in$-\widetilde{V}_\lambda$,

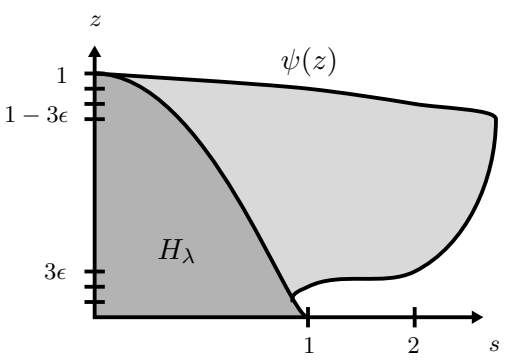
(4) a sequence of positive handleslides in $-\widetilde{V}_\lambda$, inverting the above negative handleslides

The key complications are due to the fact that there may be other negative handleslides that occur between each pair of steps.

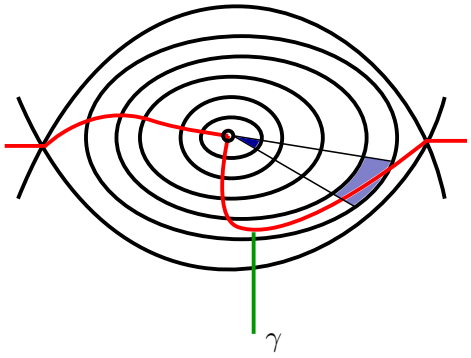
- While we can modify the primitive κ_λ as in Lemma 8.6, this gives the same ω-area distribution on both $\widetilde{V}_\lambda$ and $-\widetilde{V}_\lambda$. In order to meet our ω-area estimates for the intermediate negative handleslides, we need to satisfy a lower bound on the negative ω-area for Step (2) and but also satisfy a smaller upper bound on the negative ω-area in between Steps (2) and (3).
- Lemma 7.18 requires that $\mu_{\lambda+1}$ and ξ_∂ meet convexly in order to modify ω. The negative shear in Step (3) will disrupt this in an elementary neighborhood of the handleslide arc. Since we need to smooth ω immediately after Step (3), our modification of κ_λ must be disjoint from the handle slide arc.

The first complication can be easily dealt with, in the same way as Case (2abc) above, since between Steps (2) and (3) the foliations induced by ξ_∂ agree with the foliations $\mathcal{F}''_{X,\lambda}$ and $\overline{\mathcal{F}}''_{Y,\lambda}$, which meet the primitives $\mu_\lambda, \mu_{\lambda+1}$ convexly.

We deal with the second complication using the following local model.



(A) Graph of enlarging H_λ to $s = \psi(z)$.



(B) Diagram of $\ker(\widetilde{\beta}_\lambda)$ and the negative Giroux graph (red) on $\widetilde{V}_\lambda$.

FIGURE 21. Enlarging H_λ near a concave maximum to add symplectic area.

Recall that with our orientation conventions (Subsection 3.1.2), negative elliptics correspond to convex maxima of $\widetilde{\beta}_\lambda$ along the compressing disk. We assume that we have polar Morse coordinates (s, θ, z) such that $\widetilde{\beta}_\lambda = dz$ and the boundary of H_λ is the sublevel set of

$$r = z + \phi(s, \theta) - 1$$

where $\phi(s, \theta) = s^2$ for $s \in [0, 1 - \epsilon]$ for some $\epsilon > 0$ and ϕ is increasing in s for all $s \in [0, 1]$. Without loss of generality, we assume that there are two hyperbolic points in the level set $\{z = 0\}$ located at $(\pm 1, 0, 0)$. Let $W = \{(z, \theta) \in (0, 1] \times [0, 2\pi]\}$ be a neighborhood of the negative separatrix. Along V_λ, we have polar coordinates (z, θ) (with the origin at $z = 1$) and orientation given by $dz \wedge d\theta$ (for $z < 1$.). In this local model, we assume that

$$\kappa_\lambda = \epsilon' s^2 d\theta$$

then choose a modification to κ'_λ as above that is almost-positive.

In cases (2def), we refer to one of $\widetilde{V}_\lambda, -\widetilde{V}_\lambda$ as the *positive shear* side and the other as the *negative shear* side, according to whether the first handleslide on that side is positive or negative. For example, in (2d) above, the positive shear side is $\widetilde{V}_\lambda$ and the negative shear side is $-\widetilde{V}_\lambda$. In the following discussion, we will assume that $\widetilde{V}_\lambda$ is the negative shear side, while $-\widetilde{V}_\lambda$ is the positive shear side; the other case follows by replacing θ with $-\theta$ in our polar coordinates.

Fix some $\epsilon > 0$. We have rectangles $R \coloneqq [2\epsilon, 1-\epsilon] \times [0, \pi/2]$ and $U \coloneqq [2\epsilon, 3\epsilon] \times [\pi/6, \pi/3]$ in W. Now, we have a foliation

$$\mathcal{F}''_{X,\lambda} = \ker(A(z,\theta)dz + B(z,\theta)d\theta)$$

With respect to the area form $dz \wedge d\theta$, it is directed by the vector field $v = B\partial_z - A\partial_\theta$ so is (upward) gradient-like if and only if $B > 0$. We can choose v so that the unstable separatrices of the negative hyperbolics are as in Figure 21b. In particular, this requires that $A < 0$ where the unstable separatrix passes through the region $U \subset R$. We assume that the handle slide arc γ intersects the negative separatrix at some $z > 3\epsilon$ and is disjoint from R.

Proposition 8.7. *Given any $C_- < 0 < C_+$, there exists an enlargment of ω to ω' supported in $W \times [-\delta', \delta']$, modifications $\rho'_\lambda, \rho'_{\lambda+1}$ of the Liouville vector fields $\rho_\lambda, \rho_{\lambda+1}$ along $W \times \{-\delta'\}, W \times \{+\delta'\}$, respectively, and a contact structure $(W \times [-\delta', \delta'], \xi_\partial)$ dominated by ω' such that*

(1) if $\mu_\lambda, \mu_{\lambda+1}$ are tight, the contact forms $\mu'_\lambda = \omega'(\rho'_\lambda, -)$ and $\mu'_{\lambda+1} = \omega'(\rho'_{\lambda+1}, -)$ are tight
(2) the foliations $ker(\mu'_\lambda), ker(\mu'_{\lambda+1})$ meet $ker(\xi_\partial)$ convexly along $W \times \{-\delta'\}, W \times \{+\delta'\}$, respectively;
(3) ω' is almost-positive (resp. almost-negative) along $W \times \{+\delta'\}$ (resp. $W \times \{-\delta'\}$).
(4) over $U \times \{+\delta''\}$ we have $\omega' > C_+ \Omega$
(5) in $U \times \{-\delta'\}$ we have $\omega' < C_- \Omega$

Let $\Omega = zdz \wedge d\theta$ be an area form on W. Then,

(6) On the positive shear side, for any $M > \pm C_\pm$, there exists an enlargment of ω' to ω'', supported in $R \times [\delta'/2, \delta']$ (resp. $R \times [-\delta', -\delta'/2]$) such that

$$\pm C_\pm \Omega < \pm\omega' < M\Omega$$

(7) Let M be a constant such that the original ω satisfies

$$-M\Omega < \omega < M\Omega$$

On the negative shear side, there exists an enlargement of ω' to ω'', supported in $R \times [\delta'/2, \delta']$ (resp. $R \times [-\delta', -\delta'/2]$), such that

$$-M\Omega < \omega'' < M\Omega$$

On the positive shear side, there exists an enlargement of ω'' to ω''', suppported in $R \times [\delta'/2, \delta']$ (resp. $R \times [-\delta', -\delta'/2]$) such that

$$-M\Omega < \omega''' < M\Omega$$

Moreover, the enlargements in (4) and (5) preserve Property (1) in their support and Property (2) everywhere in $W \times \{\pm\delta'\}$.

Proof. We enlarge H_λ as follows. For $z \in [\epsilon, 1]$, we have that V_λ is cutout by the equation

$$s = \sqrt{1-z}$$

Choose a function $\psi(z) \geq \sqrt{1-z}$ such that

$$\psi(2\epsilon) = \psi(1-\epsilon) = 1 \qquad \psi(3\epsilon) = \psi(1-2\epsilon) = 2 \qquad \psi' = \begin{cases} > 0 & \text{for } z \in [2\epsilon, 1-3\epsilon) \\ < 0 & \text{for } z \in (1-3\epsilon, 1) \end{cases}$$

We enlarge H_λ to locally be the sublevel set $\{s \leq \psi(z)\}$ (see Figure 21a) and extend $\widetilde{\beta}_\lambda = dz$ and κ'_λ as well. By adding $c_0 s^2 d\theta$ to κ_λ (with c_0 determined by ψ), we can assume that if every flowline of $\ker(\mu_\lambda)$ and $\ker(\mu_{\lambda+1})$ passes through R exactly once.

Second, let $g(s)$ be a cutoff function such that $g(1) = 0$ and $g(2) = 1$ and let $c(\theta)$ be a smooth bump function supported in $[0, \pi/2]$ that equals 1 on $[\pi/6, \pi/3]$. For a positive constant C, we can add

$$\zeta = Cg(s)c(\theta)\, d\theta$$

to our primitive κ'_λ. As a result, we have

$$\omega = d\kappa'_\lambda + d\zeta = Cc(\theta)g'\, ds \wedge d\theta + O(\epsilon')$$

Setting $ds = \psi' dz$, we see that restricting ω to V_λ yields

$$\omega|_{V_\lambda} = \psi' \left(Ccg'\right) \, dz \wedge d\theta + O(\epsilon')$$

Therefore, the sign of ψ' determines the sign of ω-area contributed by $d\zeta$; the former is negative in a 3ϵ-neighborhood of the local maximum point and positive for $z \in [2\epsilon, 1-3\epsilon)$. Along V_λ, the modification from κ_λ to κ'_λ is supported for (z,θ) in the rectangle $R \coloneqq [2\epsilon, 1-\epsilon] \times [0, \pi/2]$.

Along $\widetilde{V}_\lambda$ and $-\widetilde{V}_\lambda$, respectively, we have (cf. Section 3.2.2)

$$\mu_\lambda = (2-\delta')dz + \kappa'_\lambda + \zeta \qquad\qquad \mu_{\lambda+1} = (-1-\delta')dz + \kappa'_\lambda + \zeta$$

The pair $(\mu_\lambda, \mathcal{F}''_{X,\lambda})$ meet convexly along $\widetilde{V}_\lambda$ if

$$(2-\delta')B - A(Cgc) > 0 \tag{8.5}$$

which holds for $A < 0$ or A positive and sufficiently small. However, the pair $(\mu_{\lambda-1}, \overline{\mathcal{F}}''_{X,\lambda})$ meet convexly along $-\widetilde{V}_\lambda$ if

$$A(Cgc) + (1+\delta')B > 0 \tag{8.6}$$

If we fix the foliation $\mathcal{F}''_{X,\lambda}$ (and therefore A, B), this inequality will not hold everywhere once C gets too large.

To remedy this, we modify $\mu_{\lambda+1}$ to some a different primitive $\mu'_{\lambda+1}$; this corresponds to replacing the inward-pointing Liouville vector field $\rho_{\lambda+1}$ along $Y^*_{\lambda+1}$ with a different Liouville vector field $\rho'_{\lambda+1}$.

Let $f(s)$ be a smooth, increasing function supported on $[1-\epsilon, \infty)$ and satisfying $f' = 1$ for $s \in [1,2]$.

$$\mu'_{\lambda+1} = \mu_\lambda + \zeta + Mf'ds$$

for $M > 0$. This defines a new contact form since

$$\mu'_{\lambda+1} \wedge d\mu'_{\lambda+1} = \mu_{\lambda+1} \wedge d\mu_{\lambda+1} > 0$$

Moreover, it remains a tight contact structure on $H_\lambda \times \{+\delta'\}$. The foliation $\ker(\mu'_\lambda)$ remains gradient-like for z since $ds = \psi' dz$. Hence the two characteristic foliations induced by $\mu_{\lambda+1}, \mu'_{\lambda+1}$ on $-\widetilde{V}_\lambda$ are topologically conjugate.

We can then replace Equation 8.6 with

$$A(Cgc) + (1+\delta')B + Mf'\psi' > 0 \tag{8.7}$$

This inequality will be satisfied for $M \gg 0$ and $2\epsilon \leq z \leq 3\epsilon$. However, this just moves the problem around as it will be not be true for $z > 1-3\epsilon$ where $\psi' < 0$.

To fix this, we note that the disk $D \coloneqq \{z > 1-3\epsilon\} \subset -\widetilde{V}_\lambda$ is ω-symplectic. Hence, we can apply Lemma 7.11 to modify ξ_∂. In particular, we can choose a foliation $\overline{\mathcal{F}}'''$ on D that is gradient-like and meets $\ker(\mu'_\lambda)$ convexly and a 1-parameter family $\overline{\mathcal{F}}'''_t$ of gradient-like foliations that interpolates between $\overline{\mathcal{F}}''_{X,\lambda}$ and $\overline{\mathcal{F}}'''$ on D. In an arbitrary neighborhood of $-\widetilde{V}_\lambda$, we can replace ξ_∂ with another contact structure ξ'_∂ dominated by ω that realizes the family $\overline{\mathcal{F}}'''_t$.

The enlargements in (6) and (7) follow by applying Lemma 7.18. □

8.6. **Completion.** The final two steps to complete the proof are to plumb in symplectizations of (B^3, ξ_{std}) then fill in the strongly concave interior boundary components to obtain a weak fillings.

Proposition 8.8. *There exists*

(1) a symplectic structure $(\widehat{\mathcal{S}}, \widetilde{\omega})$ on the capped spine, and
(2) a 2-plane field (S^3, ξ_∂) dominated by $\widetilde{\omega}$

such that

(3) each interior boundary component Y_λ of the capped spine is strongly concave,
(4) the induced contact structure on each interior boundary component is tight, and
(5) if $k \geq 0$, then ξ_∂ is a contact structure.

Proof. Using the plumbing construction of Proposition 6.10, we can plumb in a tight (B^3, ξ_{std}) along each of the three corners of $\nu(\mathcal{S})$ to obtain a symplectic structure on the capped spine with three strongly concave interior boundary components. The interior contact structure is obtained by gluing three tight contact structures along 2-spheres: the interior boundary of the spine is tight by Theorem 1.9, the (B^3, ξ_{std}) piece is tight by assumption, and the interpolating contact structure is tight by Proposition 3.9. Moreover, since they are glued along 2-spheres, the resulting contact structure is tight [Col97]. □

We can now finish the main theorem.

Proof of Theorem 1.4. Each interior boundary component is homeomorphic to either S^3 or $\#_l S^1 \times S^2$ for some $l > 0$. Each of these 3-manifolds has a unique tight contact structure [Eli92, Col97]; this contact structure admits a strong symplectic filling via a Weinstein 1-handlebody. Therefore, we can fill in each of the interior boundary components and obtain a symplectic structure $(\widetilde{B}, \omega)$ that domaintes (S^3, ξ_∂). A posteriori, we can normalize the Gompf invariant and use Proposition 10.19 to conclude that

$$\Theta_G = -2 - 4k$$

□

9. Contact Category

Honda introduced the *contact category* $\mathcal{C}(\Sigma, F)$ of a compact surface. Various flavors of the contact category were studied by Mathews [Mat10], Honda-Tian [HT22], and Cooper [Coo24]. We introduce a new variant, the *strong contact category* of a surface $\mathcal{SC}(\Sigma, S, \sigma)$, whose objects are characteristic foliations instead of dividing sets. By formally inverting morphisms, we also obtain the *contact groupoid* and *strong contact groupoid* of a surface. We then focus on the topological properties of these groupoids.

9.1. **Convex surfaces.** An (oriented) surface $S \subset (Y, \xi)$ is *convex* if there exists a transverse vector field whose flow preserves the contact structure. After a C^∞-small perturbation, we can assume S is convex and has a finite number of tangencies with the contact structure. These can be divided into *positive* and *negative* singularities, according to whether the surface orientation agrees or disagrees with orientation on contact planes. These can further be decomposed into *elliptic* and *hyperbolic*, according to the index of $\alpha|_S$, viewed as a section of T^*S, where α is any contact 1-form defining ξ.

In a tubular neighborhood $\nu(S)$ we can write a contact form as

$$\alpha = udv + \beta$$

where v is a normal coordinate, u is a function, and β is a 1-form. The restriction $\beta|_S$ is a 1-form whose kernel is the *characteristic foliation* $\mathcal{F}$ of the surface.

Definition 9.1. A multicurve Γ on S *divides* a singular foliation $\mathcal{F}$ if

(1) Γ separates S into two components S_+ and S_-
(2) Γ is transverse to the singular foliation $\mathcal{F}$, and
(3) there exists an area form ω on S and a vector field v directing the singular foliation $\mathcal{F}$ such that
   (a) $\mathcal{L}_v\omega > 0$ on S_+ and $\mathcal{L}_v\omega < 0$ on S_-,
   (b) v points transversely our of S_+ into S_- along Γ.

Lemma 9.2. *Let $\mathcal{F}$ be a singular foliation and let Γ be a multicurve. If*

(1) Γ separates S into two components S_+ and S_- such that
   (a) S_+ contains all positive singularities and repelling periodic orbits,
   (b) S_- contains all negative singularities and attracting periodic orbits
(2) Γ is transverse to $\mathcal{F}$

Then Γ divides $\mathcal{F}$.

Theorem 9.3 (Giroux)**.** *Let S be an orientable surface in (Y, ξ) and let $\mathcal{F}$ be the characteristic foliation on S.*

(1) S is convex if and only if $\mathcal{F}$ admits dividing curves,
(2) if $S = S^2$, then $(\nu(S), \xi)$ is a tight contact structure if and only if the dividing set Γ is connected.

Let Σ be a convex surface with singular foliation $\mathcal{F}$. Given an area form Ω on Σ, we obtain a vector field v directing the characteristic foliation by setting

$$\Omega(v, -) = \alpha|_\Sigma$$

The integral curves of v are precisely the leaves of the singular foliation $\mathcal{F}$. Each positive hyperbolic point p_+ has two stable separatrices, each of which is (the closure) of an integral curve from a positive elliptic point to p_+; while each negative hyperbolic point p_- has two unstable separatrices, each of which is (the closure) of an integral curve from p_- to a negative elliptic point.

Definition 9.4. Suppose that the singular foliation $\mathcal{F}$ is Morse-Smale. The *Giroux graph* G of a convex surface is the 1-complex obtained as the union of:

(1) all singularities of the characteristic foliation,
(2) all periodic orbits of the characteristic foliation,
(3) the stable separatrices of the positive hyperbolic points,
(4) the unstable separatrices of the negative hyperbolic points

The *positive Giroux graph* is the union of the connected components of the Giroux graph containing positive singularities and repelling periodic orbits, while the *negative Giroux graph* is its complement, consisting of the union of components containing negative singularities and attracting periodic orbits.

Remark 9.5. The terminology 'graph' can be a misnomer in general. Separatrices of hyperbolic points can accumulate on periodic orbits, so the Giroux graph is not necessarily homeomorphic to a CW-complex of dimension 1.

9.2. **Bypasses via dividing set.** Let Σ be a convex surface with dividing set Γ in a contact 3-manifold (Y,ξ).

Definition 9.6. A *bypass arc* with respect to the dividing set $\Gamma \subset \Sigma$ is a properly embedded arc $\gamma : [0,1] \to \Sigma$ such that

(1) $\gamma(0), \gamma(1) \in \Gamma$, and
(2) γ intersects Γ transversely exactly once in its interior.

See Figure 22.

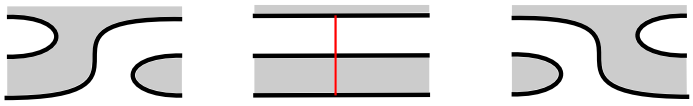


FIGURE 22. *(Center)* A bypass arc γ with respect to the dividing set. *(Left)* The dividing set after carving a bypass along γ. *(Right)* The dividing set after attaching a bypass along γ.

Given a bypass arc, there exists a contact structure $(\Sigma \times [0,1], \xi_\gamma)$ such that

(1) the surfaces $\Sigma_z = \Sigma \times \{z\}$ are convex for all $z \neq 1/2$,
(2) for $z < 1/2$, the dividing set Γ_z is isotopic to $\Gamma_0 = \Gamma$,
(3) for $z > 1/2$, the dividing set Γ_z is obtained from Γ by the local move described in the right of Figure 22..

Definition 9.7. If (Y,ξ) is a contact structure with convex boundary (Σ,Γ), then we can form the union

$$(Y,\xi') = (Y,\xi) \cup_\Sigma (\Sigma \times [0,1], \xi_\gamma)$$

We refer to (Y,ξ') as *attaching a bypass along* γ.

Moreover, there exists a contact structure $(\Sigma \times [0,1], \xi_{\gamma^*})$ such that

(1) the surfaces $\Sigma_z = \Sigma \times \{z\}$ are convex for all $z \neq 1/2$,
(2) for $z > 1/2$, the dividing set Γ_z is isotopic to $\Gamma_1 = \Gamma$,
(3) for $z < 1/2$, the dividing set Γ_z is obtained from Γ by the local move described in the left of Figure 22..

Definition 9.8. If there exists a decomposition

$$(Y,\xi) = (Y,\xi') \cup_\Sigma (\Sigma \times [0,1], \xi_{\gamma^*})$$

(which may not exist in general), we refer to (Y,ξ') as *carving out a bypass along* γ and γ as a *dual bypass arc.*

Definition 9.9. Let (M,ξ) be a contact structure with convex boundary. An *attaching/carving sequence* is a sequence of contact structures

$$(M,\xi) \to (M,\xi^{(1)}) \to \cdots \to (M,\xi^{(n)})$$

such that each $(M,\xi^{(i+1)})$ is obtained from $(M,\xi^{(i)})$ by either attaching a bypass or carving out a bypass.

9.3. **The contact category.** Let Σ be a surface with nonempty boundary. A decoration is a set of signed points $F \subset \partial\Sigma$, where the signs alternate along $\partial\Sigma$. The *positive boundary* $\partial\Sigma^+$ is the closure of the subset of $\partial\Sigma \setminus F$ such that $\partial(\partial\Sigma^+) = F$ as a set of oriented points and the *negative boundary* $\partial\Sigma^-$ denote the closure of its complement.

Definition 9.10. The *contact category* $\mathcal{C}(\Sigma, F)$ of a decorated surface with boundary is the pre-additive category where

(1) $\mathfrak{Ob}(\mathcal{C}(\Sigma, F))$: the objects are oriented 1-manifolds Γ with $\partial\Gamma = F$, up to isotopy rel boundary;
(2) $\mathrm{Hom}_{\mathcal{C}}(\Gamma_0, \Gamma_1)$: the morphisms are isotopy classes of contact structures $(\Sigma \times [0,1], \xi)$ such that
   (a) $\Sigma \times \{0,1\}$ are convex surfaces divided by Γ_0, Γ_1, respectively,
   (b) $\partial(\Sigma \times \{t\})$ is ξ-Legendrian for all $t \in [0,1]$.
   (c) Two morphisms $\xi, \xi' \in \mathrm{Hom}_{\mathcal{C}}(\Gamma_0, \Gamma_1)$ are equivalent if they are isotopic through contact structures satisfying (a) and (b) above.

In addition

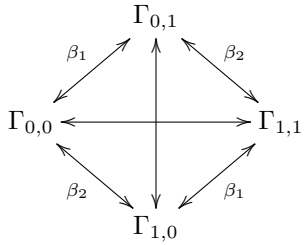
(3) The identity morphism $\xi_{id} \in \mathrm{Hom}_{\mathcal{C}}(\Gamma, \Gamma)$ is the isotopy class of the vertically-invariant contact structure $(\Sigma \times [0,1], \xi_I)$ with dividing set Γ on each $\Sigma_t = \Sigma \times \{t\}$ for all $t \in [0,1]$.
(4) The composition of two morphisms $\xi_1 \in \mathrm{Hom}_{\mathcal{C}}(\Gamma_0, \Gamma_1)$ and $\xi_2 \in \mathrm{Hom}_{\mathcal{C}}(\Gamma_1, \Gamma_2)$ is the contact structure $(\Sigma \times [0,2], \xi) = (\Sigma \times [0,1], \xi_1) \cup (\Sigma \times [0,2], \xi_2)$ obtained by concatenation.
(5) Overtwisted dividing sets (i.e. containing contractible components) are identified with the zero object and overtwisted contact structures in $\mathrm{Hom}(\Gamma_0, \Gamma_1)$ are identified with the zero morphism.

Remark 9.11. Honda-Tian [HT22] do not identify isotopic dividing sets in their definition of the contact category, as there can be nontrivial isotopies of the dividing sets on S^2 and T^2 or if there is a contractible component. Since these cases are not relevant to our purposes, we identify isotopic dividing sets.

By formally inverting all morphisms in the contact category, we obtain the *contact groupoid* of the marked surface (Σ, F).

Definition 9.12. The *contact groupoid* $\mathcal{G}(\Sigma, F)$ of a decorated surface with boundary is the groupoid such that

(1) $\mathfrak{Ob}(\mathcal{G}(\Sigma, F))$: the objects are the nonzero objects of $\mathcal{C}(\Sigma, F)$. In particular, they are isotopy classes of tight dividing sets on Σ with oriented boundary F.
(2) $\mathrm{Mor}(\mathcal{G}(\Sigma, F))$. The isomorphism sets are determined by the following generators and relations:
   (a) *Generators.* For each tight bypass $\Gamma_0 \xrightarrow{\beta} \Gamma_1$, we obtain an isomorphism $\beta \in \mathrm{Hom}(\Gamma_0, \Gamma_1)$.
   (b) *Relations.* Let β_1, β_2 be disjoint bypass arcs that do not form a pinwheel (see [HKM05]). Then we have a commutative diagram



9.3.1. *Boundary sum and stabilization.* Let (Σ_1, F_1) and (Σ_2, F_2) be two decorated surfaces with boundary. Fix two points x_1, x_2 in $\partial\Sigma_1, \partial\Sigma_2$, respectively, that are both in either the positive or negative boundaries. Then $(\Sigma_1 \natural \Sigma_2, F_1 \cup F_2)$ is a decorated surface, where the boundary sum is defined by attaching a 1-handle to $\Sigma_1 \coprod \Sigma_2$ at the points x_1 and x_2. We then obtain a *boundary sum functor*

$$B_{x_1,x_2} : \mathcal{C}(\Sigma_1, F_1) \times \mathcal{C}(\Sigma_2, F_2) \to \mathcal{C}(\Sigma_1 \natural \Sigma_2, F_1 \cup F_2)$$

Now, we restrict to the case where $\Sigma_2 = D^2$ and F_2 consists of two points. The category $\mathcal{C}(D^2, F_2)$ has a unique nonzero object Γ, which is a single arc connecting the two points, and a single nonzero automorphism,

which is the vertically-invariant contact structure with dividing set Γ. Let $y_\pm$ denote points in the positive and negative boundaries of the decorated surface (D^2, F_2). For a point x in the positive (resp. negative) boundary of (Σ, F), we obtain the *stabilization functor*

$$Stab_x^\pm = B_{x,y_\pm}(-,\Gamma) : \mathcal{C}(\Sigma, F) \to \mathcal{C}(\Sigma, F \cup F_2)$$

9.3.2. *The monodromy and shift functors.* Let $\phi : \Sigma \to \Sigma$ be a homeomorphism that fixes F setwise. Pulling back via this map, we obtain the *monodromy functor*

$$M_\phi : \mathcal{C}(\Sigma, F) \to \mathcal{C}(\Sigma, F)$$

where

(1) $M_\phi(\Gamma) = \phi^{-1}(\Gamma)$ is the pullback of the dividing set Γ under the homeomorphism, and
(2) $M_\phi(\xi) = (\Sigma \times [0,1], \phi^*(\xi)) \in \mathrm{Hom}(\phi(\Gamma_0), \phi(\Gamma_1))$ is the pullback of $\xi \in \mathrm{Hom}(\Gamma_0, \Gamma_1)$

Now, let $A_1, \dots, A_k$ denote the boundary components of $\partial\Sigma$ and let $n_i = \frac{1}{2}\#(F \cap A_i)$. We can identify $\nu(A_i) \cong S^1 \times [-\delta, 0]$ such that

$$F \cap A_i = \left\{ \frac{1 \cdot \pi}{n_i} \times \{0\}, \dots, \frac{2n_i \cdot \pi}{n_i} \times \{0\} \right\}$$

Let $\sigma_F : \Sigma \to \Sigma$ be the homeomorphism consisting of a fractional Dehn twist along each boundary component, according to the intersection numbers $\{n_i\}$. In particular, we can locally define σ_F near A_i by setting

$$\sigma_F(\theta, t) = \left(\theta - \frac{(\delta + t)\pi}{\delta n_i}, t \right)$$

for $-\delta \le t \le 0$ and extending by the identity over the interior of Σ. The *shift functor*

$$S := M_{\sigma_F} : \mathcal{C}(\Sigma, F) \to \mathcal{C}(\Sigma, F)$$

is defined by using σ_F to pull back dividing sets and contact structures.

9.3.3. *Fibered Legendrian links.* Let $L \subset (Y, \xi)$ be a fibered Legendrian link (not necessarily supporting the contact structure ξ), let Σ be a fiber surface, and let ϕ denote the monodromy of the open book decomposition of Y given by the fibered link L.

Suppose that Σ can be made convex with dividing set Γ, (which occurs if and only if the difference between the contact framing and the surface framing along each component of L is nonpositive). Then we can cut Y along Σ and restrict ξ to obtain a contact structure $(\Sigma \times [0,1], \xi)$. Then Y is obtained as the quotient

$$Y \setminus \nu(L) = (\Sigma \times [0,1], \xi)/(x, 1) \sim (\widehat{\phi}(x), 0)$$

where $\widehat{\phi}$ is the composition of the monodromy ϕ, which is the identity near $\partial\Sigma$, and a fractional Dehn twist along each boundary component, where the fraction is determined by the contact framing. Specifically, let $-n_i \le 0$ denote the twisting number of ξ relative to Σ along the component L_i of L. The boundary of a tubular neighborhood of L_i can be made convex with two dividing curves of slope $-\frac{n_i}{1}$, which from the perspective of the surface Σ corresponds to a fractional Dehn twist $-\frac{1}{n_i}$.

Consequently, the dividing set for $(\Sigma \times [0,1], \xi)$ is Γ along $\Sigma \times \{0\}$ and $\widehat{\phi}^{-1}(\Gamma)$ along $\Sigma \times \{1\}$. We can therefore interpret ξ as a morphism

$$\Gamma \xrightarrow{\xi} M_{\widehat{\phi}}(\Gamma) = S \circ M_\phi(\Gamma) \tag{9.1}$$

in the contact category $\mathcal{C}(\Sigma, \partial\Gamma)$.

9.4. **Strong contact category.** The objects in the contact category are dividing sets up to isotopy. However, to get finer-grained structure, we work instead with the characteristic foliations themselves.

Let Σ be a compact surface, possibly with boundary. Fix a finite collection $S \subset \Sigma$ of points.

Definition 9.13. A *singularity coloring* is a map

$$\sigma : S \to \{e_+, e_-, h_+, h_-\}$$

A singularity coloring is *admissible* if

(1) σ maps $S \cap \partial\Sigma$ to $\{e_+, e_-\}$ and these alternate along the boundary, and
(2) $\chi(\Sigma) = (E_+ + E_-) - (H_+ + H_-)$, where

$$E_\pm = \#\sigma^{-1}(e_\pm) - \frac{n}{2}$$
$$H_\pm = \#\sigma^{-1}(h_\pm) - \frac{n}{2}$$
$$n = \frac{1}{2}\#(S \cap \partial\Sigma)$$

The *Euler number* of a singularity coloring is defined as

$$e(\sigma) = (E_+ + H_+) - (E_- + H_-)$$

Let $\mathcal{MS}(\Sigma, S, \sigma)$ denote the set of all Morse-Smale pairs (f, g) on Σ such that

(1) $S = \mathrm{Crit}(f)$,
(2) the local maxima of f occur at the points colored by e_+,
(3) the local minima of f occur at the points colored by e_-, and
(4) the index-1 critical points of f occur at the points colored by h_+ and h_-.

Given a Morse-Smale pair (f, g), let $\mathcal{F}_{f,g}$ be the singular foliation induced by the flow of $-\nabla^g f$.

Definition 9.14. Let $\mathcal{F}$ be a Morse-Smale foliation.

(1) A *handle slide arc* is a map $\gamma : [0, 1] \to \mathcal{F}$ transverse to $\mathcal{F}$ such that $\gamma(0)$ lies in a stable separatrix of a hyperbolic point h_0 and $\gamma(1)$ lies in an unstable separatrix of another hyperbolic point h_1.
(2) By shearing the foliation along γ, we obtain a 1-parameter family of foliations $\mathcal{F}_t$ for $t \in [0, 1]$ where
   (a) $\mathcal{F}_t$ is Morse-Smale for $t \in [0, 1] \smallsetminus \{1/2\}$, and
   (b) $\mathcal{F}_{1/2}$ has a heteroclinic orbit from h_1 to h_0.

   We say that $\mathcal{F}_1$ is obtained from $\mathcal{F}_0 = \mathcal{F}_1$ by a *handleslide along* γ

Definition 9.15. A contact structure $(\Sigma \times [0, 1], \xi)$ has *fixed singularities* at $S \subset \Sigma$ if for any contact form $\alpha = h dz + \beta$ we have $\beta^{-1}(0) = S \times [0, 1]$.

Definition 9.16. The *strong contact category* $\mathcal{SC}(\Sigma, S, \sigma)$ is the category where

(1) $\mathfrak{Ob}(\mathcal{SC}(\Sigma, S, \sigma))$. The objects are singular foliations $\mathcal{F}_{f,g}$ for $(f, g) \in \mathcal{MS}(\Sigma, S, \sigma)$, up to topological isotopy fixing the singular points.
(2) $\mathrm{Mor}(\mathcal{F}_0, \mathcal{F}_1)$. The morphism sets are tight contact structures $(\Sigma \times [0, 1], \xi)$ with fixed singularities at $S \subset \Sigma$ such that
   (a) $\mathcal{F}_i$ is the induced characteristic foliation on $\Sigma \times \{i\}$ for $i = 0, 1$,
   (b) $\partial(\Sigma \times \{t\})$ is ξ-Legendrian for all $t \in [0, 1]$,
   (c) Two morphisms $\xi, \xi' \in \mathrm{Hom}_{\mathcal{SG}}(\mathcal{F}_0, \mathcal{F}_1)$ are equivalent if they are isotopic through contact structures with fixed singularities at S and satisfying (a) and (b) above.

In addition,

(3) The identify morphism $\xi_{id} \in \mathrm{Hom}_{\mathcal{SG}}(\mathcal{F}, \mathcal{F})$ is the vertically-invariant contact structure $(\Sigma \times [0, 1], \xi_{\mathcal{F}})$ whose characteristic foliation on each $\Sigma_t = \Sigma \times \{t\}$ is $\mathcal{F}$.
(4) The composition of two morphisms $\xi_1 \in \mathrm{Hom}_{\mathcal{SC}}(\mathcal{F}_0, \mathcal{F}_1)$ and $\xi_2 \in \mathrm{Hom}_{\mathcal{SC}}(\mathcal{F}_1, \mathcal{F}_2)$ is the contact structure $(\Sigma \times [0, 2], \xi) = (\Sigma \times [0, 1], \xi_1) \cup (\Sigma \times [0, 2], \xi_2)$ obtained by concatenation.
(5) Overtwisted characteristic foliations are identified with the zero object and overtwisted contact structures in $\mathrm{Hom}_{\mathcal{SG}}(\mathcal{F}_0, \mathcal{F}_1)$ are identified with the zero morphism.

There is a 'forgetful functor'

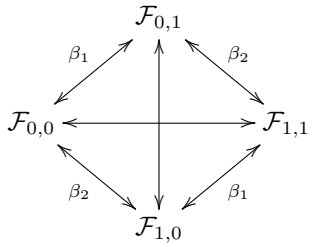
$$\Gamma : \mathcal{SC}(\Sigma, S, \sigma) \to \mathcal{C}(\Sigma, F) \tag{9.2}$$

that sends a characteristic foliation $\mathcal{F}$ to its dividing set $\Gamma(\mathcal{F})$. The dividing set can be obtained as the boundary of tubular neighborhood of the negative Giroux graph, which is the closure of the unstable separatrices of the negative hyperbolic points.

As with the contact category, we obtain a groupoid by formally inverting all morphisms in the strong contact category.

Definition 9.17. The *strong contact groupoid* $\mathcal{SG}(\Sigma, S, \sigma)$ is the groupoid such that

(1) $\mathfrak{Ob}(\mathcal{SG})$: The objects are the nonzero objects of $\mathcal{SC}(\Sigma, S, \sigma)$. In particular, they are (topological) isotopy classes of Morse-Smale foliations
(2) $\mathrm{Mor}(\mathcal{SG}(\Sigma, S, \sigma))$. The isomorphism sets are determined by the following generators and relations
   (a) *Generators.* For each handleslide $\mathcal{F}_0 \xrightarrow{\beta} \mathcal{F}_1$, we obtain an isomorphism $\beta \in \mathrm{Hom}(\mathcal{F}_0, \mathcal{F}_1)$.
   (b) *Relations.* Let β_1, β_2 be disjoint handleslides that do not form a pinwheels (see [HKM05]). Then we have a commutative diagram



We have a groupoid functor

$$\Gamma : \mathcal{SG}(\Sigma, S, \sigma) \to \mathcal{G}(\Sigma, F) \tag{9.3}$$

Let $\mathcal{F} \in \mathcal{SG}(\Sigma, S, \sigma)$, let γ be a handle slide arc, and let $\mathcal{F}' \in \mathcal{SG}(\Sigma, S, \sigma)$ be the foliation obtained by handle sliding along γ. We obtain a morphism

$$\gamma : \mathcal{F} \to \mathcal{F}'$$

in the strong contact category given by the trace of the 1-parameter family $\mathcal{F}_t$ obtained by shearing along γ.

Definition 9.18. A handle slide arc γ is *trivial* if

$$\Gamma(\mathcal{F}) \xrightarrow{\Gamma(\gamma)} \Gamma(\mathcal{F}')$$

is the identity in the contact groupoid.

9.5. **Contact category of the disk.** We now restrict to $\Sigma = D^2$ and F consists of $2n > 0$ points along boundary. We write $\mathcal{C}(D^2, n)$ for $\mathcal{C}(D^2, F)$. The nonzero objects of $\mathcal{C}(D^2, n)$ consist of *chord diagrams*, which are pairwise disjoint arcs in the disk with boundary in F. Given a chord diagram Γ, we get a decomposition

$$D^2 = R_+(\Gamma) \cup R_-(\Gamma)$$

where $\partial(R_\pm(\Gamma) = \Gamma \cup (\partial D^2)^\pm$. We can further partition $\mathcal{C}(D^2, n)$ according to the *rotation number*

$$r(\Gamma) := \chi(R_+(\Gamma)) - \chi(R_-(\Gamma))$$

If $r(\Gamma) \neq r(\Gamma')$, then $\mathrm{Hom}(\Gamma, \Gamma')$ consists only of the zero morphism. Therefore, we have a decomposition

$$\mathcal{C}(D^2, n) = \coprod \mathcal{C}(D^2, n, r)$$

where $r \in \{1-n, 3-n, \ldots, n-3, n-1\}$. By formally inverting morphisms in $\mathcal{C}(D^2, n, r)$ we obtain the *contact groupoid of the disk* $\mathcal{G}(D^2, n, r)$

The morphisms of $\mathcal{C}(D^2, n, r)$ are generated by nontrivial *bypass moves*.

Lemma 9.19. *Let Γ, Γ' be objects in $\mathcal{C}(D^2, n, r)$.*

(1) *(Uniqueness) There is at most one nonzero* $\xi \in Hom(\Gamma, \Gamma')$.
(2) *(Factorization into bypasses) Let* $\xi \in Hom(\Gamma, \Gamma')$ *be nonzero. There exists a sequence* $\Gamma_0 = \Gamma, \Gamma_1, \dots, \Gamma_k = \Gamma'$ *such that*
   (a) *there is a nonzero* $\xi_i \in Hom(\Gamma_i, \Gamma_{i+1})$ *given by a nontrivial bypass attachment, and*
   (b) *the composition*

$$Hom(\Gamma_0, \Gamma_1) \times \cdots \times Hom(\Gamma_{k-1}, \Gamma_k) \to Hom(\Gamma_0, \Gamma_k)$$

   sends $(\xi_1, \dots, \xi_k)$ *to* ξ.

Lemma 9.20. *Let* $S(\Gamma)$ *denote the shift functor on* $\mathcal{C}(D^2, n, r)$.

(1) [Mat10] *If* $\Gamma \neq \Gamma'$, *then the composition*

$$Hom(\Gamma, \Gamma') \times Hom(\Gamma', \Gamma) \to Hom(\Gamma, \Gamma)$$

   is trivial.
(2) [HT22, Lemma 3.3.1.3] $Hom(\Gamma, \Gamma')$ *is nontrivial if and only if* $Hom(\Gamma', S(\Gamma))$ *is nontrivial.*
(3) [HT22, Lemma 3.3.1.4] *If* $Hom(\Gamma, \Gamma')$ *is nontrivial, then the composition*

$$Hom(\Gamma, \Gamma') \times Hom(\Gamma', S(\Gamma)) \to Hom(\Gamma, S(\Gamma))$$

   is nontrivial.

Proposition 9.21. *Let* $\xi \in Hom_{\mathcal{C}}(\Gamma, \Gamma')$.

(1) ξ *is equivalent to a composition of a sequences of bypasses:*

$$\Gamma = \Gamma_0 \xrightarrow{\beta_1} \Gamma_1 \xrightarrow{\beta_2} \cdots \xrightarrow{\beta_k} \Gamma_k = \Gamma'$$

(2) *any two sequences of bypass attachments whose composition is* ξ *can be related by the following two operations:*
   (a) *Far commutativity: Given two bypasses with disjoint attachment arcs, the order of attachment can be reversed.*
   (b) *Weak equivalence: Adding or removing a trivial bypass.*

Proof. We give a sketch of the proof, following [HT22, Section 3.2], because it will be relevant later. Other proofs include [Tia18, BHH24, LSV26].

Claim. Fix a collar neighborhood $\nu(\partial D) \cong S^1 \times (-\epsilon, 0]$ in which Γ_0, Γ_1 are cylindrical. Then ξ can be realized by a sequence of bypass attachments in $\nu(\partial D)$.

To prove the claim, we proceed by induction on $n = \#\Gamma$, the number of arcs in the dividing set. If $n = 1, 2$ there is a unique dividing set with a given rotation number, hence the only tight morphism is the identity. Now assume the statement is true for $n - 1 \geq 2$. Choose a boundary-parallel arc in Γ_1, connecting the points $x_i, x_{i+1} \in \partial D^2$. Push the arc ζ_n from x_i to x_{i+2} in ∂D^2 into $\nu(\partial D^2)$. We can decompose ξ into the composition

$$\Gamma_0 \xrightarrow{\zeta_n} \Gamma_0' \xrightarrow{\xi'} \Gamma_1$$

for some Γ_0' and ξ'. Moreover, we can choose a disk $D' \subset D'' \subset D$ such that $\partial D''$ intersects the dividing sets Γ_0', Γ_1 in $2n - 2$ points and such that these dividing sets agree on $D \setminus D''$. By induction, we can assume that ξ' can be realized by a sequence of bypass attachments in $\nu(\partial D'') \subset \nu(\partial D)$. This proves the claim.

Now, let $(\beta_1, \dots, \beta_k)$ be a given sequence of bypasses realizing ξ. Then after attaching these bypasses, the sequence $(\zeta_n, \dots, \zeta_1)$ becomes a sequence of trivial bypasses. However, since this second sequence can be attached in an arbitrary neighborhood of ∂D, the two sequences are disjoint and we can attach $(\zeta_n, \dots, \zeta_1)$ first, followed by $(\beta_1, \dots, \beta_k)$. In this ordering, the original bypasses all become trivial. Consequently, up to far commutativity and weak equivalence, every bypass sequence is equivalent to $(\zeta_n, \dots, \zeta_1)$. □

9.5.1. *Universal morphism.* The proof of Proposition 9.21 turns on the fact that we can define two 'universal' bypass sequences to realize a tight morphism $\Gamma_0 \to \Gamma_1$ that depend only on the (co)domain of the morphism.

Lemma 9.22. *Let Γ be a tight dividing set on D^2 with $k+1$ components.*

(1) there exists a sequence $B^+(\Gamma) = \{\gamma_1, \dots, \gamma_k\}$ of bypass arcs in a tubular neighborhood of ∂D^2, depending only on Γ, such that if $Hom(\Gamma', \Gamma)$ is nonzero, the morphism can be obtained by attaching bypasses along $B^+(\Gamma)$ to Γ'.

(2) there exists a sequence $B^-(\Gamma) = \{\gamma_1, \dots, \gamma_k\}$ of bypass arcs in a tubular neighborhood of ∂D^2, depending only on Γ, such that if $Hom(\Gamma, \Gamma')$ is nonzero, the morphism can be obtained by carving bypasses along $B^-(\Gamma)$ from Γ'.



(A) $B^+(\Gamma)$ (B) $B^-(\Gamma)$

FIGURE 23. An example of the bypass arcs that generated the universal morphisms to *(A)* and from *(B)* a given dividing set and the resulting dividing curves. Longer arcs become bypass arcs after the nested bypasses are attached.

9.5.2. *Realizing the universal morphism via foliations.* Recall that a morphism $\Gamma_0 \to \Gamma_1$ in the contact category is *elementary* if it can be obtained by attaching bypasses along a disjoint collection of arcs with respect to Γ_0. Not all morphisms are elementary ([Mat10, Lemma 3.17]) and the universal bypass sequences are almost never elementary. However, it can be realized by a sequence of $n-1$ 'long bypasses' along long handleslide arcs with disjoint endpoints. A *long handleslide arc* (of length k) is an arc $\gamma : [0,1] \to \Sigma$ such that $\gamma([0,1])$ intersects the separatrices at $t = \frac{i}{2k-1}$ for $i = 0, \dots, 2k-1$, alternating between stable and unstable. It is the union of $\binom{k+1}{2}$ handleslides. See Figure 20a. The innermost bypasses in the universal morphism can be realized by honest handleslides (Figure 24a). Once these have been performed, the longer bypasses can be realized but first require handlesliding in the characteristic foliation.

9.5.3. *Opposite cateogry.* The opposite category $\mathcal{C}(D^2, n, r)^{op}$ is equivalent to $\mathcal{C}(D^2, n, -r)$. We define the contravariant 'mirror' functor

$$- : \mathcal{C}(D^2, n, r) \longrightarrow \mathcal{C}(D^2, n, -r)$$

where $\overline{\Gamma}$ is the chord diagram consisting of an arc from x_{-i+2} to x_{-j+2} (with indices mod $2n$) precisely when Γ contains the arc from x_i to x_j. Given a morphism $\xi \in \mathrm{Hom}(\Gamma_0, \Gamma_1)$, we denote the mirror morphism $\overline{\xi} \in \mathrm{Hom}(\overline{\Gamma}_1, \overline{\Gamma}_0)$ by

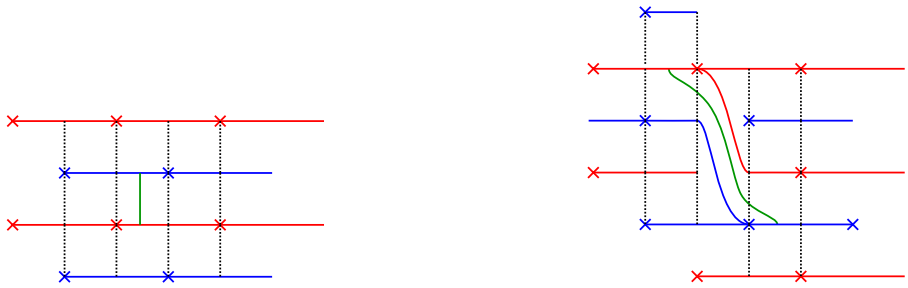
$$\overline{\Gamma_0 \xrightarrow{\xi} \Gamma_1} = \overline{\Gamma}_1 \xrightarrow{\overline{\xi}} \overline{\Gamma}_0$$

Moreover, we have a gluing functor

$$D : \mathcal{C}(D^2, n, r) \times \mathcal{C}(D^2, n, -r) \to \mathcal{C}(S^2)$$

obtained by gluing these contact structures together along $\partial D^2 \times [0,1]$. We use the notation $D(\xi_1, \xi_2) = \xi_1 \# \xi_2$ for contact structures and $D(\Gamma_1, \Gamma_2) = \Gamma_1 \# \Gamma_2$ for dividing sets.

Lemma 9.23. *$\overline{\Gamma}_1 \# \Gamma_2$ is a nonzero object in $\mathcal{C}(S^2)$ if and only if $Hom(\Gamma_1, \Gamma_2)$ is nontrivial. In particular, $\Gamma_1 \# \overline{\Gamma}_1$ is nontrivial.*



(A) An innermost bypass is an honest handleslide bypass move.

(B) Nested bypasses can be realized by long handleslides.

FIGURE 24. Realizing the universal morphism at the level of foliations via handleslides. The positive Giroux graph is red, the negative Giroux graph is blue, the remaining separatrices are dotted.

Proof. Suppose that $\xi \in \mathrm{Hom}(\overline{\Gamma}_1, \Gamma_2)$ is tight. Then after smoothing the corners on $(D^2 \times [0,1], \xi)$, the dividing set on the boundary 2-sphere is connected and isotopic to $\Gamma_1 \# \Gamma_2$, hence the latter is nontrivial in $\mathcal{C}(S^2)$. Conversely, if $\Gamma_1 \# \Gamma_2$ is nontrivial, we can construct a tight contact structure $(D^2 \times [0,1], \xi)$, which is unique up to isotopy rel boundary. □

9.6. **Strong contact category of the disk.** To define the strong contact category of the disk, we assume $D^2 = \mathbb{D}$ is the unit disk in $\mathbb{C}$. Let $\zeta = e^{\frac{\pi i}{n}}$ and $\eta = e^{\frac{2\pi}{n-1}}$. Define

$$S_0 = \{\zeta, \zeta^2, \dots, \zeta^{2n}, \frac{1}{2}\eta, \dots, \frac{1}{2}\eta^{n-1}\} \subset \mathbb{C}$$

Take the singularity coloring

$$\sigma_{0,r}(\zeta^i) = \begin{cases} e_+ & \text{if } i \text{ even} \\ e_- & \text{if } i \text{ odd} \end{cases} \qquad \sigma_{0,r}\left(\frac{1}{2}\eta^i\right) = \begin{cases} h_+ & \text{if } 1 \le i \le \frac{1}{2}(n-1+r) \\ h_- & \text{if } \frac{1}{2}(n-1+r) < i \le n-1 \end{cases}$$

We then inductively define

$$S_{k+1} = \frac{1}{2}S_k \cup \left\{\frac{3}{4}\zeta, \zeta, \dots, \frac{3}{4}\zeta^{2n}, \zeta^{2n}\right\}$$

and extend the singularity coloring by setting $\sigma_{k+1,r} = \sigma_{k,r}$ on S_k and

$$\sigma_{k+1,r}\left(\frac{3}{4}\zeta^i\right) = \begin{cases} h_+ & \text{if } i \text{ even} \\ h_- & \text{if } i \text{ odd} \end{cases} \qquad \sigma_{k+1}(\zeta^i) = \begin{cases} e_+ & \text{if } i \text{ even} \\ e_- & \text{if } i \text{ odd} \end{cases}$$

We let $\mathcal{C}(n,r,k) = \mathcal{SC}(\mathbb{D}, S_k, \sigma_{k,r})$ be the strong contact category determined by $S_k, \sigma_{k,r}$ and let $\mathcal{SG}(n,r,0)$ be the corresponding strong contact groupoid.

We define the stabilization functor

$$\Sigma : \mathcal{SC}(n,r,k) \to \mathcal{SC}(n,r,k+1)$$

as follows. Given a singular foliation $\mathcal{F}$ on $\mathbb{D}$, we map it homeomorphically onto $\mathbb{D}_{\frac{1}{2}}$ by scaling. We then extend across the annulus $\mathbb{D} \smallsetminus \mathbb{D}_{\frac{1}{2}}$ as follows:

(1) for i even, the point $\frac{3}{4}\zeta^i$ is a positive hyperbolic:
   (a) the unstable separatrices connect to $\frac{1}{2}\zeta^{i-1}$ and $\frac{1}{2}\zeta^{i+1}$, and
   (b) the stable separatrices connect to ζ^i and $\frac{1}{2}\zeta^i$.

(2) for i odd, the point $\frac{3}{4}\zeta^i$ is a negative hyperbolic:
   (a) the unstable separatrices connect to ζ^i and $\frac{1}{2}\zeta^i$, and
   (b) the stable separatrices connect to ζ^{i-1} and ζ^{i+1}.

This functor is faithful and gives an ascending inclusion

$$\mathcal{SG}(n,r,0) \subset \mathcal{SG}(n,r,1) \subset \cdots \subset \mathcal{SG}(n,r,k) \subset \cdots$$

and taking the direct limit we obtain the *strong contact groupoid* of the disk

$$\mathcal{SG}_{n,r} = \lim_{\to} \mathcal{SG}(n,r,k)$$

The forgetful functors $\{\Gamma : \mathcal{SG}(n,r,k) \to \mathcal{G}(D^2,n,r)\}$ extends to a functor

$$\Gamma : \mathcal{SG}_{n,r} \to \mathcal{G}(D^2,n,r)$$

9.7. **The Nerve of $\mathcal{G}(D^2,n,r)$.** Recall that the nerve $N\mathcal{C}$ of a category consists of

(1) a 0-simplex for each object,
(2) a 1-simplex for each morphism,
(3) a k-simple for each k-fold composition $\Gamma_0 \to \cdots \to \Gamma_k$ of morphisms

Therefore, we can investigate whether a given path is homotopic in the nerve to a path obtained by composing morphisms.

In this subsection, we compute the fundamental group of $N\mathcal{G}(D^2,n,r)$, the nerve of the contact groupoid of the disk.

Proposition 9.24. *The fundamental group of $N\mathcal{G}(D^2,n,r)$ satisfies*

(1) $\pi_1(N\mathcal{G}(D^2,n,r)) \cong \langle 1\rangle$ if $n-|r|=1$,
(2) $\pi_1(N\mathcal{G}(D^2,n,r))) \cong \mathbb{Z}$ if $n-|r|>1$.

Moreover, let Γ_0 be any chord diagram and let β denote a nontrivial bypass arc for Γ. The bypass triangle

$$\Delta_\beta : \Gamma_0 \xrightarrow{\beta} \Gamma_0' \xrightarrow{\beta'} \Gamma_0'' \xrightarrow{\beta''} \Gamma_0$$

generates $\pi_1(N\mathcal{G}(D^2,n,r),\Gamma_0)$.

Proof. If $n-|r|=1$, then there exists exactly one chord diagram, consisting of n boundary-parallel arcs. Consequently, the nerve $N\mathcal{G}(D^2,n,r)$ consists of a single vertex, hence its fundamental group is trivial. If $n-|r|>1$, then we can always find a nontrivial bypass arc β and nontrivial bypass triangle Δ_β. Since the Hopf invariant of the bypass is nontrivial, the subgroup $\langle\Delta_\beta\rangle$ is infinite-cyclic (Corollary 9.27). Therefore, we need to identify the set of homotopy classes of paths in $N\mathcal{G}(D^2,n,r)$ with the set of homotopy classes of 2-plane fields on $D^2\times[0,1]$.

To complete the statement, we induct on n. We construct simplicial inclusions

$$N\mathcal{G}(D^2,n-1,r-1) \hookrightarrow N\mathcal{G}(D^2,n,r) \qquad \text{and} \qquad N\mathcal{G}(D^2,n-1,r+1) \hookrightarrow N\mathcal{G}(D^2,n,r)$$

Then the group $\pi_1(N\mathcal{G}(D^2,n,r))$ is generated by $\pi_1(N\mathcal{G}(D^2,n-1,r-1)), \pi_1(N\mathcal{G}(D^2,n-1,r+1))$ and the infinite-cyclic subgroup generated by a specific nontrivial bypass arc β_0 (Proposition 9.32). By induction, we can assume all three subgroups are either trivial or infinite-cyclic and generated by a nontrivial bypass arc. Finally, each of these bypass arcs generates the same infinite-cyclic subgroup, because rotated bypasses generate the same cyclic subgroup (Lemma 9.33). □

Remark 9.25. The proposition, appropriately stated, should hold in more generality for all surfaces using the general version of Proposition 9.21, proven in [Tia18, BHH24, LSV26], combined with Eliashberg's classification of overtwisted contact structures (see also [Hua13]).

9.7.1. *Paths and 2-plane fields.* The first step is to prove that a nontrivial bypass triangle generates an infinite-cyclic subgroup of $\pi_1(N\mathcal{G}(D^2,n,r))$.

Lemma 9.26. *Let Γ_0,Γ_1 be chord diagrams.*

(1) Every path γ in nerve $N\mathcal{G}(D^2,n,r)$ determines a 2-plane field ξ_γ on $D^2\times[0,1]$ such that Γ_0 divides the characteristic foliation on $D^2\times\{0\}$ and Γ_1 divides the characteristic foliation on $D^2\times\{1\}$.
(2) If $\gamma\sim\gamma'$ are homotopic paths, then $\xi_\gamma,\xi_{\gamma'}$ are homotopic rel boundary as 2-plane fields.

Proof. First, for each dividing set Γ we can choose a vertically-invariant contact structure $(D^2 \times [0,1], \xi = \ker(hdz + \beta))$. Second, for each tight cobordism ξ from Γ_0 to Γ_1, we can choose ξ to have the form

$$(D^2 \times [0,1], \xi = \ker(h_z dz + \beta_z))$$

such that ξ agrees with the vertically-invariant contact structures near $z = 0, 1$. Third, each edge in $N\mathcal{G}(D^2, n, r)$ has a canonical orientation; to obtain the opposite orientation we use the *reverse* of the contact structure. That is, to represent the formal inverse of ξ, we use the 2-plane field

$$(D^2 \times [0,1], \xi^r = \ker(h_{1-z} dz + \beta_{1-z}))$$

Therefore, given any sequence of morphisms $\xi_i \in \mathrm{Hom}(\Gamma_i, \Gamma_{i+1})$, we can obtain a representative of their composition by concatenation.

Given some $\xi = \ker(h_z + \beta_z)$ representating a nontrivial morphism in $\mathrm{Hom}(\Gamma_0, \Gamma_1)$, the compositions $\xi^r * \xi$ has the form

$$\xi^r * \xi = \ker(h_{\phi(z)} + \beta_{\phi_z})$$

where $\phi : [0,1] \to [0,1]$ is a function satisfying $\phi(0) = \phi(1) = 0$. Using a homotopy ϕ_t such that $\phi_0 = \phi$ and ϕ_1 is the constant map, we obtain a homotopy from $\xi^r * \xi$ to the vertically-invariant contact structure in $\mathrm{Hom}(\Gamma_0, \Gamma_0)$. Similarly, we can construct a homotopy from $\xi * \xi^r$ to the vertically-invariant contact structure in $\mathrm{Hom}(\Gamma_1, \Gamma_1)$. □

A bypass arc for Γ is *nontrivial* if it intersects three distinct components of the dividing set.

Corollary 9.27. *Let β be a nontrivial bypass. The subgroup $\langle \Delta_\beta \rangle$ is an infinite-cyclic subgroup of $\pi_1(N\mathcal{G}(D^2, n, r))$*

Proof. Let $(D^2 \times [0,1], \xi_n)$ be the 2-plane field corresponding to Δ_β^n. The relative homotopy invariant satisfies $d_3(\xi_n, \xi_0) = n$; hence $\xi_n, \xi_{n'}$ are distinct for $n \neq n'$. □

9.7.2. *Generating $\pi_1(N\mathcal{G}(D^2, n, r))$.* To proceed by induction, we first identify specific subcomplexes $N^\pm \subset N\mathcal{G}(D^2, n, r)$ isomorphic to $N\mathcal{G}(D^2, n, r \mp 1)$. This is inspired by the approach of Mathews [Mat10].

Let $F_n = \{x_1, \dots, x_{2n}\}$ in cyclic order around ∂D and such that the sign of x_i is $(-1)^i$. We can define a faithful functor

$$\Sigma_\pm : \mathcal{C}_{n-1,e} \to \mathcal{C}_{n,r\pm 1}$$

as follows. Define a map $\Sigma_+ : F_{n-1} \to F_n$ as $\sigma_+(y_i) = x_{i+2}$ and $\sigma_- : F_{n-1} \to F_n$ as $\sigma_-(y_i) = y_{i+1}$ The chord diagram $\Sigma_\pm(\Gamma)$ contains the arc from $\sigma_\pm(y_i)$ to $\sigma_\pm(y_j)$ if and only if Γ contains the arc from y_i to y_j. In addition, $\Sigma_+(\Gamma)$ contains the arc from x_1 to x_2 and $\Sigma_-(\Gamma)$ contains the arc from x_1 to x_{2n}.

We therefore have simplicial inclusions

$$\Sigma_+ : N\mathcal{G}(D^2, n-1, r-1) \to N\mathcal{G}(D^2, n, r) \qquad \Sigma_- : N\mathcal{G}(D^2, n-1, r+1) \to N\mathcal{G}(D^2, n, r)$$

Let $N^\pm = \Sigma_\pm(N\mathcal{G}(D^2, n-1, r \mp 1))$ denote their images. Define

$$N^m := N\mathcal{G}(D^2, n, r) \setminus st(N^+ \cup N^-)$$

to be the complement of the star of $N^+ \cup N^-$.

Given any $\Gamma \in V^m$, we have a bypass arc β_0 that is a pushoff of the arc from x_{2n} to x_2 in the boundary. This generates a bypass triangle

$$\Gamma \to \Gamma_- \to \Gamma_+ \to \Gamma$$

For example, let Γ_0 denote be the middle chord diagram in Figure 25. The bypass arc β_0 generates the bypass triangle

$$\Delta_0 : \Gamma_0 \xrightarrow{\beta_0} \Gamma_{0,-} \xrightarrow{\beta_0'} \Gamma_{0,+} \xrightarrow{\beta_0''} \Gamma_0$$

.

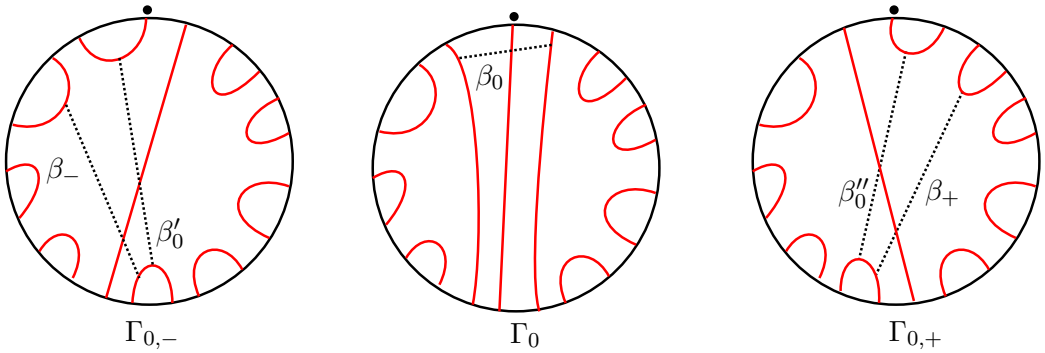
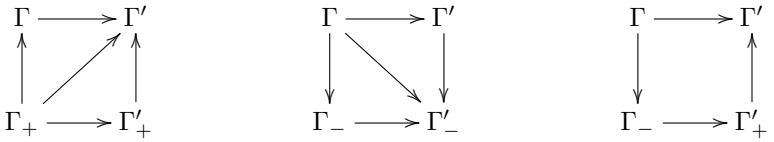


FIGURE 25. The three chord diagrams corresponding the bypass exact triangle Δ_0 along β_0.

Lemma 9.28. *Every nontrivial morphism $\Gamma \to \Gamma'$ is contained in one of the following three commutative diagrams*



Proof. Take a contact structure $(D^2 \times [0,1], \xi)$ representing the morphism $\xi \in \mathrm{Hom}(\Gamma, \Gamma')$ and round the edges. The dividing set on the boundary is the union of Γ' and the mirror $\overline{\Gamma}$, with a shear as in Figure 26. In particular, the basepoints are not identified. Note that the fundamental bypass arcs β_0, β_0' in Figure 26 form a *pinwheel* [HKM05], hence attaching bypasses along both β_0 and β_0' is overtwisted. We therefore have three cases:

(1) We can attach a bypass along β_0. The resulting tight contact structure is a morphism $\Gamma_+ \to \Gamma'$. We can then carve out a bypass along β_0', giving a morphism Γ_+ to Γ'_+.
(2) We can attach a bypass along β_0'. The resulting tight contact structure is a morphism $\Gamma \to \Gamma'_-$. We can then carve out a bypass along β_0, giving a morphsim $\Gamma_- \to \Gamma'_-$,
(3) We can carve out bypasses along β_0 and β_0'. The resulting tight contact structure is a morphism Γ_- to Γ'_+

These three cases give the three commutative diagrams in the contact category. □

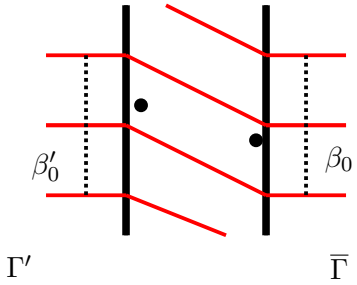


FIGURE 26. The dividing set on $S^2 = D^2 \cup D^2$ when gluing Γ' to $\overline{\Gamma}$.

Corollary 9.29. *Up to homotopy, every path in $N\mathcal{G}(D^2, n, r)$ is a composition of the following four types of paths:*

(a) paths in N_-,

(b) paths in N_+,
(c) a bridge $\Gamma_+ \to \Gamma \to \Gamma_-$ for some $\Gamma \in N^m$,
(d) a morphism $\Gamma_- \to \Gamma_+$ for some $\Gamma \in N^m$.

Lemma 9.30. *N^m is path connected.*

Proof. We will show that there is a sequence of bypasses moves that turn an arbitrary Γ into Γ_0. Consider a chord diagram Γ. Cutting along the based arc, we can split into two disks D_1, D_2 with n_1, n_2 arcs of Γ, respectively, and such that $n_1 + n_2 + 1 = n$. By induction, there is a path between Γ and some Γ' of the form in Figure 27; since the based arc is fixed, we can assume this path is in N^m. In particular, Γ' has four sets of boundary-parallel arcs of cardinalities $n_1^-, n_1^+, n_2^-, n_2^+$. Note that the basepoint diagram Γ_0 is characterized by $n_1^+ = n_2^- = 0$. Therefore, we need to redistribute the boundary-parallel arcs. Attaching a bypass as in Figure 28a then finding a path as before, we can decrease n_1^+ by one and increase n_2^+ by one. Then once $n_1^+ = 0$, we can carve a bypass as in Figure 28b and find a path to decrease n_2^- by one and increase n_1^- by one. □

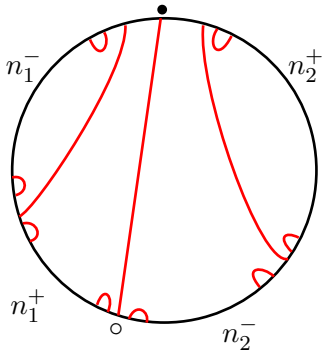
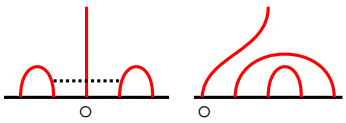
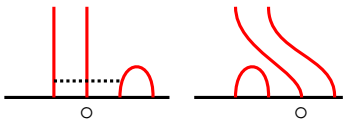


FIGURE 27



(A) Attaching a bypass along this arc decreases n_1^+ by 1.



(B) Carving a bypass along this arc decreases n_2^- by 1.

FIGURE 28. Redistributing the boundary-parallel arcs

Lemma 9.31. *Let $\Gamma \in N^m$.*

(a) There is a homotopy of paths in $N\mathcal{G}(D^2, n, r)$

$$(\Gamma_+ \to \Gamma \to \Gamma_-) \sim (\Gamma_+ \xrightarrow{\gamma_+} \Gamma_{0,+} \to \Gamma_0 \to \Gamma_{0,-} \xrightarrow{\gamma_-} \Gamma_-)$$

where $\gamma_\pm$ are paths in $N_\pm$,

(b) There is a homotopy of paths in $N\mathcal{G}(D^2, n, r)$

$$(\Gamma_- \to \Gamma_+) \sim (\Gamma_- \xrightarrow{\gamma_-} \Gamma_{0,-} \to \Gamma_{0,+} \xrightarrow{\gamma_+} \Gamma_+)$$

where $\gamma_\pm$ are paths in $N_\pm$.

Proof. If $\Gamma \xrightarrow{\beta} \Gamma'$ is a bypass edge in N^m, then we have a homotopy

$$(\Gamma_+ \to \Gamma \to \Gamma_-) \sim (\Gamma_+ \xrightarrow{\beta_+} \Gamma'_+ \to \Gamma' \to \Gamma'_- \xrightarrow{\beta_-^{-1}} \Gamma_-)$$

where $\Gamma_\pm \xrightarrow{\beta_\pm} \Gamma'_\pm$ are the corresponding bypass edges in $N^\pm$. By the previous lemma, N^m is path-connected so we can concatenate such homotopies to obtain the desired homotopy. □

Finally, the final proposition follows by combining Corollary 9.29 and Lemma 9.31.

Proposition 9.32. *The fundamental group $\pi_1(N\mathcal{G}(D^2, n, r), \Gamma_0)$ is generated by the subgroups*

$$\beta_0 * \pi_1(N_-, \Gamma_{0,-}) * (\beta_0)^{-1} \qquad \langle \Delta_0 \rangle \qquad (\beta_0'')^{-1} * \pi_1(N_+, \Gamma_{0,+}) * \beta_0''$$

9.7.3. *Rotated bypasses.*

Lemma 9.33. *Let β_1, β_2 be nontrivial bypasses on Γ such that β_1 be a left rotation of β_2. Then the bypass triangles $\Delta_{\beta_1}, \Delta_{\beta_2}$ are homotopic in $\pi_1(N\mathcal{G}(D^2, n, r))$.*

Proof. We can choose a neighborhood of $\beta_1 \cup \beta_2$ that intersects Γ in four segments as in Figure 29. We have five subdiagrams A, B, C, D, E as in the Figure, where

$$\begin{aligned} \Delta_{\beta_1} &: A \to D \to E \to A \\ \Delta_{\beta_2} &: A \to B \to C \to A \end{aligned}$$

Furthermore, we have the factorizations

$$\begin{aligned} (A \to B) &\sim (A \to D \to B) \\ (E \to A) &\sim (E \to C \to A) \\ (D \to C) &\sim (D \to B \to C) \\ &\sim (D \to E \to C) \end{aligned}$$

Therefore

$$\begin{aligned} \Delta_{\beta_2} &= A \to B \to C \to A \\ &\sim A \to D \to B \to C \to A \\ &\sim A \to D \to E \to C \to A \\ &\sim A \to D \to E \to A \\ &= \Delta_{\beta_1} \end{aligned}$$

□

9.7.4. *Completing the proof of Proposition 9.24.* If $(n, r) = (3, 0)$, then there are exactly three chord diagrams, each with a unique nontrivial bypass. Therefore, the nerve consists of three vertices and three edges connecting them in sequence; hence it is homeomorphic to S^1.

From Proposition 9.32, we have a generating set for $\pi_1(N\mathcal{G}(D^2, n, r))$. By induction and Corollary 9.27, we know that each generator is nontorsion if nontrivial; therefore the proposition follows if we can identify the nontrivial generators of these infinite cyclic subgroups. We can accomplish this via bypass rotation using Lemma 9.33 on the bypass arcs identified in Figure 25.

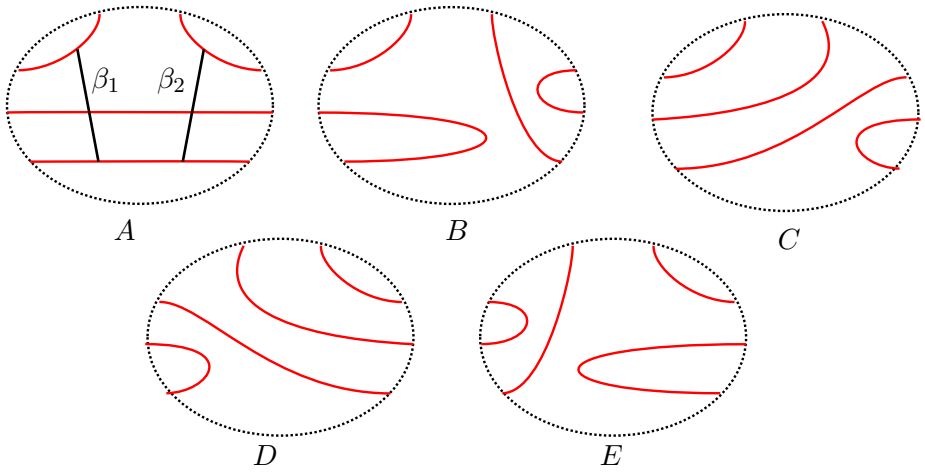


FIGURE 29. The dividing sets corresponding to rotated bypasses.

By induction the bypass triangle along β_+ on the chord diagram $\Gamma_{0,+}$ generates $\pi_1(N_+, \Gamma_{0,+})$. Since β_+ is a right rotation of β_0'', the loops Δ_{β_+} and $\Delta_{\beta_0''}$ are homotopic by Lemma 9.33.

$$\begin{aligned}
\Delta_0 &= \beta_0 * \beta_0' * \beta_0'' \\
&\sim ((\beta_0'')^{-1} * \beta_0'') * \beta_0 * \beta_0' * \beta_0'' \\
&= (\beta_0'')^{-1} * (\beta_0'' * \beta_0 * \beta_0') * \beta_0'' \\
&= (\beta_0'')^{-1} * \Delta_{\beta_0''} * \beta_0'' \\
&\sim (\beta_0'')^{-1} * \Delta_{\beta_+} * \beta_0''
\end{aligned}$$

An identical argument shows that Δ_0 is homotopic to $\beta_0 * \Delta_{\beta_-} * (\beta_0)^{-1}$ since $\Delta_{\beta_0'}$ is a right rotation of β_-.

Note that if $n - 1 - |r| = 2$, then exactly one of β_+ or β_- exists.

9.8. **Homotopy Lifting.** Recall that we have a 'forgetful functor'

$$\Gamma : \mathcal{SG}_{n,r} \to \mathcal{G}(D^2, n, r)$$

which induces a simplicial map

$$\Gamma : N\mathcal{SG}_{n,r} \to N\mathcal{G}(D^2, n, r)$$

on the nerves of these categories. We show that this map is a fibration of groupoids and in particular, satisifes a version of the homotopy lifting property for paths.

By adapting the proof of Proposition 9.21, we first show that Γ satisifes the homotopy lifting property for points.

Proposition 9.34. *Let $\zeta : \Gamma_0 \to \Gamma_1$ be a tight morphism in the contact category $\mathcal{C}(n, r)$.*

(1) If $\mathcal{F}_0$ is a lift of Γ_0 to $\mathcal{SC}(n, r, k)$, there exists a stabilzation $\Sigma^l : \mathcal{SC}(n, r, k) \to \mathcal{SC}(n, r, k + l)$ and lift

$$\widetilde{\zeta} : \Sigma^l(\mathcal{F}_0) \to \mathcal{F}_1$$

to $\mathcal{SC}(n, r, k + l)$.

(2) If $\mathcal{F}_1$ is a lift of Γ_1 to $\mathcal{SC}(n, r, k)$, there exists a stabilization $\Sigma^l : \mathcal{SC}(n, r, k) \to \mathcal{SC}(n, r, k + l)$ and a lift

$$\widetilde{\zeta} : \mathcal{F}_0 \to \Sigma^l(\mathcal{F}_1)$$

to $\mathcal{SC}(n, r, k + l)$.

Proof. The key claim in the proof of Proposition 9.21 is that $\xi : \Gamma_0 \to \Gamma_1$ can be realized by a sequence of bypass attachments in a tubular neighborhood of ∂D^2. After stabilizing the characteristic foliation sufficiently many times, which adds canceling pairs of elliptic/hyperbolic singularities near the boundary, we can realize this bypass sequence by a sequence of handleslides of the characteristic foliation, where these handleslides are contained in a tubular neighborhood of the boundary. The second statement is equivalent to the first statement in the mirror category $\mathcal{SC}(n, -r, k)$. □

Corollary 9.35. *Let $\zeta : \Gamma_0 \to \Gamma_1$ be a morphism in the contact groupoid $\mathcal{G}(D^2, n, r)$. For every $\widetilde{\mathcal{F}}_0 \in \Gamma^{-1}(\Gamma_0)$, there exists a lift*

$$\widetilde{\zeta} : \widetilde{\mathcal{F}}_0 \to \widetilde{\mathcal{F}}_1$$

In particular

$$\Gamma : \mathcal{G}(D^2, n, r) \to \mathcal{C}_{n,r}$$

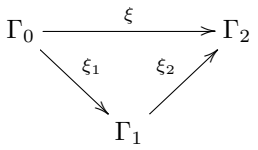
is a fibration of groupoids.

Proof. The morphism ζ in $\mathcal{G}(D^2, n, r)$ is a finite composition of morphisms $\{\zeta_i\}$ and their formal inverses in the contact category $\mathcal{C}_{n,r}$. By the previous proposition, given a lift of $\zeta_1 \circ \cdots \circ \zeta_i$ to some $\mathcal{G}(n, r, k_i)$, we can extend this lift to $\zeta_1 \circ \cdots \circ \zeta_{i+1}$ after possibly lifting to $\mathcal{G}(n, r, k_{i+1})$. □

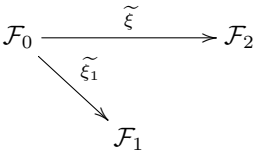
Since Γ is a fibration of groupoids, the induced map $\Gamma : N\mathcal{G}(D^2, n, r) \to N\mathcal{C}_{n,r}$ on nerves is a Serre fibration and therefore satisifes the homotopy lifting property. To spare the reader too much abstraction, we can explicitly describe the homotopy lifting property for paths when the homotopy is given by a single 2-simplex.

Proposition 9.36. *Let ζ_1, ζ_2 be a sequential pair of bypasses, yielding the following commutative diagram in the contact groupoid of the disk*

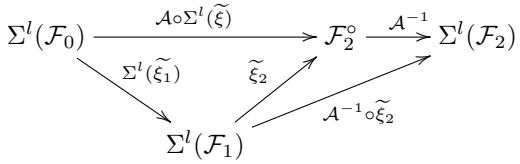
(9.4)



(1) Given a lift

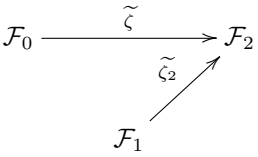


to $\mathcal{SG}(n, r, k)$, there exists some stabilization $\Sigma^l : \mathcal{SG}(n, r, k) \to \mathcal{SG}(n, r, k+l)$ and a lift $\mathcal{A} : \Sigma^l(\mathcal{F}_2) \to \mathcal{F}_2^\circ$ of the identity morphism such that

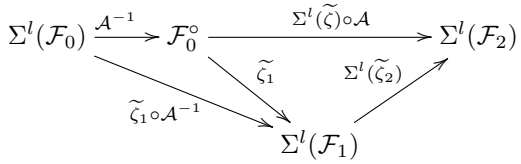


lifts Diagram 9.4

(2) Given a lift



to $\mathcal{SG}(n,r,k)$, there exists some stabilization $\Sigma^l : \mathcal{SG}(n,r,k) \to \mathcal{SG}(n,r,k+l)$ and a lift $\mathcal{A} : \mathcal{F}_0^\circ \to \Sigma^l(\mathcal{F}_0)$ of the identity morphism such that



lifts Diagram 9.4.

Proof. Returning the proof of Proposition 9.21, observe that the bypass sequence constructed in the Claim depends *only on the codomain*. Therefore, given two tight morphisms

$$\Gamma_0 \xrightarrow{\xi_0} \Gamma_2 \qquad \Gamma_1 \xrightarrow{\xi_1} \Gamma_2$$

we can realize both by the same sequence of bypass attachments near ∂D, even if $\Gamma_0 \neq \Gamma_1$. Therefore, after stabilizing the characteristic foliations, we obtain morphisms

$$\Sigma^l(\mathcal{F}_1) \to \mathcal{F}_2^\circ \qquad\qquad \Sigma^l(\mathcal{F}_2) \to \mathcal{F}_2'^\circ$$

that (a) are each the identity on some subdisk $D' \subset D$ and (b) agree on $D \smallsetminus D'$. Both $\widetilde{\xi}, \widetilde{\xi}_1$ are supported in D' and now correspond to a sequence of trivial bypasses. Hence we obtain the required lift

$$\mathcal{A} : \Sigma^l(\mathcal{F}_2) \to \mathcal{F}_2'^\circ \xrightarrow{\widetilde{\xi}_1\circ\widetilde{\xi}^{-1}} \mathcal{F}_2^\circ$$

of the identity. □

Corollary 9.37 (Homotopy Lifting Property for paths)**.** *Let γ_0, γ_1 be paths in $\mathcal{G}(D^2,n,r)$ from Γ to Γ' and let F be a homotopy from γ_0 to γ_1. If $\widetilde{\gamma}_0$ is a lift of γ_0 to $\mathcal{SC}(n,r,k)$, there exists some stabilization $\Sigma^l : \mathcal{SG}(n,r,k) \to \mathcal{SG}(n,r,k+l)$ such that F lifts to a homotopy from $\Sigma^l(\widetilde{\gamma}_0)$ to some $\widetilde{\gamma}_1$. in $\mathcal{SG}(n,r,k+l)$.*

Proof. A triangulated disk realizing the homotopy consists of a finite number of 2-simplices. By repeatedly applying the previous proposition, we can lift the homotopy across each 2-simplex in turn. If only one edge of the 2-simplex has been lifted, we can apply Proposition 9.34 and find a lift of a second edge. With two lifted edges, we apply Proposition 9.36 to lift the 2-simplex. □

Finally, objects in the strong contact groupoid are considered *up to topological isotopy*. However, for our application to gluing weak fillings in Section 8, we need to consider characteristic foliations on a fixed parameterized surface. In both Proposition 9.34 and Proposition 9.36, homotopy lifting is achieved by stabilization then modifying bypass sequences; topological isotopy is never required. Hence we can state a stronger version of the homotopy lifting property for paths.

Proposition 9.38 (Homotopy Lifting property for sequences of elementary handle-slides)**.** *Let γ_0, γ_1 be paths in $\mathcal{G}(D^2,n,r)$ from Γ to Γ' and let F be a homotopy from γ_0 to γ_1. Let*

$$\widetilde{\gamma}_0 : \mathcal{F}_{0,0} \to \mathcal{F}_{0,1} \to \cdots \to \mathcal{F}_{0,k_0-1} \to \mathcal{F}_{0,m_0}$$

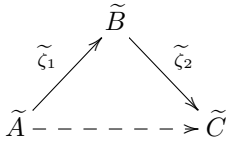
be a sequence of Morse-Smale foliations that lifts γ_0 to $\mathcal{SC}(n,r,k)$, such that $\mathcal{F}_{0,i+1}$ is obtained from $\mathcal{F}_{0,i}$ via an elementary handle slide. Then there exists some stabilization $\Sigma^l : \mathcal{SG}(n,r,k) \to \mathcal{SG}(n,r,k+l)$, a sequence

$$\widetilde{\gamma}_1 : \mathcal{F}_{0,0} = \mathcal{F}_{1,0} \to \mathcal{F}_{1,1} \to \cdots \to \mathcal{F}_{1,m_1-1} \to \mathcal{F}_{1,m_1} = \mathcal{F}_{0,m_0}$$

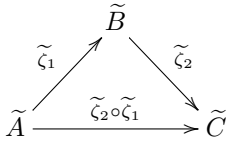
of Morse-Smale foliations connected by elementary handle slides that lifts γ_1, and a homotopy $\widetilde{F}$ from $\Sigma^l(\widetilde{\gamma}_0)$ to some $\widetilde{\gamma}_1$ in $\mathcal{SG}(n,r,k+l)$ that lifts F.

Proof. It suffices to prove the proposition when lifting a single 2-simplex. Up to symmetry and reversing all of the arrows, there are four cases to consider.

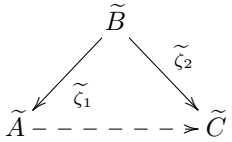
Case 1: The lifted edges of the 2-simplex can be composed.



In this case, our third edge lifts to the composition of the first two.

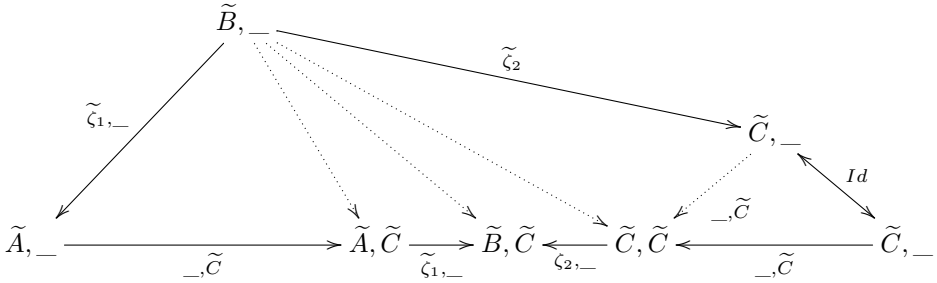


Case 2: The lifted edges of the 2-simplex cannot be composed.

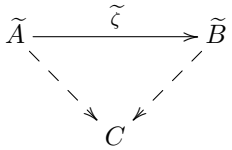


We will assume that B is the initial object and C the terminal object of this 2-simplex. The other cases are obtained by either reversing the arrows or exchanging A and C.

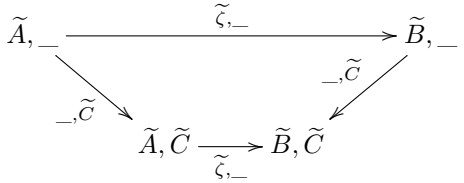
Starting out, we have three foliations $\widetilde{A}, \widetilde{B}, \widetilde{C}$ on the disk and two morphisms $\widetilde{\zeta}_1, \widetilde{\zeta}_2$ on the disk. We can shrink this and assume they occur on the disk $\mathbb{D}_{1/2}$ of radius $1/2$. Over the annular region $\mathbb{D}_1 \setminus \mathbb{D}_{1/2}$, we have the universal morphism to C. We let $\widetilde{A}, _$; $\widetilde{B}, _$; and $\widetilde{C}, _$ denote the original foliations and $\widetilde{\zeta}_1, _$ and $\widetilde{\zeta}_2, _$ the original morphisms in $\mathbb{D}_{1/2}$. Since C is the terminal object, we have new foliations $\widetilde{A}, \widetilde{C}$; $\widetilde{B}, \widetilde{C}$ and $\widetilde{C}, \widetilde{C}$ obtained by applying the universal morphism on the outside annulus, all of which represent the dividing set C. We let $_, \widetilde{C}$ denote a lift of the universal morphism to C. We then have the following diagram of lifts that gives a morphism from $\widetilde{A}, _$ to $\widetilde{C}, _$.



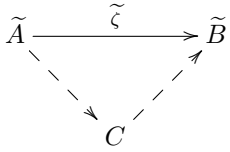
Case 3. One edge and two vertices are lifted, with the third vertex either terminal or initial in the 2-simplex.



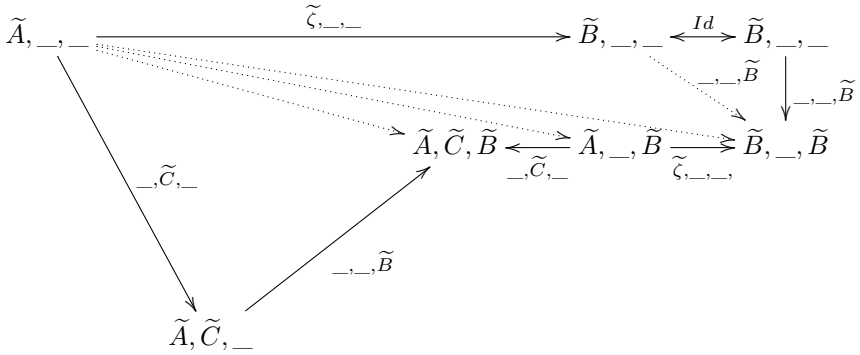
As in the previous case, we can realize $\widetilde{A}, \widetilde{B}$ and $\widetilde{\zeta}$ in $\mathbb{D}_{1/2}$ and the universal morphism to C in the outer annulus $\mathbb{D}_1 \setminus \mathbb{D}_{1/2}$. Using the same notation as in Case 2, we have the diagram



Case 4. One edge and two vertices are lifted, with the third vertex intermediate between the initial and terminal vertices of the 2-simplex.



This case is the most complicated. We realize the original foliations on $\mathbb{D}_{1/2}$, the universal morphism to C on the middle annulus $\mathbb{D}_{3/4} \setminus \mathbb{D}_{1/2}$, and the universal morphism to B on the outer annulus $\mathbb{D}_1 \setminus \mathbb{D}_{3/4}$.



□

10. Strong minimal twisting on S^2 and the Gompf invariant

This final section is, strictly speaking, ancillary to the main result of this paper. However, it may be useful for understanding the relationship between the Gompf invariant and the tomography of a contact structure, which is crucial for understanding the main construction of this paper. In Giroux's terminology, *tomography* refers to the 'movie' of characteristic foliations $\mathcal{F}_z$ printed by a contact structure $(\Sigma \times [0,1], \xi)$ on the leaves $\Sigma \times \{x\}$ of the product foliation. The 3-dimensional contact structure can be reconstructed from the 1-parameter family of these 2-dimensional cross-sectional slices.

We prove two main results:

(1) We compute the effect of a bypass triangle on the Gompf invariant, which was originally computed by Huang [Hua14].
(2) We prove a 'Homotopy implies isotopy' result for strongly minimally-twisting contact structures on S^2; namely that if a strongly minimally twisting contact structure $(S^2 \times [0,1], \xi)$ is homotopic rel boundary to the tight contact structure, then it is in fact isotopic.

10.1. **Gompf invariant.** A *Hamiltonian structure* on a 3-manifold is a closed, nondegenerate 2-form ω. A *homotopy* of Hamiltonian structures is a 1-parameter family $\{\omega_t\}$ of Hamiltonian structures. McDuff proved that Hamiltonian structures satisfy a h-principle [McD87]. Specifically, if ω_0 and ω_1 are Hamiltonian structures that can be connected by a path of nondegenerate 2-forms, there exists a path of Hamiltonian structures connecting them. The *characteristic line field* of a Hamiltonian structure on Y is the line field $\ker(\omega) \subset TY$. Via the Thom-Pontryagin construction, a framing of TY determines a 1-1 correspondence between homotopy classes of line fields on Y and the set of homotopy classes of maps from Y to S^2.

There exists various concrete interpretations of the Thom-Pontryagin construction. Given two line fields l_1, l_2, the obstruction to homotoping l_2 to l_1 is exactly captured by two obstruction classes

$$d_2(l_1, l_2) \in H^2(Y; \mathbb{Z}) \qquad d_3(l_1, l_2) \in H^3(Y; \mathbb{Z})$$

Consequently, if Y is an integer homology 3-sphere, the only nontrivial obstruction is the d_3-invariant, which is an integer.

When l is the kernel line field to a Hamiltonian structure induced on the boundary of a symplectic 4-manifold (X, ω), the relative invariant d_3 can be replaced by the absolute Gompf invariant. Any metric g determines an ω-compatible almost-complex structure J on X. The 2-plane field $\xi = TY \cap J(TY)$ on Y is dominated by ω, hence it is transverse to the kernel line field of ω. The Gompf invariant [Gom98] of ξ is defined as

$$\Theta_G(\partial X, \omega) \coloneqq \Theta_G(\xi) = c_1(X, J)^2 - 2\chi(X) - 3\sigma(X)$$

The Hirzebruch signature theorem implies that this integer is independent of the bounding almost-complex 4-manifold X and only depends on (Y, ξ). Furthermore, if Y separates (X, ω) into two pieces $(X_1, \omega) \cup (X_2, \omega)$, then

$$\Theta_G(\partial X_1, \omega) = -\Theta_G(\partial X_2, \omega)$$

Proposition 10.1. *If (X, ω) is a symplectic manifold with boundary and X is an integer homology 4-ball, then*

$$\Theta_G(\partial X, \omega) = -2$$

In particular, the Gompf invariant of (S^3, ξ_{std}), the standard tight contact structure on S^3, is -2.

Consequently, if $Y = S^3$ embeds as a separating hypersurface in (X, ω), then at most one component of $X \setminus Y$ can be a symplectic filling of (S^3, ξ_{std}).

Proof. The 2nd cohomology of X vanishes, hence $c_1^2 = \sigma(X) = 0$, and the Euler characteristic of X is $\chi(X) = 2$. □

10.2. **Combings.** We review some background material on homotopy classes of nonvanishing vector fields on 3-manifolds, covering material in Gompf [Gom98], Kirby-Melvin [KM99], Neumann-Rudolph [NR90], Dufresne [Duf05], and particularly Lescop [Les15].

Let M be an integer homology 3-sphere. A *combing* on M is a nonvanishing vector field X on M. Two combings X_0, X_1 are *homotopic* if there exists a path X_t for $t \in [0,1]$ of combings connecting them. A combing X determines a homotopy class of 2-plane field $X^\perp$, where $X \oplus X^\perp$ span TM. Since M is an integral homology 3-sphere, the Euler class $e(X^\perp) \in H^2(M;\mathbb{Z})$ vanishes and so the 2-plane bundle $X^\perp$ is trivializable. Furthermore, since $H^1(M;\mathbb{Z}) = 0$, this trivialization is unique up to homotopy. Therefore, the combing X extends to a trivialization $\tau : TM \cong M \times \mathbb{R}^3$ that is unique up to homotopy.

Let $\tau : TM \cong M \times \mathbb{R}^3$ be a trivialization of the tangent bundle of M. By definition, a combing X on M is a section of the tangent bundle. Composing with the trivialization and the projection onto the unit sphere, we get a map

$$X : M \to TM \to M \times \mathbb{R}^3 \to \mathbb{R}^3 \to S^2 \tag{10.1}$$

which by abuse of notation we also call X.

Let Y, Y_2, Y_3 be the frame induced by the trivializaton $\tau : TM \to M \times \mathbb{R}^3$. Each of these vector fields is a combing of M. In particular, Y corresponds to the constant map $Y : M \to (1,0,0) \in S^2$. Furthermore, we can identify Y_2, Y_3 with a basis for the tangent space $T_{(\pm 1,0,0)}S^2$. Generically, the map $X : M \to S^2$ is transverse to the points $(\pm 1, 0, 0)$. Therefore the preimages

$$L_+ = X^{-1}((1,0,0)) \qquad L_- = X^{-1}((-1,0,0))$$

are smooth links in M. An orientation on M induces orientations on $S^2, M \times S^2$, and the image $X(M)$. Therefore $L_\pm$ inherit orientations as the transverse intersection of smooth, oriented submanifolds. Furthermore, the frame $\{Y_2, Y_3\}$ on $T_{(\pm 1,0,0)}S^2$ pullback to framings on the normal bundles to the links $L_{\pm 1}$.

Remark 10.2. Suppose that X and Y_1 are homotopic combings. The link L_- is exactly the obstruction to the straightline homotopy $X_t = tX + (1-t)Y_1$ being nonvanishing for all $t \in [0,1]$.

Proposition 10.3 ([Duf05, Les15])**.** *Suppose that M is an integer homology 3-sphere. The combing X is homotopic to Y if and only if*

$$lk(L_+, L_-) = 0$$

The linking number of L_+ and L_- exactly captures the relative Gompf invariant of a pair of combings.

Proposition 10.4 ([Les15])**.** *Let X, Y be combings. The Gompf invariants satisfy the relations*

$$\Theta_G(X) - \Theta_G(Y) = 4lk(L_+, L_-)$$

More generally, given three combings X, Y, Z on an integer homology 3-sphere M, we can define links

$$\begin{array}{ccc} L_{X=Y} & L_{Y=Z} & L_{Z=X} \\ L_{X=-Y} & L_{Y=-Z} & L_{Z=-X} \end{array}$$

by using the trivializations of TM determined by X, Y, Z. These satisfy the relation

$$lk(L_{X=Y}, L_{X=-Y}) + lk(L_{Y=Z,Y=-Z}) + lk(L_{Z=X,Z=-X})$$

Consequently, if we view Z as a background combing, we obtain the relation

$$\begin{aligned} \Theta_G(X) - \Theta_G(Y) &= (\Theta_G(X) - \Theta_G(Z)) - (\Theta_G(Y) - \Theta_G(Z)) \\ &= 4\left(lk(L_{X=Z}, L_{X=-Z}) - lk(L_{Y=Z}, L_{Y=-Z})\right) \end{aligned}$$

10.3. **Tomography and the homotopy obstruction.** We can use an auxiliary open book decomposition (B,π) on M to reinterpret the Gompf invariant. Via the Thurston-Winkelnkemper construction, an open book decomposition (B,π) on M induces a positive contact structure $(M,\xi_{B,\pi})$.

Let (ψ,r,θ) be coordinates on a connected component $\nu(B_i)\cong S^1\times D^2$ of a tubular neighborhood of the binding. We choose the coordinates such that

$$\mu=(0,\epsilon,\theta)\qquad \lambda=(\psi,\epsilon,0)$$

the curves are a meridian and page-framed longitude, respectively. Let $h(r)$ be a cutoff function that equals 1 for $r\leq\epsilon$ and 0 for $r\geq 2\epsilon$. Then $(M,\xi_{B,\pi})$ is homotopic to the 2-plane field defined by the 1-form

$$\beta=r^2d\psi+h(r)d\theta$$

near each binding component and extending it by

$$\beta=\phi(r)d\psi=\phi(r)d\pi$$

for some over the complement of the tubular neighborhood of the binding. In particular, away from the binding the 2-plane field is tangent to the pages of the open book decomposition.

We now consider 2-plane fields ξ that are contact near the binding.

Lemma 10.5. *Let (B,π) be an open book decomposition on M, let ξ be a 2-plane field on M that is positive contact in a tubular neighborhood of B. Let $\mathcal{T}_+$ and $\mathcal{T}_-$ be the traces of the positive and negative singularity set of ξ with respect to the open book decomposition (B,π). Then we can homotope $\xi_{B,\pi}$ so that*

$$L_+=B\cup\mathcal{T}_+\qquad L_-=\mathcal{T}_-$$

Moreover, $\mathcal{T}_+$ is oriented so that an elliptic (resp. hyperbolic) singularity on a page P is a positively (resp. negatively) oriented intersection point with the page P; the link $\mathcal{T}_-$ is oriented so that an elliptic (resp. hyperbolic) singularity on a page P is a negatively (resp. positively) oriented intersection point with the page P.

Proof. The identification of the links L_+ and L_- with the specified links is immediate. To compute the orientation, choose local coordinates (x,y,z) near p in M so that $z=\pi$ and (x,y) are coordinates on the page. Then $\{\partial_x,\partial_y,\partial_z\}$ determine a local trivialization of TM.

The sign of the intersection is locally constant; therefore we can assume that orthogonal complement to ξ is spanned by the vector field

$$X=\epsilon\partial_z+x\partial_x+(\sigma y)\partial_y$$

where $\epsilon,\sigma\in\{\pm1\}$. The tangency is positive if $\epsilon=1$ and negative otherwise; it is elliptic if $\sigma=+1$ and hyperbolic otherwise. As in Line 10.1, we can view X as a map $M\mapsto S^2\subset\mathbb{R}^3$ defined by

$$X:(x,y,z)\mapsto\frac{1}{||X||}\langle 1,x,\sigma y\rangle$$

The pair $\{\partial_x,\partial_y\}$ is a basis for N_pL_ϵ; it is also a positively-oriented basis for $T_{(1,0,0)}S^2$ and a negatively-oriented basis for $T_{(-1,0,0)}S^2$. With these coordinates

$$dX=\begin{bmatrix}1&0\\0&\sigma\end{bmatrix}$$

which has determinant σ. Therefore the sign of $\epsilon\sigma$ determines the sign of $\{\partial_x,\partial_y\}$ as a basis for N_pL_ϵ, which is the sign of the intersection of L_ϵ with the page of the open book decomposition. □

Corollary 10.6. *The relative Gompf invariant of ξ with $\xi_{B,\pi}$ is given by the*

$$\Theta_G(\xi)-\Theta_G(\xi_{B,\pi})=4\left(lk(B,\mathcal{T}_-)+lk(\mathcal{T}_+,\mathcal{T}_-)\right)$$

10.4. **Strong minimal twisting on the sphere.** The strong minimal-twisting condition constrains the tomography of a contact structure on $S^2 \times [0,1]$.

We start with a basic lemma.

Lemma 10.7. *Let $\mathcal{F}_0$ be a singular foliation on S^2 printed by a contact structure with connected dividing set. Fix two distinguished points, one positive elliptic e_0^+ and one negative elliptic e_0^-.*

There exists unique pairings ζ_+, ζ_- of the remaining positive (resp. negative) elliptic and hyperbolic singularities in the Giroux graph.

Proof. To prove, we induct on the number of singularities. Suppose there are $n+1$ positive (full) elliptic singularities. If n is 0, the positive Giroux graph consists of just e_0^+ and the claim is trivially true. Now suppose the claim is true for $n-1$. Since the dividing set is connected, the positive Giroux graph is a tree, rooted at e_0^-. AllS other 1-valent vertices of the tree are elliptic points. Each of these is connected to a unique 2-valent hyperbolic point. If $n > 0$, there is some 1-valent elliptic point; pair it with its unique hyperbolic point. This elliptic/hyperbolic pair is in elimination position. Removing these two vertices and their incident edges from the Giroux graph reduces the number of positive elliptic points by 1. By induction, there is a unique pairing of the remaining positive singularities.

The same argument proves that we can uniquely pair up the negative singularities. □

We will see below (Lemma 10.18) that a strongly minimally-twisting contact structure on $S^2 \times [0,1]$ can be overtwisted. However, we can prove that if ξ is tight near $z = 0, 1$, no periodic orbits can occur. This simplifies the study of the tomography $\mathcal{F}_z$ of ξ.

Lemma 10.8. *Let $(S^2 \times [0,1], \xi)$ be a 2-plane field with characteristic foliations $\mathcal{F}_z$. After a C^∞-small perturbation of ξ, we can assume that if $\{\gamma_i\}$ is a sequence of nondegenerate periodic orbits in $\mathcal{F}_{z_i}$, there is a subsequence $\{\gamma_{i_j}\}$ that converges to*

(1) a nondegenerate periodic orbit,
(2) a degenerate periodic orbit,
(3) a homoclinic orbit

Proof. This follows from Sotomayor's description of generic 1-parameter families of vector fields on surfaces [Sot74]. Nondegenerate periodic orbits are structurally stable, hence persist under small changes in the characteristic foliation. Among the codimension-1 phenomena in a C^∞-generic 1-parameter family, only a degenerate periodic orbit or a homoclinic orbit correspond to the birth/death of a periodic orbit. □

Lemma 10.9. *Let $(S^2 \times [0,1], \xi)$ be a strongly minimally-twisting contact structure with characteristic foliations $\mathcal{F}_z$. After a C^∞-small perturbation of ξ, (which remains minimally twisting), we can assume that*

(1) if γ_1, γ_2 are a pair of either periodic or homoclinic orbits in $\mathcal{F}_{z_1}, \mathcal{F}_{z_2}$, respectively, then either
  (a) γ_1 and γ_2 are disjoint in S^2, or
  (b) $z_1 = z_2$ and $\gamma_1 = \gamma_2$
(2) Let $\mathcal{P} \subset S^2$ denote the union of all periodic orbits for all $z \in [0,1]$. Then $\mathcal{P}$ is the disjoint union of open annuli and $\partial\overline{\mathcal{P}}$ consists of homoclinic orbits and nondegenerate periodic orbits.

Proof. To prove statement (1), note that if $z_1 = z_2$, then clearly either $\gamma_1 = \gamma_2$ or they are disjoint. Now suppose, without loss of generality, that $z_2 > z_1$. Then γ_1 and γ_2 intersect transversely and every point in $\gamma_1 \pitchfork \gamma_2$ has negative sign; therefore their algebraic and geometric intersection numbers agree. However, both periodic orbits are nullhomologous, so their algebraic intersection is zero.

If γ_{z_0} is a nondegenerate periodic orbit in $\mathcal{F}_{z_0}$, then we obtain a smooth family of periodic orbits γ_z in $\mathcal{F}_z$ for $z \in (z_0 - \delta, z_0 + \delta)$ for small $\delta > 0$. Since these must be pairwise disjoint, they trace out an open tubular neighborhood of γ_{z_0} in S^2. However, a nondegenerate orbit $\gamma_0 \in \mathcal{F}_0$ or $\gamma_1 \in \mathcal{F}_1$ will have a neighborhood only on one side.

If γ_{z_0} is a degenerate periodic orbit corresponding to the birth of two nondegenerate periodic orbits, then we obtain two families $\gamma_z^\pm$ of periodic orbits for $z \in (z_0, z_0 + \delta)$ for small $\delta > 0$. Each of these limit to γ_{z_0} as

$z \to z_0$. Again, by pairwise disjointness, these trace out two open annuli $\mathcal{P}_{\gamma,\pm}$, one on either side of γ, such that

$$\overline{\mathcal{P}}_{\gamma,+} \cap \overline{\mathcal{P}}_{\gamma,-} = \gamma$$

Finally, suppose that γ_{z_0} is homoclinic orbit that births a nondegenerate periodic orbit. Suppose γ_{z_9} starts and ends at a saddle point h and separates S^2 into two components. We say that the *exterior* of γ is the component containing the other two separatrices of h; while the *interior* is the other component. Then we have a single family γ_z of nondegenerate periodic orbits for $z \in (z_0, z_0 + \delta)$ for small $\delta > 0$. This family are pairwise disjoint and are contained in the interior of γ_{z_0}; therefore they foliate a 1-sided neighborhood of the homoclinic orbit. □

Lemma 10.10. *Let $(S^2 \times [0,1], \xi)$ be a generic, strongly minimally-twisting contact structure with characteristic foliations $\mathcal{F}_z$. Suppose that $\mathcal{F}_{z_0}$ contains a homoclinic orbit γ. Then the exterior of γ contains either*

(1) a periodic orbit, or
(2) at least one source and at least one sink.

Proof. Since ξ is generic on the 2-sphere, all foliations satisfy the Poincare-Bendixson property. Moreover, if $\mathcal{F}_{z_0}$ has a homoclinic orbit, we can assume it has no other saddle connections. Apart from the homoclinic orbit, at the saddle point h we have one stable separatrix, whose α-limit is either a source or a repelling periodic orbit, and one unstable separatrix, whose ω-limit is either a sink or an attracting periodic orbit. □

Lemma 10.11. *Let $(S^2 \times [0,1], \xi)$ be a strongly minimally-twisting contact structure that is tight near $z = 0$ and $z = 1$. Then $\mathcal{F}_z$ has no periodic orbits for any $z \in [0,1]$.*

Proof. Since ξ is tight near $z = 0, 1$, the foliations $\mathcal{F}_0, \mathcal{F}_1$ do not contain periodic orbits. If $\mathcal{P}$, the subset of periodic orbits, is nonempty, then its boundary consists of homoclinic orbits.

We can choose two 'innermost' homoclinic orbits γ_1, γ_2 that separate S^2 into two disks D_1, D_2, where D_i is the exterior of γ_i, and an annulus A that contains all nondegenerate periodic orbits. Then since D_1, D_2 do not contain periodic orbits, they each contain at least one source and at least one sink. Therefore, there exists some nondegenerate periodic orbit $\gamma \in \mathcal{F}_{z_0}$ that splits S^2 into two hemispheres N and S (where γ is oriented as the boundary of S) each of which contains at least one positive and one negative singularity of the characteristic foliation $\mathcal{F}_z$ for all $z \in [0,1]$. Since the positive and negative Giroux graphs of $\mathcal{F}_0, \mathcal{F}_1$ are connected, they must intersect γ.

Let $e^+(N), h^+(N), e^+(S), h^+(S)$ denote the number of positive elliptic and hyperbolic singularities in each hemisphere of S^2. Define

$$\sigma^+(N) = e^+(N) - h^+(N) \qquad \sigma^+(S) = e^+(S) - h^+(S)$$

Since $\sigma^+(N) + \sigma^+(S) = 1$, one of them is nonpositive. If $\sigma^+(S) \leq 0$, since there exists at least one positive elliptic in S, there exists at least one positive hyperbolic in S. There must be a flowline in both $\mathcal{F}_0$ and $\mathcal{F}_1$ from a positive elliptic point in N to a positive hyperbolic point in S. However, this violates the strong minimal-twisting condition, as every flowline in $\mathcal{F}_z$ for $z > z_0$ that crosses γ must flow from S to N. Similarly, if $\sigma^+(N) \leq 0$, then there must be a flowline in both $\mathcal{F}_0$ and $\mathcal{F}_1$ from a positive elliptic point in S to a positive hyperbolic point in N. This again violates the strong minimal-twisting condition as every flowline in $\mathcal{F}_z$ for $z < 1/2$ that crosses γ must flow from N to S. □

Corollary 10.12. *Let $(S^2 \times [0,1], \xi)$ be a strongly minimally-twisting contact structure that is tight near $z = 0$ and $z = 1$. Suppose that Γ divides $\mathcal{F}_0, \mathcal{F}_1$ and is transverse to $\mathcal{F}_z$ for all $z \in [0,1]$. Then ξ is tight.*

Proof. By Lemma 10.11, there are no periodic orbits for any $\mathcal{F}_z$. Then since the singular points are fixed and Γ is transverse to $\mathcal{F}_z$, it divides the foliation. □

Corollary 10.13. *Let $z \in (0,1)$. Let $\gamma_z : [0,1] \to \Sigma$ be an oriented arc in $\mathcal{F}_z$. Then there cannot exist*

(1) *an oriented arc* $\gamma_0 : [0,1] \to \Sigma$ *of* $\mathcal{F}_0$ *such that*
$$\gamma_0(0) = \gamma_z(1) \qquad \gamma_0(1) = \gamma_z(0)$$
whose interior is disjoint from γ_z,
(2) *an oriented arc* $\gamma_1 : [0,1] \to \Sigma$ *of* $\mathcal{F}_1$ *such that*
$$\gamma_1(0) = \gamma_z(1) \qquad \gamma_1(1) = \gamma_z(0)$$
whose interior is disjoint from γ_z,

Proof. In both cases, the arc $\gamma_{0/1}$ is a (dual) bypass arc and we can isotope ξ through strongly minimally twisting contact structures so that $\mathcal{F}_{1/2}$ has a periodic orbit in its characteristic foliation, which contradicts Lemma 10.11. □

Lemma 10.14. *Let* $(S^2 \times [0,1], \xi)$ *be a strongly minimally-twisting contact structure that is tight near* $z = 0$ *and* $z = 1$. *Then after a* C^∞*-small perturbation there exists a finite set* $Z = \{z_1, \dots, z_k\} \subset (0,1)$ *such that*

(1) *if* $z \in [0,1] \setminus Z$ *the foliation* $\mathcal{F}_z$ *admits dividing curves*
(2) *each* $\mathcal{F}_{z_i}$ *has a single retrograde saddle-saddle connection from a negative hyperbolic point to a positive hyperbolic point.*

In particular, $(S^2 \times [0,1], \xi)$ *is the union of* k *bypass attachments.*

Proof. Since S^2 is planar, the Poincare-Bendixson property automatically holds if the singularity set is finite. In 1-parameter family, there is an open, dense set $C \subset [0,1]$ such that $\mathcal{F}_z$ is Morse-Smale if $z \in C$. The complement consists of two sets of critical values Σ_{sc}, where $\mathcal{F}_z$ contains a saddle connection, and Σ_d, where $\mathcal{F}_z$ contains a degenerate periodic orbit or homoclinic orbit. An infinite sequence of saddle-saddle connections may accumulate on a degenerate periodic orbit or homoclinic orbit. Thus $\overline{\Sigma}_{sc} \setminus \Sigma_{sc} \subset \Sigma_d$. However, by the previous lemma Σ_d is empty so Σ_{sc} is closed, hence compact and consists of finitely-many points. We then let $Z \subset \Sigma_{sc}$ be the subset of retrograde (i.e. from negative to positive) saddle-saddle connections. □

Remark 10.15. We can extend all of these results to minimally-twisting contact structures $(D^2 \times [0,1], \xi)$ where

(1) ξ is tight near $z = 0, 1$,
(2) ξ is vertically-invariant and near $\partial D^2 \times [0,1]$, and
(3) $\partial D^2 \times \{z\}$ is Legendrian, with no retrograde saddle-saddle connections along the boundary

by fixing a vertically-invariant extension of ξ across $S^2 \setminus D^2$, which is tight near $z = 0, 1$, then perturbing to be minimally-twisting.

10.5. **Homotopy classes of strongly minimally-twisting contact structures on S^2.** We first give a general description of the homotopy clsss of a 2-plane field on $S^2 \times [0,1]$, relative to the tight contact structure.

Proposition 10.16. *Let* $(S^2 \times [0,1], \xi)$ *be a 2-plane field and let* $\mathcal{F}_z$ *denote the family of characteristic foliations. Suppose that*

(1) ξ *is a tight contact structure near* $z = 0, 1$ *with* $S^2 \times \{0,1\}$ *convex,*
(2) *there exists points* $n, s \in S^2$ *such that* ξ *has a positive (resp, negative) elliptic tangencies to* $S^2 \times \{z\}$ *at* n *(resp.* s*) for all* $z \in [0,1]$.

For $i = 0, 1$, *let* $\zeta^i_\pm \subset S^2 \times \{i\}$ *be the unique collection of arcs pairing singularities of* $\mathcal{F}_i$ *in elimination position in* $S^2 \setminus \{n, s\}$.

Then
$$\Theta_G(\xi) = -2 + 4\, lk(L_+, L_-)$$
where
$$L_\pm = \zeta^0_\pm \cup T_\pm \cup \zeta^1_\pm$$
and T_+ *and* T_- *are traces of the positive and negative singularities of the characteristic foliations* $\mathcal{F}_z = \xi|_{S^2 \times \{z\}}$ *as* z *varies in* $[0,1]$.

The links $L_\pm$ *are oriented consistent with the orientation of* $\zeta^1_\pm$ *as arcs in the characteristic foliation.*

Proof. Recall that $\Theta_G(\xi_{std}) = -2$, so we simply need to apply the Lescop formula (Proposition 10.4) to the appropriate obstruction links.

First, since ξ is vertically-invariant near $z = 0, 1$, we slightly enlarge to obtain $(S^2 \times [-\delta, 1+\delta], \xi)$. Using the elimination lemma along the multiarcs $\zeta^0_\pm$, there is an ambient isotopy supported in $[-\delta, 0]$ so that ξ has only two tangencies with $S^2 \times \{-\delta\}$; similarly, we can use $\zeta^1_\pm$ to direct an ambient isotopy in $[1, 1+\delta]$ so that ξ only has two tangencies with $S^2 \times \{1+\delta\}$. These isotopies can be chosen C^0-small and the traces of the tangencies are C^0-close to

$$\widehat{T}_\pm = e_0^\pm \times [-\delta, 1+\delta] \cup \zeta^0_\pm \cup T_\pm \cup \zeta^1_\pm$$

Since $S^2 \times \{-\delta, 1+\delta\}$ are tight and convex, we get a contact structure

$$(S^3, \xi') = (B^3, \xi_{std}) \cup (S^2 \times [0,1], \xi) \cup (B^3, \xi_{std})$$

by capping off each end with the unique tight contact structure on B^3. In Darboux coordinates, we have that

$$\alpha = dz + xdy - ydx$$

and we can identify the basepoints $e_0^\pm$ with the points $(0, 0, \pm 1)$.

In order to compute the relative homotopy obstruction, we homotope ξ_{std} to $\widehat{\xi}$ until it is positively tangent to the S^2-leaves, except in some arbitrary neighborhood of $e_0^- \times [-\delta, 1+\delta]$. Let $r = x^2 + y^2 + z^2$ and consider the family of 2-plane fields

$$\alpha_t = \alpha + t(x^2 + y^2)dr$$

for $t \in [0, \infty)$. Then for $t > 0$, we have that

$$\alpha \wedge \alpha_t = 0$$

if and only if $x = y = 0$. Moreover, by letting t grow large, we can ensure that the 2-plane field $\xi_t = \ker(\alpha_t)$ is arbitrarily close to $\ker(dr)$, outside some fixed small neighborhood of the z-axis. We choose this homotopy over each B^3 and extend across $S^2 \times [-\delta, 1+\delta]$ to obtain $\widehat{\xi}$.

We now consider where ξ' and $\widehat{\xi}_{std}$ are linearly dependent. The two 2-plane fields agree along the arcs $e_0^\pm \times [-\delta, 1+\delta]$ as well as the z-axis in B^3. The union of these four arcs is a split unknot in S^3 that is irrelevant to the linking number. The two 2-plane fields are positively tangent along $\widehat{T}_+$, the trace of the remaining positive singularities, and negatively tangent along $\widehat{T}_-$, the trace of the remaining negative singularities. Therefore these are the obstruction links necessary to compute the relative Gompf invariant.

To compute the sign, we first remark that the pairing arcs $\zeta^i_\pm$ have two distinct orientations: one as flowlines of the characteristic foliations and the other as segments of the oriented links $L_\pm$. In the characteristic foliation, an arc of ζ^i_+ is oriented from a positive elliptic to a positive hyperbolic, while an arc of ζ^i_- is oriented from a negative hyperbolic to a negative elliptic.

By Lemma 10.5, a positive elliptic singularity is a positive intersection of L_+ with the page. This means that along $S^2 \times \{1\}$, the positive pairing arc is oriented from positive elliptic to positive hyperbolic, which agrees with its orientation as an arc in the characteristic foliation; however along $D^2 \times \{0\}$, the orientations disagree. Similarly, by Lemma 10.5, a negative elliptic singularity is a negative intersection of L_- with the page. This means that along $D^2 \times \{1\}$, the negative pairing arc is oriented from negative hyperbolic to negative elliptic, which agrees with its orientation as an arc in the characteristic foliation; however along $D^2 \times \{0\}$, the orientations disagree. Summarizing, the orientation on $L_\pm$ agrees with the orientation of the characteristic foliation along $\Sigma \times \{1\}$ and disagrees along $\Sigma \times \{0\}$. □

If $(S^2 \times [0,1], \xi)$ is strongly minimally-twisting and tight near $z = 0, 1$, then it becomes much easier to compute the relative homotopy obstruction $\Theta_G(\xi, \xi_{std})$. The arcs $T_\pm$, which trace out the singularites as z varies, project to a collection of points. Therefore, $L_\pm$ projects to $\zeta^0_\pm \cup \zeta^1_\pm$ and the linking number $lk(L_+, L_-)$ can be determined diagrammatically from the algebraic intersection numbers of these arcs.

Lemma 10.17. *If $(S^2 \times [0,1], \xi)$ is strongly minimally-twisting, then all diagrammatric crossings of L_+ and L_-, when projected onto D^2, are negative. Therefore*

$$lk(L_+, L_-) \leq 0$$

Moreover, if $lk(L_+, L_-) = 0$, then

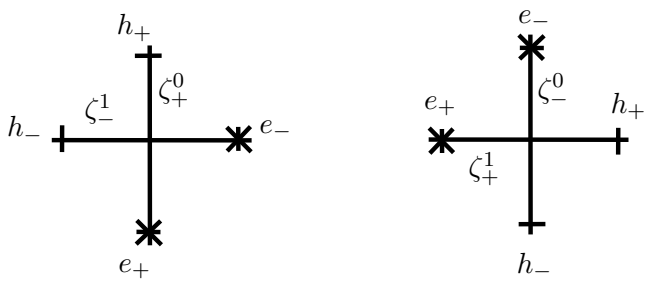


FIGURE 30. (Left) A negative pairing arc ζ_-^1 in $\Sigma \times \{1\}$ crosses over a positive pairing arc ζ_+^0 in $\Sigma \times \{0\}$. (Right) A positive pairing arc ζ_+^1 in $\Sigma \times \{1\}$ crosses over a negative pairing arc ζ_-^0 in $\Sigma \times \{0\}$.

(1) ζ_+^1 *and* ζ_-^0 *are disjoint, and*
(2) ζ_-^1 *and* ζ_+^0 *are disjoint*

Proof. To compute the linking number, we only need to consider diagrammatic crossings between arcs of opposite sign. An intersection of ζ_+^0 and ζ_-^1 is a point where L_- crosses over L_+, while an intersection of ζ_-^0 and ζ_+^1 is a point where L_+ crosses over L_-. Furthermore, the strong minimal twisting condition implies that any such transverse intersection must be as in Figure 30.

According to the orientation criteria in the previous proposition, both ζ_+^0 and ζ_-^1 are oriented from the hyperbolic singularity to the elliptic singularity, while ζ_-^0 and ζ_+^1 are both oriented from the elliptic singularity to the hyperbolic singularity. In both cases, the sign of the crossing is negative.

Since the linking number can be computed by a signed count of crossings in a planar projection, this implies that the linking number is nonpositive. Moreover, the linking number is 0 if and only if the projections of the positive and negative pairing arcs are disjoint from each other. (Note that there might still be intersections between ζ_+^1 and ζ_+^0 or between ζ_-^1 and ζ_-^0, as these do not affect the linking number computation.) □

10.6. **Bypass triangles.** We take a brief detour and use the tomographical method of Proposition 10.16 to compute the homotopical effect of attaching a bypass triangle. Recall that a *bypass triangle* is a contact structure on $\Sigma \times [0,1]$ obtained by a sequence of three bypass attachments along a convex surface as in Figure 31. Let $(\Sigma \times [0,1], \xi_I)$ denote a vertically-invariant contact structure and $(\Sigma \times [0,1], \xi_{\mathcal{B}})$ the contact structure obtained by attaching a bypass triangle. Huang computed that $d_3(\xi_I, \xi_{\mathcal{B}}) = -1$ [Hua14], which is equivalent to the formula

$$\Theta_G(\xi_{\mathcal{B}}) = \Theta_G(\xi_I) - 4$$

relating the Gompf invariants.

All bypass triangles are contactomorphic, so we can assume that it occurs in a Darboux ball along the convex Seifert disk of a maximal Legendrian unknot.

Lemma 10.18. *A bypass triangle can be realized by a strongly minimally-twisting contact structure* $(D^2 \times [0,1], \xi)$.

Proof. If η_0 is a trivial bypass arc, then attaching a bypass along η_1 increases the dividing set by 2 components, then the bypass along η_2 decreases the number of components by 2. Consequently, we only need to consider the two nontrivial bypasses. In Figure 32, we exhibit a tight characteristic foliation and two bypass arcs that realize the two nontrivial bypasses. The foliation has two elliptic singularities along the boundary, one positive and one negative, and we introduce two pairs of singularities in elimination position, one positive pair and one negative pair. The bypass arcs are disjoint, hence we can realize the bypass triangle by a strongly minimally-twisting contact structure. □

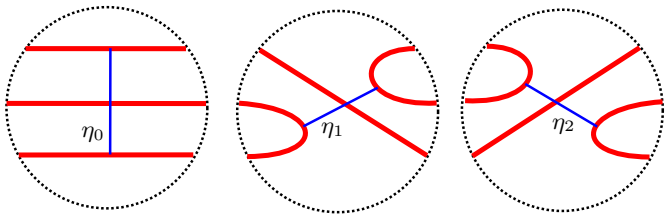


FIGURE 31. A bypass triangle consists of three successive bypass attachments along arcs η_0, η_1, η_2. After the third bypass, the dividing set is isotopic to the original dividing set.

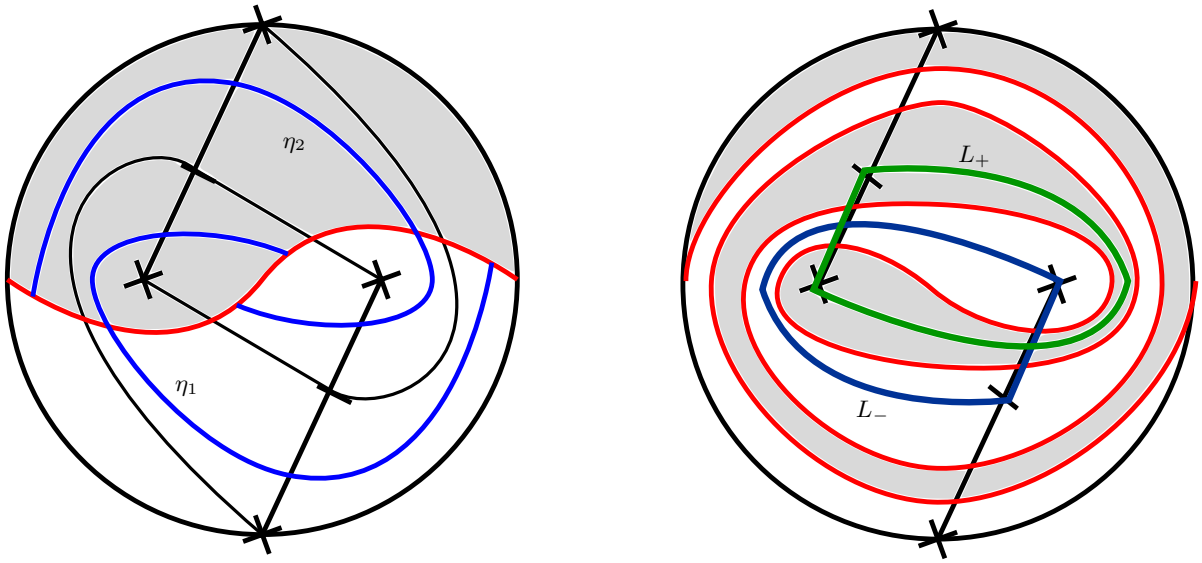


FIGURE 32. *(Left)* Attaching bypasses along η_1 and η_2 is equivalent to attaching a bypass triangle along a trivial bypass arc. *(Right)* After attaching the bypasses, the new dividing set Γ' is connected but not isotopic to the original dividing set Γ in the complement of the singularity set. The diagrammatic projection of the obstruction links L_+ and L_-, which connect the interior singularities, has two crossings and is therefore a Hopf link.

Proposition 10.19. *Let $(\Sigma \times [0,1], \xi_I)$ be a vertically-invariant contact structure and let $(\Sigma \times [0,1], \xi_{\mathcal{B}})$ be the contact structure obtained by attaching a bypass triangle to ξ_I. Then*

$$\Theta_G(\xi_{\mathcal{B}}) = \Theta_G(\xi_I) - 4$$

Equivalently,

$$d_3(\xi_I, \xi_{\mathcal{B}}) = -1$$

Proof. Without loss of generality, we can assume that our bypass triangle is realized by the strongly minimally-twisting contact structure on $D^2 \times [0,1]$ from Lemma 10.18.

The obstruction link L_+ consists of

(1) the elimination arc ζ_0^+ in $D^2 \times \{2\delta\}$ connecting the positive elliptic/hyperbolic pair, (perturbed into $D^2 \times [\delta, 2\delta]$)
(2) the traces of the two positive singularities in $D^2 \times [2\delta, 1-2\delta]$,
(3) the elimination arc ζ_1^+ in $D^2 \times \{1-2\delta\}$ (perturbed into $D^2 \times [1-2\delta, 1-\delta]$)

Similarly, the obstruction link L_- consists of

(1) the elimination arc ζ_0^- in $D^2 \times \{2\delta\}$ connecting the negative elliptic/hyperbolic pair (perturbed into $D^2 \times [\delta, 2\delta]$),
(2) the traces of the two negative singularities in $D^2 \times [2\delta, 1-2\delta]$,
(3) the elimination arc ζ_1^- in $D^2 \times \{1-2\delta\}$ (perturbed into $D^2 \times [1-2\delta, 1-\delta]$)

The union $L_+ \cup L_-$ is now clearly a Hopf link (see Figure 32). Moreover, since we have constructed the bypass triangle as a strongly minimally twisting contact structure, we can apply Lemma 10.17 and conclude that the oriented crossings in Figure 32 has negative sign. This implies that the obstruction link is the *negative* Hopf link so $lk(L_+, L_-) = -1$. The proposition now follows from Proposition 10.4. □

10.7. **Homotopy implies isotopy.** We are now ready to prove the key proposition.

Proposition 10.20. *If $(S^2 \times [0,1], \xi)$ is a strongly minimally-twisting contact structure, tight near $z = 0, 1$, but ξ is overtwisted, then we can choose basepoints $e_0^\pm$ such that*

(1) $lk(L_+, L_-)$ is negative, and
(2) the Gompf invariant satisfies $\Theta_G(\xi) < \Theta_G(\xi_{std})$

In particular, if $\Theta_G(\xi) = \Theta_G(\xi_{std})$, then it is tight and isotopic rel boundary to ξ_{std}.

Proof. Since ξ is tight near $z = 0, 1$, then after a C^∞-perturbation, the boundary surfaces are convex with connected dividing set. By Lemma 10.14, the contact structure is the concatenation of n bypasses at points $0 < z_1 < \cdots < z_n < 1$. For all regular z-values, we can perturb and assume that the surface is convex. If for all regular z-values, the dividing set is connected, then ξ is tight. Therefore, if ξ is overtwisted, there exists some regular z-value at which the dividing set is disconnected. Without loss of generality, we focus on the final nontrivial bypass attachment, which reduces $\#\Gamma$ from 3 concentric curves to a connected curve. We can assume this occurs at $z = 3/4$ and that there are no trivial bypasses for $3/4 < z < 1$.

For $z = 3/4 - \epsilon$ and small $\epsilon > 0$, the positive Giroux graph consists of two components, one tree and one component with a single cycle; similarly for the negative Giroux graph. By Lemma 10.11, we can assume neither cycle is periodic orbit; instead they are comprised of separatrices. Choose the basepoint e_0^- to be an elliptic point that is a vertex in the negative cycle. This negative cycle decomposes S^2 into two components, one containing the positive tree and the other containing the positive cycle. We can choose e_0^+ in the positive tree so that there is a flowline in the characteristic foliation from e_0^+ to e_0^-.

Let C_+ and C_- denote the cycles in the positive and negative Giroux graphs, respectively. These bound (open) disks $D_+ \subset D_-$ in the complement of the basepoints $e_0^\pm$. We can further decompose the cycles $C_\pm = C_\pm^l \cup C_\pm^r$ according the orientation on the separatrices. Let $C_\pm^l$ be the separatrices oriented consistently with the orientations on $\partial D_\pm$ and $C_\pm^r$ the separatrices oriented inconsistently.

At the bypass, we have a retrograde saddle connection from a negative hyperbolic point h_- in C_- to a positive hyperbolic point h_+ in C_+. Up to isotopy, we can identify the bypass arc ρ with endpoints on C_+ and C_- as follows. Since $C_+ = \partial D_+$ is contained within D_-, then the bypass arc connects separatrices that are oriented *inconsistently* with the boundary orientation on $D_\pm$. That is, the bypass arc connects the stable separatrix u_r of h_+ in C_+^r to the unstable separatrix s_r of h_- in C_-^r.

Without loss of generality, we can assume that the foliations on $\mathcal{F}_{3/4-\epsilon}$ and $\mathcal{F}_1$ are C^∞-close outside an arbitrary neighborhood of the bypass arc ρ. In particular, the separatrices u_r, s_r from h_+ and h_- are replaced by some u', s', but otherwise the remaining edges in the Giroux graph are the same.

We now identify the pairing multiarc ζ_+^1. Before, the bypass, we can inductively and pair up all of the 1-valent positive elliptics with a unique positive hyperbolic, until we can assume that the positive Giroux graph consists of the basepoint e_0^+ and the cycle C_+. The bypass replaces s_l, whose α-limit is some elliptic point e, with some s', whose α-limit is a positive elliptic in the complement of the disk D_-. The vertex e is now 1-valent and is uniquely paired with a hyperbolic point by a separatrix of C_+^l. Proceeding inductively around C_+, we see that the elliptic points are paired with the hyperbolics by the arcs of C_+^l. Consequently, C_+^l is a subset of ζ_+^1.

Finally, note that D_+ contains the tree component of the negative Giroux graph. Therefore, it contains one more negative elliptic than negative hyperbolic. The multi-arc ζ_-^0 must pair at least one negative elliptic

within D_+ to a negative hyperbolic in its complement. So there is an oriented arc in ζ_-^0 from the exterior of D_+ to its interior. By minimal twisting, it must cross $C_+ = \partial D_+$ along some arc oriented consistent with the boundary orientation. Therefore, we have some intersection point of ζ_-^0 with $C_+^l \subset \zeta_+^1$ and the lemma follows from Lemma 10.17 □

Corollary 10.21 (Cross-section realization)**.** *Let $(S^2 \times [0,1], \xi)$ be strongly minimally-twisting, tight near $z = 0, 1$ and satisfying $\Theta_G(\xi) = \Theta_G(\xi_{std})$.*

Suppose $\mathcal{F}_z = ker(\beta_z)$ and let β be a 1-form as in Lemma 6.8 satisfying

$$\beta \wedge \beta_0 \geq 0 \qquad \beta_1 \wedge \beta \geq 0$$

Then β defines a tight characteristic foliation.

Proof. By Lemma 6.8, there is an isotopy of $(S^2 \times [0,1], \xi)$ through strongly minimally-twisting contact structures realizing $\ker(\beta)$ as the characteristic foliation of a cross-section. Since ξ is tight, then it induces a tight characteristic foliation on every cross-section. □

10.8. **Reverse.** Let $(\Sigma \times [0,1], \xi)$ be a contact structure. Suppose that

$$\lambda = gdz + \beta_z$$

is a contact form for ξ. The *mirror* negative contact structure is the 2-plane field defined as the kernel of the 1-form

$$\lambda^m = -gdz + \beta_{1-z}$$

It is obtained by pulling ξ back by the orientation-reversing diffeomorphism $\Phi(z) = 1 - z$.

The *reverse* of ξ is the 2-plane field ξ^r defined by the kernel of

$$\lambda^r = gdz + \beta_{1-z}$$

The family $\mathcal{F}_z = \ker(\beta_z)$ of singular foliations on Σ can be interpreted as a 'movie' of foliations that encode. The movies of singular foliations for the mirror ξ^m and the reverse ξ^r are identical and given by playing the movie for ξ in reverse. However, they differ as 2-plane fields because they have opposite coorientations at the singularities of the foliations.

Lemma 10.22. *Let $(S^2 \times [0,1], \xi)$ be a 2-plane field satisfying the hypotheses of Proposition 10.16. Let ξ^m denote its mirror and let ξ^r denote its reverse. Furthermore, let $L_\pm^r$ denote the obstruction links for ξ^r and $L_\pm^m$ the obstruction links for ξ^m. Then*

$$lk(L_+^m, L_-^m) = lk(L_+^r, L_-^r) = -lk(L_+, L_-)$$

Proof. The orientation-reversing map $\Phi : z \mapsto 1 - z$ sends ξ to ξ^m and $\Phi(L_\pm) = L_\mp^m$. Since this is orientation-reversing and the linking number is symmetric, we have that

$$lk(L_+^m, L_-^m) = Lk(L_-^m, L_+^m) = -lk(L_+, L_-)$$

The 2-plane fields ξ^m to ξ^r have the same singularity sets but with opposite coorientations. Furthermore, by Lemma 10.5, flipping the coorientation flips the orientations on the obstruction links. Therefore, as oriented links

$$L_\pm^r = -L_\mp^m$$

so

$$lk(L_+^r, L_-^r) = lk(-L_+^m, -L_-^m) = (-1)^2 lk(L_+^m, L_-^m)$$

□

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